



Software 6 Management Guide

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Westermo Network Technologies AB

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1 Introduction

This document describes the functionality and features of the *Software 6*. The *Software 6* is the firmware controlling the operation of the *RT 11n* and *RT 11ac* family products.

The *RT 11n* and *RT 11ac* family products are wireless communication devices for demanding industrial applications. The *RT 11n* and *RT 11ac* family devices can operate at 2.4 and 5GHz WLAN bands depending on installation country limitations.

The devices can generally operate either as Access Point or Station. The operation is compatible with commercial IEEE 802.11 WLAN devices allowing co-existence with standard WLAN devices.

Software 6 delivers a complete set of functionality including:

- layer-2 basic switching, VLAN, etc.
- layer-3 routing, firewall, etc.
- higher-level services such as DHCP, DNS, Firewall, etc.

1.1 Supported Products

This document applies to the following product variants:

- RT-220
- RT-320 and RT-310
- Ibex-RT-370 and Ibex-RT-280
- Ibex-RT-610
- Ibex-RT-630 and Ibex-RT-330

1.2 Connector Labels

The following tables show the relation between labels found next to the connectors and names used within the firmware and its user interfaces.

Device	Connector Label	Description	Firmware Name	LED
RT-220	X1	Ethernet 0	eth0	X1
	X2	Ethernet 1	eth1	X2
	A1, A2	Radio 0	radio0	
RT-320	ETH1	Ethernet 0	eth0	X1
	ETH2	Ethernet 1	eth1	X2
	A1, A2, A3	Radio 0	radio0	
RT310	ETH1	Ethernet 0	eth0	X1
	ETH2	Ethernet 1	eth1	X2
	A1, A2, A3	Radio 0	radio0	
RT-370	ETH0	Ethernet 0	eth0	X1
	ETH1	Ethernet 1	eth1	X2
	A1, A2, A3	Radio 0	radio0	
	A4	Radio 1	radio1	
RT-280	X1	Ethernet 1 (fiber optical)	eth1	X1
	X2	Ethernet 2 (fiber optical)	eth2	X2
	X3	Ethernet 0	eth0	(No LED)
	A1, A2, A3	Radio 0	radio0	
RT-610	X1	Ethernet 0	eth0	X1
	X2	Ethernet 1	eth1	X2
	A1, A2, A3, A4	Radio 0	radio0	
	B1, B2	Radio 1	radio1	
RT-630	X1	Ethernet 0	eth0	X1
	X2	Ethernet 1	eth1	X2
	A1, A2, A3	Radio 0	radio0	
	A4, A5	LTE	wwan0	
	A6	GNSS		
RT-330	X1	Ethernet 0	eth0	X1
	X2	Ethernet 1	eth1	X2
	A1, A2	LTE	wwan0	
	A3	GNSS		

Table 1.1: Connector Labels, Firmware Names and LEDs

1.3 Supported Features

Feature	RT-320	RT-220	RT-370, RT-280	RT-310	RT-610	RT-330	RT-630
Web-Based Management (Web Interface)	yes	yes	yes	yes	yes	yes	yes
Simple Network Management Protocol (SNMP)	yes	yes	yes	yes	yes	yes	yes
Configuration Files	yes	yes	yes	yes	yes	yes	yes
Command Line Interface (CLI)	yes	yes	yes	yes	yes	yes	yes
Web Application Programming Interface (WebAPI)	yes	yes	yes	yes	yes	yes	yes
Factory Settings and Reset	yes	yes	yes	yes	yes	yes	yes
Technical Support File	yes	yes	yes	yes	yes	yes	yes
Status Indication	yes	yes	yes	yes	yes	yes	yes
Monitoring TRAPs	yes	yes	yes	yes	yes	yes	yes
Alarm Handling	yes	yes	yes	yes	yes	yes	yes
System Firmware	yes	yes	yes	yes	yes	yes	yes
Technical Preview	yes	yes	yes	yes	yes	yes	yes
Network Configuration	yes	yes	yes	yes	yes	yes	yes
Ethernet Configuration	yes	yes	yes	yes	yes	yes	yes
Wireless Configuration	yes	yes	yes	yes ¹	yes	no	yes
Cellular Network Configuration	no	no	no	no	no	yes	yes
Tunnel Endpoint Configuration	yes	yes	yes	yes	yes	yes	yes
IPsec Configuration	yes	yes	yes	yes	yes	yes	yes
OpenVPN Configuration	yes	yes	yes	yes	yes	yes	yes
Wireguard Configuration	yes	yes	yes	yes	yes	yes	yes
Wireless Network Access Point	yes	yes	yes	yes	yes	no	yes
Wireless Data Rate Control	yes	yes	yes	yes	no	no	yes
802.11s Mesh	yes	yes	yes	no	no	no	yes
Bridge Mode (4addr)	yes	yes	yes	no	yes	no	yes
Layer 2 NAT Mode	yes	yes	yes	no	yes	no	yes
Wireless MAC Address Overwrite	yes	yes	yes	yes	yes	no	yes
Wireless Security	yes	yes	yes	yes	yes	no	yes
IP Routing	yes	yes	yes	yes	yes	yes	yes
VLAN	yes	yes	yes	yes	yes	yes	yes
Mobility	yes	yes ²	yes	no	no	yes	yes
Quality of Service (QoS)	yes	yes	yes	yes	no	no	yes
Common Address Redundancy Protocol (CARP)	yes	yes	yes	yes	yes	no	yes

¹Client (STA) mode not supported

²Only Fast Association is supported

Feature	RT-320	RT-220	RT-370, RT-280	RT-310	RT-610	RT-330	RT-630
Network Link Monitor (NLM)	yes	yes	yes	yes	yes	no	yes
DNS/DHCP Server	yes	yes	yes	yes	yes	yes	yes
Service Indicators and Counters	yes	yes	yes	yes	yes	yes	yes
Logging Features	yes	yes	yes	yes	yes	yes	yes
Wireless Link Monitor	yes	yes	yes	yes	no	no	yes
Inter-Carriage Link (ICL)	yes	yes	yes	no	no	no	yes
Public Wireless Network (PWN)	no	no	no	no	yes	no	no
RSTP	yes	yes	yes	yes	yes	no	yes
Wireless Standalone	yes	yes	yes	yes	yes	no	yes
Wireless Manager (NWM)	no	no	yes	no	no	no	no
Area Frequency Management (AFM)	no	no	yes	no	no	no	no
Interference Detection Function (IDF)	no	no	yes	no	no	no	no
Http Report	no	no	yes	no	no	no	no
Firewall	yes	yes	yes	yes	yes	yes	yes
Global Navigation Satellite System	no	bo	no	no	no	yes	yes

1.4 Installation Country, Product Use

Installation country regulatory limits and operating parameters are controlled by the devices software driver. The Country Code limits are valid for client (STA) and Access Point operation modes.

All devices supports ETSI and FCC based country, among others:

Country Code	Frequency Range	Notes
Europe(EU)	2400-2483.5 MHz, 5150-5350 MHz and 5470-5725 MHz	Operation according to ETSI limitations.
United States(USA)	2400-2483.5 MHz, 5150-5350 MHz, 5470-5725 MHz and 5750-5850 MHz	Operation according to FCC limitations.

Each country has its own regulatory requirements which must be met to import the products. Please refer to section 5 for more information or contact your support for more Information on regulatory requirements.

1.5 Delivery Content

The products are delivered without connection cables, and any plugs, adapters etc. The delivery includes one dust cap for one Ethernet interface.

1.5.1 Important Safety Notes



Danger! Do not use damaged equipment and/or accessories such as damaged power cord.



Danger! Never try to open the device. There are no serviceable parts inside! By trying to open the device you will be exposed to a risk of death or injury.



Warning! Never unplug equipment from the electrical outlet by holding the cord only, always disconnect the cable by applying force directly to the plug.

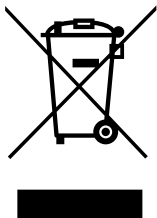


Warning! Do not operate the device in any other environmental conditions than it is designed for.



Warning! Before attaching the power cable to the device, please make sure you have antennas or terminators attached to the antenna connectors.

1.6 Information on Disposal of Old Electronic Equipment



This symbol on the product indicates that this product should not be treated as household waste when disposing it. Instead it shall be handed over to an applicable collection point for the recycling of electrical and electronic equipment. By ensuring this product is disposed correctly, you will help prevent potential negative consequences to the environment and human health, which could otherwise be caused by inappropriate disposal of this product.

1.7 License and Copyright

License and copyright for included Free/Libre Open Source Software

This product includes software developed by third parties, including Free/Libre Open Source Software (FLOSS). The specific license terms and copyright associated with the software are included in each software package respectively. Please contact your support for more information.

Upon request, the applicable source code will be provided. A nominal fee may be charged to cover shipping and media. Please direct any source code request to your normal sales or support channel.

2 Installation

Installation step	Description
Fixing	Fix the products in their use environment, ensuring that the fixing environment complies with the installation environment constraints. Ensure correct system grounding based on customer's electrical installation.
Antennas	Install the antenna interfaces according to customer's requirements. Unused antenna ports shall be terminated.
Ethernet	Connect the Ethernet interfaces.
Power Feed	Connect the power cable first to the device and then to the power plug. Verify that the LED indicators shows correct power up procedure.
Configuration	Configure the device.

3 Quick Start

This section provides a simple guide to get started quickly with the device.

3.1 Starting the product for the first time



Warning! Before connecting the power cable to the device, make sure antennas or terminators have been attached to the antenna connectors.

When booting the device for the first time, the factory default configuration is used. With this configuration, the device is prepared to operate as an Access Point with a layer-2 bridge. The WLAN interface (which is disabled) and the two Ethernet ports belong to the same bridge.

The default IP settings for the device are:

- IP address: 192.168.1.20
- Netmask: 255.255.255.0
- Default gateway: 0.0.0.0

Before connecting the device to the LAN, make sure to change its IP settings according to the network topology.

3.2 Setting the IP address

The device can be configured to either use a static IP address or obtain a dynamic one via DHCP.

3.2.1 Setting the IP address via Web Interface

To configure the IP settings via Web Interface the PC needs to be located in the same IP subnet as the device, i.e. the PC should be assigned with an IP address in the 192.168.1.0/24 network. For example:

- PC IP address: 192.168.1.2
- PC Netmask: 255.255.255.0

Open the web browser and enter **https://192.168.1.20** in the address field. When the login page requests username and password, enter the factory default credentials:

- Username: webadmin
- Password: admin

After successful login the "First Time Setup" page is displayed, asking for the country code of the current location. Select the country code and proceed by clicking on the "Save" button.

Wait some time before being redirected to **Configuration - Quick Setup** (see [Configuration - Quick Setup](#) for more details) where the current settings are displayed. In section System change the **Hostname** and **IP Address** (in CIDR notation) as needed. Or set the **Default Gateway**, where a value of "0.0.0.0" means that there is no Default Gateway in use.

Click the Apply button to apply the settings.

Wait some time before accessing the Web interface from the above set **IP address**.

3.3 Troubleshooting

The DC LED is dark:

The device is not powered. If you power the device using PoE (Power over Ethernet) please note that only connector X2 supports PoE.

The OPR LED is solid orange:

The device is not connected to an Access Point. If you expect connection please check your wireless configuration.

The ERR LED is solid orange:

Check the System Messages in the Web Interface under Status - View Log - System Messages. For more information on System Messages please refer to [Alarm Handling](#).

The ERR LED is solid red:

Critical hardware failure. The device must be repaired. For more information on System Messages please refer to [Alarm Handling](#).

The OPR LED is dark:

Critical hardware failure. The device must be repaired.

Unspecified problem:

- Check the System Messages in the Web Interface under Status - View Log - System Messages.
- Check the System Log (Syslog) in the Web Interface under Status - View Log - System Log.
- Download [Technical Support File](#) and send it together with the support request to your support.

4 Administration

The operation parameters in configuration let you choose the needed functionality. The configuration files are stored in the device. The devices are delivered with [Default settings](#). These settings define a set of parameters for typical device use.

Following maintenance and configuration interfaces are supported by the software:

1. **Web-based management:** Web interface offers fast access to basic functions and status information - see [Web-Based Management \(Web Interface\)](#).
2. **Simple Network Management Protocol :** is the main interface for maintenance and configuration. Access is done via remote host application - see [Simple Network Management Protocol \(SNMP\)](#).
3. **Configuration Files:** can be used to backup the configuration of a device. Later it can be used to restore the previous configuration - see [Configuration Files](#).
4. **Command Line Interface (CLI):** The CLI offers you the same scope of operation as with SNMP over Secure Shell (ssh) or Telnet - see [Command Line Interface \(CLI\)](#).
5. **Web Application Programming Interface (WebAPI):** The WebAPI offers you a programmable maintenance and configuration interface - see [Web Application Programming Interface \(WebAPI\)](#).



Important: During configuration phases write operations to the flash are performed. To avoid loss of data, it is necessary to avoid power loss during these procedures.



Important: If power is switched off during configuration phase, the device may fail to boot due to invalid configuration or it may fall back to the factory default configuration.

4.1 Web-Based Management (Web Interface)

Graphical user interface (HTTPS) can be accessed with web-browser via the IP address of the device:

- Default IP: 192.168.1.20
- Default user name: webadmin
- Default password: admin

4.1.1 Configuration - Quick Setup

Quick Setup allows for quick and easy configuration of the device, for example to use the device as Access Point.

Configuration of the following parameters is possible:

- System parameters (like Hostname and IP address)
- for WLAN devices only
 - WLAN country code
 - WLAN interfaces
- for LTE devices only
 - Enable/disable DNS/DHCP
 - Enable/disable GNSS
 - Cellular interface

If some advanced configuration was set which cannot be handled by Quick Setup, a warning message will be displayed describing the options on how to proceed. Normally it is recommended to not use Quick Setup in these cases.

Use the **Configuration - Advanced** page for configuring more advanced features.

4.2 Simple Network Management Protocol (SNMP)

The SNMP interface can be used for configuring the device and also for monitoring the device via Network Management System (NMS). The device includes an SNMP agent and a SNMP trap daemon. The SNMP agent is answering requests from the SNMP client. The SNMP trap daemon is responsible for sending events and exception conditions to the NMS.

As graphical SNMP tool we recommend iReasoning MIB browser. It is supported on many OS-platforms and it is easy to use. However, it is not free for professional use (see <http://www.ireasoning.com/mibbrowser.shtml> for details).

For command line and scripting purpose you can use Net-SNMP, which is open source and free to use (see <http://www.net-snmp.org/> for details).

SNMP exposes management data in the form of variables of the managed system. These variables describe the system configuration and status. They can be queried and changed by managing applications.

SNMP gives you many options for configuring and monitoring the device. To get access to the configuration and other parameters use the following settings for the SNMP client:

Parameter	Default
cfgSnmpdVersion	v2c
cfgSnmpdComAdmin (Read-Write Community)	admin-community

All configuration parameters of the device are defined and described in the Management Information Base (MIB). In case of the *Software 6* this is done through the [WESTERMO-SW6-MIB](#) file and others, which is part of the delivered software package.

Further, also hardware (serial number, etc.), firmware (version, etc.) and operation status (wireless, network, etc.) can be retrieved from the device using SNMP.

4.2.1 Supported Commands

Command	Description
GET	For reading parameters
SET	For modification of parameters
TRAP	For notifications. TRAPs can only be sent if the address of the NMS is configured (includes NOTIFICATIONS, INFORM) NOTE: TRAP messages are sent using SNMP Version 2c.

4.2.2 Availability

Access	Description
Using SNMP in custom application	For most programming languages there exists an SNMP library which can be used. This enables the integration of dedicated SNMP requests into an application without the need for additional external tools.
Accessing the SNMP parameters via command line interface (CLI)	For developing purposes, the command line tools from the Net-SNMP project has been used.
Using commercial NMS	Integrate a MIB into a full size NMS, like HP OpenView, Castle Rock SNMPc.

4.2.3 Example: Change and permanently save a Parameter through SNMP command line tool

The following information must be available for SNMP tool to access the SNMP capable device:

- MIB information
- IP address of the device
- Admin Community string if SET operations are required (admin-community)

If a parameter is changed through SNMP, the value is only stored in volatile memory. To permanently store the parameter, the change must be applied. For an example see the following steps:

1. Change the parameter

```
snmpset -v 2c -c admin-community 192.168.11.238 WESTERMO-SW6-MIB::cfgSysHostname.0 s testname
```

2. Verify the change

```
snmpget -v 2c -c admin-community 192.168.11.238 WESTERMO-SW6-MIB::cfgSysHostname.0
```

3. Apply the change

```
snmpset -v 2c -c admin-community 192.168.11.238 WESTERMO-SW6-MIB::rpcCfgApply.0 i 1
```

With the [rpcCfgApply](#) command above the apply process is started. Depending on the changes this will take several seconds to finish. To observe the apply process the parameter [rpcCfgApply](#) should be polled with *snmpget* until a 0 is read. Please refer to [rpcCfgApply](#) for more information.

4.2.4 Supported MIBs

4.2.4.1 Device specific MIBs

- [WESTERMO-SW6-MIB](#)
- [WESTERMO-SW6-BRIDGE-MIB](#)
- [WESTERMO-SW6-FIREWALL-MIB](#)
- [WESTERMO-SW6-GNSS-MIB](#)
- [WESTERMO-SW6-ICL-MIB](#)
- [WESTERMO-SW6-NWM-MIB](#)
- [WESTERMO-SW6-PWN-MIB](#)

4.2.4.2 Standard MIBs

Note: Not all entries in the standard MIBs are supported.

- OID root .1.0.8802.1.1.2 (LLDP-MIB)
- OID root .1.3.6.1.2.1 (MIB-2, RFC1213, HOST-RESOURCES, BRIDGE, ETHERLIKE, IF-MIB, etc.)
- OID root .1.3.6.1.4.1.2021 (UCD-SNMP-MIB)

4.3 Configuration Files

All relevant parameters to operate a device in a clearly defined manner can be exported or imported by means of a configuration file. This can be achieved by using one of the following options:

- The Web-Interface provides functionality for exporting and importing configuration files on the **System - Maintenance** page.
- An SNMP manager can be used to trigger a transfer of a configuration file from or to a specified URL, see [setCfgFileUrl](#), [setCfgFileFormat](#) and [rpcCfgFile](#).

- In the configuration mode of the CLI, the `export` and `import` commands can be used to export and import configurations respectively, refer to Section [4.4.2](#).
- The WebAPI provides special endpoints in order to transfer configuration files from or to the device. Section [15.2.1](#) describes how to use these endpoints.

Products that are operated with Software 6 know two types of configuration files:

1. The SNMP-styled file is a flat list of the configuration parameters in an SNMP-based format. Because the parameter names are identical to ones used in the MIBs (e.g. [WESTERMO-SW6-MIB](#)), the documentation of each parameter can be found there.
2. The CLI-styled file reflects the parameters of the device in a hierarchical structure in the way they are used in the CLI.

Since both configuration files are formatted in plain text, it is possible to modify them with a simple text editor.

4.4 Command Line Interface (CLI)

The CLI is available either through Secure Shell (SSH) or Telnet. It can be configured with the parameters listed in Table [4.1](#). For further information on CLI configuration, in particular on security considerations, refer to Section [6.3.1](#) on page [149](#).

Parameter	Default
<code>cfgCliEnabled</code>	enabled
<code>cfgCliUsername</code>	admin
<code>cfgCliPassword</code>	admin
<code>cfgCliSshEnabled</code>	enabled
<code>cfgCliSshAddress</code>	0.0.0.0:22
<code>cfgCliTelnetEnabled</code>	disabled
<code>cfgCliTelnetAddress</code>	0.0.0.0:23

Table 4.1: CLI Parameters and Their Default Values

The CLI is basically divided into two modes: the execution and the configuration mode. Right after a successful login to the CLI the execution mode is active. In this mode it provides a selection of important maintenance tools. The `help` command offers explanations and examples for each tool.

Table [4.2](#) lists all commands and tools of the CLI.

Command	Short Description
apply	Apply all pending configuration changes
changes	Show a list of changed configuration parameters
configure	Manage the global configuration
date	Display the current system time
dmesg	Print the kernel ring buffer
get	Show the value of a configuration parameter
gpsmon	GPS packet monitor and control utility
grep	Print lines matching a pattern
help	Show a list of all CLI commands
ip	Show or manipulate network devices
iperf	Perform network throughput tests
iw	Show or manipulate wireless devices
logread	Show system log messages
ping	Ping network hosts
reset	Reset configuration parameters
revert	Revert all pending changes
set	Set the value of a configuration parameter
ssh	A secure shell client
tcpdump	Dump traffic on a network
traceroute	Traceroute network hosts
watch	Execute a program periodically
wg	Wireguard management tool

Table 4.2: Overview of commands and tools of the CLI

The following examples illustrate how to access the CLI by using an SSH connection, use the `help` command, enter the configuration mode and change the current configuration.

Suppose the CLI of a Software 6 device is accessible at 192.168.1.20 with the user name `myself`, use the following command to connect via SSH and enter the password when prompted. The default user name and password can be found in Table 4.1.

```
$ ssh myself@192.168.1.20
myself@192.168.1.20's password:
      _ _ _ _ _      _ _ _ _ _      _ _ _ _ _
    . ' _ _ _ |  | _ _ _ |  | _ _ _ |  Software 6 - Command Line Interface (CLI)
  /  . '   \ _ |  | |   |  | |   |
 | |   |   |   |  | |   _   | |   |   Enter 'help' to get assistance on commands
 \ ' _ _ _ . ' \ _ | | _ _ / |  _ | |   and tools. In order to leave the CLI,
   ' _ _ _ . '   | _ _ _ _ _ |  | _ _ _ |   enter 'exit'.

myself@Rmodem:/$
```

When using the CLI for the first time, consider using the `help` command to get an overview of all

supported tools, as shown in Table 4.2. To get an explanation and examples on a specific tool enter help and the name of the tool.

```
$ help ping
USAGE: ping [OPTIONS] HOST

Send ICMP ECHO_REQUEST packets to network hosts.

OPTIONS:
  -h          Show a more detailed help text.

EXAMPLES:
  * Ping 192.168.1.1 until the ping command is terminated by pressing CTRL-C:
    ping 192.168.1.1

$
```

4.4.1 Configuration Mode

The code extract below describes how to enter the configuration mode, change parameters, show the made changes and how to apply them.

```
$ configure
Rmodem(config)> system
Rmodem(config-system)> hostname
Rmodem
Rmodem(config-system)> hostname newhost
Rmodem(config-system)> timezone CET-1CEST,M3.5.0,M10.5.0/3
Rmodem(config-system)> changes
system.hostname: 'Rmodem' => 'newhost'
system.timezone: 'UTC' => 'CET-1CEST,M3.5.0,M10.5.0/3'
Rmodem(config-system)> apply
Please wait, applying a new configuration may take up to 90s ...
Apply done.
newhost(config-system)> quit
$
```

4.4.2 Export and Import Configurations with the CLI

The following example shows how configurations can be imported from or exported to a TFTP server.

```
$ configure
Rmodem(config)> export tftp://192.168.1.2/backup.cfg
transferred 13766 bytes to 'tftp://192.168.1.2/backup.cfg'.
```

```
Rmodem(config)> import tftp://192.168.1.2/mydevice.cfg
transferred 13769 bytes from 'tftp://192.168.1.2/mydevice.cfg'.
Applying the imported configuration may take up to 90s ...
Successfully imported new configuration.
mydevice(config)> exit
$
```

Note: The CLI only supports exporting or importing the CLI-styled configuration file format.

4.5 Web Application Programming Interface (WebAPI)

The WebAPI is split in two main parts:

- File upload (import) / download (export) part
- Remote Procedure Call (RPC) part

For detailed information see [WebAPI Detailed Specification](#).

4.5.1 File upload / download

The file upload / download is implemented as simple http upload / download (multi-part) interface. The top URL is `/cgi-bin/luci/api/files`. Please refer to [WebAPI Detailed Specification](#) for examples.

4.5.2 Remote Procedure Call (RPC)

- The remote procedure calls are implemented as JSON RPC (version 2.0) API. The top URL is `/cgi-bin/luci/api`.
- A JSON object with method name and parameters in a http(s) POST request sent to the device is answered with a JSON object in response. Please refer to [WebAPI Detailed Specification](#) for examples.
- For a detailed JSON-RPC 2.0 specification please see <http://www.jsonrpc.org/specification>.

4.5.2.1 Key elements

- JSON-RPC is a stateless, light-weight remote procedure call (RPC) protocol.
- An rpc call is represented by sending a Request object to a Server. The Request object has the following members:
 - `method`: Method is a string containing the name of the method to be invoked.
 - `params`: (Optional) Params is a structured value that holds the parameter values to be used during the invocation of the method.
 - `id`: Id is an identifier established by the client that must contain a String or a Number
- Response object
 - When a rpc call is made, the server is going to reply with a Response. The Response is expressed as a single JSON object with the following members:
 - `result`:
 - * The result is only present if the RPC call was successful
 - * The value of this member is determined by the method invoked
 - `error`:
 - * The error is only present if the RPC call in case of an error
 - * The value of this member is an Error Object
 - `id`:
 - * It is the same value as the value of the id member of the corresponding request object
 - * If there was an error in detecting the id in the Request object, then it will be null
- Error Object
 - The error object is an object with the following members:
 - * `code`: Code is a Number (integer) which indicates the error type that occurred
 - * `message`: Message is a String providing a short description of the error.

* data: (Optional) Data is a Primitive or Structured value that contains additional information about the error.

4.5.2.2 Error Codes

Code	Message	Description
-32700	Parse error	Invalid JSON was received by the server. An error occurred on the server while parsing the JSON text.
-32600	Invalid Request	The JSON sent is not a valid Request object.
-32601	Method not found	The method does not exist or is not available.
-32602	Invalid params	Invalid method parameter(s).
-32603	Internal error	Internal JSON-RPC error.
-32000	Error	General error code. Message and (optionally) data member will contain further information about the error.
-32001	Import error	File import error code. Message and (optionally) data member will contain further information about the error.
-32002	FW upgrade error	Firmware upgrade error code. Message and (optionally) data member will contain further information about the error.
-32003	FW upgrade running error	This denotes that a firmware upgrade is running and therefore no further WebAPI call is available.

4.5.2.3 Difference to JSON RPC specification

Following functions specified by JSON RPC are not supported:

- Notifications (methods without "id" member)
- Batch requests

4.5.3 Device configuration

There are two ways on how to configure the device using the Web API:

1. Using device configuration file import
2. Setting MIB elements

In both cases the process is as following:

1. Set configuration items (either by importing a configuration file or by setting MIB elements)

2. Apply the changes by setting MIB item [rpcCfgApply.0](#).

- The rpc call starts the apply process, where the device checks if the configuration is valid and if so stores it in persistent memory
- Please note that the rpc call returns immediately after the apply process has been started

3. Check if changes have been applied by polling MIB item [rpcCfgApply.0](#). Please refer to [rpcCfgApply](#) for details.

Please note that the process is similar as for when using SNMP interface or CLI.

4.6 Factory Settings and Reset

The *Software 6* is delivered with a default device configuration. A Factory Reset will reset the device configuration (including administrator password and https/802.1x certificates) to its default state ([Default settings](#)).

The following list shows the most important default configuration values which are needed to access the device. More default configuration values are listed in [Default settings](#).

Network

<i>IP Address Mode</i>	static
<i>IP Address</i>	192.168.1.20, Factory IP address to access the device
<i>IP Subnet Mask</i>	255.255.255.0

Basic Configuration

<i>SNMP</i>	Enabled
-------------	---------

A Factory Reset can be issued by [rpcSysFactoryReset](#) or by using the Factory Reset Plug as described in the following.

4.6.1 Factory Reset using Factory Reset Plug

1. Power off the device
2. Plug-in the FACTORY RESET PLUG in one of the ethernet ports
3. Power on the device
4. Wait until Operation (orange) and Failure (red) LED are constantly on

5. Remove the FACTORY RESET PLUG within 15 seconds: Operation (orange) and Failure (red) LED will blink three times
6. Wait (approx. 1 minute) until device is rebooted, then you can access the device using the factory default IP

4.7 Technical Support File

The *Technical Support File* is a compressed tar file with collected system information, including log files and system configuration. It is accessible using the Web Interface. After logging in, go to *System* and then *Support*.

Download Technical Support File

Click this button to create and download the Technical Support File.

Warning: Passwords and other sensitive information may be included in the report.

The device checks during boot-up if any Kernel Log(s) are available. It will report this issue with a system message (see [Code: \[323 \]](#)). Before resetting the Kernel Log(s) you should first download the Technical Support file and send it to your support contact.

Reset Kernel Logs

This action is **irreversible** and will take approx. 20 seconds.

4.8 Status Indication

4.8.1 LED Indicators

The *RT 11n* and *RT 11ac* family products are equipped with five LED indicators:

DC	indicates whether or not power is connected to the device.
OPR	is flashing during boot up and afterwards shows a solid green when the device has booted. When wlan0 is configured in client (STA) mode, the LED shows a solid orange if not connected and solid green if connected.
ERR	indicates errors and warnings which require the intervention of the operator. When an error occurs, the Status LED show a solid red where in case of warning the LED is orange. In the error case the device has detected a serious hardware failure und must be repaired. In the warning case please check the Troubleshooting chapter of this manual.
X1/X2	show link status and activities on the respective ethernet interfaces.

4.8.2 Firmware Update

When a firmware update is in progress, the **OPR** LED is flashing orange and the **ERR** LED is flashing red at the same time until device reboot. **IMPORTANT:** One **MUST NOT** interrupt the power source until normal operation state is reached.

4.8.3 Factory Reset

The **OPR** LED is solid orange and **ERR** LED is solid red when a factory reset is detected. Both LEDs are blinking when the factory reset is in process. See also [Factory Settings and Reset](#).

4.9 Monitoring TRAPs

SNMP traps enable the device to notify the management station of significant events by way of an unsolicited SNMP message. The remote host can listen to SNMP traps in order to get notified about important events (like handoff notification) of the device. For a description how to use this feature see [SNMP Trap](#).

4.10 Alarm Handling

System warnings are indicated by the **ERR** LED (solid orange). System errors are indicated by the **ERR** LED (solid red). Warning and error messages are logged on the device and can be seen in the Web Interface (Status - View Log - System Messages). A detailed description of the single error codes can be found in [Message Codes](#).

For additional error analysis, the system log is available in the Web Interface (Status - View Log - System Log).

4.11 System Firmware

System firmware consists of two types of firmware images in a single binary: 1. bootloader image and 2. primary firmware image. The primary firmware image contains the main system software and the features described in this document. In rare cases it is required to update the bootloader, but this is handled by the system firmware update process.

4.11.1 Upgrading System Firmware

The system firmware can be upgraded via Web Interface, Web API, CLI or SNMP.



Warning! Ensure that the device is always powered during the firmware upgrade is in progress.



Warning! Although a system firmware upgrade with 'keep' configuration is supported, it is recommended to backup configuration before doing the system firmware upgrade.

The firmware file format differs between firmware version < 6.9 and ≥ 6.9 and needs to be considered:

- Firmware versions < 6.9 accepts only firmware files with extension ***.bin**
- Firmware versions ≥ 6.9 accepts only firmware files with extension ***.img**

Please contact your support if you need ***.bin** firmware files to downgrade from ≥ 6.9 to a firmware version < 6.9 or if you need ***.img** to upgrade from a firmware version < 6.9 to ≥ 6.9 .

4.11.1.1 Upgrading Firmware via Web Interface

In order to upgrade the firmware via Web Interface, your PC needs to be located in the same IP subnet as the device. When the device is using the default IP address (192.168.1.20), then the connected PC might be configured as follows:

- PC IP address: 192.168.1.1

- PC Netmask: 255.255.255.0

Open your web browser and enter the IP address of the device (default **https://192.168.1.20**) in the browser's address field. You will be asked to enter a username (default: 'webadmin') and a password (default: 'admin'). Then click on **Login** button.

Once logged in choose menu entry **System - Maintenance**. In section **Flash Firmware** you can browse your PC file system for the firmware image by clicking the **Browse...** button.

Concerning device configuration, the following three options are available:

1. **Reset to default configuration:** Device configuration will be reset to its default values.
2. **Keep the current configuration:** Device configuration will not be changed. If the new firmware image provides new configuration items, these will be initialized with their default values.
3. **Apply custom config after upgrade:** See more detailed description below.

After selecting the image file (and the custom configuration file if you have chosen **Apply custom config after upgrade**), proceed to uploading and flashing by clicking the **Flash Firmware** button. A verify page will be displayed with checksum and size information. Click **Proceed** in order to finally start the firmware upgrade process on the device.

When the **Apply custom config after upgrade** is selected, then the following steps will be processed:

1. Firmware image will be uploaded to the device.
2. Device will be upgraded using new firmware image.
3. Device configuration will be reset to new firmware image default values.
4. Provided custom configuration will then be applied. In order to assert that this custom configuration is valid for the new firmware image, it should best have been exported from a device already using the new firmware image.

4.11.1.2 Upgrading Firmware via CLI/SNMP

The following steps can be done either by using a TFTP or an HTTP server. This server must be reachable from the device.

1. Define a valid URL which is accessible by the device. This is done by changing the firmware URL parameter `setFwFileUrl` in the settings section.

2. Optional: If you want to reset the device to factory settings, set [setFwKeepConfig](#) to 'reset(0)'. If you want to use a *custom config* (see paragraph above), define a valid URL in [setCfgFileUrl](#).
3. Writing 'flash(2)' to [rpcFwFlash](#) parameter will download and validate the new firmware file. Writing 'flashWithConfig(3)' to [rpcFwFlash](#) parameter will download and validate the new firmware file and the custom config file.
4. If the downloaded file is considered a valid firmware for this device, it will be flashed to its file system.
5. Reading [rpcFwFlash](#) parameter will report the status of the firmware flash process. A value of 'flash_error(-2)' denotes that the flash process failed during write of the firmware file to the file system. A return value of 'download_error(-1)' indicates an error while the firmware was downloaded or validated. A value of 'flash(2)' means that the device is writing the firmware to the file system.

4.12 Technical Preview

The Technical Preview allows access to features that are not yet officially released. It can be enabled or disabled on the **System - Support** page of the Web Interface or by using [setTechPreviewEnabled](#) via SNMP.

If the Technical Preview is activated additional configuration parameters are available on the **Configuration - Advanced** page.

Important: Configuration parameters provided by the Technical Preview



- are only exported if it is enabled,
- only persist after a firmware upgrade if it is enabled, and
- are subject to future changes or may be removed completely.

4.13 Certificate Store

The *Certificate Store (CS)* is responsible to handle all Certificates (CRT), Private Keys (KEY) and Certificate Revocation Lists (CRL) used by services on the device. These services are:

- Webserver (HTTPS)
- 802.1X

- OpenVPN

The files are stored in non-volatile memory. A factory reset or a firmware upgrade without *keep config* deletes all files.

All CRT, KEY and CRL files may be imported and exported in PEM or DER format.

To import an encrypted key the respective password has to be set before importing the key.

The import process validates the key, and verifies that the configured password is correct. A key file which had no password before the import, will be encrypted with the configured password. Key files are always stored in encrypted format. Thus an exported keyfile is encrypted.

After a new file has been installed, the appropriate service needs to be restarted.

4.13.1 Certificate Handling

SNMP

Files may be imported, exported or deleted by the following process:

1. Define the URL of where the file is located with [setCrtFileUrl](#)
2. Select the file to be manipulated with [setCrtFileSelector](#)
3. Select the file format by setting [setCrtFileFormat](#). For PKCS#12 containers this setting is ignored.
4. Start the process by setting [rpcCrtFile](#) to the desired action (import(1), export(2) or delete(3))
5. Observe the process progress by polling [rpcCrtFile](#).

When working with encrypted key files, the process is slightly different. The password has to be set before the above flow is applicable.

An example flow for OpenVPN via SNMP is:

1. Set the key password with [cfgVpnOpenvpnKeyPassword](#)
2. Ensure the instance is not enabled [cfgNetOpenvpnEnabled](#)
3. Apply the config.
4. Follow the above instructions to import the key via SNMP.

5. After the key has been imported, configure and enable the instance.

An example flow for OpenVPN via Web Interface is:

1. Navigate to Configuration -> Advanced -> Network -> OpenVPN Interface.
2. Create an OpenVPN instance by clicking the "Add Row" button.
3. Ensure that the instance is not enabled and set the "Bridge" to -2.
4. Navigate to the VPN tab and set the "Key Password".
5. Press the apply button.
6. Follow the above instructions to import the key via SNMP.
7. After the key has been imported, configure and enable the instance.

4.13.2 Web Server (HTTPS)

The Web Server for the *Web Interfaces* uses TLS by default. Therefore the following files are required:

- HTTPS Certificate
- HTTPS Key
 - Importing an encrypted key is not supported.

4.13.3 802.1X

The following files are supported for 802.1X (WAP-EAP encryption):

- **Client Certificate (CRT):** Client x509 Certificate
- **Client Key (KEY):** Client Private Key matching the Client x509 Certificate
 - For encrypted keys set [cfgWlan802dot1xClientKeyPassword](#) before import
- **Root Certificate Authority (CA):** Authentication Server X509 Certificate

- **Subordinate Certificate Authority (CA2):** Subordinate / Intermediate Authentication Server X509 Certificate
- **Root Certificate Revocation List (CRL):** CRL issued by the Root CA
- **Subordinate Certificate Revocation List (CRL2):** CRL issued by the Subordinate / Intermediate CA
- **PKCS#12 Container:** Container file which include the CRT, KEY and maybe CA/CA2
 - For an encrypted key file set [setCrtFilePkcs12Passphrase](#) before import

Important notes:

- Only the Client Certificate and Client Key will be imported from a PKCS#12 Container.

Subordinate CA (CA2) and Subordinate CRL (CRL2) are optional. When using CA2, then the following combination of CRLs are supported:

Case	CRL	CRL2	Description
1	-	-	ok - no CRL checking
2	-	x	invalid/not supported; do not use!
3	x	-	ok - revoked Client/Server Certificates are rejected
4	x	x	ok - revoked CA2 and revoked Client/Server Certificates are rejected

Table 4.3: *Supported CRL/CRL2 combinations*

- "-" meaning CRL/CRL2 is not available, "x" meaning CRL/CRL2 is available.
- Case 2 (i.e. having CRL2 but no CRL) is not supported! In this case, CRL2 will be ignored!
- Case 3 is indirectly supported: When using a CRL and an empty CRL2 (which makes case 3 to case 4).

4.13.3.1 Certificate Refresh

The CS provides two storage locations:

1. The *current* is the actually used certificate.
2. The *next* is the new certificate installed by a refresh.

The desired storage location may be chosen during import (see [Certificate Handling](#)).

A refresh of the files may be triggered by calling [rpcCrtRefresh](#) or by rebooting the device. Refreshing the files does the following steps:

1. The *next* Certificate Bundle (see [Certificate Bundle](#)) in the CS will be checked for consistency. If found to be consistent:
2. Certificates in CS storage location *next* are moved to CS storage location *current*.
3. WLAN interfaces will be restarted.

4.13.3.2 Certificate Bundle

A *Certificate Bundle* contains at least a CA Certificate, a Client Key and a Client Certificate. In such a bundle the file in the storage location *next* will be used if available, otherwise the one from the storage location *current* will be used.

The following example illustrates how the *next* files are handled:

Certificate	<i>current</i>	<i>next</i>	use
CA	x		<i>current</i>
CA2	x	x	<i>next</i>
Client	x	x	<i>next</i>
Client Key	x	x	<i>next</i>
CA CRL	x		<i>current</i>
CA2 CRL	x	x	<i>next</i>

Table 4.4: *Certificate Bundle Example*

Therefore the Certificate Bundle for the example in table 4.4 is: CA "current", CA2 "next", Client "next", Client Key "next", CA CRL "current" and CA2 CRL "next".

4.13.4 OpenVPN

The following files are available for the OpenVPN instances:

- Client Certificate
- Client Key
 - For encrypted keys set [cfgVpnOpenvpnKeyPassword](#) before import
- CA Certificate
- Static Key

4.13.5 IPsec VPN

The following files are available for the IPsec instances:

- CA Certificate
- Left Certificate
- Right Certificate
- Left Signature Key: The left participant's public key
- Right Signature Key: The right participant's public key
- Left Client Key: The left participant's private key

4.14 System

The hostname of the the device is specified with `cfgSysHostname`. Together with `cfgSysDomain` it defines the Fully Qualified Domain Name (FQDN).

Configure the domain to which the device belongs with `cfgSysDomain`. It is by default set to `none` which means the device does not belong to any domain. The device can resolve names of other devices which are in the same domain as itself without specifying the FQDN but only the hostname. DHCP-clients which get a domain provided by DHCP-option 15 (Domain Name) will never override the value provided here, not even when it is set to `none`. Domain names provided via DHCP-option 15 will instead be added to the search domain list configurable via `cfgSysSdSearch`. Refrain from setting this to `local` as it will interfere with Multicast DNS (mDNS) operation and lead to problems. See RFC2606 (<https://www.ietf.org/rfc/rfc2606.txt>) and RFC6762 appendix G (<https://tools.ietf.org/html/rfc6762#appendix-G>).

One can specify up to 6 domains in the `cfgSysSearchdomainTable`. These will be searched when looking up a hostname. Be aware that entries which are dynamically added via DHCP-option 15 (Domain Name) and via DHCP-option 119 (Domain Search) count against this limit as well.

The `cfgSysNameserverTable` allows to specify a prioritised list of nameservers with which the device may resolve hostnames. There are 2 types of entries:

- static entries
- dynamic entries

The order of `cfgSysNameserverTable` defines the priority of the nameserver. The lower the index, the higher the priority. Entries are tried sequentially with a timeout of 3 seconds.

Static entries allow to specify a static server IP as nameserver in `cfgSysNsServer`.

Dynamic entries allow to reference a network interface, by setting `cfgSysNsDhcpInterface`, on which a DHCP-client is running. The cellular network and OpenVPN interfaces may be referenced as well, since they may provide dynamic DNS information.

Nameserver entries from DHCP-clients that are not explicitly listed are put to the end of the list.

4.15 Network Configuration

4.15.1 Interfaces and Bridges

Software 6 supports several kind of interfaces (Ethernet, WLAN, VLAN). These interfaces can be assigned to a bridge, but they do not need to be on a bridge. An interface is assigned to a bridge by the Bridge configuration parameter. For example an Ethernet interface is assigned to a bridge by [cfgNetEthBridge](#), a WLAN interface by [cfgNetWlanBridge](#) and a VLAN interface by [cfgNetVlanBridge](#). Interfaces with the same bridge index are assigned to the same bridge.

The following example shows how to assign two interfaces (eth0 and wlan0) to bridge with index 0 and one interface (eth1) which is not assigned to a bridge:

Configuration File Example: Interface configuration

```
# Config.Format = raw
# eth0 to bridge 0
WESTERMO-SW6-MIB::cfgNetEthBridge.0 = 0
# eth1 not in a bridge
WESTERMO-SW6-MIB::cfgNetEthBridge.1 = -1
# wlan0 to bridge 0
WESTERMO-SW6-MIB::cfgNetWlanBridge.0 = 0
```

Bridges with an index ≥ 100 are special bridges which forward link local traffic. This can be used for wireless links in 4addr mode which should act as a cable-replacement. Note: Such a bridge may only contain 2 interfaces!

The network interface configuration parameters are described in [cfgNetEthernetTable](#) and [cfgNetWlanTable](#).

4.15.2 VLAN Interfaces

VLAN interfaces can be created on top of Ethernet and WLAN interfaces by [cfgNetVlanParent](#) directly or the more common use case is to create VLAN interfaces assigned to a bridge by [cfgNetVlanBridge](#). Untagged frames are internally handled by using VID 0.

The following example shows the how to create a VLAN assigned to bridge with index 0 and a VLAN on top of an interface (eth0):

Configuration File Example: VLAN configuration

```
# Config.Format = raw
# VLAN interface with VID 22 assigned to bridge 0
WESTERMO-SW6-MIB::cfgNetVlanEnabled.0 = 1
```

```
WESTERMO-SW6-MIB::cfgNetVlanBridge.0 = 0
WESTERMO-SW6-MIB::cfgNetVlanVid.0 = 22
# VLAN interface with VID 33 on top of eth0
WESTERMO-SW6-MIB::cfgNetVlanEnabled.0 = 1
WESTERMO-SW6-MIB::cfgNetVlanBridge.0 = -1
WESTERMO-SW6-MIB::cfgNetVlanParent.0 = eth0
WESTERMO-SW6-MIB::cfgNetVlanVid.0 = 33
```

The VLAN configuration parameters are described in [cfgNetVlanTable](#).

4.15.3 MAC VLAN Interfaces

The MAC VLAN configuration parameters are described in [cfgNetMacVlanTable](#).

4.15.4 IP Addresses

The IP configuration is separated from the interface configuration, so that there is a single way of how to assign IP addresses to interfaces. Multiple IP addresses can be assigned to the interfaces. As long as an interface is not part of a bridge, you can add an IP address directly to this (Ethernet or WLAN) interface. To assign an IP address to a bridge, one has to create a VLAN interface on that bridge. All IP addresses are configured using Classless Inter-Domain Routing (CIDR) notation (e.g 192.168.1.20/24).

The following example shows how to assign an IP address an interface (eth0) and another to VLAN 22 in bridge with index 0 (br0.vlan22):

Configuration File Example: IP address configuration

```
# Config.Format = raw
# Static IP address on eth0
WESTERMO-SW6-MIB::cfgNetIpEnabled.0 = 1
WESTERMO-SW6-MIB::cfgNetIpProto.0 = 0
WESTERMO-SW6-MIB::cfgNetIpAddr.0 = 192.168.1.22/24
WESTERMO-SW6-MIB::cfgNetIpInterface.0 = eth0
# Static IP address on br0.vlan22 (VLAN 22 in bridge with index 0)
WESTERMO-SW6-MIB::cfgNetIpEnabled.0 = 1
WESTERMO-SW6-MIB::cfgNetIpProto.0 = 0
WESTERMO-SW6-MIB::cfgNetIpAddr.0 = 192.168.2.22/24
WESTERMO-SW6-MIB::cfgNetIpInterface.0 = br0.vlan22
```

The IP configuration parameters are described in [cfgNetIpTable](#).

4.15.5 Network Configuration Examples

Example 1: Access Point Network Configuration

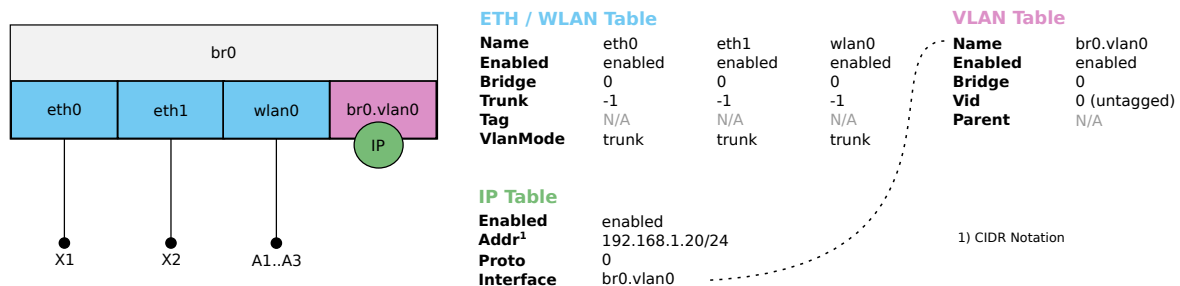


Figure 4.1: Access Point Network Configuration Example

Example 2: Client (STA) Network Configuration

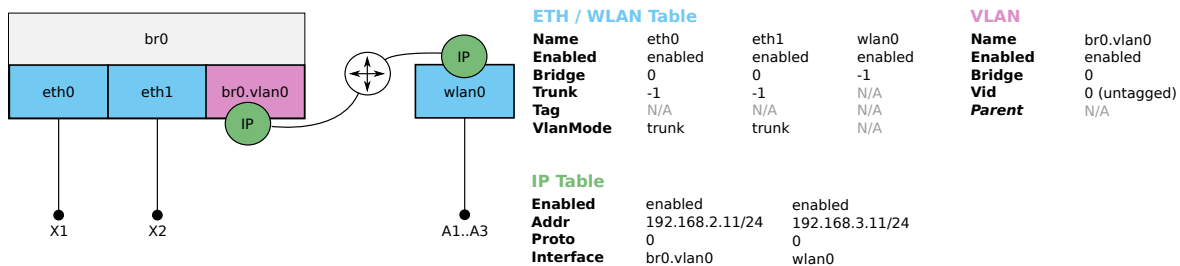


Figure 4.2: Client (STA) Network Configuration Example

Example 3: VLAN Network Configuration

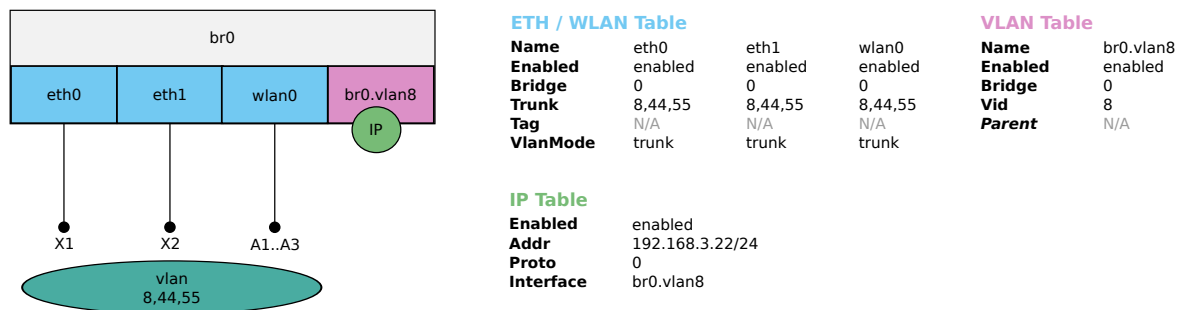


Figure 4.3: VLAN Network Configuration Example

4.16 Ethernet Configuration

All Ethernet and one WLAN port on the device are enabled and bridged by default. Ethernet interfaces auto-negotiate speed (10/100/1000 Mbit/s) and duplex mode (half/full) to the best common mode when a physical link is established.

It is possible to disable auto-negotiation of speed and duplex mode by [cfgNetEthAutoneg](#) and [cfgNetEthSpeed](#).

4.17 Wireless Configuration

The wireless configuration can be divided into three levels:

- **Network WLAN ([cfgNetWlanTable](#))** - such as bridge, VLAN mode, etc.
- **WLAN Device - physical ([cfgWlanDeviceTable](#))** - such as frequency and output power
- **WLAN Interface - logical ([cfgWlanInterfaceTable](#))** - such as operating mode, SSID, encryption and allowed bitrates, up to 16 logical WLAN interfaces can be configured

4.17.1 WLAN Device physical radio configuration

The parameters in [cfgWlanDeviceTable](#) define the physical level configuration parameters for each WLAN radio available in the product. The number of available physical radios is depending on the product variant, for example:

- RT-220, RT-310 and RT-320 include one radio: radio0 (for communication)
- RT-370 includes two radios: radio0 (for communication), radio1 (monitoring only)
- RT-610 includes two radios: radio0 (communication in 5 GHz band), radio1 (communication in 2.4 GHz or 5 GHz band)

Physical device configurations of a radio are common to all logical/virtual WLAN Interfaces created on top of it.

The most important configurations parameters are explained in the following paragraphs.

4.17.1.0.1 Bandwidth sets the Wireless Bandwidth Mode by specifying the bandwidth of the channel. Please refer to [cfgWlanDevBandwidth](#) for more information.

4.17.1.0.2 Modulation sets the modulation mode of the physical wireless radio device. Please refer to [cfgWlanDevModulation](#) for more information.

Legacy modes (a or g) are recommended and only usable in application with low throughput requirements.

4.17.1.0.3 Output Power and Antenna Gain are defining the resulting output power of the physical radio device. [cfgWlanDevPower](#) can be used to limit the combined EIRP power of all active TX antenna chains ([cfgWlanDevTxAntenna](#)) in dBm including the antenna gain ([cfgWlanDevAntennaGain](#)). The antenna gain is the value of the installed antenna in dBi. If multiple antennas with different gains are connected, the value of the antenna with the highest gain shall be configured.

Note that the maximum combined EIRP is limited by local regulations for each country. Please go to the page *Application -> Regulatory Domain Manager* in the Web Interface. (HTTP) to get more information about the maximum achievable RF output power of your device.

4.17.1.0.4 Distance sets the maximum distance ([cfgWlanDevDistance](#)) in meters a client can be away from the access point. Even though the distance is set in meters, the slot time settings change in 450m steps. In order to maximize the MAC layer efficiency it is recommended to keep this setting as low as possible.

4.17.1.0.5 Transmission Retries can be set with [cfgWlanDevShortRetry](#) and [cfgWlanDevLongRetry](#). The short retry defines the number of retransmissions of the RTS frame if there is no CTS received from the AP, whereas the long retry defines the number of retransmissions for unicast data frames not ACKed by the receiver.

In wireless environments retries on physical level are inevitable. Generally it is recommended to set lower values for UDP traffic and higher values for TCP. If the used channel is overloaded, excessive retransmissions can introduce a negative feedback-loop reducing available bandwidth even further.

4.17.1.0.6 TX and RX Antennas are configured in with [cfgWlanDevTxAntenna](#) and [cfgWlanDevRxAntenna](#). The configuration is a bitmask to enable/disable the chains. Depending on the product variant, there are either two, three or four chains available.

The number of enabled chains should match the number of antennas used.

Note, that for example with 11n modulation and with one chain enabled only, the 1-stream bitrates ([cfgWlanInterfaceBitrates](#)) MCS0 to MCS7 can be used. With two chains enabled, the 1- and 2-stream bitrates ([cfgWlanInterfaceBitrates](#)) MCS0 to MCS15 can be used.

Note, that enabling additional antennas will automatically reduce the output power of each chain as explained in Paragraph [Output Power and Antenna Gain](#).

Also note that if only one antenna is used, the used antenna must be the Antenna 1 (chain 0 i.e. 001).

4.17.2 WLAN Interface configuration

Up to 16 logical interfaces can be configured and enabled simultaneously in [cfgWlanInterfaceTable](#). This table configures the logical WLAN Interfaces which can be added on top of the physical WLAN device radios which were explained in [WLAN Device physical radio configuration](#).

The most important logical WLAN interface configuration parameters are explained in the following paragraphs.

4.17.2.0.1 Operating Mode of the WLAN interface can be set with [cfgWlanInterfaceMode](#). The supported modes are AP, STA, MONITOR and MESH.

AP is the standard Access Point / Infrastructure mode. This mode is typically used at the infrastructure side in stationary installations. One access point supports up to 200 simultaneous client connections. The AP interface is usually bridged to the Ethernet or to a VLAN directly.

STA is the station/client mode. This mode is typically used at the mobile part of the system. Stations provide a router functionality between the WLAN and Ethernet interface and therefore usually keep the bridging disabled.

The MONITOR mode allows use the interface in a fully passive monitoring mode. It is used to listen to WLAN traffic or scan for wireless interference and off-channel radar signals. Please consult the support for details, if you want to use this mode.

The MESH mode allows to connect multiple devices without the need of an Access Point. Refer to [802.11s Mesh](#) for more information.

4.17.2.0.2 Service Set Identifier (SSID) of the wireless interface is set with [cfgWlanInterfaceSsid](#). SSID is the arbitrary name of the wireless network this interface is part of.

4.17.2.0.3 Encryption can be set with [cfgWlanInterfaceEncryption](#). Six encryption modes are currently supported: open(0), psk(3), eap(6), sae(7), owe(8) and saepsk(9). It is highly recommended that at least WPA2 encryption is used. The password for the encryption can be set with [cfgWlanInterfacePassword](#). Refer to [Wireless Security](#) for more information.

4.17.2.0.4 Bitrate Limitations can be set with [cfgWlanInterfaceBitrates](#). This setting can be used to set fixed MCS index or range for 802.11n rates. Set to -1 to disable (leave on auto). It is also possible

to enter multiple indices divided by a space which are then used in auto rate. This entry is only active when an n-mode is set in [cfgWlanDevModulation](#), and is ignored if g-rates or a-rates are used.

4.17.2.0.5 Enabling Wireless Multimedia Extensions (WME) for the interface is possible with [cfgWlanInterfaceWmeEnabled](#). When using legacy rates (a-rates and g-rates), this is optional. When using n-rates, this always has to be enabled. WME uses all parameters in the [cfgWlanWmeTable](#) whose [cfgWlanWmled](#) is set to the value in [cfgWlanInterfaceWmeParameter](#). For more information about WME tables please check [Quality of Service \(QoS\)](#).

4.17.2.0.6 MAC Address Access Control List (ACL) mode for the interface can be set in [cfgWlanInterfaceMacaddrAcl](#). The available modes are:

- 0: Accept unless deny filter - accepts every MAC unless it is on the list defined in [cfgWlanAclBlackTable](#).
- 1: Deny unless accept filter - denies every MAC unless it is on the list defined in [cfgWlanAclWhiteTable](#).
- 2: Use RADIUS to accept/deny clients.

ACL can only be set if the operating mode of the interface is set to AP.

4.17.2.0.7 Scan Channels in station mode are configured with the [cfgWlanFreqTable](#) frequency lists. Each of those lists can hold up to 24 entries. Unused entries at the end of the list must be set to 0. In order to assign a frequency list to a WLAN interface, the list index is set in [cfgWlanInterfaceScanList](#). As factory default the index 0 ([cfgWlanFreqTable.0](#)) includes all 2.4Ghz center frequencies and the index 1 ([cfgWlanFreqTable.1](#)) includes all the 5Ghz center frequencies. In order to scan full country code all frequencies of the frequency list must be set to 0.

4.17.2.1 SSID Hide Feature

Service Set Identifier (SSID) specifies the name of a WLAN network. When people want to connect to a WLAN network, they normally check which WLAN networks are available in their neighbourhood, and then they choose the one they want to connect to.

If an Access Point provider does not want that random persons connect to their network, they will normally configure the Access Point so that the SSID is hidden. Then people not knowing the name of the WLAN network (i.e. its SSID) will not try to connect to this WLAN network by accident; but only people who know the name of the WLAN network will connect.

There are two MIB elements available to configure the SSID Hide Feature:

- [cfgWlanIfaceIgnoreBroadcastSsid](#)
- [cfgWlanIfaceUseVendorSsid](#)

Setting [cfgWlanIfaceIgnoreBroadcastSsid](#) to enabled(1) specifies that the SSID will not be announced in the beacon of the Access Point (AP). Additionally, probe request frames that do not specify the full SSID will not be answered by this AP. As a consequence, Clients/Stations need to know the SSID to be able to connect to this WLAN network.

When [cfgWlanIfaceIgnoreBroadcastSsid](#) is enabled, a passively scanning Client/Station (forced or because of DFS) have no way of detecting the AP it tries to find. For being able to connect to the WLAN network in these two cases, [cfgWlanIfaceUseVendorSsid](#) must be used:

- On an AP: set [cfgWlanIfaceUseVendorSsid](#) to enabled(1) so that the hidden SSID is added as vendor element in the AP beacon.
- And on a STA: set [cfgWlanIfaceUseVendorSsid](#) to enabled(1) to allow the Client/Station to use the vendor element in the AP beacon.

This parameter is supported on 802.11n products only.

4.18 Cellular Network Configuration

Cellular communication is only available on specially designed products, see also section 1.3. The configuration is divided into different sections:

- **Interface Configuration** ([cfgNetWwanEnabled](#)),
- **SIM Slot Configuration** ([cfgCellSimTable](#)),
- **SIM Profile Configuration** ([cfgCellSimProfileTable](#)), and
- **Connection Management** ([cfgCellConnectionManagement](#)).

4.18.1 Interface Configuration

The cellular network interface `wwan0`, typically labeled Wireless Wide Area Network (WWAN) at system level, is disabled by default. In order to use cellular communication, the corresponding network interface must be enabled using [cfgNetWwanEnabled](#).

4.18.2 SIM Slot Configuration

At least one SIM slot of the cellular network interface `wwan0` must be assigned to a SIM profile. Each value of the SIM slots `cfgCellSimSlot1` and `cfgCellSimSlot2` refers to the index of a SIM profile in the `cfgCellSimProfileTable`. In case one of the slots is not used, set its reference to `-1` in order to deactivate the slot. The primary SIM slot can be selected by `cfgCellSimPrimarySlot`. First connection attempts are made with the primary SIM. If the SIM card cannot be unlocked, increasing the unlock timeout `cfgCellSimUnlockTimeout` will enable support for older and slower SIM cards. This value affects the startup speed and the performance of the SIM rotation, keep it low if possible. Changes become effective after restarting the device.

4.18.3 SIM Profile Configuration

The various parameters for a SIM card are organised in SIM profiles. The `cfgCellSimProfileTable` contains the respective profiles, which are then assigned to the SIM slots. If the SIM card is locked by a PIN, `cfgCellSimProfilePinEnabled` must be enabled to unlock it with the PIN defined in `cfgCellSimProfilePin`. The Access Point Name (APN) `cfgCellSimProfileApn` is set to `auto` by default, so that the SIM card selects it by itself, the credential parameters are ignored. Defined APN and credentials configuration may be required for roaming.

Note: Roaming is enabled by default and can be deactivated using `cfgCellSimProfileRoaming`.

4.18.4 Connection Management Configuration

The Connection Management controls which SIM slot is to be used. If SIM Rotation `cfgCellConnMgmtSimRotationEnabled` is enabled, the Connection Management swaps from the primary to the secondary slot and vice versa as soon as the existing connection is considered lost. The connection loss detection mode is selected by `cfgCellConnMgmtMonMode`.

- `signal(0)` for monitoring the strength and quality of the cellular signal, and
- `remoteHosts(1)` for monitoring the cellular network traffic.

The evaluation of the connection based on the `signal(0)` is hardware dependent and considered slow compared to the `remoteHosts(1)` option. The advantage of monitoring the connection at network level is that an interrupted data flow is detected much faster depending on settings of the detection algorithm. On the downside, constant network traffic is needed and the monitoring is dependent on at least one remote host, see `cfgCellConnMgmtMonRemoteTable`.

The algorithms for detecting a connection loss can be adjusted with the monitor period `cfgCellConnMgmtMonPeriod` and the monitor count `cfgCellConnMgmtMonCount`. The monitor period defines the time in seconds between two consecutive evaluations of the connection. Whereas, the monitor

count defines the needed amount of consecutive failed connection tests before it is considered down. These parameters only apply for monitoring an established connection.

The loss of connection triggers the Connection Management to change the SIM slot and to set up a new connection. If no connection could be set up within a certain time, which is hardware dependent and can take up to one minute, the Connection Management changes the SIM slots connection could be established successfully. However, the monitoring of the new connection does not start until it is up and running.

4.18.4.1 Monitored Remotes

If `cfgCellConnMgmtMonMode` is set to `remoteHosts(1)`, at least one enabled entry in the `cfgCellConnMgmtMonRemoteTable` is required. The connection attempts are made one after another according to the order of the table entries. As soon as one remote host is available the cellular connection is considered up. The address of the remote host is defined by `cfgCellConnMgmtMonRemoteAddress` and `cfgCellConnMgmtMonRemoteType` determines how the availability the remote host is tested.

4.18.5 Cellular Router as Gateway

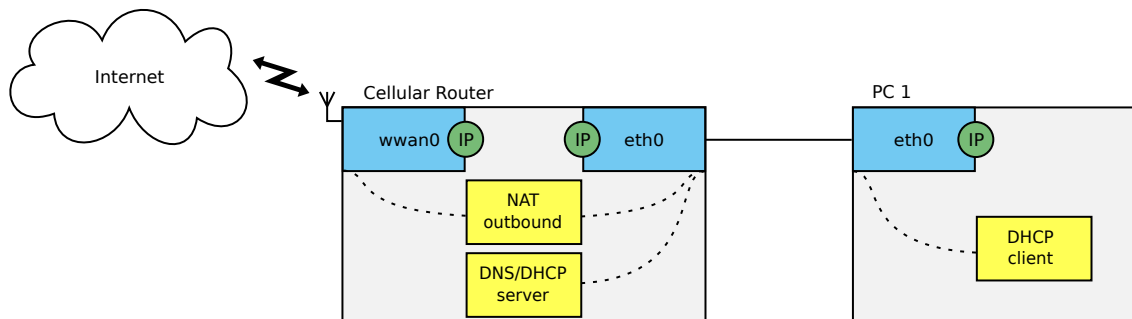


Figure 4.4: Cellular Router as Internet Gateway

The configuration example in Listing 4.1 defines a Cellular Router as a gateway with an enabled DHCP server and an outbound NAT, which is by default set up as described in section 4.47.1.2. Connect a PC with a DHCP client to one of the Ethernet interfaces of the Cellular Router. The SIM card must be inserted in slot 1. The APN is selected automatically.

```

WESTERMO-SW6-MIB::cfgSysHostname.0 = MR-Gateway
WESTERMO-SW6-MIB::cfgNetWwanEnabled.0 = INTEGER: 1
WESTERMO-SW6-MIB::cfgCellSimSlot1.0 = INTEGER: 0
WESTERMO-SW6-MIB::cfgCellSimSlot2.0 = INTEGER: -1
WESTERMO-SW6-MIB::cfgDhcpGlobalEnabled.0 = 1

```

Listing 4.1: Configuration of a Cellular Router as Gateway

Note: The parameter ([cfgCellSimSlot1](#)) is set to 0, which represents the index of the default SIM profile.

4.18.6 Cellular Network Status

The general status of each cellular network interface can be requested by using the entries of the [swCellTable](#). Additionally there are SNMP traps that indicate the link status of the corresponding network interface, see [Code: \[360 \]](#) 360 and 361.

4.19 Tunnel Endpoint Configuration

Tunnel Endpoint (TEP) Interfaces are defined in the [cfgNetTunnelEndPointTable](#). Tunnels enable stateless encapsulation of IP or Ethernet frames.

Supported protocols are:

Protocol	Encapsulation Overhead	Bridgeable
GRE	24 bytes	no
GRETAP	38 bytes	yes

4.19.1 Interface Configuration

Each Tunnel Endpoint interface in the table [cfgNetTunnelEndPointTable](#) is managed by a separate tunnel instance. These instances are created and enabled via the table [cfgVpnTunnelEndPointTable](#). The indices of both tables refer to the same instance.

[cfgNetTunnelEndPointTable](#) is used to configure generic network related parameters like the MTU of the interface, bridge assignment, or vlan.

[cfgVpnTunnelEndPointTable](#) is used to configure tunnel specific parameters, like the receive/transmit key or address of the remote peer.

4.19.2 Generic Routing Encapsulation (GRE)

The GRE protocol allows running IPv4 and IPv6 frames over IPv4.

Every inner IP frame is prepended with a GRE header and then encapsulated within an outer IPv4 frame. The resulting overhead is 24 bytes (4 bytes GRE header + 20 bytes outer IPv4 header).

GRE is often used in conjunction with IPsec. GRE is not encrypted and IPsec does not allow multicast traffic. Thus the combination of both result in an encrypted IPsec tunnel with a multicast capable GRE tunnel on top.

The example in Figure 4.5 showcases a simple scenario where the WLAN Router creates a routed tunnel to the Backbone Router. The traffic between them traverses multiple routed networks (the Wireless Network and the Routed Backbone).

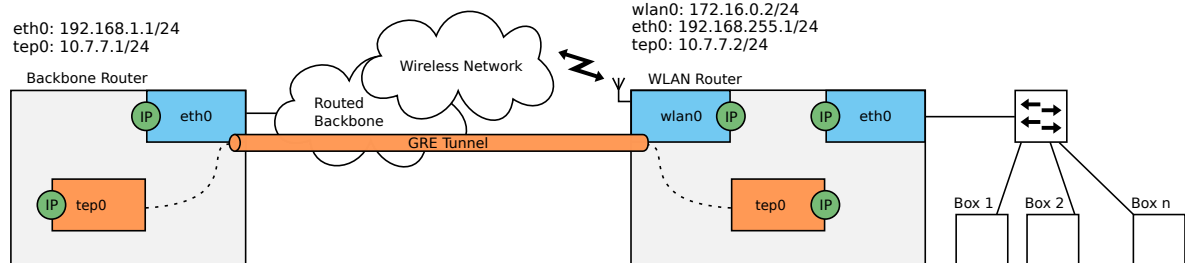


Figure 4.5: GRE in a Wireless Network

The relevant configuration items for the WLAN Router in Figure 4.5 to create a GRE tunnel, are described in the Listing 4.2:

```
WESTERMO-SW6-MIB::cfgNetTepEnabled.0 = 1
WESTERMO-SW6-MIB::cfgNetTepBridge.0 = -1
WESTERMO-SW6-MIB::cfgVpnTepTunnelType.0 = 0
WESTERMO-SW6-MIB::cfgVpnTepSource.0 = 0.0.0.0
WESTERMO-SW6-MIB::cfgVpnTepDestination.0 = 192.168.1.1

WESTERMO-SW6-MIB::cfgNetIpEnabled.3 = 1
WESTERMO-SW6-MIB::cfgNetIpAddr.3 = 10.7.7.2/24
WESTERMO-SW6-MIB::cfgNetIpInterface.3 = tep0
```

Listing 4.2: TEP Interface Configuration for GRE

Because a TEP interface of type GRE may not be bridged, `cfgNetTepBridge` is set to -1. The source address `cfgVpnTepSource` is set to 0.0.0.0, this lets the routing process decide which source address is used for the tunnel.

`cfgVpnTepTos` may be used to specify the TOS of the GRE frames. When it is set to `inherit` the TOS value of the outer IP header is used. The key ID's `cfgVpnTepRxKeyId` and `cfgVpnTepTxKeyId`

identify the header frame to allow parallel operation of multiple tunnels. Therefore, the RX and TX key ID's of the two endpoints must match. The same applies to GRE TAP.

4.19.3 Generic Routing Encapsulation Tap (GRETAP)

The GRE protocol may operate in TAP mode, allowing it to run Ethernet frames over IPv4.

Every inner Ethernet frame is prepended with a GRE header and then encapsulated within an outer IPv4 frame. The resulting overhead is 38 bytes (4 bytes GRE header + 14 bytes inner Ethernet header + 20 bytes outer IPv4 header).

This allows bridging of TEP interfaces and enables usage of VLANs over the tunnel.

The scenario in Figure 4.6 showcases the WLAN Router which creates a bridged tunnel to the Backbone Router. The traffic between them traverses multiple routed networks (the Wireless Network and the Routed Backbone). The bridged devices connected to the WLAN Router are in the same subnet as tep0 of the Backbone Router.

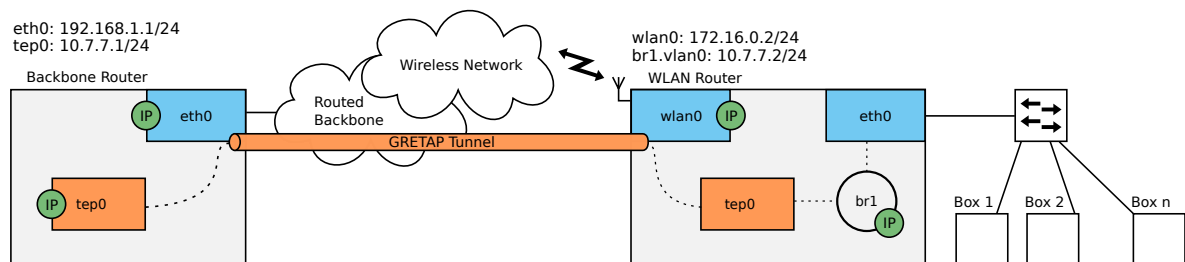


Figure 4.6: GRE TAP in a Wireless Network

The essential configuration items for the WLAN Router in Figure 4.6 to create a GRE TAP tunnel, are described in the Listing 4.3:

```
WESTERMO-SW6-MIB::cfgNetTepEnabled.0 = 1
WESTERMO-SW6-MIB::cfgNetTepBridge.0 = 1
WESTERMO-SW6-MIB::cfgVpnTepTunnelType.0 = 1
WESTERMO-SW6-MIB::cfgVpnTepSource.0 = 172.16.0.2
WESTERMO-SW6-MIB::cfgVpnTepDestination.0 = 192.168.1.1

WESTERMO-SW6-MIB::cfgNetEthBridge.0 = 1

WESTERMO-SW6-MIB::cfgNetVlanEnabled.2 = 1
WESTERMO-SW6-MIB::cfgNetVlanBridge.2 = 1

WESTERMO-SW6-MIB::cfgNetIpEnabled.3 = 1
WESTERMO-SW6-MIB::cfgNetIpAddr.3 = 10.7.7.2/24
```

```
WESTERMO-SW6-MIB::cfgNetIpInterface.3 = br1.vlan0
```

Listing 4.3: *TEP Interface Configuration for GRETAP*

To bridge eth0 together with tep0 on br1, set both [cfgNetTepBridge](#) and [cfgNetEthBridge](#) to 1. To allow creation of an IP address on br1, a VLAN interface br1.vlan0 is added to br1. The source address [cfgVpnTepSource](#) is set to 172.16.0.2, binding the source to a specific address.

4.20 IPsec Configuration

Software 6 provides virtual private network (VPN) support via IPsec VPN. The device can act as a VPN gateway in mainly in NETWORK-NETWORK scenarios.

IPsec VPN tunnels can be used to securely connect hosts and networks over the Internet. IPsec offers high level security and best performance over all VPNs due to HW acceleration.

4.20.1 Interface Configuration

Each IPsec interface in the table [cfgNetIpsecTable](#) is managed by a separate instance of IPsec. These instances are created and enabled via the table [cfgVpnIpsecTable](#). The indices of both tables refer to the same instance.

[cfgNetIpsecTable](#) is used to configure generic network related parameters like the MTU of the interface.

[cfgVpnIpsecTable](#) is used to configure IPsec specific parameters, like the used encryption or address of remote peers.

4.20.2 Introduction to IPsec VPNs

A common use case for IPsec VPNs is to connect two networks via a secure tunnel over the Internet. We refer to this scenario as NETWORK-NETWORK VPNs. It is accomplished by having two VPN gateways, one at each site, negotiate and establish a secure tunnel, and to forward all traffic between the two networks through this tunnel. Creating VPN tunnels establishes a secure overlay network on top of the regular Internet connections.

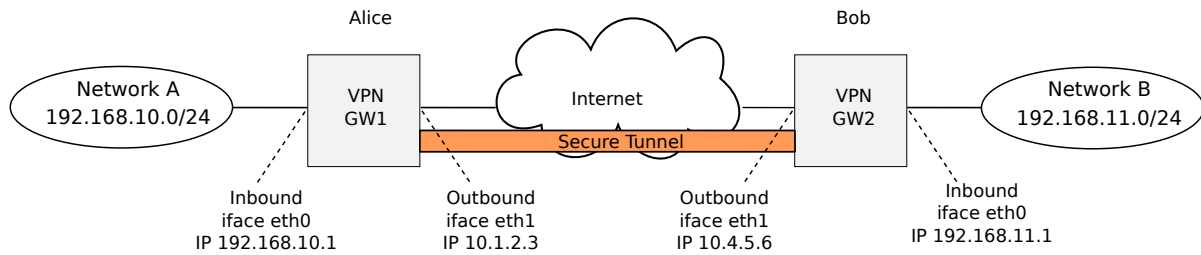


Figure 4.7: IPsec VPN Sample Topology

Fig. 4.7 shows a sample IPsec VPN topology used to help describe IPsec terminology and to illustrate some IPsec VPN configurations (see 4.20.6).

Explanation of some VPN related terminology:

- **Peers:** The two VPN gateways (Alice and Bob) are referred to as IPsec peers. The peers constitute the end-points of the secure tunnel. One of the peers will take the role of tunnel initiator and the other takes the responder role.
- **Initiator and Responder:** The VPN initiator is the peer that is responsible for initiating the tunnel establishment by contacting the other peer - the responder. A device is able to act both as responder and as initiator. Use `cfgVpnIpsecAuto = "start"` to configure a device as VPN initiator. In some cases, both peers can be configured as initiator at the same time.
- **NAT-traversal, Peer IP addresses:** In order to act as a responder, Alice must be assigned a public (routable) IP address on its interface towards the Internet. Thus, Alice generally cannot be located behind a NAT gateway, since the initiator (Bob) would not be able to initiate the tunnel. Bob will need to know Alice's IP address (or domain name) in order to know where to send the tunnel establishment messages. If Alice is assigned a fixed IP address, Bob can choose between using Alice's IP address or her domain name. But if Alice gets her address dynamically (e.g., via DHCP), Bob should use her domain name to establish the contact. The initiator (Bob) does not need to be assigned a public IP address. Bob is able to establish the tunnel even if he is located behind a NAT gateway. Furthermore, it is not mandatory for Alice to know Bob's IP address beforehand. It is possible to configure the VPN tunnel such that Bob could connect to the Internet at various locations and still be able to establish the VPN tunnel. This is commonly referred to as Bob being a road warrior. The IKEv2 protocol includes NAT traversal (NAT-T) in the core standard. There is no need for any special configuration. If a NAT situation is detected, the client switches to UDP port 4500 (if it used port 500 initially) to send IKE_AUTH requests and UDP encapsulation will be activated for IPsec SAs.
- **Local and Remote Subnet:** Each peer will define what traffic should be allowed to pass through the established tunnel. Each peer will define the local and remote subnet, and all traffic between these subnets is sent securely through the tunnel. To secure all traffic between networks "A" and "B", Alice would define 192.168.10.0/24 as local subnet, and 192.168.11.0/24 as remote subnet in the tunnel configuration. Bob would do the opposite, i.e., define 192.168.11.0/24 as local subnet, and 192.168.10.0/24 as remote subnet. More

advanced settings for the local and remote subnet parameters are possible, e.g., it is possible to configure the tunnel so that all traffic from Network B is sent through the tunnel (i.e., not only the traffic heading for Network A).

- **Outbound Interface:** The outbound interface denotes the interface, and implicitly the IP address, a VPN gateway uses to tunnel the traffic through, and to communicate with its peer. In fig. 4.7 Alice's outbound interface would be her interface towards the Internet (and the same goes for Bob). By default, the outbound interface is set to the interface leading to the default gateway.

4.20.3 Authenticated Keying using Internet Key Exchange

As part of the IPsec VPN tunnel establishment, Alice and Bob will use the IKE (Internet Key Exchange) protocol to authenticate each other and create necessary session keys to protect the data traffic. *Software 6* supports IKE version 1 (IKEv1) and IKE version 2 (IKEv2). It supports IKE with authentication through pre-shared keys (PSK) or certificates (RSA signature keys using X.509 certificates). In the IKE handshake, Alice and Bob identify themselves and use their configured PSK or certificates to authenticate each other. When configuring an IPsec tunnel, the identities of the peers should be defined in `cfgVpnIpsecLeftId` and `cfgVpnIpsecRightId`.

See <https://wiki.strongswan.org/projects/strongswan/wiki/IdentityParsing> for a description of types and binary encoding of identity strings.

It is recommended to use IKEv2 (`cfgVpnIpsecKeyExchange = ikev2(2)`) whenever possible.

Limitations when using IKEv1:

- Using IKEv1 aggressive Mode with pre-shared keys (PSK) is the least secure option. In this particular scenario, it is possible for an attacker to gather all necessary information in order to mount an off-line dictionary (brute force) attack on the PSK. Therefore: IKEv1 aggressive mode with PSK authentication is **not** recommended.
- Due to limitations in IKEv1, main mode with PSK can only be used with IP address (or %any) as identity (`cfgVpnIpsecLeftId` and `cfgVpnIpsecRightId`).

Both for the IKE (`cfgVpnIpsecIke`) and ESP (`cfgVpnIpsecEsp`) handshakes the user can specify which cryptographic protocols to use.

Perfect Forward Secrecy (PFS) is a must have for every IPsec tunnel. Using PFS will make the endpoints negotiate a new key for each IKE and/or IPsec Security Association (SA) upon rekey or reauthentication event. This protects the confidentiality of the traffic, if the IKE shared secret is leaked. Note, that the keys of the first SA of a new IKEv2 connection are derived from the IKE shared secret. However, subsequent SAs will use new keys if PFS is used. PFS is enabled by appending a DH group to the ESP cipher settings (`cfgVpnIpsecEsp`).

4.20.4 Dead Peer Detection

The connectivity through an established IPsec tunnel may be broken unexpectedly, for example one of the peers go down or is disconnected, or if some kind of routing, NAT or firewall problem occurs on the path between them. Dead Peer Detection (DPD) can be used to discover and manage such situations. In DPD the peers exchange keep-alive messages to monitor if the remote peer is still reachable. If a peer determines connectivity to be broken, appropriate actions should be taken. Use [cfgVpnIpsecDpdAction](#), [cfgVpnIpsecDpdDelay](#) and [cfgVpnIpsecDpdTimeout](#) to configure DPD.

4.20.5 Route Based VPNs

Generally IPsec processing is based on policies. After regular route lookups are done, the OS kernel consults its Security Policy Database (SPD) for a matching policy and if one is found that is associated with an IPsec Security Association (SA), the packet is processed (e.g. encrypted and sent as ESP packet).

VirtualTunnelInterface (VTI) devices act like a wrapper around existing IPsec policies. This means you cannot just route arbitrary packets to a VTI device to get them tunneled, the established IPsec policies have to match too. However, you can negotiate 0.0.0.0/0 traffic selectors on both ends to allow tunneling anything that's routed via VTI device.

Enable VTI for Route Based IPsec VPN by setting [cfgVpnIpsecGlbVirtualTunnelInterface](#) = enabled(1) and by configuring routes ([cfgRouteTableEnabled](#), [cfgRouteTableDestinationNetwork](#), [cfgRouteTableSource](#), [cfgRouteTableInterface](#)).

4.20.6 IPsec Configuration Example

Fig. 4.7 shows a sample IPsec VPN topology used to illustrate IPsec VPN configurations.

We have two VPN gateways, Alice (VPN GW1) and Bob (VPN GW2), which are used to establish a secure VPN tunnel between the central office network (192.168.10.0/24) and the branch office network (192.168.11.0/24).

4.20.6.0.1 Example of using IPsec VPN with PSK

This section illustrates configuration parameters for configuring IPsec VPNs using IKE authentication with pre-shared key (PSKs).

Configuration of the two gateways then looks like this:

Configuration File Example for VPN GW1: IPsec PSK Authentication

```
# Config.Format = raw
# Static IP address for inbound iface
WESTERMO-SW6-MIB::cfgNetIpEnabled.0 = 1
WESTERMO-SW6-MIB::cfgNetIpProto.0 = 0
WESTERMO-SW6-MIB::cfgNetIpAddr.0 = 192.168.10.1/24
WESTERMO-SW6-MIB::cfgNetIpInterface.0 = eth0
# IPsec alias for outbound iface
WESTERMO-SW6-MIB::cfgNetIpEnabled.3 = 1
WESTERMO-SW6-MIB::cfgNetIpProto.3 = 0
WESTERMO-SW6-MIB::cfgNetIpAddr.3 = "10.1.2.3"
WESTERMO-SW6-MIB::cfgNetIpInterface.3 = eth1
# IPsec tunnel configuration
WESTERMO-SW6-MIB::cfgNetIpsecEnabled.0 = 1
WESTERMO-SW6-MIB::cfgVpnIpsecType.0 = 0
WESTERMO-SW6-MIB::cfgVpnIpsecAuto.0 = 1
WESTERMO-SW6-MIB::cfgVpnIpsecLeftAuth.0 = psk
WESTERMO-SW6-MIB::cfgVpnIpsecRightAuth.0 = psk
WESTERMO-SW6-MIB::cfgVpnIpsecPassword.0 = TopSecretPw1!
WESTERMO-SW6-MIB::cfgVpnIpsecLeft.0 = 10.1.2.3
WESTERMO-SW6-MIB::cfgVpnIpsecLeftId.0 = GW1
WESTERMO-SW6-MIB::cfgVpnIpsecLeftSubnet.0 = 192.168.10.0/24
WESTERMO-SW6-MIB::cfgVpnIpsecRight.0 = 10.4.5.6
WESTERMO-SW6-MIB::cfgVpnIpsecRightId.0 = GW2
WESTERMO-SW6-MIB::cfgVpnIpsecRightSubnet.0 = 192.168.11.0/24
WESTERMO-SW6-MIB::cfgVpnIpsecKeyExchange.0 = 2
WESTERMO-SW6-MIB::cfgVpnIpsecIke.0 = aes128-sha1-modp1024!
WESTERMO-SW6-MIB::cfgVpnIpsecEsp.0 = aes128-sha1-modp1024!
```

Configuration File Example for VPN GW2: IPsec PSK Authentication

```
# Config.Format = raw
# Static IP address for inbound iface
WESTERMO-SW6-MIB::cfgNetIpEnabled.0 = 1
WESTERMO-SW6-MIB::cfgNetIpProto.0 = 0
WESTERMO-SW6-MIB::cfgNetIpAddr.0 = 192.168.11.1/24
WESTERMO-SW6-MIB::cfgNetIpInterface.0 = eth0
# IPsec alias for outbound iface
WESTERMO-SW6-MIB::cfgNetIpEnabled.3 = 1
WESTERMO-SW6-MIB::cfgNetIpInterface.3 = eth1
WESTERMO-SW6-MIB::cfgNetIpProto.3 = 0
WESTERMO-SW6-MIB::cfgNetIpAddr.3 = 10.4.5.6
# IPsec tunnel configuration
WESTERMO-SW6-MIB::cfgNetIpsecEnabled.0 = 1
WESTERMO-SW6-MIB::cfgVpnIpsecType.0 = 0
WESTERMO-SW6-MIB::cfgVpnIpsecAuto.0 = 1
WESTERMO-SW6-MIB::cfgVpnIpsecLeftAuth.0 = psk
```

```
WESTERMO-SW6-MIB::cfgVpnIpsecRightAuth.0 = psk
WESTERMO-SW6-MIB::cfgVpnIpsecPassword.0 = TopSecretPw1!
WESTERMO-SW6-MIB::cfgVpnIpsecLeft.0 = 10.4.5.6
WESTERMO-SW6-MIB::cfgVpnIpsecLeftId.0 = GW2
WESTERMO-SW6-MIB::cfgVpnIpsecLeftSubnet.0 = 192.168.11.0/24
WESTERMO-SW6-MIB::cfgVpnIpsecRight.0 = 10.1.2.3
WESTERMO-SW6-MIB::cfgVpnIpsecRightId.0 = GW1
WESTERMO-SW6-MIB::cfgVpnIpsecRightSubnet.0 = 192.168.10.0/24
WESTERMO-SW6-MIB::cfgVpnIpsecKeyExchange.0 = 2
WESTERMO-SW6-MIB::cfgVpnIpsecIke.0 = aes128-sha1-modp1024!
WESTERMO-SW6-MIB::cfgVpnIpsecEsp.0 = aes128-sha1-modp1024!
```

4.20.6.0.2 Example of using IPsec VPN with Public Key

This section illustrates configuration parameters for configuring IPsec VPNs using IKEv2 authentication with Public Key Signature Authentication.

Configuration of the two gateways then looks like this:

Upload following certificates to Gateway 1:

1. Alice's public key
2. Alice's private key
3. Bob's public key

Upload following certificates to Gateway 2:

1. Bob's public key
2. Bob's private key
3. Alice's public key

Configuration File Example for VPN GW1: IPsec Public Key Authentication without CA

```
# Upload following certificates:
# Config.Format = raw
# Static IP address for inbound iface
WESTERMO-SW6-MIB::cfgNetIpEnabled.0 = 1
WESTERMO-SW6-MIB::cfgNetIpProto.0 = 0
WESTERMO-SW6-MIB::cfgNetIpAddr.0 = 192.168.10.1/24
WESTERMO-SW6-MIB::cfgNetIpInterface.0 = eth0
# IPsec alias for outbound iface
```

```
WESTERMO-SW6-MIB::cfgNetIpEnabled.3 = 1
WESTERMO-SW6-MIB::cfgNetIpProto.3 = 0
WESTERMO-SW6-MIB::cfgNetIpAddr.3 = 10.1.2.3
WESTERMO-SW6-MIB::cfgNetIpInterface.3 = eth1
# IPsec tunnel configuration
WESTERMO-SW6-MIB::cfgNetIpsecEnabled.0 = 1
WESTERMO-SW6-MIB::cfgVpnIpsecType.0 = 0
WESTERMO-SW6-MIB::cfgVpnIpsecAuto.0 = 1
WESTERMO-SW6-MIB::cfgVpnIpsecLeftAuth.0 = pubkey
WESTERMO-SW6-MIB::cfgVpnIpsecRightAuth.0 = pubkey
WESTERMO-SW6-MIB::cfgVpnIpsecLeft.0 = 10.1.2.3
WESTERMO-SW6-MIB::cfgVpnIpsecLeftId.0 = C=CH, O=Westermo, CN=Alice
WESTERMO-SW6-MIB::cfgVpnIpsecLeftSubnet.0 = 192.168.10.0/24
WESTERMO-SW6-MIB::cfgVpnIpsecRight.0 = 10.4.5.6
WESTERMO-SW6-MIB::cfgVpnIpsecRightId.0 = C=CH, O=Westermo, CN=Bob
WESTERMO-SW6-MIB::cfgVpnIpsecRightSubnet.0 = 192.168.11.0/24
WESTERMO-SW6-MIB::cfgVpnIpsecKeyExchange.0 = 2
WESTERMO-SW6-MIB::cfgVpnIpsecIke.0 = aes128-sha1-modp1024!
WESTERMO-SW6-MIB::cfgVpnIpsecEsp.0 = aes128-sha1-modp1024!
```

Configuration File Example for VPN GW2: IPsec Public Key Authentication without CA

```
# Config.Format = raw
# Static IP address for inbound iface
WESTERMO-SW6-MIB::cfgNetIpEnabled.0 = 1
WESTERMO-SW6-MIB::cfgNetIpProto.0 = 0
WESTERMO-SW6-MIB::cfgNetIpAddr.0 = 192.168.11.1/24
WESTERMO-SW6-MIB::cfgNetIpInterface.0 = eth0
# IPsec alias for outbound iface
WESTERMO-SW6-MIB::cfgNetIpEnabled.3 = 1
WESTERMO-SW6-MIB::cfgNetIpInterface.3 = eth1
WESTERMO-SW6-MIB::cfgNetIpProto.3 = 0
WESTERMO-SW6-MIB::cfgNetIpAddr.3 = 10.4.5.6
# IPsec tunnel configuration
WESTERMO-SW6-MIB::cfgNetIpsecEnabled.0 = 1
WESTERMO-SW6-MIB::cfgVpnIpsecType.0 = 0
WESTERMO-SW6-MIB::cfgVpnIpsecAuto.0 = 1
WESTERMO-SW6-MIB::cfgVpnIpsecLeftAuth.0 = pubkey
WESTERMO-SW6-MIB::cfgVpnIpsecRightAuth.0 = pubkey
WESTERMO-SW6-MIB::cfgVpnIpsecLeft.0 = 10.4.5.6
WESTERMO-SW6-MIB::cfgVpnIpsecLeftId.0 = C=CH, O=Westermo, CN=Bob
WESTERMO-SW6-MIB::cfgVpnIpsecLeftSubnet.0 = 192.168.11.0/24
WESTERMO-SW6-MIB::cfgVpnIpsecRight.0 = 10.1.2.3
WESTERMO-SW6-MIB::cfgVpnIpsecRightId.0 = C=CH, O=Westermo, CN=Alice
WESTERMO-SW6-MIB::cfgVpnIpsecRightSubnet.0 = 192.168.10.0/24
WESTERMO-SW6-MIB::cfgVpnIpsecKeyExchange.0 = 2
```

```
WESTERMO-SW6-MIB::cfgVpnIpsecIke.0 = aes128-sha1-modp1024!
WESTERMO-SW6-MIB::cfgVpnIpsecEsp.0 = aes128-sha1-modp1024!
```

4.20.6.0.3 Example of using IPsec VPN with CA Certificate

This section illustrates configuration parameters for configuring IPsec VPNs using IKEv2 authentication with CA Authentication.

Configuration of the two gateways then looks like this:

Upload following certificates to Gateway 1:

1. CA Certificate
2. Alice's Certificate (signed by CA)
3. Alice's private key

Upload following certificates to Gateway 2:

1. CA Certificate
2. Bob's Certificate (signed by CA)
3. Bob's private key

Configuration File Example for VPN GW1: IPsec Public Key Authentication with CA

```
# Upload following certificates:
# Config.Format = raw
# Static IP address for inbound iface
WESTERMO-SW6-MIB::cfgNetIpEnabled.0 = 1
WESTERMO-SW6-MIB::cfgNetIpProto.0 = 0
WESTERMO-SW6-MIB::cfgNetIpAddr.0 = 192.168.10.1/24
WESTERMO-SW6-MIB::cfgNetIpInterface.0 = eth0
# IPsec alias for outbound iface
WESTERMO-SW6-MIB::cfgNetIpEnabled.3 = 1
WESTERMO-SW6-MIB::cfgNetIpProto.3 = 0
WESTERMO-SW6-MIB::cfgNetIpAddr.3 = 10.1.2.3
WESTERMO-SW6-MIB::cfgNetIpInterface.3 = eth1
# IPsec tunnel configuration
WESTERMO-SW6-MIB::cfgNetIpsecEnabled.0 = 1
WESTERMO-SW6-MIB::cfgVpnIpsecType.0 = 0
WESTERMO-SW6-MIB::cfgVpnIpsecAuto.0 = 1
WESTERMO-SW6-MIB::cfgVpnIpsecLeftAuth.0 = pubkey
```



```
WESTERMO-SW6-MIB::cfgVpnIpsecRightAuth.0 = pubkey
WESTERMO-SW6-MIB::cfgVpnIpsecLeft.0 = 10.1.2.3
WESTERMO-SW6-MIB::cfgVpnIpsecLeftId.0 = C=CH, O=Westermo, CN=Alice
WESTERMO-SW6-MIB::cfgVpnIpsecLeftSubnet.0 = 192.168.10.0/24
WESTERMO-SW6-MIB::cfgVpnIpsecRight.0 = 10.4.5.6
WESTERMO-SW6-MIB::cfgVpnIpsecRightId.0 = C=CH, O=Westermo, CN=Bob
WESTERMO-SW6-MIB::cfgVpnIpsecRightSubnet.0 = 192.168.11.0/24
WESTERMO-SW6-MIB::cfgVpnIpsecKeyExchange.0 = 2
WESTERMO-SW6-MIB::cfgVpnIpsecIke.0 = aes128-sha1-modp1024!
WESTERMO-SW6-MIB::cfgVpnIpsecEsp.0 = aes128-sha1-modp1024!
```

Configuration File Example for VPN GW2: IPsec Public Key Authentication with CA

```
# Config.Format = raw
# Static IP address for inbound iface
WESTERMO-SW6-MIB::cfgNetIpEnabled.0 = 1
WESTERMO-SW6-MIB::cfgNetIpProto.0 = 0
WESTERMO-SW6-MIB::cfgNetIpAddr.0 = 192.168.11.1/24
WESTERMO-SW6-MIB::cfgNetIpInterface.0 = eth0
# IPsec alias for outbound iface
WESTERMO-SW6-MIB::cfgNetIpEnabled.3 = 1
WESTERMO-SW6-MIB::cfgNetIpInterface.3 = eth1
WESTERMO-SW6-MIB::cfgNetIpProto.3 = 0
WESTERMO-SW6-MIB::cfgNetIpAddr.3 = 10.4.5.6
# IPsec tunnel configuration
WESTERMO-SW6-MIB::cfgNetIpsecEnabled.0 = 1
WESTERMO-SW6-MIB::cfgVpnIpsecType.0 = 0
WESTERMO-SW6-MIB::cfgVpnIpsecAuto.0 = 1
WESTERMO-SW6-MIB::cfgVpnIpsecLeftAuth.0 = pubkey
WESTERMO-SW6-MIB::cfgVpnIpsecRightAuth.0 = pubkey
WESTERMO-SW6-MIB::cfgVpnIpsecLeft.0 = 10.4.5.6
WESTERMO-SW6-MIB::cfgVpnIpsecLeftId.0 = C=CH, O=Westermo, CN=Bob
WESTERMO-SW6-MIB::cfgVpnIpsecLeftSubnet.0 = 192.168.11.0/24
WESTERMO-SW6-MIB::cfgVpnIpsecRight.0 = 10.1.2.3
WESTERMO-SW6-MIB::cfgVpnIpsecRightId.0 = C=CH, O=Westermo, CN=Alice
WESTERMO-SW6-MIB::cfgVpnIpsecRightSubnet.0 = 192.168.10.0/24
WESTERMO-SW6-MIB::cfgVpnIpsecKeyExchange.0 = 2
WESTERMO-SW6-MIB::cfgVpnIpsecIke.0 = aes128-sha1-modp1024!
WESTERMO-SW6-MIB::cfgVpnIpsecEsp.0 = aes128-sha1-modp1024!
```

4.20.6.0.4 Example of using IPsec VPN with PSK as route-based VPN

This section illustrates configuration parameters for configuring route-based IPsec VPNs using PSK authentication.

Configuration of the two gateways then looks like this:

Configuration File Example for VPN GW1: Route-based IPsec VPN with PSK Authentication

```
# Config.Format = raw
# Static IP address for inbound iface
WESTERMO-SW6-MIB::cfgNetIpEnabled.0 = 1
WESTERMO-SW6-MIB::cfgNetIpProto.0 = 0
WESTERMO-SW6-MIB::cfgNetIpAddr.0 = 192.168.10.1/24
WESTERMO-SW6-MIB::cfgNetIpInterface.0 = eth0
# IPsec alias for outbound iface
WESTERMO-SW6-MIB::cfgNetIpEnabled.3 = 1
WESTERMO-SW6-MIB::cfgNetIpProto.3 = 0
WESTERMO-SW6-MIB::cfgNetIpAddr.3 = 10.1.2.3/24
WESTERMO-SW6-MIB::cfgNetIpInterface.3 = eth1
# IPsec VTI
WESTERMO-SW6-MIB::cfgVpnIpsecGlbVirtualTunnelInterface.0 = 1
WESTERMO-SW6-MIB::cfgNetIpsecEnabled.0 = 1
WESTERMO-SW6-MIB::cfgVpnIpsecType.0 = 0
WESTERMO-SW6-MIB::cfgVpnIpsecAuto.0 = 1
WESTERMO-SW6-MIB::cfgVpnIpsecLeftAuth.0 = psk
WESTERMO-SW6-MIB::cfgVpnIpsecRightAuth.0 = psk
WESTERMO-SW6-MIB::cfgVpnIpsecPassword.0 = TopSecretPw1!
WESTERMO-SW6-MIB::cfgVpnIpsecLeft.0 = 10.1.2.3
WESTERMO-SW6-MIB::cfgVpnIpsecLeftId.0 = GW1
WESTERMO-SW6-MIB::cfgVpnIpsecLeftSubnet.0 = 192.168.10.0/24
WESTERMO-SW6-MIB::cfgVpnIpsecRight.0 = 10.4.5.6
WESTERMO-SW6-MIB::cfgVpnIpsecRightId.0 = GW2
WESTERMO-SW6-MIB::cfgVpnIpsecRightSubnet.0 = 192.168.11.0/24
WESTERMO-SW6-MIB::cfgVpnIpsecKeyExchange.0 = 2
WESTERMO-SW6-MIB::cfgVpnIpsecIke.0 = aes128-sha256-modp3072!
WESTERMO-SW6-MIB::cfgVpnIpsecEsp.0 = aes128-sha256!
# Add route to 192.168.11.0/24 over ipsec0 with src 10.1.2.3
WESTERMO-SW6-MIB::cfgRouteTableEnabled.0 = 1
WESTERMO-SW6-MIB::cfgRouteTableDestinationNetwork.0 = 192.168.11.0/24
WESTERMO-SW6-MIB::cfgRouteTableSource.0 = 10.1.2.3
WESTERMO-SW6-MIB::cfgRouteTableInterface.0 = ipsec0
```

Configuration File Example for VPN GW2: Route-based IPsec VPN with PSK Authentication

```
# Config.Format = raw
# Static IP address for inbound iface
WESTERMO-SW6-MIB::cfgNetIpEnabled.0 = 1
WESTERMO-SW6-MIB::cfgNetIpProto.0 = 0
WESTERMO-SW6-MIB::cfgNetIpAddr.0 = 192.168.11.1/24
WESTERMO-SW6-MIB::cfgNetIpInterface.0 = eth0
# IPsec alias for outbound iface
WESTERMO-SW6-MIB::cfgNetIpEnabled.3 = 1
```

```

WESTERMO-SW6-MIB::cfgNetIpProto.3 = 0
WESTERMO-SW6-MIB::cfgNetIpAddr.3 = 10.4.5.6/24
WESTERMO-SW6-MIB::cfgNetIpInterface.3 = eth1
# IPsec VTI
WESTERMO-SW6-MIB::cfgVpnIpsecGlbVirtualTunnelInterface.0 = 1
WESTERMO-SW6-MIB::cfgNetIpsecEnabled.0 = 1
WESTERMO-SW6-MIB::cfgVpnIpsecType.0 = 0
WESTERMO-SW6-MIB::cfgVpnIpsecAuto.0 = 1
WESTERMO-SW6-MIB::cfgVpnIpsecLeftAuth.0 = psk
WESTERMO-SW6-MIB::cfgVpnIpsecRightAuth.0 = psk
WESTERMO-SW6-MIB::cfgVpnIpsecPassword.0 = TopSecretPw1!
WESTERMO-SW6-MIB::cfgVpnIpsecLeft.0 = 10.4.5.6
WESTERMO-SW6-MIB::cfgVpnIpsecLeftId.0 = GW2
WESTERMO-SW6-MIB::cfgVpnIpsecLeftSubnet.0 = 192.168.11.0/24
WESTERMO-SW6-MIB::cfgVpnIpsecRight.0 = 10.1.2.3
WESTERMO-SW6-MIB::cfgVpnIpsecRightId.0 = GW1
WESTERMO-SW6-MIB::cfgVpnIpsecRightSubnet.0 = 192.168.10.0/24
WESTERMO-SW6-MIB::cfgVpnIpsecKeyExchange.0 = 2
WESTERMO-SW6-MIB::cfgVpnIpsecIke.0 = aes128-sha256-modp3072!
WESTERMO-SW6-MIB::cfgVpnIpsecEsp.0 = aes128-sha256!
# add route to 192.168.10.0/24 over ipsec0 with src 10.4.5.6
WESTERMO-SW6-MIB::cfgRouteTableEnabled.0 = 1
WESTERMO-SW6-MIB::cfgRouteTableDestinationNetwork.0 = 192.168.10.0/24
WESTERMO-SW6-MIB::cfgRouteTableSource.0 = 10.4.5.6
WESTERMO-SW6-MIB::cfgRouteTableInterface.0 = ipsec0

```

4.21 OpenVPN Configuration

OpenVPN may operate as a client or server. *Software 6* currently only supports the client role, see [cfgVpnOpenvpnMode](#).

OpenVPN interfaces are either of type tun or tap. For routed (tun) or bridged (tap) configurations, see [cfgVpnOpenvpnDevType](#). Both sides of the connection must use the same DevType. If the interface is of type tun, it may not be part of a bridge.

4.21.1 Interface Configuration

Each OpenVPN interface in the table [cfgNetOpenvpnTable](#) is managed by a separate instance of OpenVPN. These instances are created and enabled via the table [cfgVpnOpenvpnTable](#). The indices of both tables refer to the same instance.

[cfgNetOpenvpnTable](#) is used to configure generic network related parameters like the MTU of the interface, bridge assignment, or vlan.

`cfgVpnOpenvpnTable` is used to configure OpenVPN specific parameters, like the used encryption or address of the server.

4.21.2 OpenVPN Configuration Example

OpenVPN, WireGuard, IPsec and GRE have been introduced to supplement cellular networks in *Software 6*.

This following example depicts the usage of OpenVPN in a cellular network.

The goal is to provide private access from the Cellular Router to PC 1, see Figure 4.8. An encrypted tunnel is created to travers public networks such as the Internet or private transfer networks of cellular providers.

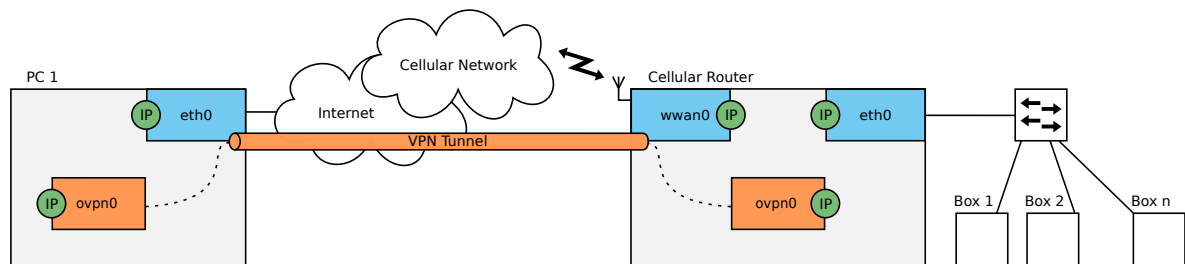


Figure 4.8: *OpenVPN in a Cellular Network*

In this example, PC 1 acts as OpenVPN server and the Cellular Router as a client. Multiple clients may connect to the server.

The focus of this example is OpenVPN. A working cellular network is assumed. Section 4.18 explains how to configure cellular networks.

A working OpenVPN instance requires:

- Installed key material
- A configured and enabled OpenVPN interface
- A configured OpenVPN instance

4.21.2.1 Installing the Key Material

The key material is provided by the administrator of the OpenVPN server. Assuming that the tunnel between PC 1 and the Cellular Router is secured by the Transport Layer Security (TLS) protocol, the key material consists of:

- The Certificate of the Certificate Authority (CA)
- The Client Certificate
- The Private Key

Section 4.13.1 describes how to install key material.

4.21.2.2 Configuring the OpenVPN Interface

The configuration of an OpenVPN interface is defined by the MIB entries of the [cfgNetOpenvpnTable](#).

The required entries for the example in Figure 4.8 are in the Listing 4.4:

```
WESTERMO-SW6-MIB::cfgNetOpenvpnEnabled.0 = 1
WESTERMO-SW6-MIB::cfgNetOpenvpnBridge.0 = -1
```

Listing 4.4: *OpenVPN Interface Configuration*

The OpenVPN interface and instance are created by setting [cfgNetOpenvpnEnabled](#) to `enabled(1)`. The OpenVPN interface `ovpn0` in the example in Figure 4.8 is of type `tun`. The interface may not be bridged, thus [cfgNetOpenvpnBridge](#) is set to `-1`. The remaining entries may be left with their default values.

No IP address needs to be configured. The server provides the IP address for the client.

4.21.2.3 Configuring the OpenVPN Instance

The OpenVPN instance, similar to its corresponding interface, is defined in the SNMP table [cfgVpnOpenvpnTable](#).

The required entries for the example in Figure 4.8 are in the Listing 4.5:

```
WESTERMO-SW6-MIB::cfgVpnOpenvpnMode.0 = 0
WESTERMO-SW6-MIB::cfgVpnOpenvpnDevType.0 = 0
WESTERMO-SW6-MIB::cfgVpnOpenvpnRemote.0 = 192.168.1.1
WESTERMO-SW6-MIB::cfgVpnOpenvpnRPort.0 = 1194
```

```
WESTERMO-SW6-MIB::cfgVpnOpenvpnKeyType.0 = 0
WESTERMO-SW6-MIB::cfgVpnOpenvpnKeyPassword.0 = password4key
WESTERMO-SW6-MIB::cfgVpnOpenvpnAuth.0 = SHA256
WESTERMO-SW6-MIB::cfgVpnOpenvpnCipher.0 = AES-256-CBC
WESTERMO-SW6-MIB::cfgVpnOpenvpnCompress.0 = 2
```

Listing 4.5: *OpenVPN Instance Configuration*

The Cellular Router acts as client, which is configured by using [cfgVpnOpenvpnMode](#). The interface type must be set to `tunnel(0)`, see [cfgVpnOpenvpnDevType](#). The Entries [cfgVpnOpenvpnRemote](#) and [cfgVpnOpenvpnRPort](#) define the network address of the OpenVPN server to which the client connects to. The Entry [cfgVpnOpenvpnRemote](#) accepts IPv4 addresses, as well as hostnames. For hostnames to work, a Domain Name Server (DNS) needs to be configured, see [cfgSysName-serverTable](#). Port number 1194 is the default IANA (Internet Assigned Numbers Authority) assigned port for OpenVPN.

The installed key material for the OpenVPN instance is selected by the key type, see [cfgVpnOpenvpnKeyType](#). The server uses TLS, thus the key type is `asymmetric(0)`. If the private key is encrypted, the key password must be configured using [cfgVpnOpenvpnKeyPassword](#).

Starting with OpenVPN 2.4, Negotiable Crypto Parameters (NCP) is enabled by default. It is used to automatically set a cipher and authentication algorithm. However, it is recommended to always set the values specified by the server administrator in [cfgVpnOpenvpnAuth](#) and [cfgVpnOpenvpnCipher](#), since the server may run an older version. Similar applies to the compression algorithm. In order to receive the compression settings automatically from the server, [cfgVpnOpenvpnCompress](#) may be set to `allowPush(1)`. However, an older server version will not send the expected settings, leaving the OpenVPN tunnel to fail.

The remaining entries of the OpenVPN instance configuration may be left with their default values.

4.22 Wireguard Configuration

WireGuard does not operate in a Client/Server fashion in the traditional sense. Each participant is a peer with a private key and a public key. One or multiple remote peers may be defined for each local peer. A peer accepts traffic from a remote peer when the inbound traffic matches the configured private key.

WireGuard is a pure L3 tunnel, thus it may not be bridged. If a L2 transparent tunnel is required, GRE in TAP mode may be used on top of WireGuard. WireGuard uses UDP as transport protocol.

WireGuard provides excellent roaming capabilities, since it does not need to re-establish its connection-state with its peers when moving from one network to another.

4.22.1 Interface Configuration

Each WireGuard interface in the table `cfgNetWireguardTable` is managed by a separate instance of WireGuard. These instances are created and enabled via the table `cfgVpnWireguardTable`. The indices of both tables refer to the same instance.

`cfgNetWireguardTable` is used to configure generic network related parameters like the MTU of the interface.

`cfgVpnWireguardTable` is used to configure WireGuard specific parameters, like the private key or address of remote peers.

4.22.2 Wireguard Configuration Example

OpenVPN, WireGuard, IPsec and GRE have been introduced to supplement cellular networks in *Software 6*.

This following example depicts the usage of WireGuard in a cellular network.

The goal is to provide private access from the Cellular Router to PC 1, see Figure 4.9. An encrypted tunnel is created to travers public networks such as the Internet or private transfer networks of cellular providers.

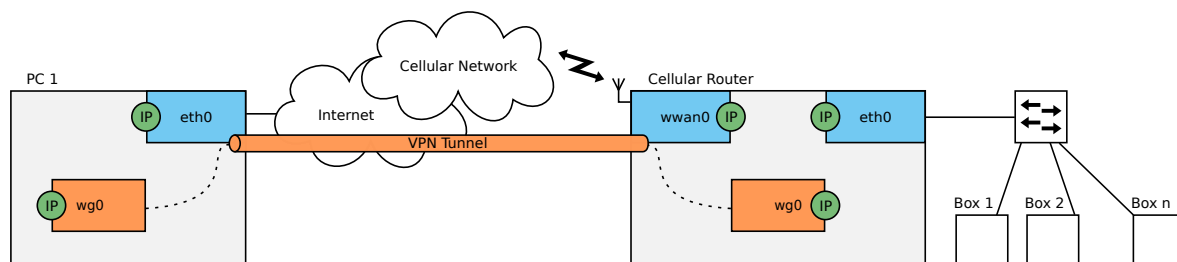


Figure 4.9: *WireGuard in a Cellular Network*

In this example, PC 1 acts as WireGuard peer that only receives incoming connections (responder) and the Cellular Router that actively establishes the connection (initiator). Multiple initiators may connect to the responder.

The focus of this example is WireGuard. A working cellular network is assumed. [Section 4.18](#) explains how to configure cellular networks.

A working WireGuard instance requires:

- A configured and enabled WireGuard interface
- A configured WireGuard instance

4.22.2.1 Configuring the WireGuard Interface

The configuration of an WireGuard interface is defined by the entries of the table [cfgNetWireguardTable](#).

The required entries for the example in Figure 4.9 are in the Listing 4.6:

```
WESTERMO-SW6-MIB::cfgNetWgEnabled.0 = 1
WESTERMO-SW6-MIB::cfgNetIpEnabled.3 = 1
WESTERMO-SW6-MIB::cfgNetIpAddr.3 = 10.0.99.2/24
WESTERMO-SW6-MIB::cfgNetIpInterface.3 = wg0
```

Listing 4.6: WireGuard Interface Configuration

The WireGuard interface and instance are created by setting [cfgNetWgEnabled](#) to `enabled(1)`. The remaining entries may be left with their default values.

Additionally the WireGuard interface requires an IP address.

4.22.2.2 Configuring the WireGuard Instance

The WireGuard instance, similar to its corresponding interface, is defined in the table [cfgVpnWireguardTable](#).

The required entries for the example in Figure 4.9 are in the Listing 4.7:

```
WESTERMO-SW6-MIB::cfgVpnWgPrivateKey.0 4CwLw8p8UGdv6baTN1dMxhxVq+
    ↪ m779v12IjDpSFecW8=

WESTERMO-SW6-MIB::cfgVpnWgPEnabled.0 1
WESTERMO-SW6-MIB::cfgVpnWgPPeer.0 dVbkEFbV4mvXkA/lbRxiJH6/cUPg8h+vFRluZN5PGRI=
WESTERMO-SW6-MIB::cfgVpnWgPEndpoint.0 1.2.3.4:51820
WESTERMO-SW6-MIB::cfgVpnWgPAllowedIps.0 10.0.99.0/24
WESTERMO-SW6-MIB::cfgVpnWgPPersistentKeepalive.0 25
```

Listing 4.7: WireGuard Instance Configuration

[cfgVpnWgPrivateKey](#) is the key with which incoming traffic is decrypted. This key should never be given to anyone. Based on the private key a public key is automatically generated and can be read from [cfgVpnWgPublicKey](#). The public key has to be installed on the remote peer.

The remote peers public key has to be installed in [cfgVpnWireguardPeersTable](#) at [cfgVpnWgPPeer](#). The Cellular Router acts as initiator, which is configured by setting [cfgVpnWgPEndpoint](#) to an address and port. Port number 51820 is the preferred port for a WireGuard responder.

This example is intended to travers the internet and a cellular network, it is probable that NAT is involved somewhere on the path from initiator to responder. When traversing NAT it is recommended to set [cfgVpnWgPPersistentKeepalive](#) to some low value to ensure traffic can flow in both directions. A sane value is 25s.

Finally [cfgVpnWgPAllowedIps](#) is used to specify which subnets are allowed to be received through the tunnel, and which destinations to send into the tunnel.

The remainig entries of the WireGuard instance configuration may be left with their default values.

4.23 Wireless Network Access Point

For an initial configuration of an Access Point which is used as public wireless network AP configure at least the following items:

- SSID ([cfgWlanIfaceSsid](#)) - example: "wpwnap"
- Wireless password ([cfgWlanIfacePassword](#))
- Wireless mode ([cfgWlanDevModulation](#)) - example: 2.4 GHZ = ng(10), 5 GHz = na(12)
- Wireless bandwidth ([cfgWlanDevBandwidth](#)) - example: ht20(0) or ht40Plus(1)
- Optionally set system hostname ([cfgSysHostname](#)) - example: "AP2G" or "AP5G-HT40+"

4.23.1 Configuration File Example: Access point 2.4GHz

Configuration File Example: Access point 2.4GHz

```
# Config.Format = raw
WESTERMO-SW6-MIB::cfgSysHostname.0 = AP2G
WESTERMO-SW6-MIB::cfgWlanIfaceSsid.0 = wpwnap
```

4.23.2 Configuration File Example: Access point 5GHz

Configuration File Example: Access point 5GHz HT40+


```
# Config.Format = raw
WESTERMO-SW6-MIB::cfgSysHostname.0 = AP5G-HT40+
WESTERMO-SW6-MIB::cfgWlanDevModulation.0 = 12
WESTERMO-SW6-MIB::cfgWlanDevBandwidth.0 = 1
WESTERMO-SW6-MIB::cfgWlanDevFrequency.0 = 5500
WESTERMO-SW6-MIB::cfgWlanIfaceScanList.0 = 1
```

For a more sophisticated public wireless network AP configuration (i.e. WiFi Hotspot) please refer to [Public Wireless Network \(PWN\)](#).

4.24 Wireless Data Rate Control

Wireless technology supports a wide range of bitrates for optimal throughput in any environment. The bitrate depends mainly on the signal level seen by the wireless receiver.

Usually the best bitrate for the actual environment is evaluated and chosen by the rate controller. For some applications, especially with fast changing environments, (e.g. mobility applications) the task to find the optimal bitrate must be optimized. *Software 6* supports fine grade optimization for bitrates.

4.24.1 Reduced Set of Bitrates

With [cfgWlanIfaceBitrates](#) and [cfgWlanDevHtCapabilities](#) the number of possible bitrates can be reduced to allow the rate controller for faster adaption to a changing environment.

4.24.2 QMRR

QMRR supports Multi Rate Retry per wireless queue (VO, VI, BK, BE) and is configured with [cfgWlanDevQmrrString](#).

QMRR logging into Syslog can be enabled by [cfgWlanGlbLinkmonitorQmrrlogging](#) and [cfgWlanGlbLinkmonitorInterval](#)

Format: QMRR|<kernel time>|<queue>|<mode>|<guard>|<rate>|<success>|<attempts>

- mode: HT20|HT40 (there can be further modes like CCK)
- guard: LGI|SGI (there can be further guards like SP)
- rate: MCS0, MCS1, ..., MCS31 (there can be further rates like 1.0M)

4.25 802.11s Mesh

802.11s Mesh is an official IEEE standard for Wireless Mesh operation. Contrary to other proprietary Mesh implementations it is open and allows interoperability with other vendors which are 802.11s compliant. It allows to connect multiple devices without the need for an Access Point. Each node is a Mesh Point and transmits its own Mesh beacons. When two Mesh Points see each others beacons they automatically associate with each other and exchange keys when configured for encryption.

To configure a Mesh Point set [cfgWlanifaceMode](#) to 3. Mesh only supports two types of encryption (see [cfgWlanifaceEncryption](#)): open(0) (no encryption) and sae(7).

When a Mesh Points starts up, it takes the frequency list specified by [cfgWlanifaceAcsList](#) and scans these frequencies for the configured Mesh SSID ([cfgWlanifaceBssid](#)). If it finds an already existing Mesh on any of the scanned frequencies, it joins it. Should it not find an already existing one, it startes beaconing a new Mesh on the frequency specified in [cfgWlanDevFrequency](#).

802.11s Mesh has 6 address fields in its header. Because of this it is possible to bridge a Mesh interface.

A Mesh can only work on a single frequency at the same time, thus operation on DFS frequencies is not recommended.

4.26 Bridge Mode (4addr)

Bridge mode or WDS (Wireless Distribution System) mode is a non-standard extension to the wireless 802.11 standard using a 4-address-format to allow transparent Ethernet bridging on the client (STA).

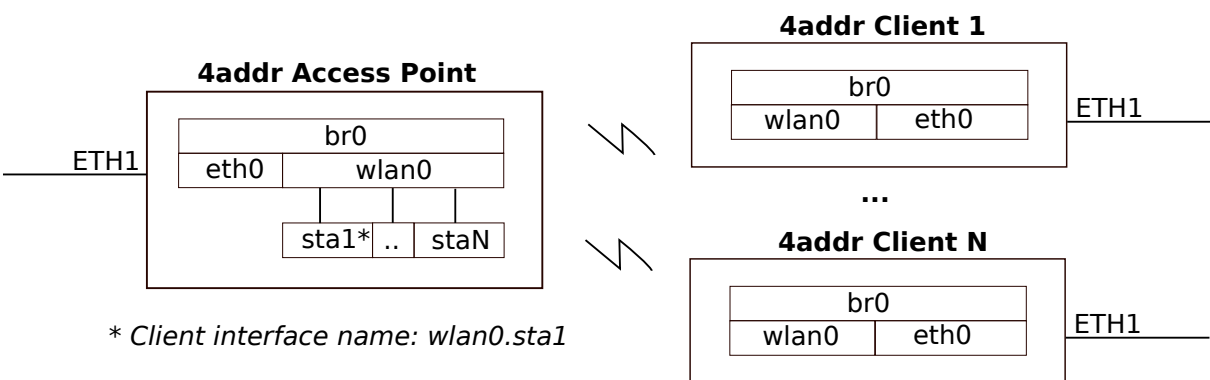


Figure 4.10: 4addr bridge mode setup

Configuration File Example: Access point

```
# Config.Format = raw
```

```
WESTERMO-SW6-MIB::cfgWlanIface4addr.0 = 1
```

Configuration File Example: STA/client

```
# Config.Format = raw
WESTERMO-SW6-MIB::cfgNetIpAddr.0 = 192.168.1.30/24
WESTERMO-SW6-MIB::cfgWlanIfaceMode.0 = 1
WESTERMO-SW6-MIB::cfgWlanIface4addr.0 = 1
```



Important: [Layer 2 NAT Mode](#) instead of Bridge Mode (4addr) is recommended in combination with [Mobility](#).

4.27 Layer 2 NAT Mode

As the 4addr bridge mode (section [Bridge Mode \(4addr\)](#)), the Layer 2 NAT mode allows transparent IP bridging on the client (STA).



Important: This only works for IP traffic and not for generic L2 frames.

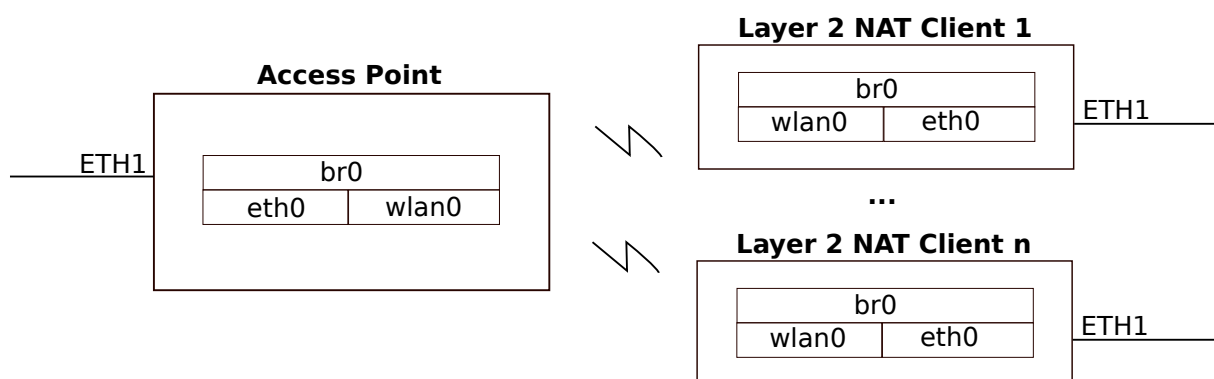


Figure 4.11: *Layer 2 NAT bridge mode setup*

There is no special Layer 2 NAT configuration necessary on the AP side. For the client (STA) the configuration is as following.

Configuration File Example: STA/client

```
# Config.Format = raw
WESTERMO-SW6-MIB::cfgNetIpAddr.0 = 192.168.1.30/24
WESTERMO-SW6-MIB::cfgWlanInterfaceMode.0 = 1
WESTERMO-SW6-MIB::cfgWlanInterfaceL2nat.0 = 1
```

The Layer 2 NAT mode is recommended if layer 3 (IP) transparency is required in combination with [Mobility](#) (fast handover). With Layer 2 NAT the same handover performance is achieved as with client (STA) in routing mode, which is the default mode. Whereas with Bridge Mode (4addr) the handover performance is about 10 times slower. Therefore, if Layer 3 transparency and fast handover are required, it is recommended to use Layer 2 NAT instead of 4addr bridge mode.



Important: Layer 2 NAT bridging mode cannot be used in combination with DHCP. Devices on the Ethernet side of the client (STA) cannot obtain DHCP leases.

4.28 Wireless MAC Address Overwrite

It is possible to overwrite the MAC address of the wireless interface in Access Point and client (STA) mode.

This feature might be useful in a mobility application with several Access Point along the track. The MAC Address could encode for example the index of the Access Point which makes it easier to configure static neighbour lists.

Another use case might be for applications where the MAC address of the client (STA) shall be the same as of the device attached to the modem. The user can configure the MAC address of the attached wired equipment as the source MAC address of the wireless interface. The cloned address is then used for all wireless communication.

Configuration File Example: Configure new MAC for the wireless radio

```
# Config.Format = raw
WESTERMO-SW6-MIB::cfgWlanInterfaceBssid.0 = STRING: 00:11:22:33:44:55
```

4.29 Wireless Security

4.29.1 WPA Encryption

The WPA (Wifi Protected Access) security standard is implemented according to IEEE 802.11i. It uses CCMP (Counter Mode Cipher Block Chaining Message Authentication Code Protocol) for message confidentiality and integrity. CCMP is an enhanced data cryptographic encapsulation mechanism designed for data confidentiality and based upon the Counter Mode with CBC-MAC (CCM mode) of the Advanced Encryption Standard (AES) standard.

WPA Personal uses shared passwords and Enterprise is based on certificates (TLS) and passwords when used with TTLS or PEAP.

WPA3 Personal and Enterprise are new Wi-Fi security standards developed by WifiAlliance. WPA3 Personal introduces SAE (Simultaneous Authentication of Equals), which is an authentication protocol with stronger security protections against attacks such as offline dictionary attacks, key recovery, and message forging. WPA3 Enterprise is equivalent of WPA2 with strongest ciphers used. The WPA3 Personal Transition Mode allows both SAE and WPA2 PSK. It can be used in scenarios where not all the clients support SAE.

Only WPA2/IEEE 802.11i is supported. WPA/IEEE 802.11i/D3.0 which uses TKIP encryption protocol is considered not secure and is not available.

Wi-Fi Enhanced Open introduces Opportunistic Wireless Encryption (OWE). It provides encryption and privacy on open, non-password-protected networks in areas.

The following configuration items are supported:

- [cfgWlanfaceEncryption](#) to configure the wireless encryption mode. Available modes are open(0), psk(3) for WPA2 Personal, eap(6) for WPA2/3 Enterprise, sae(7) for WPA3 Personal, owe(8) for Opportunistic Wireless Encryption and saepsk(9) for WPA3 Personal Transition Mode
- [cfgWlanfacePassword](#) to configure the WPA2 or WPA3 Personal passphrase (pre-shared key)

To further increase security, the Management Frame Protection (specified in IEEE 802.11w) can be enabled (see section [Management frame protection \(MFP, 802.11w\)](#)). MFP is mandatory for WPA3 Personal and Enterprise.

Encryption mode	cfgWlanIfaceEncryption	cfgWlan802dot1xCiphers
OPEN	open(0)	none
WPA2 Personal (PSK)	psk(3)	none
WPA3 Personal (SAE)	sae(7)	none
WPA2 Enterprise	eap(6)	none
WPA3 Enterprise	eap(6)	ECDHE-RSA-AES256-GCM-SHA384 ECDHE-ECDSA-AES256-GCM-SHA384 ECDHE-RSA-AES256-SHA384
Wi-Fi OWE	owe(8)	none
WPA3 Transition Mode	saepsk(9)	none

Table 4.5: Overview of the encryption modes

4.29.1.1 Overview of the encryption modes

4.29.2 Port-based Network Access Control (802.1X)

IEEE 802.1X is an IEEE Standard for port-based Network Access Control (PNAC). It provides an authentication and authorization mechanism to devices wishing to attach to a LAN or WLAN.

IEEE 802.1X authentication involves three parties:

supplicant client device that wishes to attach to the LAN or WLAN (STA)

authenticator network device such as an Ethernet switch or wireless access point through which the supplicant is connected to the network (AP)

authentication server a host running software supporting the RADIUS and EAP protocols (AS)

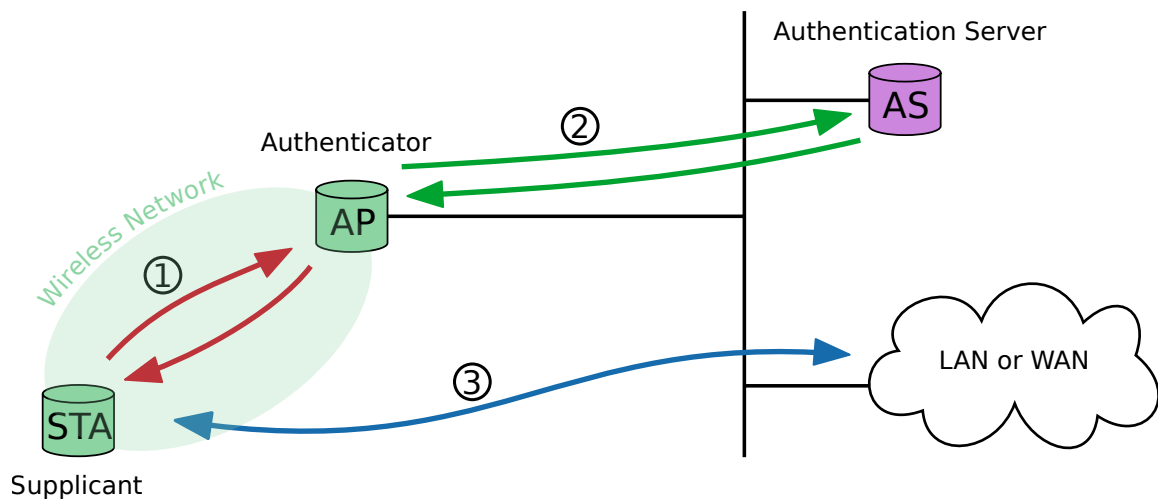


Figure 4.12: Supplicant authentication using IEEE 802.1X

The supplicant is not allowed to access through the authenticator to the protected side of the network ③ until the supplicant's identity has been validated and authorized. The authenticator acts like a security guard to a protected network. With IEEE 802.1X port-based authentication, the supplicant provides credentials to the authenticator ①. Accepted credentials can be user name/password or a digital certificate. The authenticator forwards the credentials to the authentication server for verification ②. If the authentication server determines the credentials as valid, the supplicant (client device) is allowed to access resources located on the protected side of the network.

The following example shows how to configure SW6 devices as supplicant (STA) and authenticator (AP).

Requirements

- Configured authentication server¹
- PKI to generate digital certificates, valid client certificate files
- Time server to share time information between the hosts in the network

Wireless access point configuration

The following configuration example configures a device as authenticator:

Configuration File Example: authenticator

```
# Config.Format = raw
```

¹FreeRADIUS 4 is used for internal testing, find the project under <http://freeradius.org/> (February 2017)

```
# basic AP configuration
WESTERMO-SW6-MIB::cfgSysTimezone.0 = CET-1CEST,M3.5.0,M10.5.0/3
WESTERMO-SW6-MIB::cfgNetIpAddr.0 = 192.168.3.22/24
WESTERMO-SW6-MIB::cfgWlanDevModulation.0 = 12
WESTERMO-SW6-MIB::cfgWlanDevFrequency.0 = 5180
WESTERMO-SW6-MIB::cfgWlanIfaceSsid.0 = radiustesting
WESTERMO-SW6-MIB::cfgWlanIfaceEncryption.0 = 6
WESTERMO-SW6-MIB::cfgWlanGlblCountry.0 = EU
WESTERMO-SW6-MIB::cfgNtpServer1.0 = 192.168.2.2
# IEEE 802.1X related configuration
WESTERMO-SW6-MIB::cfgWlan802dot1xAuthSrvEnabled.0 = 1
WESTERMO-SW6-MIB::cfgWlan802dot1xAuthSrvIpAddr.0 = 192.168.3.2
WESTERMO-SW6-MIB::cfgWlan802dot1xAuthSrvSharedSecret.0 = superSharedSecret
```

The authenticator does not need any certificate information. The shared secret 'superSharedSecret' has to match the authentication server client configuration.

The authenticator will forward incoming client authentication requests to the primary authentication server address (192.168.3.2). More than one authentication server address can be defined by adding more records to the authentication server configuration table.

Configuration File Example: authenticator, second authentication server address

```
# Config.Format = raw
# basic AP configuration
WESTERMO-SW6-MIB::cfgWlan802dot1xAuthSrvEnabled.1 = 1
WESTERMO-SW6-MIB::cfgWlan802dot1xAuthSrvIpAddr.1 = 192.168.3.2
WESTERMO-SW6-MIB::cfgWlan802dot1xAuthSrvSharedSecret.1 = superSharedSecret
```

If the primary authentication server is not reachable, an exponential back off is implemented:

- timeout of first attempt is 3 seconds
- after each failure the timeout is increased to maximum timeout of 120 seconds
- the authenticator gives up after 10 attempts
- after the fourth attempt, a backup server (secondary, if configured) will be attempted

If none of the configured authentication server is reachable, the supplication will not be authorized to communicate over the access point (authenticator) - no communication is possible.

Wireless station configuration

If encrypted private keys are used for the client, the pass phrase to unlock the key has to be configured before the private key upload. If no matching pass phrase is provided, the client private key file will be rejected by the device.

Use the MIB element [cfgWlan802dot1xClientKeyPassword](#) to define the pass phrase and [rpcCfgApply](#) to apply the changes.

If the private key is not locked by a pass phrase (unencrypted), this configuration parameter is ignored.

Use the web interface to upload the certificate files: *System -> Certificate Manager -> 802.1X*. Each file is verified before it is stored on the device. The mandatory information can be provided in this files:

client.crt client certificate (public key, PEM or X509 format)

client.key client private key matching the client X509 certificate (RSA)

ca.crt authentication server X509 certificate (public key, PEM or X509 format)

The client private key file may also contain the full CA certificate chain information in PEM format: it can contain client private key and CA certificate chain information (one or multiple certificates) in one single file. Make sure that in this case no separate client certificate file is provided.

Alternatively SNMP can be used to upload the mandatory certificate information to the supplicant. The MIB element [setCrtFileSelector](#) is used to define what kind of file should be uploaded and for which interface it should be used. The following file classes are defined:

10x class 100 is used for client certificate files where x is the interface index

20x class 200 is used for client private key files where x is the interface index

30x class 300 is used for CA certificate files where x is the interface index

Client certificate file upload via TFTP:

- WESTERMO-SW6-MIB::[setCrtFileSelector](#).0 100
- WESTERMO-SW6-MIB::[setCrtFileUrl](#).0 tftp://192.168.2.2/client.crt
- WESTERMO-SW6-MIB::[rpcCrtFile](#).0 1

Client private key file upload via TFTP:

- WESTERMO-SW6-MIB::[setCrtFileSelector](#).0 200
- WESTERMO-SW6-MIB::[setCrtFileUrl](#).0 tftp://192.168.2.2/client.key
- WESTERMO-SW6-MIB::[rpcCrtFile](#).0 1

CA certificate file upload via TFTP:

- WESTERMO-SW6-MIB::[setCrtFileSelector.0](#) 300
- WESTERMO-SW6-MIB::[setCrtFileUrl.0](#) tftp://192.168.2.2/ca.crt
- WESTERMO-SW6-MIB::[rpcCrtFile.0](#) 1

The following configuration example configures a device as supplicant.

Configuration File Example: supplicant

```
# Config.Format = raw
# basic STA configuration
WESTERMO-SW6-MIB::cfgSysTimezone.0 = CET-1CEST,M3.5.0,M10.5.0/3
WESTERMO-SW6-MIB::cfgNetWlanBridge.0 = -1
WESTERMO-SW6-MIB::cfgNetIpEnabled.1 = 1
WESTERMO-SW6-MIB::cfgNetIpAddr.0 = 192.168.2.11/24
WESTERMO-SW6-MIB::cfgNetIpAddr.1 = 192.168.3.11/24
WESTERMO-SW6-MIB::cfgWlanDevModulation.0 = 12
WESTERMO-SW6-MIB::cfgWlanIfaceMode.0 = 1
WESTERMO-SW6-MIB::cfgWlanIfaceSsid.0 = radiustesting
WESTERMO-SW6-MIB::cfgWlanIfaceEncryption.0 = 6
WESTERMO-SW6-MIB::cfgWlanIfaceScanList.0 = 2
WESTERMO-SW6-MIB::cfgWlanGlblCountry.0 = EU
WESTERMO-SW6-MIB::cfgNtpServer1.0 = 192.168.2.2
# IEEE 802.1X related configuration
WESTERMO-SW6-MIB::cfgWlan802dot1xClientKeyPassword.0 = superKeySecret
```

If no certificates are uploaded before applying WPA2-EAP as encryption mode ([cfgWlanIfaceEncryption](#)), the configuration apply will fail. NTP has to be enabled on the station (time sync on the access point is optional), make sure that the station shares the same time information and time zone configuration as the authentication server. This is later important for the successful certificate validation.

After applying the described configuration on supplicant (STA) and authenticator (AP), the STA should be successfully authorized. The STA authorization status can be verified using the STA's web interface: (*Status -> Wireless Connections -> Station Dump*).

4.29.3 Ciphers

Supported authentication and encryption algorithms (cipher suites).

The TLSv1.2 cipher suites are supported and recommended to use:

ADH-AES128-GCM-SHA256	ADH-AES128-SHA256
ADH-AES256-GCM-SHA384	ADH-AES256-SHA256
AES128-CCM	AES128-CCM8
AES128-GCM-SHA256	AES128-SHA256
AES256-CCM	AES256-CCM8
AES256-GCM-SHA384	AES256-SHA256
DHE-DSS-AES128-GCM-SHA256	DHE-DSS-AES128-SHA256
DHE-DSS-AES256-GCM-SHA384	DHE-DSS-AES256-SHA256
DHE-PSK-AES128-CCM	DHE-PSK-AES128-CCM8
DHE-PSK-AES128-GCM-SHA256	DHE-PSK-AES256-CCM
DHE-PSK-AES256-CCM8	DHE-PSK-AES256-GCM-SHA384
DHE-PSK-CHACHA20-POLY1305	DHE-RSA-AES128-CCM
DHE-RSA-AES128-CCM8	DHE-RSA-AES128-GCM-SHA256
DHE-RSA-AES128-SHA256	DHE-RSA-AES256-CCM
DHE-RSA-AES256-CCM8	DHE-RSA-AES256-GCM-SHA384
DHE-RSA-AES256-SHA256	DHE-RSA-CHACHA20-POLY1305
ECDHE-ECDSA-AES128-CCM	ECDHE-ECDSA-AES128-CCM8
ECDHE-ECDSA-AES128-GCM-SHA256	ECDHE-ECDSA-AES128-SHA256
ECDHE-ECDSA-AES256-CCM	ECDHE-ECDSA-AES256-CCM8
ECDHE-ECDSA-AES256-GCM-SHA384	ECDHE-ECDSA-AES256-SHA384
ECDHE-ECDSA-CHACHA20-POLY1305	ECDHE-PSK-CHACHA20-POLY1305
ECDHE-RSA-AES128-GCM-SHA256	ECDHE-RSA-AES128-SHA256
ECDHE-RSA-AES256-GCM-SHA384	ECDHE-RSA-AES256-SHA384
ECDHE-RSA-CHACHA20-POLY1305	PSK-AES128-CCM
PSK-AES128-CCM8	PSK-AES128-GCM-SHA256
PSK-AES256-CCM	PSK-AES256-CCM8
PSK-AES256-GCM-SHA384	PSK-CHACHA20-POLY1305
RSA-PSK-AES128-GCM-SHA256	RSA-PSK-AES256-GCM-SHA384
RSA-PSK-CHACHA20-POLY1305	TLS_AES_128_GCM_SHA256
TLS_AES_256_GCM_SHA384	TLS_CHACHA20_POLY1305_SHA256

The cipher string format explained for ECDHE-RSA-AES256-GCM-SHA384:

ECDHE key exchange algorithm, Elliptic Curve Diffie-Hellman

RSA authentication algorithm, Rivest-Shamir-Adleman

AES256-GCM cipher algorithm, Advanced Encryption Standard with strength and mode (Galois Counter Mode)

SHA384 MAC or PRF algorithm, Secure Hash Algorithm with strength

The list of supported cipher suites may change over time. New cipher suites will be added, weak cipher suites will be removed. Following cipher suites are recommended and are maintained over long term:

- ECDHE-RSA-AES128-GCM-SHA256

- ECDHE-RSA-AES256-GCM-SHA384
- DHE-RSA-AES128-GCM-SHA256
- DHE-RSA-AES256-GCM-SHA384
- ECDHE-RSA-AES256-SHA384

With `cfgWlan802dot1xCiphers` the cipher suite(s) to be used can be configured according to the syntax of OpenSSL.

Examples

- Define cipher suites with RSA key exchange or authentication:
`cfgWlan802dot1xCiphers.0 = RSA`
- Define a single cipher suite:
`cfgWlan802dot1xCiphers.0 = DHE-RSA-AES256-GCM-SHA384`
- Provide an ordered list of cipher suites:
`cfgWlan802dot1xCiphers.0 = DHE-RSA-AES256-GCM-SHA384:RSA-AES-128GCM-SHA256`
- Provide a list of cipher suites which is to sort in order of encryption algorithm key length:
`cfgWlan802dot1xCiphers.0 = DHE-RSA-AES256-GCM-SHA384:RSA-AES-128GCM-SHA256:@STRENGTH`

The OpenSSL tool *ciphers* can be used as a test tool to determine the appropriate cipherlist (e.g. *openssl ciphers RSA*)

Please note the the configuration will fail to apply if the value set for `cfgWlan802dot1xCiphers` provides an empty list (no cipher suite).

4.29.4 Management frame protection (MFP, 802.11w)

The management frame protection (MFP) protects the device from denial of service attacks. By default, the management frames are not protected from being used to attack the device. The MFP 802.11w adds a field in the frame to authenticate the frame sender. If the device receives a management frame from an incorrect sender, it will discard the frame.

The following configuration items are supported:

- `cfgWlanInterface80211w` to enable the Management Frame Protection (MFP) mechanism.
- `cfgWlanInterface80211wMaxTimeout` to configure the SA query max timeout
- `cfgWlanInterface80211wRetryTimeout` to configure the SA query retry timeout

4.30 IP Routing

The devices not only provide switch functionality but are also able to route data packages.

4.30.1 Static Routing

Using static routing the devices can specify the next hop router to use to reach a given IP subnet, or add additional (directly attached) subnets to a local interface.

The example below shows a static route to the 192.168.11.0/24 subnet using 192.168.1.2 as gateway. A detailed description of all routing related SNMP commands can be found in [cfgRouting](#).

Configuration File Example: Static route

```
# Config.Format = raw
WESTERMO-SW6-MIB::cfgRouteTableEnabled.0 = 1
WESTERMO-SW6-MIB::cfgRouteTableDestinationNetwork.0 = 192.168.11.0/24
WESTERMO-SW6-MIB::cfgRouteTableGateway.0 = 192.168.1.2
WESTERMO-SW6-MIB::cfgRouteTableSource.0 = 192.168.1.20
```

4.31 VLAN

The *Software 6* has built in capability for virtual LANs (VLAN). The devices can be easily integrated into existing network environments where VLANs are in use.

The VLAN configuration is located under [cfgNetVlanTable](#).

Untagged frames are internally handled by using VID 0.

4.31.1 Multi SSID and VLAN

The devices support multiple SSIDs on a single radio. It can broadcast up to 64 wireless networks with different names (i.e SSIDs). When using Multi SSID, users could also assign different VLAN ID to different wireless network. This makes it possible to get a device work with switches which has VLAN assigned for different access level and authority.

In the following example the Ethernet interfaces (eth0 and eth1) are configured so that either eth0 or eth1 are physically connected, but not both at the same time. The device shall be accessible via Ethernet (eth0 or eth1) for administrative purposes.

The process of configuring VLANs to separate SSIDs and making these networks accessible via Ethernet is as follows:

1. Setup 1st Ethernet interface (eth0) in VLAN 8, 103 and 203 and add it to the bridge (br0)
2. Setup redundant 2nd Ethernet interface (eth1)
3. Setup WLAN physical interface: frequency, power etc.
4. Setup 1st WLAN interface (wlan0) in VLAN 103 and add it to the bridge (br0)
5. Setup 2nd WLAN interface (wlan1) in VLAN 203 and add it to the bridge (br0)
6. Setup Administrative VLAN (VID 8)
7. Setup IP for Administrative VLAN (VID 8)

An example configuration of this kind is shown in Figure 4.13 below.

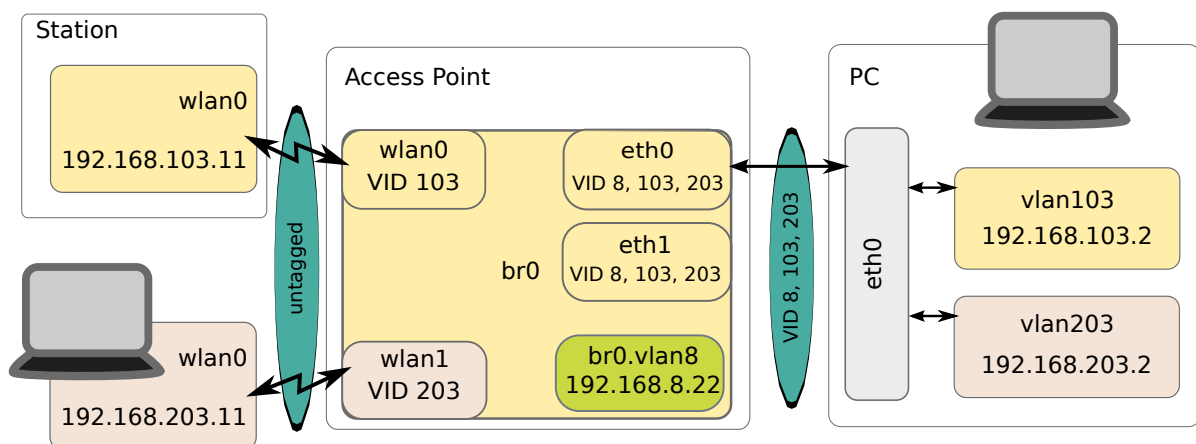


Figure 4.13: Multi SSID and VLAN example setup

Configuration File Example: Multiple SSID and VLAN

```
# Config.Format = raw
# 1. Setup 1st ethernet interfaces(eth0):
WESTERMO-SW6-MIB::cfgNetEthTrunk.0 = 8,103,203
# 2. Setup 2nd (redundant) ethernet interface (eth1):
WESTERMO-SW6-MIB::cfgNetEthTrunk.1 = 8,103,203
# 4. Setup 1st WLAN interface (wlan0, VID 103):
WESTERMO-SW6-MIB::cfgNetWlanTag.0 = 103
WESTERMO-SW6-MIB::cfgNetWlanVlanMode.0 = 1
WESTERMO-SW6-MIB::cfgWlanIfaceSsid.0 = ssid103
```

```
# 5. Setup 2nd WLAN interface (wlan1, VID 203):  
WESTERMO-SW6-MIB::cfgNetWlanEnabled.1 = 1  
WESTERMO-SW6-MIB::cfgNetWlanBridge.1 = 0  
WESTERMO-SW6-MIB::cfgNetWlanTag.1 = 203  
WESTERMO-SW6-MIB::cfgNetWlanVlanMode.1 = 1  
WESTERMO-SW6-MIB::cfgWlanIfaceDevice.1 = 0  
WESTERMO-SW6-MIB::cfgWlanIfaceSsid.1 = ssid203  
# 6. Setup Administrator VLAN (VID 8):  
WESTERMO-SW6-MIB::cfgNetVlanVid.0 = 8  
# 7. Setup IP for Administrative VLAN:  
WESTERMO-SW6-MIB::cfgNetIpAddr.0 = 192.168.8.22/24  
WESTERMO-SW6-MIB::cfgNetIpInterface.0 = br0.vlan8
```

4.32 Mobility

4.32.1 Fast Association

Fast Association bundles the optimization features which allows the client (STA) to find and connect to the best Access Point as fast as possible.

When the client (STA) is not connected it scans for available Access Point with the same SSID and then connects to the Access Point with the best RSSI level (Signal). The scan and re-connect process works as follows:

1. Loop through all frequencies defined in the [cfgWlanIfaceScanList](#) and
 - a) tune the radio to a frequency on the [cfgWlanIfaceScanList](#)
 - b) observe the channel for a few milliseconds. This is necessary to update the NAV (network allocation vector)
 - c) send a Probe Request frame with SSID specified
 - d) collect all Probe Responses for within a time window of a few milliseconds
2. Select from the Access Point with the same [cfgWlanIfaceSsid](#) the one with the best RSSI level (Signal)
3. Connect to the selected Access Point

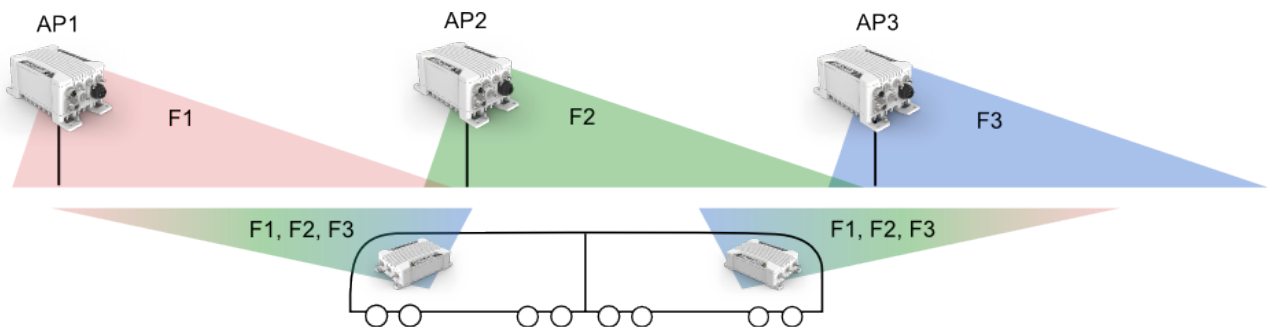
Once the client (STA) successfully connected, the client observes the Beacon frames received from its Access Point. As long as the filtered RSSI level of the Beacon frames is above a configurable threshold ([cfgWlanHoScanningLevel](#)), the client stays connected. If the RSSI level is below the

threshold or if too many Beacon frames are lost (`cfgWlanInterfaceBeaconMiss`), the client will disconnect and scan for a better Access Point.

Tip: For applications where the access points are located further apart so that no continuous coverage can be achieved, the `cfgWlanHoScanningLevel` should be set to 0 so that the client is always connected to the access point as long as there is one available.

4.32.2 Inter-AP Roaming

Inter-AP Roaming assumes continuous coverage. Meaning that on the edge of an Access Point coverage area there must be always certain overlapping with the next Access Point's coverage area. In a typical mobile to ground application, on ground (trackside or infrastructure) the device is configured in Access Point mode where the mobile device is in client (STA) mode.



Once the client is connected to an Access Point, it receives a list of neighbor Access Points as possible roaming candidates. The neighbor list contains the current operating frequency and BSSID of the neighbor Access Points. On the edge of the coverage area of the currently associated Access Point, the client initiates the scanning process to search for the next Access Point. Depending on the configured `cfgWlanHoProfile` the client scans for new Access Points while being still connected (background scanning) or it disconnects before starting to scan (foreground scanning). Which profile to use depends on the actual needs of the application. The scan and re-connect process works as follows:

1. Loop through the list of neighbor Access Points received from the Access Point, sorted by frequency
 - a) tune the radio to the current frequency
 - b) observe the channel for a few milliseconds. This is necessary to update the NAV (network allocation vector)
 - c) send a directed Probe Request frame to all neighbor BSSIDs on the current frequency
 - d) collect all Probe Responses

2. Select from the neighbor Access Points the one with the best RSSI level (Signal)
3. Connect to the selected Access Point

As a fall-back, if no neighbor list is available or no neighbour is a valid roaming candidate, the client uses the same scan and re-connect process as described in [Fast Association](#).

The client (STA) decides based on the received RSSI (Signal) and Distance of its Access Point when to roam. Either RSSI or Distance can initiate a handover independently. The RSSI is recorder for all Beacon frames received from the AP while the distance is periodically measured by ranging frames (see [cfgWlanHoDistanceMeasurementPeriod](#)). The low / high thresholds for the RSSI level and the near / far thresholds for the Distance can be configured on client and additionally on the Access Point. If the thresholds are configured on the Access Point these values are applied and used by the client for the current Access Point. This allows to tune the optimal roaming point for the coverage area of each Access Point cell separately.

The list of neighbors must be configured on each Access Point (see [cfgWlanNeighbourTable](#)) and the neighbour reports must be enabled (see [cfgWlanIfaceNeighbourReport](#)) when the so called Static Neighbour List (SNL) is configured. Inter-AP Roaming supports also Dynamic Neighbor Lists (DNL). DNL is required as soon as the operating frequency of the Access Points is not fix. This is the case if the Access Point changes its frequency due to a radar event (DFS) or due to interferences. Please refer to [Area Frequency Management \(AFM\)](#) for more information.

4.32.3 Handoff Filters

The client (STA) decides when to roam based on Beacon RSSI of its Access Point or based on Distance measurements to its Access Point. Filters are applied to both, RSSI and Distance, before comparing with the thresholds.

For Beacon RSSI the filter is configured on the client (STA) with [cfgWlanHoFilterLongX](#) and [cfgWlanHoFilterLongY](#). In combination with the [cfgWlanIfaceBeaconInterval](#), which is configured on the Access Point, the RSSI filter characteristic can be optimally tuned for the application.

The filtered RSSI is calculated through a IIR low pass filter:

$$RSSI_{AVG} = \frac{1}{X + Y} (X * RSSI + Y * RSSI_{AVG}) \quad (4.1)$$

For Distance measurements the filter is configured on the STA with [cfgWlanHoDistanceFilterX](#) and [cfgWlanHoDistanceFilterY](#). In combination with the [cfgWlanHoDistanceMeasurementPeriod](#), which is configured on the STA, the Distance filter characteristic can be optimally tuned for the application.

The filtered Distance is calculated through a IIR low pass filter, the measurements are pre-filtered by

a plausibility check:

$$Distance_{AVG} = \frac{1}{X + Y} (X * Distance_{AVG} + Y * Distance) \quad (4.2)$$

4.32.4 Mobility Logging

For debugging purposes and verification of the system setup it is very important to have the possibility to log signal levels, distance measurements and the roaming process. To serve these needs the client (STA) provides several handoff debug flags (see [cfgWlanDbgTable](#) and [setWlanDbgTable](#)) which allows to log received RSSI for each beacon, distance information for each ranging frame and handoff information in the Syslog (see [Logging Features](#)).

4.32.4.1 Important Messages for Handoff

Key	Format	Syslog	SNMP Trap
cfgWlanDbgHandoff	Handoff (Message Code: [434])	yes	yes
cfgWlanDbgBeaconrssi	RSSI (Message Code: [430])	yes	no
cfgWlanDbgBeaconfiltered	Filtered RSSI (Message Code: [431])	yes	no
cfgWlanDbgReports	Reports	yes	no
cfgWlanDbgRange	Distance	yes	no

4.32.4.1.1 Handoff

Format:

```
NOTICE 434 <iface> Handoff:<reason>|<count>|<prev bssid>|<cur bssid>|<ssid
↳ mgmt>|<ho\_time>|offchan_scan_time|
```

- **|Detection Reason Code|:**

- 0 No reason (after restart)
- 2 Low RSSI
- 3 Beacon miss (connection loss)
- 4 Low Ack
- 5 High RSSI

6 Near Distance

7 Far Distance

- **|count|**: Number of handoffs since boot up
- **|prev_bssid|**: BSSID (MAC) of the previous access point
- **|cur_bssid|**: BSSID (MAC) of the current access point
- **|ssid_mgmt|**: Reserved
- **|ho_time|**: Handoff time measured between the disassociated and associated in milliseconds.
- **|offchan_scan_time|**: Handoff time measured for offchannel scanning.

4.32.4.1.2 RSSI

Format:

```
NOTICE 430 TS|<uptime>|<bssid>|RSSI_BCN|<rssi>
```

The RSSI is shown for all wireless Beacons received by the radio.

4.32.4.1.3 Filtered RSSI

Format:

```
NOTICE 431 TS|<uptime>|<bssid>|RSSI_BCN|<rssi>|<cur_filer_rssi>|<  
↳ short_filter_rssi>|<long_filter_rssi>
```

The Filtered RSSI is shown only for the wireless Beacons of the Access Point the client(STA) is connected to.

4.32.4.1.4 Reports

The Reports ([cfgWlanDbgReports](#)) are dumped in Syslog (three lines for each report) and contains information about RSSI levels and other counters. The interval between the Reports depends on [cfgWlanIfaceBeaconInterval](#)

Format:

```
REP0|<uptime>|<mac>|<completed>|<sretries>|<lretries>|<xretries>
REP1|<uptime>|<mac>|<acompleted>|<aretries>|<axretries>|<expthruput>
REP2|<uptime>|<mac>|<cnt>|<seq>|<rssi>|<rssi0>|<rssi1>|<snr>|<crc>|<precrc
↳>|<postcrc>
```

Example:

```
REP0|42351592|00:14:5a:03:49:2e|5809|82838|7185|16
REP1|42351592|00:14:5a:03:49:2e|3930011|1372|495|34312
REP2|42351592|00:14:5a:03:49:2e|139746|2107|32|32|0|31|0|0|0
```

Please contact support for more information about reports.

4.32.4.1.5 Distance

The Distance measurements ([cfgWlanDbgRange](#)) are dumped in Syslog and contains distance information (raw and filtered). The interval between the Reports depends on [cfgWlanHoDistanceMeasurementPeriod](#).

Format:

```
DISTANCE|<uptime>|<mac>|<raw>|<filtered>
```

Please contact support for more information about Distance measurements.

4.32.5 Fast BSS Transition (802.11r)

IEEE 802.11r is an amendment to the IEEE 802.11 standard. Fast BSS transition (FT) (802.11r) allows continuous connectivity of wireless devices in motion with fast and secure handoff. It is working for wireless devices in the same Mobility Domain (MD).

If using 802.1X, 802.11r provides a fast and still secure handoff within an MD. Whereas if using 802.1X without 802.11r, the handoff is secure but not fast.

For a simple configuration of the R0- and R1-Key Holder List (R0KH-/R1KH-list), use wildcard entries as given in the following example configuration. The wildcard entry in the R0-KH-list means that all APs of the same MD are allowed as R0-Key Holder. And the wildcard entry in the R1-KH List means, that every AP of the same MD is allowed to request an R1-Key.

4.32.5.1 AP 802.11r Configuration

When not using wildcard entries for R0KH-/R1KH-list, then each AP of an MD needs to have one entry for all of the other APs in the same MD. See [cfgWlan802dot11rR0KHTable](#) and [cfgWlan802dot11rR1KHTable](#) for some more description.

The following configuration example configures an AP for using FT, and using wildcard entry for R0- and R1-Key Holder List:

Configuration File Example: FT AP

```
# Config.Format = raw
# basic 802.11r configuration
WESTERMO-SW6-MIB::cfgWlan802dot11rEnabled.0 = 1
WESTERMO-SW6-MIB::cfgWlan802dot11rMobilityDomain.0 = a1b2
WESTERMO-SW6-MIB::cfgWlan802dot11rPmkR0Lifetime.0 = 10000
WESTERMO-SW6-MIB::cfgWlan802dot11rPmkR1KeyHolderIdentifier.0 = 000102030405 # i.e. use
    ↳ own MAC
WESTERMO-SW6-MIB::cfgWlan802dot11rR0KHParameter.0 = 0
WESTERMO-SW6-MIB::cfgWlan802dot11rR1KHParameter.0 = 0
# 802.11r R0 key holder list wildcard entry
WESTERMO-SW6-MIB::cfgWlan802dot11rR0KHId.0 = 0
WESTERMO-SW6-MIB::cfgWlan802dot11rR0KHEnabled.0 = 1
WESTERMO-SW6-MIB::cfgWlan802dot11rR0KHDestinationMac.0 = ff:ff:ff:ff:ff:ff
WESTERMO-SW6-MIB::cfgWlan802dot11rR0KHID.0 = *
WESTERMO-SW6-MIB::cfgWlan802dot11rR0KHKey.0 = 000102030405060708090a0b0c0d0e0f
# 802.11r R1 key holder list wildcard entry
WESTERMO-SW6-MIB::cfgWlan802dot11rR1KHId.0 = 0
WESTERMO-SW6-MIB::cfgWlan802dot11rR1KHEnabled.0 = 1
WESTERMO-SW6-MIB::cfgWlan802dot11rR1KHDestinationMac.0 = 00:00:00:00:00:00
WESTERMO-SW6-MIB::cfgWlan802dot11rR1KHID.0 = 00:00:00:00:00:00
WESTERMO-SW6-MIB::cfgWlan802dot11rR1KHKey.0 = 000102030405060708090a0b0c0d0e0f
```

The 256-bit R0KH-key [cfgWlan802dot11rR0KHKey](#) (000102030405060708090a0b0c0d0e0f in the example above) must match the 256-bit R1KH-key [cfgWlan802dot11rR1KHKey](#). Please use your own key here and not the one which is given here just as an example to demonstrate the format of such a key.

PMK-R0 Key Holder ID ([cfgWlan802dot11rPmkR0KeyHolderIdentifier](#)) is configurable via [cfgWlan802dot1xNasId](#). See also section [4.29.2](#).

4.32.5.2 Client (STA) 802.11r Configuration

The following configuration example configures a STA for using FT:

Configuration File Example: FT STA

```
# Config.Format = raw
WESTERMO-SW6-MIB::cfgWlanHoProfile.0 = 2 # t2gv2(2)
WESTERMO-SW6-MIB::cfgWlan802dot11rEnabled.0 = 1
```

4.32.5.3 PMK-R0 / PMK-R1 Key Lifetime

The PMK-R0 lifetime is derived from the Session-Timeout provided by the authentication server, if available, and the PMK-R1 lifetime is derived from PMK-R0 lifetime. In addition the PMK-R0 lifetime can be configured per AP with [cfgWlan802dot11rPmkR0Lifetime](#). In this case the minimum of Session-Timeout and [cfgWlan802dot11rPmkR0Lifetime](#) is used.

It is also possible to enforce daily PMK-R0 / PMK-R1 expiration on a defined time (hour and minute). Enforce PMK-R0 / PMK-R1 expiration means that all the STAs are disconnected and the key material is deleted so that the STAs need to reconnect with full authentication over the RADIUS server.

- [cfgWlan802dot11rExpirationEnabled](#) to enable forced expiration
- [cfgWlan802dot11rExpirationTime](#) to define time (hour:minute) at which PMK-R0s and PMK-R1s expiration is daily forced

4.32.5.4 PMK-R1 push

Everytime a client initiates a connection to an Mobility Domain (MD), the involved AP receives the PMK-R0 from the RADIUS server and then derives a local PMK-R1. At this time this AP may also generate PMK-R1s for every other AP in the MD and send it to the respective AP.

This can be enabled by setting [cfgWlan802dot11rPmkR1Push](#) to 1.

Pushing PMK-R1 works only for APs which have an entry in the [cfgWlan802dot11rR1KHTable](#). Since the MAC-address of the R1 is part of the derived PMK-R1, an AP which has a wildcard configured is not able to generate the PMK-R1.

However it is still desirable to work with wildcard entries in the R0KH/R1KH tables even when using push. When push is enabled and an AP can not find the required PMK-R1 when a client roams to it, it will fall back to pull. Once an R0 AP had PMK-R1 material pulled from it, it learns dynamically the R1 which did the pull. The next time this particular R0 AP tries to push PMK-R1 material it will consider this learned entry.

4.33 Quality of Service (QoS)

The *Software 6* and devices supports Wireless Multimedia Extensions (WME) based on the IEEE 802.11e standard. WME provides basic Quality of service (QoS) features to IEEE 802.11 networks. The WME settings are configured on the Access Point only. Connecting clients are informed upon association what their QoS parameters are. Note that QoS/WME does not provide guaranteed throughput, but it is suitable for well defined applications that require QoS, such as Voice over IP (VoIP) on Wi-Fi phones (VoWLAN).

The levels of priority in Enhanced Distributed Channel Access (EDCA) are called access categories (ACs). They are used by a WMM-enabled device to control the Arbitration Inter-Frame Space (AIFS), the Contention Window Minimum (CwMin), the Contention Windows Maximum (CwMax), and the Transmit Opportunity (TXOP).

The ACs can be set with [cfgWlanWmeAc](#). The available ACs are:

- background (1)
- besteffort (2)
- video (3)
- voice (4)

All configurable parameters for ACs exist in two types: For the AP itself, and what the connecting STAs has to adapt to.

The AIFS can be set with [cfgWlanWmeAifs](#) and [cfgWlanWmeApAifs](#). A smaller AIFS increases probability of a frame getting a slot on the air to be transmitted. Higher prioritised queues should have smaller AIFS values.

The contention window (CW) can be set with [cfgWlanWmeCwMin](#), [cfgWlanWmeCwMax](#), [cfgWlanWmeApCwMin](#) and [cfgWlanWmeApCwMax](#). It has a similar function as the AIFS value, but adds some randomness between CwMin and CwMax. Queues which are expected to transmit large amounts of traffic should have a wider window with higher values to allow more randomness. Similar, queues with fewer high prioritised traffic should have a small window with low values.

To set the Transmit opportunity use [cfgWlanWmeTxOpMax](#) and [cfgWlanWmeApBurst](#). It specifies how long a given STA/AP is allowed to transmit when it has gained access to the medium.

The default values recommended by the WiFi-Alliance are:

AC	AIFS	CwMin	CwMax	TXOP
BK	7	4	10	0
BE	3	4	10	0
VI	2	3	4	94
VO	2	2	3	47

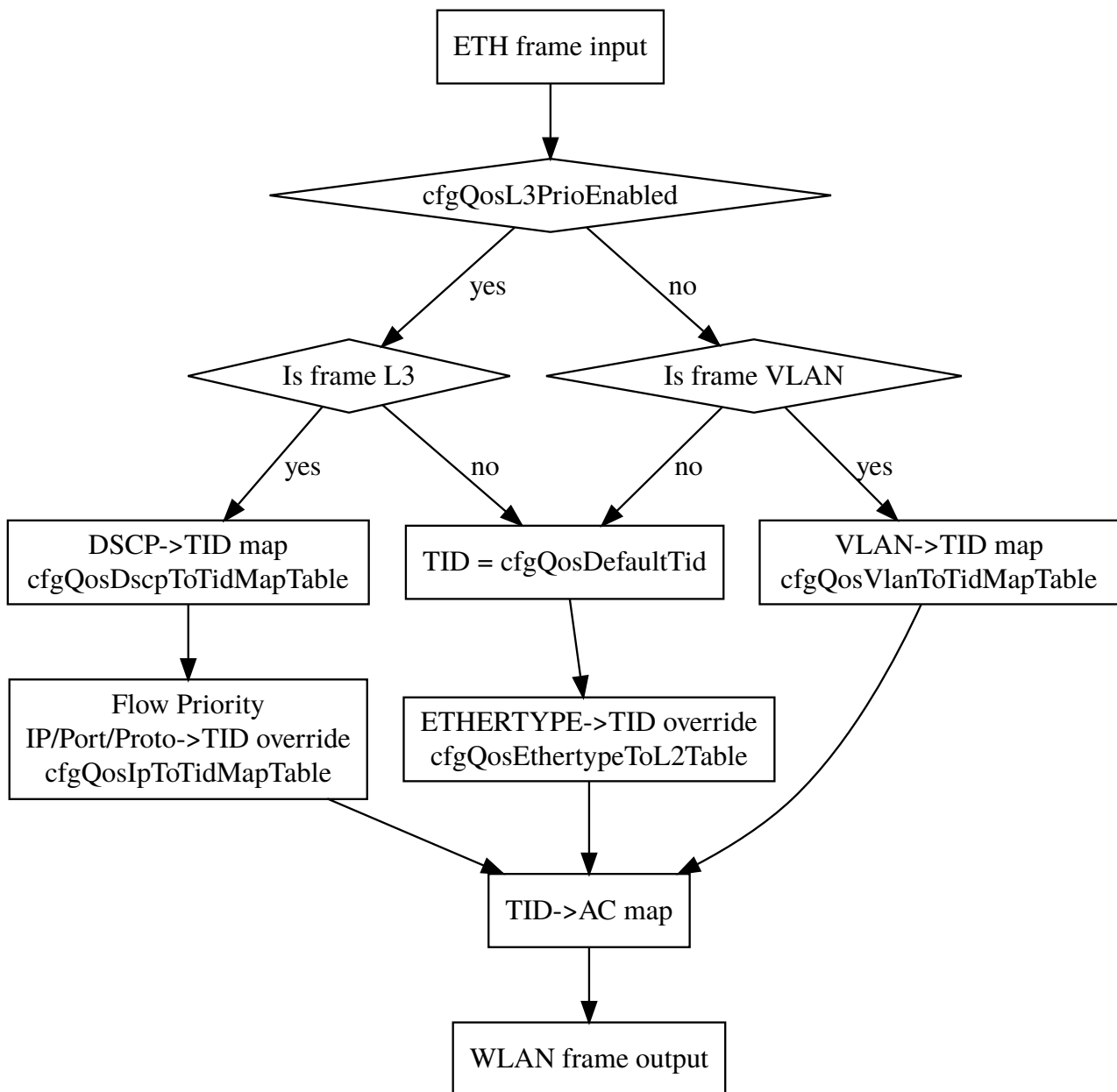
As indicated above, there are 4 hardware queues available with the names: Back Ground, Best Effort, Video and Voice.

To be able to maximise the chance of important traffic going through, it is necessary to match traffic and map it to the appropriate AC.

To achieve this, the following settings and maps are available:

1. [cfgQosL3PrioEnabled](#) - Use layer3, respectively layer2 processing.
- 2a. [cfgQosDscpToTidMapTable](#) - Map from DSCP class selector (IP TOS) to wireless priority (TID).
- 2b. [cfgQosVlanToTidMapTable](#) - Map from layer 2 priorities (802.1p) to wireless priority (TID).
3. [cfgQosIpToTidMapTable](#) - Map from IP header (Src, Dst, Proto, etc.) to wireless priority (TID).
4. [cfgQosEthertypeToL2Table](#) - Map from ethertype to wireless priority (TID).

The transmit path of frames with Qos enabled is visualised with the following diagram:



When [cfgQosL3PrioEnabled](#) is enabled, the DSCP header is mapped to the TID (Traffic Identifier) according to [cfgQosDscpToTidMapTable](#). By default it is in a 1:1 manner. The DSCP header consists of 6 bits. Only the upper most 3 bits, the class selector bits, are considered. The lower 3 bits, the drop probability bits, are ignored. Notice that with the default settings, the class selector 0, which is what most IP frames carry in their header without any configuration, maps to the Best Effort queue. With [cfgQosL3PrioEnabled](#) enabled, mapping of the DSCP header to wireless queues is always performed even if a different priority is specified in the 802.1p part of an 802.1q VLAN header. After the map from DSCP to TID, it is possible to override the selected TID based on the properties of the IP header (Protocol, Source, Destination, Port, etc.).

When [cfgQosL3PrioEnabled](#) is disabled, the 802.1p part of the 802.1q tag is mapped to the TID according to [cfgQosVlanToTidMapTable](#). By default it is in a 1:1 manner.

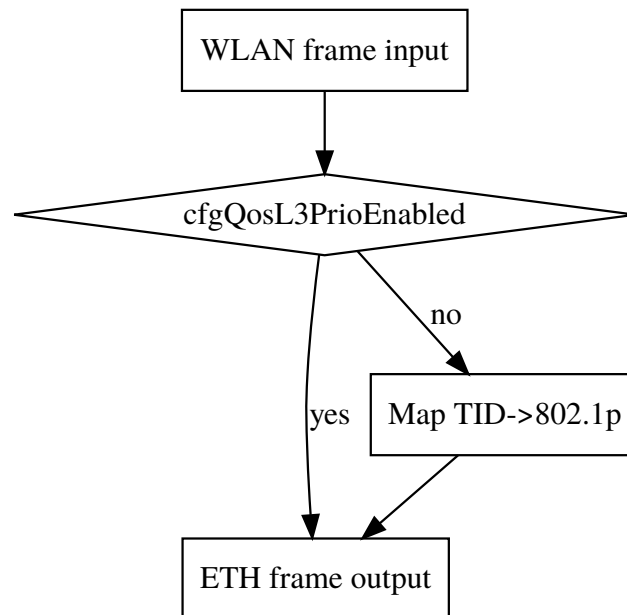
The TID is mapped to the AC classes according to the following table:

TID	AC
0	BE
1	BK
2	BK
3	BE
4	VI
5	VI
6	VO
7	VO

All frames that don't match the specified mode (e.g. non-IP frames when [cfgQosL3PrioEnabled](#) is enabled) are matched against [cfgQosEthertypeToL2Table](#). If the ethertype of the frame matches an entry in the table, the TID is set to the value specified [cfgQosEthertypeToL2Tid](#). If no match is found, then the TID is set to the value specified in [cfgQosDefaultTid](#).

The QoS configuration is described in [cfgQos](#).

The receive path of frames with Qos enabled is visualised with the following diagram:



When L3 processing is disabled, the received TID will be mapped to the 802.1p field in the 802.1q tag.

4.34 Common Address Redundancy Protocol (CARP)

CARP is a networking protocol which allows multiple devices on the same LAN to share a set of IP addresses. For *Software 6* the main use case for CARP is to provide redundant gateways in a redundant mobility application.

The CARP configuration is described in [cfgNetCarpTable](#). The list of managed IP addresses per CARP instance can be configured in the [cfgNetIpTable](#) by setting the [cfgNetIpProto](#) to *carp(5)* and the correct [cfgNetIpCarpId](#)

For redundant mobility application a WLAN demote trigger can be configured using the [Network Link Monitor \(NLM\)](#). If enabled, the CARP instance demotes itself if there is no WLAN link.

4.35 Network Link Monitor (NLM)

The NLM supports several types of monitors (phy, icmp, wlan, ..) and executes so called *actions* on monitor state change from down to up and vice versa.

Supported monitors are documented in [cfgNlmMonType](#).

Supported actions are documented in [cfgNlmMonUpAction](#) and [cfgNlmMonDownAction](#).

Full NLM configuration is described in [cfgNlm](#).

Main use case is to observe the network link from the client (STA), which acts as gateway on the vehicle, to the endpoint in the ground network in a redundant mobility application with two wireless links:

- **Backbone Monitor** - On Access Point, observe the backbone network on the ground by monitoring the linkstate and/or pinging an endpoint in the ground backbone and stop WLAN operation if the endpoint(s) is down.
- **CARP Failover** - On STA, observe the link state of the WLAN interface and trigger CARP group to demote itself if the link is down or otherwise unusable.

These two features combined provides link status check from the train gateway up to the endpoint in the ground backbone.

4.35.1 NLM Monitor Types

4.35.2 phy Monitor

The phy monitor is active when [cfgNlmMonType](#) is set to 0.

This is a polling based monitor. It checks in regular intervals ([cfgNlmMonInterval](#)) the state of a list of comma-separated ethernet phys ([cfgNlmMonInterfaces](#)). E.g. "eth0, eth1". As long as at least one phy is up, the state of the monitor is up. The state will change to down when the checks fails [cfgNlmMonCount](#) times in a row.

4.35.3 icmp Monitor

The icmp monitor is active when [cfgNlmMonType](#) is set to 1.

This is a polling based monitor. It sends icmp requests (ping) in regular intervals ([cfgNlmMonInterval](#)) to the configured IP address in ([cfgNlmMonDestination](#)). The state will change to down when the checks fails [cfgNlmMonCount](#) times in a row. The state will recover as soon as at least a single icmp response (pong) is received.

4.35.4 wlan Monitor

The wlan monitor is active when `cfgNlmMonType` is set to 2.

This is an event based monitor.

The wlan monitor consists of 3 components:

- **Long Handoff Detector** - Triggers when after disassociation no authorization event is detected within the configured time in `cfgNlmMonInterval`.
- **Scan Loop Detector** - Triggers immediately on trap 415. This happens if there is no AP or only a single AP which stays below/above the Handoff thresholds. This trap is only generated when `cfgWlanHoProfile` is set to 2 or higher.
- **Handoff Loop Detector** - Triggers if there have been `cfgNlmMonCount` number of Handoff events within `cfgNlmMonCount * (cfgNlmMonInterval + cfgNlmMonScanLoopInterval)`.

The wlan monitor recovers:

- **Long Handoff Detector** - Immediately after the next successful authorization.
- **Scan Loop Detector** - After the time of the last 415 trap event + the configured `cfgNlmMonScanLoopInterval`. While down, this is rechecked regularly in `cfgNlmMonScanLoopInterval` intervals.
- **Handoff Loop Detector** - When there are less than `cfgNlmMonCount` Handoff events within the time-window `cfgNlmMonCount * (cfgNlmMonInterval + cfgNlmMonScanLoopInterval)`. While down, this is rechecked regularly in `cfgNlmMonScanLoopInterval` intervals.

4.35.5 NLM Configuration for CARP Failover

The following NLM configuration will failover when:

- A Handoff takes longer than 300 ms (`cfgNlmMonInterval`).
- 3 consecutive scans can not find a better target (Trap 415).
- 4 Handoff happen within 13200 ms ($4 * (300 + 3000)$).

Configuration File Example: NLM Configuration for CARP Demote

```
# Config.Format = raw
WESTERMO-SW6-MIB::cfgNlmGblEnabled.0 = INTEGER: 1
```

```
WESTERMO-SW6-MIB::cfgNlmMonEnabled.0 = INTEGER: 1
WESTERMO-SW6-MIB::cfgNlmMonInterval.0 = INTEGER: 300
WESTERMO-SW6-MIB::cfgNlmMonCount.0 = INTEGER: 4
WESTERMO-SW6-MIB::cfgNlmMonType.0 = INTEGER: 2
WESTERMO-SW6-MIB::cfgNlmMonInterfaces.0 = STRING: wlan0
WESTERMO-SW6-MIB::cfgNlmMonScanLoopInterval.0 = INTEGER: 3000
# Up-/Down action for cfgNetCarpLocalInterfaceGroup.0 = 1
WESTERMO-SW6-MIB::cfgNlmMonUpAction.0 = INTEGER: 1001
WESTERMO-SW6-MIB::cfgNlmMonDownAction.0 = INTEGER: 1001
```

4.35.6 NLM Configuration for Backbone Monitor

Following NLM configuration periodically pings the endpoint with 192.168.1.1 and disables the Access Point operation on wlan0 if the endpoint does not reply.

Configuration File Example: NLM Configuration for Backbone Monitor

```
# Config.Format = raw
WESTERMO-SW6-MIB::cfgNlmGblEnabled.0 = INTEGER: 1
WESTERMO-SW6-MIB::cfgNlmMonEnabled.0 = INTEGER: 1
WESTERMO-SW6-MIB::cfgNlmMonInterval.0 = INTEGER: 1000
WESTERMO-SW6-MIB::cfgNlmMonCount.0 = INTEGER: 3
WESTERMO-SW6-MIB::cfgNlmMonType.0 = INTEGER: 1
WESTERMO-SW6-MIB::cfgNlmMonInterfaces.0 = STRING: br0.vlan0
WESTERMO-SW6-MIB::cfgNlmMonDestination.0 = IpAddress: 192.168.1.1
WESTERMO-SW6-MIB::cfgNlmMonUpAction.0 = INTEGER: 2000
WESTERMO-SW6-MIB::cfgNlmMonDownAction.0 = INTEGER: 2000
```

4.36 DNS/DHCP Server

Each server instance is able to provide DNS functionality, DHCP functionality or both.

The DNS/DHCP configuration is described in [cfgDhcp](#).

4.36.1 DNS Server

Enable or disable the DNS functionality of the server by setting [cfgDhcpDnsmasqDnsEnabled](#).

The Listen Address ([cfgDhcpDnsmasqDnsListenAddress](#)) is set by default to 'auto', which binds the DNS server to all IP addresses on interfaces used to serve DHCP clients.

To use the DNS server standalone without DHCP functionality ([cfgDhcpDnsmasqDhcpEnabled](#) disabled) there are 3 options. Set [cfgDhcpDnsmasqDnsListenAddress](#) to:

- A list of addresses to bind to.
- 'wildcard' which binds to 0.0.0.0.
- 'auto' without referencing any DHCP scopes (e.g. [cfgDhcpDnsmasqScopeParameter](#) set to -1). This will bind any address which exists on the system.

Rebind Protection ([cfgDhcpDnsmasqDnsStopDnsRebind](#)) blocks upstream responses which reply with an internal address. Such responses may be used by an attacker to probe the local network and access local ports on the device itself.

Rebind Protection blocks responses resolving to:

Network	RFC	Usage
127.0.0.0/8	RFC 1122	loopback
0.0.0.0/8	RFC 5735	"this"
10.0.0.0/8	RFC 1918	private
172.16.0.0/12	RFC 1918	private
192.168.0.0/16	RFC 1918	private
169.254.0.0/16	RFC 3927	linklocal
192.0.2.0/24	RFC 1166	test
198.51.100.0/24	RFC 5737	test
203.0.113.0/24	RFC 5737	test
255.255.255.255/32	RFC 0919, RFC 0922	broadcast

In split-DNS situation it may be desirable to allow to respond with a blocked range. The option [cfgDhcpDnsmasqDnsRebindDomainOk](#) may be used to exclude domains and its subdomains from the rebind protection.

The DNS server can be configured to ask a specific DNS server for the requested domain. This configuration can be done in the [cfgDhcpDomainOverrideTable](#).

The DNS server can responde with hard-coded IP address for a requested host. This configuration can be done in the [cfgDhcpHostOverrideTable](#).

4.36.2 DHCP Server

Enable or disable the DHCP functionality of the server by setting [cfgDhcpDnsmasqDhcpEnabled](#).

Use [cfgDhcpDnsmasqScopeParameter](#) to specify a reference to one or multiple scopes identified by [cfgDhcpScopeId](#). Multiple scopes may have the same Scope ID to be handled by a single instance. Specify the interface to provide DHCP functionality by setting [cfgDhcpScopeInterface](#).

Use [cfgDhcpScopeStart](#) and [cfgDhcpScopeLimit](#) to define the start address and the number of addresses provided by a scope. The start address is an offset from the network-address of the first IP address configured on the interface.

Use [cfgDhcpScopeAutoGateway](#) and [cfgDhcpScopeAutoDns](#) to simplify the configuration of the default gateway and DNS server. When they are enabled, the address of the interface used to serve a DHCP client will be sent in the DHCPOFFER.

Use [cfgDhcpScopeDhcpOptions](#) to configure arbitrary DHCP options. These custom DHCP options override configurations set by other fields. For example when [cfgDhcpScopeGateway](#) is set to an address, but [cfgDhcpScopeDhcpOptions](#) contains a configuration for DHCP option 3 as well, then the value set in [cfgDhcpScopeDhcpOptions](#) will take precedence.

Configuration File Example: DHCP server on br0.vlan0

```
# Config.Format = raw
WESTERMO-SW6-MIB::cfgDhcpGlobalEnabled.0 = 1
WESTERMO-SW6-MIB::cfgDhcpDnsmasqScopeParameter.0 = 0
WESTERMO-SW6-MIB::cfgDhcpScopeId.0 = 0,
WESTERMO-SW6-MIB::cfgDhcpScopeInterface.0 = br0.vlan0
WESTERMO-SW6-MIB::cfgDhcpScopeStart.0 = 100
WESTERMO-SW6-MIB::cfgDhcpScopeLimit.0 = 150
WESTERMO-SW6-MIB::cfgDhcpScopeGateway.0 = 192.168.1.1
```

4.37 Service Indicators and Counters

4.37.1 SNMP Trap

The WLAN modem provides notifications by means of SNMP Traps.

In order to receive SNMP traps on a trap server the trap daemon of the WLAN modem must be enabled ([cfgSnmpTrapEnabled](#)) and the trap destination IP address ([cfgSnmpTrapDest](#)) must be set to the IP address of the trap server.

The SNMP traps are defined and described in the WESTERMO-TRAP-MIB file, which is part of the delivered software package. The trap message string contains message codes in the form:

```
[ <prio> <code> ] <text message>
```

Please refer to chapter [Message Codes](#) for a complete list of all message codes.

All SNMP traps are also recorded in the Syslog (see [Logging Features](#))

4.37.2 Counters and Status

The WLAN modem provides status information and important counters defined and described by the WESTERMO-SW6-MIB.

The status information and counters are logically divided into

- **Hardware status and counters** ([hardware](#)) - such as product type, serial number, revision, etc
- **Software status and counters** ([software](#)) - such as firmware name, firmware revision, wireless counters etc

4.38 Logging Features

The *Software 6* supports extensive logging features into Syslog. The Syslog is available in the Web Interface / via Web API or it can be sent to a remote server (RFC 5424).

Up to four remote syslog instances can be configured. For enabling/disabling remote instances logging use [cfgLogRemoteEnabled](#). IP address and port of remote syslog instance can be configured using [cfgLogRemoteIp](#) and [cfgLogRemotePort](#). The remote syslog protocol (UDP or TCP) can be configured with [cfgLogRemoteProtocol](#)

There are three message types which are differentiated for the remote syslog:

- **standard**: This type contains all message which are not marked
- **security**: This type contains all message which are marked as security messages (LOCAL0)
- **standard**: This type contains all message which are marked as commissioning messages (LOCAL1). Please refer to [cfgWlanDbgTable](#) for more information.

For the four remote syslog servers the message types which shall be sent can be configured by [cfgLogRemoteType](#). Since the configuration of the message type is a bitfield, all combinations are possible.

Remote Syslog is essential to analyze mobility applications during commissioning (see [Mobility Logging](#)).

4.39 Wireless Link Monitor

The *Software 6* provides wireless link status message periodically to Syslog or as SNMP Trap.

In Access Point mode the link status for all connected clients is reported. In client (STA) mode the current connection to the Access Point is reported.

The feature can be enabled for each WLAN interface separately by [cfgWlanDbgLinkmonitor](#) where the reporting interval is defined by [cfgWlanGblLinkmonitorInterval](#) globally.

Format:

```
INFO 130 interface|mac|inactive time||rx bytes|rx packets|tx bytes|tx
    ↳ packets|tx retries|tx failed||signal combined|avg signal combined|
    ↳ signal ch0|avg signal ch0|signal ch1|avg signal ch1|signal ch2|avg
    ↳ signal ch2||rx bitrate mode|rx bitrate value|tx bitrate mode|tx
    ↳ bitrate value
```

One message per connection is reported.

4.40 Inter-Carriage Link (ICL)

The Inter-Carriage Link application (ICL) offers a hands-off approach to connect and bridge carriage networks. Once ICL is enabled and operational on two carriages, the application will do the following:

- Broadcast availability to other ICL capable carriages while outside detection range.
- Automatically form a link on approach.
- Stay linked while the carriages are connected and providing the highest throughput possible.
- Cleanly disconnect on departure and switch back to broadcasting availability.

The ICL application removes the need for a cable connection and is designed to be low maintenance. Using the ICL algorithm combined with suitable antennas ensures proper linking of carriages.

4.40.1 Configuration of the Inter-Carriage Link Application

The ICL application supports two configuration modes:

- A Web Interface as a configuration wizard.

- Configuration through SNMP for full customisation.

4.40.1.1 Configuration with the Web Interface

After logging in (See [Web-Based Management \(Web Interface\)](#) how to access), the *Inter-Carriage Link* application is available in the *Applications* menu.

The first step to activate the application is to enable it. Go to *Application -> Inter-Carriage Link -> Configuration*.

You should see a page like the screenshot in figure 4.14.

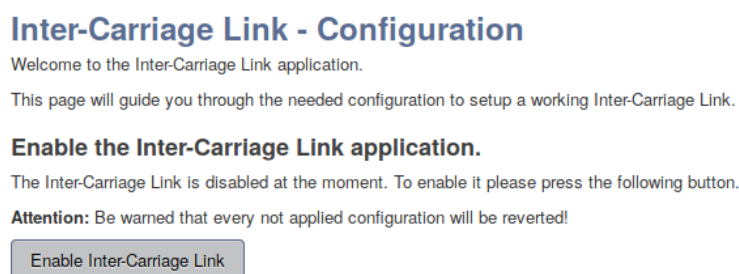


Figure 4.14: *Enable page*

By pressing the enable button the Web Interface wizard will change a number of settings for you. You will be able to adjust various parameters before starting the ICL application.

Once enabled, you will be redirected to the configuration page as shown in figure 4.15 where you will be able to customise a list of settings.

Welcome to the Inter-Carriage Link application.

This page will guide you through the needed configuration to setup a working Inter-Carriage Link.

Settings

Network

Interface assignment	eth0: Data / eth1: Admin
Admin IP	192.168.3.22/24
Admin VLAN ID	0

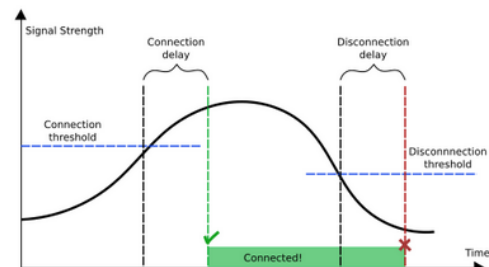
Wireless

Country Code	EU
ICL SSID	NT_ICL-
Antenna Gain (dBi)	2
Bandwidth (MHz)	bw40Plus
Frequency (MHz)	Auto 5 GHz
MAC ACL enabled	☒

Inter-Carriage Link settings

Connection threshold	-60	dBm (-90 - 0)
Connection delay	30	seconds (0 - 600)
Disconnection threshold	-65	dBm (-90 - 0)
Disconnection delay	20	seconds (1 - 600)
Cycle time	5	seconds (2 - 60)
Blacklist time	600	seconds (1 - 3600)
Suspended	☐	

Frequencies (MHz)	5180	Add
Remove all invalid frequencies		
5745 MHz		Remove
5785 MHz		Remove
5825 MHz		Remove



Apply

Figure 4.15: Configuration page

Interface assignment defines which interfaces will be configured as *data* and which as *admin* interface. Please make sure you choose the right interfaces. Otherwise you probably will not be able to connect to the device again.

Admin IP defines which IP address will be assigned to the *admin* interface.

Admin VLAN ID defines which VLAN id will be used for the *admin* interface.

ICL SSID defines which prefix will be used for the SSID. To established a link between two carriages this prefix must be the same on both sides.

Antenna Gain defines the antenna gain use for the Inter-Carriage link.

Bandwidth defines the preferred bandwidth. Options are HT20, HT40+ or HT40-. For more details, see [cfgWlanDevBandwidth](#).

Frequency defines the primary operating frequency and depends on the selected bandwidth. See

[cfgWlanDevFrequency](#) for more details.

MAC ACL enabled When the ACL is enabled only MAC addresses and ranges specified in [cfgWlanAclWhiteTable](#) are allowed to connect. By default the 3 ranges 00:14:5a:00:00:00/24, 00:07:7c:00:00:00/24 and 00:11:b4:00:00:00/24 are allowed. Use the **Configuration - Advanced** page to change these entries.

Frequency list contains all frequencies that will be scanned by the Inter-Carriage Link application. It also functions as a list of backup frequencies in case radar is detected on the current operating frequency. This list should include its own frequency the device operates on. Corresponding MIB entries are [cfgWlanFFreq0](#) to [cfgWlanFFreq23](#).

Connection threshold describes the minimum signal level a potential ICL partner needs to reach before it is considered a valid ICL partner. A higher value means the signal needs to be stronger and therefore a potential partner needs to be closer. [cfgIclConnectionThreshold](#) is the associated MIB entry.

Connection delay defines how long the Inter-Carriage Link algorithm should evaluate a potential ICL partner. If the connection delay is set to 0 the ICL algorithm will instantly connect to the very first potential partner circumventing most of the evaluation of the ICL algorithm. See [cfgIclConnectionDelay](#) for more details.

Disconnection threshold sets the signal level at which the link disconnection process will be started. For more details, see [cfgIclDisconnectionThreshold](#).

Disconnection delay defines how long the device should wait before scanning for a new partner once the old partner disconnected. It also defines how long a formerly connected partner tries to reconnect if the link is lost for any reason. [cfgIclDisconnectionDelay](#) is the corresponding MIB entry.

Cycle time sets the scan interval. Ideally the connection delay time is at least five times the cycle time to allow the ICL algorithm to properly evaluate a candidate. See [cfgIclCycleTime](#) for more details. Also consider the size of the Frequency list. A larger list requires more time spend scanning, thus the Cycle time should not be selected too short.

Blacklist time defines how long a blacklist entry will reside in the blacklist.

Suspended defines if the Inter-Carriage Link starts in the suspended state.

To start the Inter-Carriage Link application press *Apply*. If the application was already running, *Apply* will restart the service with the changed configuration.

Note Configuration with SNMP offers full customisation should the Web Interface not satisfy your configuration needs.

The status page as in figures [4.16](#), [4.17](#), [4.18](#) and [4.19](#) shows all important status information regarding the Inter-Carriage Link. This page will automatically refresh every two seconds to keep the

status up-to-date.

Inter-Carriage Link - Status

☒ Auto refresh every 2 seconds



My carriage

MAC:	00:07:7c:30:60:3f
Operation mode:	AP
Frequency (MHz):	5785
Status:	scanning
Operation:	Suspend

Figure 4.16: Status page: Scanning for an ICL partner

Inter-Carriage Link - Status

☒ Auto refresh every 2 seconds



My carriage

MAC:	00:07:7c:30:5f:af
Operation mode:	AP
Frequency (MHz):	5745
Status:	scanning
Operation:	Suspend

Candidates

MAC	Avg. Signal (dBm)	Last Signal (dBm)
00:14:5a:03:0a:bc	-31	-31

Figure 4.17: Status page: Evaluating one or more ICL partners

Inter-Carriage Link - Status

☒ Auto refresh every 2 seconds



My carriage

MAC:	00:07:7c:30:5f:af
Operation mode:	STA
Frequency (MHz):	5785
Status:	connected
Operation:	<button>Suspend</button>

Partner carriage

MAC:	00:14:5a:03:0a:bc
Signal (dBm):	-30

Force disconnect

Inter-Carriage Link - Status

☒ Auto refresh every 2 seconds



My carriage

MAC:	00:07:7c:30:60:3f
Operation mode:	
Frequency (MHz):	
Status:	suspended
Operation:	<button>Resume</button>

Figure 4.19: Status page: ICL operation suspended

Suspend

If the ICL functionality isn't needed at the moment the operation can be suspended. This can be accomplished by pressing the *Suspend* button or by setting `setIclSuspended` to 1. In this state the device will no longer broadcast availability and other partners are not able to connect.

Blacklist

Partners listed in the blacklist can't be used to establish an Inter-Carriage Link. The blacklist entries can be flushed by pressing the *Clear Blacklist* button or by setting `rpclIclClearBlacklist` to 1.

Force Disconnect

In the unlikely case the displayed connection is not the desired connection, the link can be dropped using the *Force disconnect* button or by setting [rpclclForceDisconnect](#) to 1. The current ICL partner will then be added to a blacklist to prevent reconnecting to the same device for the time configured with the *Blacklist time*.

4.41 Global Navigation Satellite System

All products that support cellular communication come with a Global Navigation Satellite System (GNSS), see section 1.3. The built-in GNSS module is monitored by the service application `gpsd`. This allows the location data to be distributed across computer networks by TCP/IP.

To enable `gpsd` use the following config example:

Configuration File Example: Enable `gpsd`

```
# Config.Format = raw
WESTERMO-SW6-GNSS-MIB::cfgGnssGpsdEnabled.0 = 1
WESTERMO-SW6-GNSS-MIB::cfgGnssGpsdAddress.0 = 0.0.0.0:2974
```

After enabling the `gpsd`, the location data is visible on the status page of the Web Interface or available via SNMP [swGnss](#). A common way to access the data is to open a TCP connection on the defined port with a `gpsd` compatible client (`gpsmon`, `xgps` or `gpspipe` on Linux).

Note: The `gpsd` can only be bound locally or globally so the value of [cfgGnssGpsdAddress](#) must be 127.0.0.1 for local or 0.0.0.0 to bind to all addresses.

4.41.1 GNSS Device Configuration

The GNSS module is able to operate 3 satellite systems simultaneously, the following are selectable:

- GPS
- Glonass
- BeiDou
- Galileo

Detailed information on how to select a valid combination can be found in [cfgGnssDevSatelliteSystems](#). The parameter [cfgGnssDevMeasurementPeriod](#) defines the overall measurement period of the GNSS

module. The effective sampling rate of each message can be set by the respective parameter [cfgGnssDevMsgsNmeaRate](#) or [cfgGnssDevMsgsUbxRate](#).

$$messageSamplingRate[Hz] = \frac{cfgGnssDevMsgsNmeaRate}{cfgGnssDevMeasurementPeriod[ms]} \times 1000 \quad (4.3)$$

Normally, there is no need to change the sampling rate for each message. It is sufficient to select the overall measurement period and set the individual sampling rate of the messages to 1.

4.41.2 NMEA Sentences

A default setting of National Marine Electronics Association (NMEA) sentences is activated by default in order to enable a location determination when the service is switched on. All NMEA sentences available through the GNSS module are provided and listed in the description of [cfgGnssDevMsgsNmeaType](#). In order to disable specific sentences, use the parameter [cfgGnssDevMsgsNmeaEnabled](#) or set the related [cfgGnssDevMsgsNmeaRate](#) to 0.

4.41.3 UBX Messages

UBX is a proprietary protocol from the company u-blox for exchanging GNSS data. All UBX messages are disabled by default. To enable a UBX message type set [cfgGnssDevMsgsUbxEnabled](#) and [cfgGnssDevMsgsUbxType](#) accordingly and set the [cfgGnssDevMsgsUbxRate](#) greater than 0. In order to disable specific messages, use the parameter [cfgGnssDevMsgsUbxEnabled](#) or set the [cfgGnssDevMsgsUbxRate](#) to 0. The accessible UBX messages are listed in the description of [cfgGnssDevMsgsUbxType](#).

4.42 Public Wireless Network (PWN)

4.42.1 Hotspot

The RT-610 product is very suitable for WiFi hotspots. It supports amongst other things dual concurrent operation in the 2.4GHz and 5GHz wireless bands, Multi-SSID mode and MU-MIMO.

With the Hotspot application as show in Figure 4.20 the configuration of a WiFi hotspot is simple.

Hotspot

Apply

Management Network

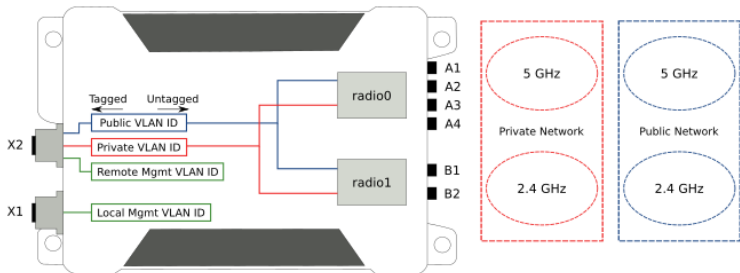
Enable Remote Mgmt ☐

Enable Local Mgmt ☒

Local Mgmt VLAN ID
0 = No VLAN, Tagged X1

Local Mgmt IP address

Default Gateway



Network Configuration

Radio Configuration

	Radio	Modulation ⓘ	Bandwidth (MHz) ⓘ	Frequency (MHz) ⓘ	Power (dBm) ⓘ	Antenna Gain (dBi) ⓘ	Antenna Mask ⓘ
5.0 GHz Band	radio0	ac ▾	bw80 ▾	Auto ▾	15	6	15(1111) ▾
2.4 GHz Band	radio1	ng ▾	bw20 ▾	2412 MHz ▾	15	6	3(0011) ▾

Public Network

Supported Bands

SSID ⓘ

Hide SSID ⓘ ☐

Isolate Clients ⓘ ☒

Encryption ⓘ

VLAN ID ⓘ 0 = No VLAN, Tagged X2

Private Network

Supported Bands

SSID ⓘ

Hide SSID ⓘ ☐

Isolate Clients ⓘ ☒

Encryption ⓘ

Encryption passphrase ⓘ

VLAN ID ⓘ 0 = No VLAN, Tagged X2

Figure 4.20: Hotspot application configuration page

After logging in (See [Web-Based Management \(Web Interface\)](#) how to access), the *Hotspot* application is available in the *Applications* menu.

4.42.2 Band Steering

Band Steering is a feature that encourages wireless clients that are capable of both 5GHz and 2.4GHz wireless bands to connect to the faster 5GHz band and leave the 2.4GHz band less-crowded for those clients who support the 2.4GHz band only. Inactive clients of a traffic overloaded 2.4GHz wireless band may be steered to the less-congested 5GHz band or vice versa. If the received signal strength indicator (RSSI) of the connected 5GHz band is very low and a stronger 2.4GHz band is available, the client will be steered to the 2.4GHz band until the 5GHz band has regained strength. Such an approach is expected to improve the wireless performance for all clients participating in the same Extended Service Set (ESS).

Since Band Steering is done within the same ESS, both bands must have the same ESS Identifier (ESSID) as well as the same security settings. That ESSID is called Matching SSID and is configurable by the MIB item [cfgWlanBsteerMatchingSsid](#). To enable Band Steering for multiple virtual access points, the matching SSID names can be defined as a comma-separated (and/or space-separated) list.

4.42.2.1 Basic Operation

Assuming that the Software 6 product is configured with the factory settings, enabling the Band Steering is fairly simple by applying the following configuraiton file.

Configuration File Example: Basic Band Steering

```
# Config.Format = raw
WESTERMO-SW6-MIB::cfgWlanIfaceSsid.0 = BSteer-Enabled
WESTERMO-SW6-MIB::cfgWlanIfaceSsid.1 = BSteer-Enabled
WESTERMO-SW6-PWN-MIB::cfgWlanBsteerEnabled.0 = 1
WESTERMO-SW6-PWN-MIB::cfgWlanBsteerMatchingSsid.0 = BSteer-Enabled
```

4.42.3 Airtime Fairness

Airtime Fairness (ATF) provides intelligent allocation of available airtime among connected stations. The following criteria are taken into account for the scheduling of airtime:

Airtime allocation based on SSID The total airtime is distributed to one or more SSIDs according to defined percentages.

Equal airtime allocation All stations connected to the same SSID get equal airtime allocation.

Unused airtime redistribution Depending on the scheduling algorithm, unused airtime is made available to stations that have already used up their quota.

4.42.3.1 Airtime Scheduling Algorithms

The ATF feature supports two mutually exclusive scheduling algorithms namely a strict and a fair one. The strict scheduling algorithm follows the configured scheduling rules rigorously and does not utilise any unused airtime. The fair algorithm on the other hand guarantees the configured airtime in congested environments and tries to redistribute unused airtime, see also [cfgWlanDevAtfSchedulingAlgorithm](#).

4.42.3.2 Configuration Based on Airtime per SSID

Listing 4.8 shows how airtime can be fairly distributed among two SSIDs. The intention is to have a private and a public SSID, whereby 80% and 20% of airtime is allocated, respectively. All SSIDs, which share airtime, must be part of the same physical device. The `cfgWlanDevAtfSchedulingAlgorithm` is set to `fair(0)` to achieve a fair distribution of airtime. This setting is only available per physical device.

```
WESTERMO-SW6-MIB::cfgWlanDevAtfSchedulingAlgorithm.0 = 1
WESTERMO-SW6-MIB::cfgWlanInterfaceSsid.0 = private
WESTERMO-SW6-MIB::cfgWlanInterfaceAtfSsidEnabled.0 = 1
WESTERMO-SW6-MIB::cfgWlanInterfaceAtfSsidAirtime.0 = 80
WESTERMO-SW6-MIB::cfgWlanInterfaceSsid.0 = public
WESTERMO-SW6-MIB::cfgWlanInterfaceAtfSsidEnabled.1 = 1
WESTERMO-SW6-MIB::cfgWlanInterfaceAtfSsidAirtime.1 = 20
```

Listing 4.8: *Airtime based on SSID*

The percentage of airtime for the respective SSIDs can be set by `cfgWlanInterfaceAtfSsidAirtime`. This percentage is only taken into account if the corresponding `cfgWlanInterfaceAtfSsidEnabled` parameter is set to `enabled(1)`. Otherwise, no airtime is reserved. The total of the allocated airtime shares must not exceed 100%, as otherwise the ATF configuration will not be accepted.

4.43 RSTP

All RSTP related configurations are described under `cfgRstpBridge` and `cfgRstpPort`.

RSTP configuration is done via the MIB `WESTERMO-SW6-BRIDGE-MIB`.

Status information is available via Web Interface (Status - RSTP) or via the standard MIB `BRIDGE-MIB` (oid `.1.3.6.1.2.1.17`).

Please note that currently only one RSTP instance is supported and this instance is `br0`.

4.44 Dynamic Frequency Selection (DFS)

All devices have a basic way to enable support for DFS:

- Wireless Standalone

802.11n capable devices with a monitor card, have 2 additional ways:

- NWM
- AFM/AFC

Wireless Standalone Wireless Standalone mode can be used on a device without monitor card. This allows operation on DFS frequencies but has the downside that if radar is detected, there is an interruption of at least 1 minute.
Refer to [Wireless Standalone](#) for detailed information on operation and configuration.

NWM The NWM is intended to be used on a single AP which has a monitor card to allow seamless operation in case of radar.
Refer to [Wireless Manager \(NWM\)](#) for detailed information on operation and configuration.

AFM/AFC The AFM/AFC is intended to be used on groups of APs which have monitor cards to allow seamless operation in case of radar.
Refer to [Area Frequency Management \(AFM\)](#) for detailed information on operation and configuration.

4.44.1 Wireless Standalone

The Wireless Standalone mode is automatically active when a DFS frequency is configured but neither NWM nor AFM/AFC are active.

In Wireless Standalone mode the communication radio is used to perform the CAC to clear the frequency for operation. The communication radio is not able to do off-channel-CAC, thus when radar is detected, it has to clear a new frequency first before operation can continue. Depending on the frequency this takes at least 1 minute; when radar is detected the interruption will be at least 1 minute. The Wireless Standalone mode also does not have access to the non-volatile memory option (cfgChMgrDfsUseNvram). Clearing of the channel is required everytime the configuration is changed.

In case radar is detected, the radio will jump to a new frequency based on the selected ACS list (see [cfgWlanIfaceAcsList](#)). When no frequency list is selected (e.g. [cfgWlanIfaceAcsList](#) set to -1) all frequencies permitted by the country code are considered for selection. Please note that this may also include frequencies which are not allowed in the installed environment, e.g indoor frequencies in an outdoor installation. Please make sure to always specify and configure a correct frequency list.

Keep in mind that once the frequency has been changed, the AP will stay on the new frequency until radar is detected again. It will not automatically revert back to it's original operating frequency once the NOP time has elapsed. This also implies that if the new frequency the AP jumped to is not a DFS frequency, the AP will stay there until it is manually reconfigured.

4.44.2 Wireless Manager (NWM)

The Wireless Manager (NWM) offers advanced features for wireless access point operation

- Usable frequency list to support advanced frequency planning
- Background channel availability check (CAC) for DFS channel to support seamless operation in case of radar detection
- Available channel list in non-volatile memory for fast recovery after reboot
- Interference reports

The Wireless Manager (NWM) is only available on RT-370 devices since it requires two radios.

The first radio (antenna ports A1, A2 and A3) is the so called communication interface providing the access point functionality. The second radio (antenna port A4) is responsible to make Channel Availability Checks on channels which are not yet available. During periods where no further CACs are required, the second radio can be used to do scan the environment for interferences as described in section [4.45](#)

Antenna port A4 is used for CAC and Off-Channel-CAC. Therefore you need to assert that you have an antenna connected at antenna port A4.

The NWM can store the list of available DFS channels in a non-volatile memory. Thus, when the channel is once marked as available it will be instantly available after a device reboot. When the non-volatile memory option is enabled (see [cfgChMgrDfsUseNvram](#)), an Operator is responsible to reset the list of available DFS channel by setting [rpcNvramFreqStatesReset.0](#) to zero at installation or re-installation of the device.

The NWM depends on following sub-features:

- Channel Manager: The Channel Manager is responsible to perform CACs on usable DFS channels. Its goal is to make all DFS channels available. Further, it proposes the wireless channel to be used by the NWM.
- Scan Worker: The Scan Worker manages the second radio of the RT-370 device. On request it performs scan work jobs like CAC or wireless interference scans. The Channel Manager makes use of the Scan Worker to do the CACs on DFS channels.

4.44.2.1 Configuration with the Web Interface

The *Wireless Manager* can be configured with the Web Interface (See [Web-Based Management \(Web Interface\)](#) how to access).

The NWM is a so called application. You will find it in the *Applications* menu.

The first step to activate the application is to enable it. Go to *Application -> Wireless Manager -> Configuration*.

You should see a page like the screenshot in figure 4.21.

Wireless Manager - Configuration

Welcome to the Wireless Manager.

This page will guide you through the needed configuration to setup a proper working Wireless Manager.

Enable the Wireless Manager

The Wireless Manager is disabled at the moment. To enable it please press the following button.

Attention: Be warned that your configuration will be overwritten!

Enable Wireless Manager

Figure 4.21: Enable page

By pressing the enable button the following settings will be changed and the NWM started.

Setting	Value
cfgNwmEnabled.0	1
cfgWlanDevModulation.0	12
cfgWlanDevFrequency.0	5260
cfgWlanDevBandwidth.0	0
cfgWlanIfaceScanList.0	0
cfgNetWlanEnabled.0	1
cfgNetWlanEnabled.1	1
cfgWlanIfaceMode.1	2
cfgChMgrUsableFrequencyList.0	0
cfgWlanFFreq0.0	5260
cfgWlanFFreq1.0	5280
cfgWlanFFreq2.0	5300
cfgWlanFFreq3.0	5320
cfgWlanFFreq4.0	5500
cfgWlanFFreq5.0	5520
cfgWlanFFreq6.0	5540
cfgWlanFFreq7.0	5560
cfgWlanFFreq8.0	5580
cfgWlanFFreq9.0	5660
cfgWlanFFreq10.0	5700

Table 4.6: Default values for the NWM

It takes some time to enable the NWM. If the NWM could be successfully started, the configuration page as in figure 4.22 should be shown.

Wireless Manager - Configuration

Welcome to the Wireless Manager.

This page will guide you through the needed configuration to setup a proper working Wireless Manager.

Settings

Wireless Manager settings

Bandwidth (MHz)

Frequency (MHz)

Frequencies (MHz)	
5180 (HT20 / HT40+)	 Add
5260 MHz	 Remove
5280 MHz	 Remove
5300 MHz	 Remove
5320 MHz	 Remove
5500 MHz	 Remove
5520 MHz	 Remove
5540 MHz	 Remove
5560 MHz	 Remove
5580 MHz	 Remove
5660 MHz	 Remove
5680 MHz	 Remove
5700 MHz	 Remove

 Apply

Figure 4.22: Configuration page

Bandwidth This define the preferred bandwidth. You can choose HT20, HT40+ or HT40-.

Frequency This define the preferred frequency to use. This depend on the selected bandwidth.

Frequency list The frequencies in this list are used as avoiding-possibility if a radar was detected on your preferred frequency.

In HT40+/HT40- you should always add the frequency and the extended frequency: For HT40+, 5300MHz you should add 5300MHz and 5320MHz.

Attention: If the frequency list is empty, all available frequencies will be used. The available frequencies depend on your country code.

To change the configuration and restart the services press *Apply*.

The status page as in figure 4.23 shows all important status informations about the NWM. The page will be automatically refreshed so you will always see the actual status.

Wireless Manager - Status

Wireless Manager status

☒ Auto refresh every 2 seconds

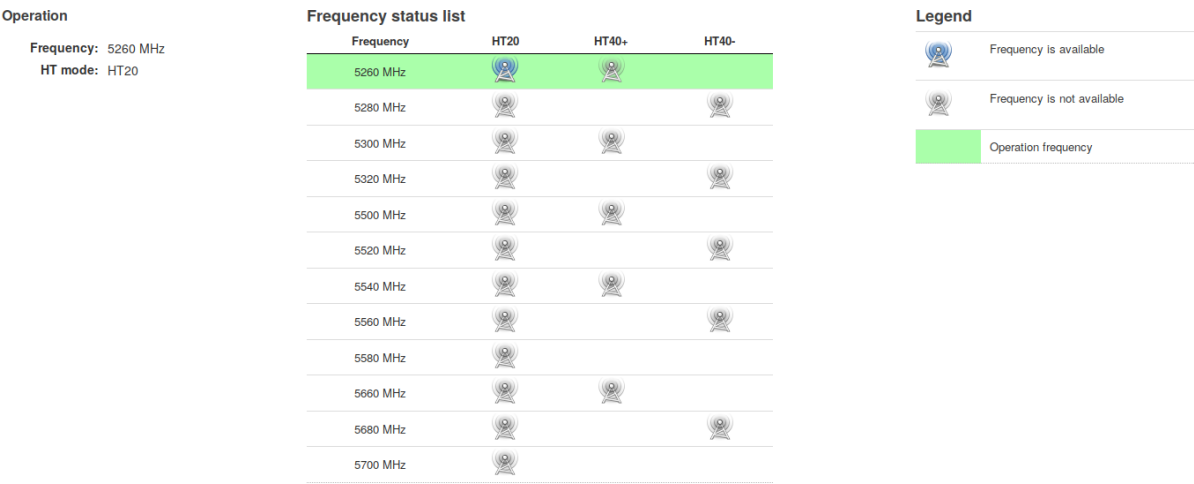


Figure 4.23: Status page

4.44.3 Area Frequency Management (AFM)

Area Frequency Management (AFM) enables communication between Access Points in a mobility application. AFM is mainly used in DFS enabled environments where frequency management over multiple Access Points is required to cope with radar escapes, i.e. when the frequency changes due to detection of a Radar pattern. The main tasks of the AFM is to update the neighbour lists of the Access Points, the so called Dynamic Neighbour List (DNL). The function of Dynamic Neighbour List is described in [Inter-AP Roaming](#)

4.44.3.1 Functional Principle

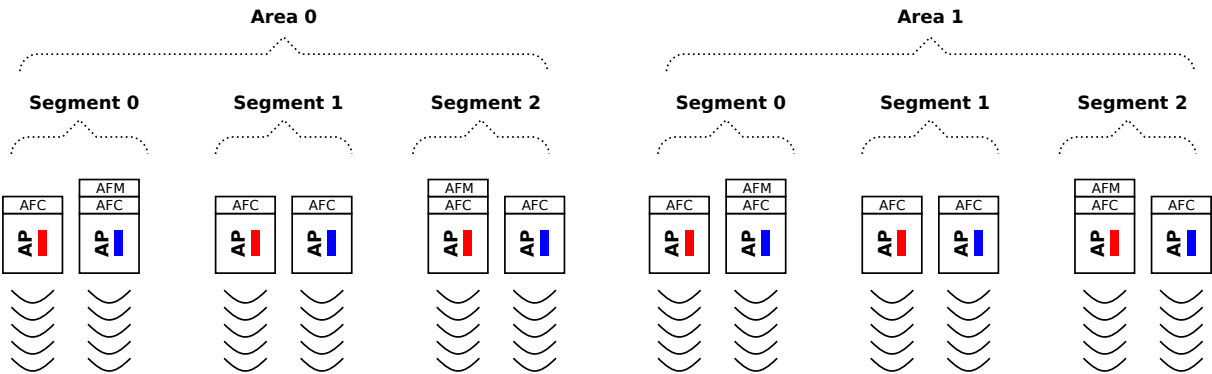


Figure 4.24: Area Frequency Management Principle

The logical structure behind Areas Frequency Management is a line. Due to performance and availability reasons the line can be split into one or more Areas, where each Area is controlled by an AFM (Area Frequency Manager) instance. The AFM instance can be run on any of the Access Point within the Area or on a device in the same IP subnet. An Area is further split into Segments, where one or more Access Points can be assigned to a Segment. The common property of a Segment is the operation frequency (i.e. the AFM assigns the same frequency to all Access Points in the Segment). Each Access Point in the Segment is controlled by an AFC (Area Frequency Client) instance. The Segments' positions within its Area are identified by their Segment index. The AFM communicates with all AFCs within his Area to exchange the required information.

An AFM also communicates with its adjacent AFMs (Area index ± 1) to exchange information about the Segments. This allows the AFMs to generate Dynamic Neighbor Lists for its AFCs which includes information from neighbour Areas. It is also possible to define a redundant AFM instance for each Area. This AFM instance will take over if the primary AFM instance fails.

The AFM computes the Dynamic Neighbour List (DNL) for each AFC in its Area. A client (STA) connected to the Access Point can request this DNL and use for the next handover. The Access Point sends updates on DNL changes. The DNL is defined by the Neighbour Offset table for each AFC. The offset is relative to the AFC on which it is defined, i.e. the delta to the Segment index. A typical DNL is shown in figure 4.25. In this example the DNL for AP5 contains information (i.e. frequency, BSSID, etc.) from AP3 to AP7.

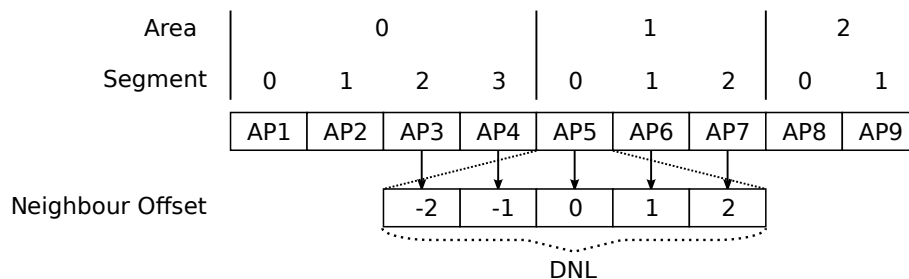


Figure 4.25: Area, Segments in relation to Neighbour Offset and DNL

The nominal frequency of a Segment is defined by the `cfgWlanDevFrequency` configured on the Access Point. If this frequency is available, it will be used as operating frequency. If an Access Point detects a Radar pattern on the operating frequency, then the AFC informs the AFM which will select another operating frequency for the Segment. The AFM selects the escape frequency based on runtime status and chosen frequency plan algorithm. Those frequencies being available at the relevant segment index are evaluated by the rule set to select the optimum escape frequency. The rule set supports preferring larger operation frequency distance between adjacent Access Points. As soon as the nominal frequency becomes available again, the AFM will set it as operating frequency again.

The first radio (antenna ports A1, A2 and A3) is the so called communication interface providing the AP functionality. The second radio (antenna port A4) is responsible to make Channel Availability Checks on channels which are not yet available. During periods where no further CACs are re-

quired, the second radio can be used to do scan the environment for interferences as described in section [Interference Detection Function \(IDF\)](#)

Antenna port A4 is used for CAC and Off-Channel-CAC. Therefore you need to assert that you have an antenna connected at antenna port A4.

4.44.3.2 Configuration

The AFM configuration is described in [cfgAfm](#), [cfgAfc](#) and [cfgChannelCleaner](#). The following configuration steps are required on the Access Points:

1. Configure WLAN in Access Point mode by [cfgWlanInterfaceMode](#).
2. Configure a common set of wireless parameters: [cfgWlanInterfaceSsid](#), [cfgWlanInterfaceEncryption](#), etc.
3. Enable the mandatory services: AFC [cfgAfcEnabled](#).
4. Define the list of usable frequencies available for the system by [cfgChanCleanUsableFrequencyList](#).
5. Define the nominal frequency plan by assigning a nominal frequency for each Access Point by [cfgWlanDevFrequency](#).
6. Define Areas:
 - a) Select one Access Point within each area which shall run the primary AFM instance by [cfgAfmEnabled](#) and [cfgAfmPrimary](#).
 - b) Select another Access Point within each area which shall run the redundant AFM instance by [cfgAfmEnabled](#) and [cfgAfmPrimary](#). Secondary AFM is optional.
 - c) Define the Area index for primary and secondary AFM instance by [cfgAfmIndex](#).
 - d) Define the Area size for primary and secondary AFM instance by [cfgAfmAreaSize](#). The Area size defines the number of Segments within an Area
 - e) For each AFM instance define the list of primary and secondary neighbour AFMs (max 4, primary/secondary to the right and primary/secondary to the left) by [cfgAfmNeighbourTable](#).
 - f) For primary and secondary AFM define the same list of AFC to be managed by [cfgAfmAfcTable](#).
7. Within each Area define Segments:

- Assign the Segment (index) to each Access Point by [cfgAfcIndex](#).
- For each AFC define its primary and secondary AFM by [cfgAfcAfmTable](#).
- For each AFC define the relative offset of neighbour Access Points which shall appear in the Dynamic Neighbour List by [cfgAfcNeighbourOffsetTable](#).

4.44.3.3 Example

Area Segment	0				1		
	0	1	2	3	0	1	2
	AP0	AP1	AP2	AP3	AP4	AP5	AP6
IP	.10	.11	.12	.13	.14	.15	.16
AFC	afc0	afc1	afc2	afc3	afc4	afc5	afc6
Primary AFM	afm0p				afm1p		
Redundant AFM		afm0r				afm1r	
Nominal Freq.	5500	5580	5700	5500	5580	5700	5500

Figure 4.26: Example AFM configuration

AFM Configuration:

Parameter	AP0	AP1	AP2	AP3	AP4	AP5	AP6
cfgAfmEnabled.0	1	1	0	0	1	1	0
cfgAfmName.0	afm0p	afm0r	-	-	afm1p	afm1r	-
cfgAfmPrimary.0	1	0	-	-	1	0	-
cfgAfmIndex.0	0	0	-	-	1	1	-
cfgAfmAreaSize.0	4	4	-	-	3	3	-
cfgAfmRedundantName.0	afm0r	afm0p	-	-	afm1r	afm1p	-
cfgAfmRedundantIp.0	.11	.10	-	-	.15	.14	-
cfgAfmNeighbourName.0	afm1p	afm1p	-	-	afm0p	afm0p	-
cfgAfmNeighbourIp.0	.14	.14	-	-	.10	.10	-
cfgAfmNeighbourName.1	afm1r	afm1r	-	-	afm0r	afm0r	-
cfgAfmNeighbourIp.1	.15	.15	-	-	.11	.11	-
cfgAfmAfcName.0	afc0	afc0	-	-	afc4	afc4	-
cfgAfmAfcIp.0	.10	.10	-	-	.14	.14	-
cfgAfmAfcName.1	afc1	afc1	-	-	afc5	afc5	-
cfgAfmAfcIp.1	.11	.11	-	-	.15	.15	-
cfgAfmAfcName.2	afc2	afc2	-	-	afc6	afc6	-
cfgAfmAfcIp.2	.12	.12	-	-	.16	.16	-
cfgAfmAfcName.3	afc3	afc3	-	-	-	-	-
cfgAfmAfcIp.3	.13	.13	-	-	-	-	-

AFC Configuration:

Parameter	AP0	AP1	AP2	AP3	AP4	AP5	AP6
cfgAfcEnabled.0	1						
cfgAfcName.0	afc0	afc1	afc2	afc3	afc4	afc5	afc6
cfgAfcIndex.0	0	1	2	3	0	1	2
cfgAfcBackupFreq.0	5180						
cfgAfcAfmName.0	afm0p	afm0p	afm0p	afm0p	afm1p	afm1p	afm1p
cfgAfcAfmIp.0	.10	.10	.10	.10	.14	.14	.14
cfgAfcAfmName.1	afm0r	afm0r	afm0r	afm0r	afm1r	afm1r	afm1r
cfgAfcAfmIp.2	.11	.11	.11	.11	.15	.15	.15
cfgAfcNeighbourOffset.0	-	-	-2	-2	-2	-2	-2
cfgAfcNeighbourOffset.1	-	-1	-1	-1	-1	-1	-1
cfgAfcNeighbourOffset.2	0	0	0	0	0	0	0
cfgAfcNeighbourOffset.3	1	1	1	1	1	1	-
cfgAfcNeighbourOffset.4	2	2	2	2	2	-	-

Common Configuration:

Parameter	AP0	AP1	AP2	AP3	AP4	AP5	AP6
cfgWlanDevModulation.0	12						
cfgWlanDevModulation.1	12						
cfgWlanDevFrequency.0	5500	5580	5700	5500	5580	5700	5500
cfgWlanIfaceMode.0	0						
cfgWlanIfaceMode.1	2						
cfgWlanIfaceNeighbourReport.0	1						
cfgWlanFFreq0.0	5180						
cfgWlanFFreq1.0	5500						
cfgWlanFFreq2.0	5520						
cfgWlanFFreq3.0	5540						
cfgWlanFFreq4.0	5560						
cfgWlanFFreq5.0	5580						
cfgWlanFFreq6.0	5660						
cfgWlanFFreq7.0	5680						
cfgWlanFFreq8.0	5700						
cfgChanCleanUsableFrequencyList	0						

4.45 Interference Detection Function (IDF)

Each wireless device (AP and STA) has the functionality to analyze the operation frequency in use. Those values can be read out at any time on any device via SNMP.

In RT-370 products, the IDF functionality supports JSON reports which can be sent to a configured URL ([cfgHttpRprtServerUrl](#)) at a configured interval ([cfgIdfInterval](#)).

IDF is (currently) only available on RT-370 devices since it uses the second radio (antenna port A4, i.e. wlan1) interface.

Please note that the second radio is also used for DFS as described in section [4.44.2](#)

Basic IDF configuration:

- Set [cfgIdfEnabled.0](#) to INTEGER: enabled(1)
- Set [cfgHttpRprtServerUrl.0](#) to STRING: http://192.168.1.1:8000/json

Configure second WLAN interface (Monitor Mode):

- Set [cfgWlanInterfaceMode.1](#) to INTEGER: monitor(2) (i.e. Monitor Mode)
- Set [cfgNetWlanEnabled.1](#) to INTEGER: enabled(1)

Example IDF task configuration (scan frequency 5500, wifi data collection):

- Set [cfgIdfScanWorkFreq.0](#) to INTEGER: 5500
- Set [cfgIdfScanWorkAction.0](#) to INTEGER: wifi(4)
- Set [cfgIdfScanWorkSeconds.0](#) to INTEGER: 1

All actions according to the IDF task list ([cfgIdfScanWorkTable](#)) are processed sequentially. The whole process is repeated endlessly. The time per action can be configured ([cfgIdfScanWorkSeconds](#)) in seconds.

For a complete list of IDF configuration elements see also [cfgIdf](#).

4.46 Http Report

The HTTP Report interface is used by several services and provides a simple way of status reporting to a standard HTTP server.

- Protocol: HTTP POST
- Content-type: application/json
- Servers URL: Can be configured by [cfgHttpRprtServerUrl](#)

4.46.1 NWM and ChannelManager Report

For more information about the current status of the NWM and the Channel Manager (see section 4.44.2) it is possible to request HTTP reports via SNMP.

Nwm 'status' report: Set [rpcNwmHttpReport.0](#) to 1

```
{
  "report": "NwmStatus",
  "epoch": 1436275841,
  "data": {
    "name": "Nwm",
    "nominal_freq": { "freq": 5300, "ext_freq": 5320, "htmode": 1 },
    "opfreq": { "freq": 5300, "ext_freq": 5320, "htmode": 1 }
  }
}
```

Nwm 'frequency state' report: Set [rpcNwmHttpReport.0](#) to 2

```
{
  "report": "NwmFreqState",
  "epoch": 1436277622,
  "data": {
    "name": "Nwm",
    "freq_state_list": [
      [ 5180, 1 ],
      [ 5200, 1 ],
      [ 5220, 1 ],
      [ 5240, 1 ],
      [ 5260, 1 ],
      [ 5280, 1 ],
      [ 5300, 1 ],
      [ 5320, 1 ]
    ]
  }
}
```

Channel Manager 'frequency state' report: Set [rpcChMgrHttpReport.0](#) to 1

```
{
  "report": "ChMgrFreqState",
  "epoch": 1436277622,
  "data": {
    "freq_state_list": [
      [ 5180, 1 ],
      [ 5200, 1 ],
      [ 5220, 1 ],

```

```
[ 5240, 1 ],
[ 5260, 1 ],
[ 5280, 1 ],
[ 5300, 1 ],
[ 5320, 1 ]
]
}
}
```

Channel Manager 'channels' report: Set [rpcChMgrHttpReport.0](#) to 2

```
{
  "report": "ChMgrChannels",
  "epoch": 1436277864,
  "data": {
    "chan_nom": { "freq": 5300, "ext_freq": 5320, "htmode": 1 },
    "proposed_chan": { "freq": 5300, "ext_freq": 5320, "htmode": 1 },
    "chans_ht20": {
      "all": [ 5180, 5200, 5220, 5240, 5260, 5280, 5300, 5320 ],
      "available": [ 5180, 5200, 5220, 5240, 5260, 5280, 5300, 5320 ],
      "htmode": 0
    },
    "chans_ht40m": {
      "all": [ 5200, 5240, 5280, 5320 ],
      "available": [ 5200, 5240, 5280, 5320 ],
      "htmode": 2
    },
    "chans_ht40p": {
      "all": [ 5180, 5220, 5260, 5300 ],
      "available": [ 5180, 5220, 5260, 5300 ],
      "htmode": 1
    }
  }
}
```

4.46.2 IDF Report

An IDF report contains the scan results of all results available in the scan result buffer at the end of the [cfgIdfInterval](#) (i.e. at the time the IDF report is sent).

Please refer to section [4.45](#) for more information on how to configure the IDF

Elements of the IDF Report are:

- data.wlan_stats will only be present if `cfgIdfScanWorkAction` = wifi(4)
- data.wlan_stats.domest_fstats, data.wlan_stats.alien_fstats and data.wlan_stats.alien_mac might not be sent in each report
- data.radar_reports will only be present if `cfgIdfScanWorkAction` = radar(3)
- data.spectral_reports will only be present if `cfgIdfScanWorkAction` = spectral(2)

General elements of the IDF Report wlan_stats element are:

- epoch: UNIX time stamp
- freq: frequency in MHz
- bandwidth: bandwidth in MHz
- window_time: window time in milliseconds
- busy_time: busy time value in milliseconds
- rx_time: rx time value in milliseconds
- tx_time: tx time value in milliseconds

Elements of the IDF Report data.wlan_stats.domest_fstats and data.wlan_stats.alien_fstats element are:

- frame_cnt: frame counter
- frame_size: minimum, average and maximum frame length counter
- frame_total: bytes counter
- rssi: minimum, average and maximum frame RSSI

Elements of the IDF Report data.wlan_stats.alien_mac (list of alien MAC addresses found) element are:

- mac: MAC address
- frame_total: frame counter
- rssi_max: maximum frame RSSI
- rssi_avg: average frame RSSI

Elements of an IDF Report radar_reports element are:

- epoch: UNIX time stamp
- freq: Frequency in MHz
- radar_detected: radar counter
- seconds: observation time

Elements of an IDF Report spectral_reports element are:

- epoch: UNIX time stamp
- freq: Frequency in MHz
- num_samples: Number of FFT data processed
- seconds: observation time
- stats: minimum, average and maximum (all in dBm) of bin0, bin1, bin3, ..., bin55

Example of an IDF Report:

```
{
  "report": "IDF",
  "epoch": 1418301089,
  "data": {
    "name": "IDF",
    "wlan_stats" : [
      {
        "epoch" : 1315522477,
        "freq" : 5500,
        "bandwidth" : 20,
        "window_time" : 60,
        "busy_time" : 15,
        "rx_time" : 12,
        "tx_time" : 0,
        "domest_fstats" : {
          "frame_cnt" : 100,
          "frame_size" : [100, 200, 300],
          "frame_total" : 20153,
          "rssi" : [ 35, 45, 50 ]
        },
        "alien_fstats" : {
          "frame_cnt" : 10,
          "frame_size" : [ 50, 300, 500 ],
```

```

        "frame_total" : 4598,
        "rssi" : [ 15, 35, 50 ]
    },
    "alien_mac" : [
        {
            "mac" : "00:00:00:00:00:01",
            "frame_total" : 2856,
            "rssi_max" : 50,
            "rssi_avg" : 50
        }
    ]
},
"radar_reports": [
    {
        "epoch": 1315528096,
        "freq": 5700,
        "radar_detected": 2,
        "seconds": 60
    }
],
"spectral_reports": [
    {
        "epoch": 1315522405,
        "freq": 5500,
        "num_samples": 14438,
        "seconds": 60,
        "stats": [
            [-159, -107, -91],
            [-154, -106, -92],
            [-157, -106, -91],
            ..
            [-157, -107, -92]
        ]
    }
]
}
}
}

```

4.47 Firewall

4.47.1 Network Address Translation (NAT)

The *Software 6* uses the well known *netfilter* to filter or mangle network traffic. The configuration is done by the *iptables* application. Most terms are based on these software components.

To use the firewall it has to be enabled. Otherwise no feature described below will work. To enable the feature the `cfgFwEnabled` flag has to be enabled.

In the following sections you can define multiple rules. Please keep in mind that the order of the rules is important! This means the rule with the index 2 will be processed before the rule with the index 3.

4.47.1.1 Port forward

Port forwarding can be used to forward network traffic to another destination. This is also known as *Destination Network Address Translation (DNAT)*.

To illustrate how to configure the port forward, we setup a port forward to a web server and a database server which are common use cases. The goal is to connect from Radio Modem 1 (RM1), through the wireless link to RM2, to the web server or the database server.

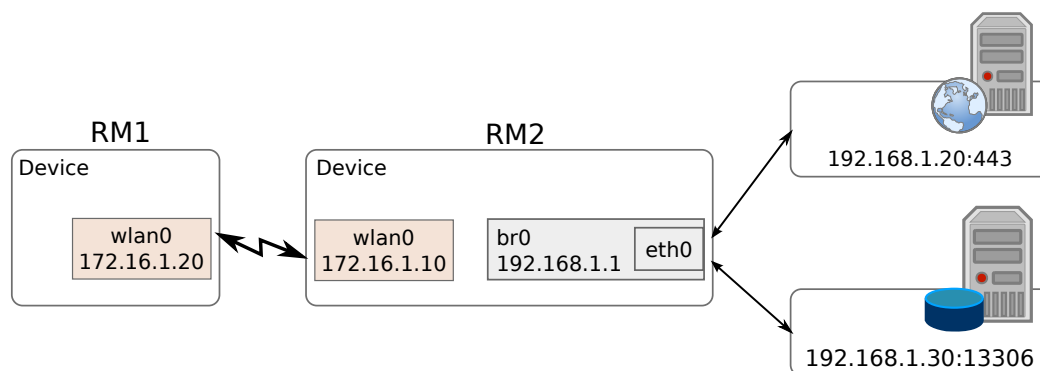


Figure 4.27: Port forward to servers behind an device

To forward the network traffic to the web server and the database server we add two new *rules* to the port forward rules table:

1. For the web server we only want to forward tcp traffic to port 443.
2. For the database server we want to forward tcp and udp traffic for the port range 2000 - 2100 to illustrate port ranges. In addition we want to accept traffic to the wlan from anywhere.

Configuration File Example: Port forward

```
# Config.Format = raw
WESTERMO-SW6-FIREWALL-MIB::cfgFwEnabled.0 = 1
# Port forward to the web server
WESTERMO-SW6-FIREWALL-MIB::cfgFwNatPrtFwdEnabled.0 = 1
WESTERMO-SW6-FIREWALL-MIB::cfgFwNatPrtFwdInterface.0 = wlan0
WESTERMO-SW6-FIREWALL-MIB::cfgFwNatPrtFwdProtocol.0 = 2
WESTERMO-SW6-FIREWALL-MIB::cfgFwNatPrtFwdDestinationAddress.0 = 172.16.1.0/24
WESTERMO-SW6-FIREWALL-MIB::cfgFwNatPrtFwdDestinationPortStart.0 = 443
WESTERMO-SW6-FIREWALL-MIB::cfgFwNatPrtFwdDestinationPortEnd.0 = -1
WESTERMO-SW6-FIREWALL-MIB::cfgFwNatPrtFwdRedirectDestinationAddress.0 = 192.168.1.20
WESTERMO-SW6-FIREWALL-MIB::cfgFwNatPrtFwdRedirectDestinationPort.0 = 443
# Port forward to the database server
WESTERMO-SW6-FIREWALL-MIB::cfgFwNatPrtFwdEnabled.1 = 1
WESTERMO-SW6-FIREWALL-MIB::cfgFwNatPrtFwdInterface.1 = wlan0
WESTERMO-SW6-FIREWALL-MIB::cfgFwNatPrtFwdProtocol.1 = 3
WESTERMO-SW6-FIREWALL-MIB::cfgFwNatPrtFwdDestinationAddress.1 = 0.0.0.0/0
WESTERMO-SW6-FIREWALL-MIB::cfgFwNatPrtFwdDestinationPortStart.1 = 2000
WESTERMO-SW6-FIREWALL-MIB::cfgFwNatPrtFwdDestinationPortEnd.1 = 2100
WESTERMO-SW6-FIREWALL-MIB::cfgFwNatPrtFwdRedirectDestinationAddress.1 = 192.168.1.20
WESTERMO-SW6-FIREWALL-MIB::cfgFwNatPrtFwdRedirectDestinationPort.1 = 13306
```

4.47.1.2 Outbound NAT

With the outbound NAT the device can control how traffic leaving the device will be translated. It's also known as Source NAT (SNAT) and used in the most home routers to rewrite the source address to the address of the WAN interface of the router so the traffic finds the way back home.

The SNAT can be done by simple masquerade, means take the address of the network interface or by defining the source address/port.

As for the port forward we use a simple example to illustrate the functionality as shown in Figure 4.28. The goal is to connect from a Laptop, through RM1 to the web interface of RM2. For this example the Laptop use RM1 as default gateway and the wlan0 interface of RM1 has a dynamic address.

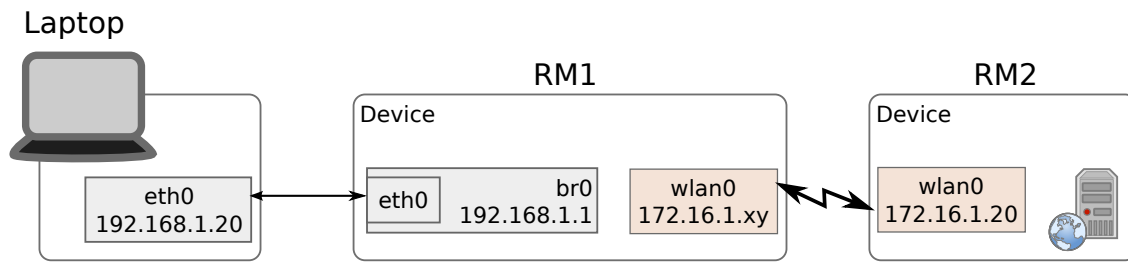


Figure 4.28: Example for outbound NAT on a device

At this point we will only describe the steps to configure the outbound NAT. Following example enables the first rule, set the output interface to *wlan0* and applies to TCP traffic only.

Configuration File Example: Outbound NAT

```
# Config.Format = raw
WESTERMO-SW6-FIREWALL-MIB::cfgFwEnabled.0 = 1
WESTERMO-SW6-FIREWALL-MIB::cfgFwNatOutEnabled.0 = 1
WESTERMO-SW6-FIREWALL-MIB::cfgFwNatOutInterface.0 = wlan0
WESTERMO-SW6-FIREWALL-MIB::cfgFwNatPrtFwdProtocol.0 = 2
```

With this configuration you should be able to connect from the laptop to 172.16.1.20, to the web interface of RM2.

For all possible configurations please see [firewall](#) in the MIB reference.

4.47.2 Filter

The iptables filter allows to create firewall rules based on interfaces, protocol, IP-addresses and ports.

There are 3 chains on which rules can be created:

- **Input**
Rules on the input chain allow/block frames which are destined to the device itself. With these rules you can e.g. block or limit access to the Web Interface or SNMP.
- **Forward**
Rules on the forward chain allow/block frames which are routed/forwarded by the device. This is useful e.g. when multiple VLANs are terminated on the device but traffic between them should not be possible. Be aware that these rules can not be applied when the frames are forwarded on L2 (switched). Use the L2 IP Filter Firewall described below for this.

- Output
Rules on the output chain allow/block frames which originate from the device.

The options `cfgFwFltDefaultPolicyInput`, `cfgFwFltDefaultPolicyForward` and `cfgFwFltDefaultPolicyOutput` define what happens with frames when no rule matches.

When there are multiple conflicting rules which have overlapping match criteria, then the rule with the lowest index will be executed.

Configuration File Example:

Input filtering, Allow SNMP from 192.168.1.0/24 and block everywhere else

```
# Config.Format = raw
WESTERMO-SW6-FIREWALL-MIB::cfgFwEnabled.0 = 1

# Set default input policy to drop
WESTERMO-SW6-FIREWALL-MIB::cfgFwFltDefaultPolicyInput.0 = 0

# Allow traffic to SNMP from the subnet 192.168.1.0/24 to 192.168.1.20
WESTERMO-SW6-FIREWALL-MIB::cfgFwFltREnabled.0 = 1
WESTERMO-SW6-FIREWALL-MIB::cfgFwFltRChain.0 = 1
WESTERMO-SW6-FIREWALL-MIB::cfgFwFltRAction.0 = 1
WESTERMO-SW6-FIREWALL-MIB::cfgFwFltRInputInterface.0 = -1
WESTERMO-SW6-FIREWALL-MIB::cfgFwFltROutputInterface.0 = -1
WESTERMO-SW6-FIREWALL-MIB::cfgFwFltRProtocol.0 = 17
WESTERMO-SW6-FIREWALL-MIB::cfgFwFltRSourceAddress.0 = 192.168.1.0/24
WESTERMO-SW6-FIREWALL-MIB::cfgFwFltRSourcePortStart.0 = -1
WESTERMO-SW6-FIREWALL-MIB::cfgFwFltRSourcePortEnd.0 = -1
WESTERMO-SW6-FIREWALL-MIB::cfgFwFltRDestinationAddress.0 = 192.168.1.20/32
WESTERMO-SW6-FIREWALL-MIB::cfgFwFltRDestinationPortStart.0 = 161
WESTERMO-SW6-FIREWALL-MIB::cfgFwFltRDestinationPortEnd.0 = -1
```

Configuration File Example:

Forward filtering, Allow HTTP from everywhere and block everything else

```
# Config.Format = raw
WESTERMO-SW6-FIREWALL-MIB::cfgFwEnabled.0 = 1

# Set default forward policy to drop
WESTERMO-SW6-FIREWALL-MIB::cfgFwFltDefaultPolicyForward.0 = 0

# Allow HTTP traffic from everywhere to everywhere
WESTERMO-SW6-FIREWALL-MIB::cfgFwFltREnabled.0 = 1
WESTERMO-SW6-FIREWALL-MIB::cfgFwFltRChain.0 = 2
WESTERMO-SW6-FIREWALL-MIB::cfgFwFltRAction.0 = 1
WESTERMO-SW6-FIREWALL-MIB::cfgFwFltRInputInterface.0 = -1
WESTERMO-SW6-FIREWALL-MIB::cfgFwFltROutputInterface.0 = -1
WESTERMO-SW6-FIREWALL-MIB::cfgFwFltRProtocol.0 = 6
WESTERMO-SW6-FIREWALL-MIB::cfgFwFltRSourceAddress.0 = -1
```

```
WESTERMO-SW6-FIREWALL-MIB::cfgFwFltSourcePortStart.0 = -1
WESTERMO-SW6-FIREWALL-MIB::cfgFwFltSourcePortEnd.0 = -1
WESTERMO-SW6-FIREWALL-MIB::cfgFwFltDestinationAddress.0 = -1
WESTERMO-SW6-FIREWALL-MIB::cfgFwFltDestinationPortStart.0 = 80
WESTERMO-SW6-FIREWALL-MIB::cfgFwFltDestinationPortEnd.0 = -1
```

Configuration File Example:

Output filtering, Block access to the tftp server at 192.168.1.3

```
# Config.Format = raw
WESTERMO-SW6-FIREWALL-MIB::cfgFwEnabled.0 = 1

# Set default output policy to accept
WESTERMO-SW6-FIREWALL-MIB::cfgFwFltDefaultPolicyOutput.0 = 1

# Block TFTP traffic to server at 192.168.1.3/32
WESTERMO-SW6-FIREWALL-MIB::cfgFwFltREnabled.0 = 1
WESTERMO-SW6-FIREWALL-MIB::cfgFwFltRChain.0 = 3
WESTERMO-SW6-FIREWALL-MIB::cfgFwFltRAction.0 = 0
WESTERMO-SW6-FIREWALL-MIB::cfgFwFltRInputInterface.0 = -1
WESTERMO-SW6-FIREWALL-MIB::cfgFwFltROutputInterface.0 = -1
WESTERMO-SW6-FIREWALL-MIB::cfgFwFltRProtocol.0 = 17
WESTERMO-SW6-FIREWALL-MIB::cfgFwFltRSourceAddress.0 = -1
WESTERMO-SW6-FIREWALL-MIB::cfgFwFltRSourcePortStart.0 = -1
WESTERMO-SW6-FIREWALL-MIB::cfgFwFltRSourcePortEnd.0 = -1
WESTERMO-SW6-FIREWALL-MIB::cfgFwFltRDestinationAddress.0 = 192.168.1.3/32
WESTERMO-SW6-FIREWALL-MIB::cfgFwFltRDestinationPortStart.0 = 69
WESTERMO-SW6-FIREWALL-MIB::cfgFwFltRDestinationPortEnd.0 = -1
```

4.47.3 L2 Filter

The L2 Filter can be used to filter frames on bridges.

To make use of the L2 Filter, it must be enabled by [cfgFwL2IpFilterEnabled](#) and the global default action must be defined by [cfgFwL2IpFilterDefaultAction](#). Since the global default action applies to all bridges, take care to not lock yourself out when the default action is 'drop'.

Up to 256 rules may be defined in the [cfgFwL2IpFilterTable](#).

The filter offers two modes of operation, selectable by [cfgFwL2IpFilterMode](#):

- IP mode (0)
- Full mode (1)

When the filter is set to *IP mode* then the filter will only apply on IP frames and will allow any non-IP frames.

For each filter rule the source and destination network/IP on which the rule matches, the bridge on which the rule is installed, the action (accept or drop) to perform and the priority of the rule can be configured.

The priority of a rule is used when multiple rules would match a frame. The rule with the highest priority will be executed for the matched frame. Multiple rules may not have the same priority since such a configuration results in undefined behaviour.

Following example of a L2 IP Filter rule drops IP frames with any source IP and 192.168.3.130 as destination IP on bridge 0. All other frames are processed normally.

Configuration File Example: L2 IP Filter

```
# Config.Format = raw
WESTERMO-SW6-FIREWALL-MIB::cfgFwL2IpFilterEnabled.0 = 1
WESTERMO-SW6-FIREWALL-MIB::cfgFwL2IpFilterMode.0 = 0
WESTERMO-SW6-FIREWALL-MIB::cfgFwL2IpFilterDefaultAction.0 = 0

WESTERMO-SW6-FIREWALL-MIB::cfgFwL2IpFiltrEnabled.0 = 1
WESTERMO-SW6-FIREWALL-MIB::cfgFwL2IpFiltrBridge.0 = 0
WESTERMO-SW6-FIREWALL-MIB::cfgFwL2IpFiltrSource.0 = 0.0.0.0/0
WESTERMO-SW6-FIREWALL-MIB::cfgFwL2IpFiltrDestination.0 = 192.168.3.130/32
WESTERMO-SW6-FIREWALL-MIB::cfgFwL2IpFiltrAction.0 = 1
```

When the mode is set to *Full mode* then the filter will process any frame.

In addition to the options of the IP mode, the Full mode offers the ability to filter on the 'IP protocol', 'Source Port' (UDP/TCP), 'Destination Port' (UDP/TCP), Source MAC, Destination MAC, Ethertype, VLAN, and Input Interface. Please refer to [cfgFwL2IpFilterTable](#) for more information.

In a setup with multiple APs it may be desirable to prevent traffic between different APs (Wifi Hotspot).

[cfgWlanIfaceApsolate](#) may be used to prevent traffic between clients on the same wireless interface.

[cfgNetWlanProtected](#) may be used to prevent traffic to flow between wlan0 and wlan1.

The following example of a ruleset may be used to prevent traffic between clients on different APs.

- Allow any frame by default
- br0: drop any unmatched frame on br0

- br0: allow frames from and to the MAC address 00:14:5a:02:11:95 of a gateway with the IP 192.168.3.1
- br0: allow ARP lookups for the gateway at 192.168.3.1
- br0: allow DHCP requests from wlan0 and wlan1
- br0: allow frames from the management interface on br0.vlan0
- br0: allow ARP lookups for the management interface at 192.168.3.22
- br0: allow IP frames to the management interface at 192.168.3.22

Layer 2 Filters

Layer 2 IP Filter ☒ Enabled

Default Action

Filter Mode

L2 IP Filter

Maximum number of rows: 256

[Add row](#)

Enabled	Bridge	Action	Priority	Source IP/Net	Destination IP/Net	Protocol	Source Port	Destination Port	Ethertype	Source MAC	Destination MAC	VLAN	Input Interface	Comment
☒	0	drop	1	0.0.0.0/0	0.0.0.0/0	-1	-1	-1	0000	00:00:00:00:00:00:00:00:00:00:00:00:00	00:00:00:00:00:00:00:00:00:00:00:00:00	-1	any	default drop
☒	0	accept	1000	0.0.0.0/0	0.0.0.0/0	-1	-1	-1	0000	00:00:00:00:00:00:00:00:00:00:00:00:00	00:00:00:00:00:00:00:00:00:00:00:00:00	-1	br0.vlan0	allow from mgmt interface (from device)
☒	0	accept	1001	0.0.0.0/0	192.168.3.22/32	-1	-1	-1	0800	00:00:00:00:00:00:00:00:00:00:00:00:00	00:00:00:00:00:00:00:00:00:00:00:00:00	-1	any	allow IP traffic to mgmt interface (to device)
☒	0	accept	1002	0.0.0.0/0	192.168.3.22/32	-1	-1	-1	0806	00:00:00:00:00:00:00:00:00:00:00:00:00	00:00:00:00:00:00:00:00:00:00:00:00:00	-1	any	allow ARP lookups for the mgmt interface
☒	0	accept	2000	0.0.0.0/32	255.255.255.255/32	17	68	67	0800	00:00:00:00:00:00:00:00:00:00:00:00:00	00:00:00:00:00:00:00:00:00:00:00:00:00	-1	wlan0	allow DHCP from wlan0
☒	0	accept	2001	0.0.0.0/32	255.255.255.255/32	17	68	67	0800	00:00:00:00:00:00:00:00:00:00:00:00:00	00:00:00:00:00:00:00:00:00:00:00:00:00	-1	wlan1	allow DHCP from wlan1
☒	0	accept	3000	0.0.0.0/0	192.168.3.1/32	-1	-1	-1	0806	00:00:00:00:00:00:00:00:00:00:00:00:00	00:00:00:00:00:00:00:00:00:00:00:00:00	-1	any	allow ARP lookups for the gateway
☒	0	accept	4000	0.0.0.0/0	0.0.0.0/0	-1	-1	-1	0000	00:00:00:00:00:00:00:00:00:00:00:00:00	00:14:5a:02:11:95:00:00:00:00:00:00:00	-1	any	allow all frames to MAC of gateway
☒	0	accept	4001	0.0.0.0/0	0.0.0.0/0	-1	-1	-1	0000	00:14:5a:02:11:95:00:00:00:00:00:00:00	00:00:00:00:00:00:00:00:00:00:00:00:00	-1	any	allow all frames from MAC of gateway

Figure 4.29: Full L2 filter to separate clients on different APs

Configuration File Example: L2 Full Filter

```
# Config.Format = raw
WESTERMO-SW6-FIREWALL-MIB::cfgFwL2IpFilterEnabled.0 = 1
WESTERMO-SW6-FIREWALL-MIB::cfgFwL2IpFilterDefaultAction.0 = 0
WESTERMO-SW6-FIREWALL-MIB::cfgFwL2IpFilterMode.0 = 1

WESTERMO-SW6-FIREWALL-MIB::cfgFwL2IpFtrEnabled.0 = 1
WESTERMO-SW6-FIREWALL-MIB::cfgFwL2IpFtrBridge.0 = 0
WESTERMO-SW6-FIREWALL-MIB::cfgFwL2IpFtrAction.0 = 1
WESTERMO-SW6-FIREWALL-MIB::cfgFwL2IpFtrPriority.0 = 1
WESTERMO-SW6-FIREWALL-MIB::cfgFwL2IpFtrSource.0 = 0.0.0.0/0
WESTERMO-SW6-FIREWALL-MIB::cfgFwL2IpFtrDestination.0 = 0.0.0.0/0
WESTERMO-SW6-FIREWALL-MIB::cfgFwL2IpFtrProtocol.0 = -1
WESTERMO-SW6-FIREWALL-MIB::cfgFwL2IpFtrSourcePort.0 = -1
WESTERMO-SW6-FIREWALL-MIB::cfgFwL2IpFtrDestinationPort.0 = -1
WESTERMO-SW6-FIREWALL-MIB::cfgFwL2IpFtrSourceMac.0 = 00:00:00:00:00:00/00:00:00:00:00:00
WESTERMO-SW6-FIREWALL-MIB::cfgFwL2IpFtrDestinationMac.0 =
    → 00:00:00:00:00:00/00:00:00:00:00:00
```

```

WESTERMO-SW6-FIREWALL-MIB::cfgFwL2IpFiltrEthertype.0 = 0000
WESTERMO-SW6-FIREWALL-MIB::cfgFwL2IpFiltrVlan.0 = -1
WESTERMO-SW6-FIREWALL-MIB::cfgFwL2IpFiltrInputInterface.0 = any
WESTERMO-SW6-FIREWALL-MIB::cfgFwL2IpFiltrComment.0 = default drop

WESTERMO-SW6-FIREWALL-MIB::cfgFwL2IpFiltrEnabled.1 = 1
WESTERMO-SW6-FIREWALL-MIB::cfgFwL2IpFiltrBridge.1 = 0
WESTERMO-SW6-FIREWALL-MIB::cfgFwL2IpFiltrAction.1 = 0
WESTERMO-SW6-FIREWALL-MIB::cfgFwL2IpFiltrPriority.1 = 1000
WESTERMO-SW6-FIREWALL-MIB::cfgFwL2IpFiltrSource.1 = 0.0.0.0/0
WESTERMO-SW6-FIREWALL-MIB::cfgFwL2IpFiltrDestination.1 = 0.0.0.0/0
WESTERMO-SW6-FIREWALL-MIB::cfgFwL2IpFiltrProtocol.1 = -1
WESTERMO-SW6-FIREWALL-MIB::cfgFwL2IpFiltrSourcePort.1 = -1
WESTERMO-SW6-FIREWALL-MIB::cfgFwL2IpFiltrDestinationPort.1 = -1
WESTERMO-SW6-FIREWALL-MIB::cfgFwL2IpFiltrSourceMac.1 = 00:00:00:00:00:00/00:00:00:00:00:00
WESTERMO-SW6-FIREWALL-MIB::cfgFwL2IpFiltrDestinationMac.1 =
    ↪ 00:00:00:00:00:00/00:00:00:00:00:00
WESTERMO-SW6-FIREWALL-MIB::cfgFwL2IpFiltrEthertype.1 = 0000
WESTERMO-SW6-FIREWALL-MIB::cfgFwL2IpFiltrVlan.1 = -1
WESTERMO-SW6-FIREWALL-MIB::cfgFwL2IpFiltrInputInterface.1 = br0.vlan0
WESTERMO-SW6-FIREWALL-MIB::cfgFwL2IpFiltrComment.1 = allow anything from the mgmt interface
    ↪ (from device)

WESTERMO-SW6-FIREWALL-MIB::cfgFwL2IpFiltrEnabled.2 = 1
WESTERMO-SW6-FIREWALL-MIB::cfgFwL2IpFiltrBridge.2 = 0
WESTERMO-SW6-FIREWALL-MIB::cfgFwL2IpFiltrAction.2 = 0
WESTERMO-SW6-FIREWALL-MIB::cfgFwL2IpFiltrPriority.2 = 1001
WESTERMO-SW6-FIREWALL-MIB::cfgFwL2IpFiltrSource.2 = 0.0.0.0/0
WESTERMO-SW6-FIREWALL-MIB::cfgFwL2IpFiltrDestination.2 = 192.168.3.22/32
WESTERMO-SW6-FIREWALL-MIB::cfgFwL2IpFiltrProtocol.2 = -1
WESTERMO-SW6-FIREWALL-MIB::cfgFwL2IpFiltrSourcePort.2 = -1
WESTERMO-SW6-FIREWALL-MIB::cfgFwL2IpFiltrDestinationPort.2 = -1
WESTERMO-SW6-FIREWALL-MIB::cfgFwL2IpFiltrSourceMac.2 = 00:00:00:00:00:00/00:00:00:00:00:00
WESTERMO-SW6-FIREWALL-MIB::cfgFwL2IpFiltrDestinationMac.2 =
    ↪ 00:00:00:00:00:00/00:00:00:00:00:00
WESTERMO-SW6-FIREWALL-MIB::cfgFwL2IpFiltrEthertype.2 = 0800
WESTERMO-SW6-FIREWALL-MIB::cfgFwL2IpFiltrVlan.2 = -1
WESTERMO-SW6-FIREWALL-MIB::cfgFwL2IpFiltrInputInterface.2 = any
WESTERMO-SW6-FIREWALL-MIB::cfgFwL2IpFiltrComment.2 = allow IP traffic to the mgmt interface (
    ↪ to device)

WESTERMO-SW6-FIREWALL-MIB::cfgFwL2IpFiltrEnabled.3 = 1
WESTERMO-SW6-FIREWALL-MIB::cfgFwL2IpFiltrBridge.3 = 0
WESTERMO-SW6-FIREWALL-MIB::cfgFwL2IpFiltrAction.3 = 0
WESTERMO-SW6-FIREWALL-MIB::cfgFwL2IpFiltrPriority.3 = 1002
WESTERMO-SW6-FIREWALL-MIB::cfgFwL2IpFiltrSource.3 = 0.0.0.0/0
WESTERMO-SW6-FIREWALL-MIB::cfgFwL2IpFiltrDestination.3 = 192.168.3.22/32
WESTERMO-SW6-FIREWALL-MIB::cfgFwL2IpFiltrProtocol.3 = -1
WESTERMO-SW6-FIREWALL-MIB::cfgFwL2IpFiltrSourcePort.3 = -1

```

```

WESTERMO-SW6-FIREWALL-MIB::cfgFwL2IpFiltrDestinationPort.3 = -1
WESTERMO-SW6-FIREWALL-MIB::cfgFwL2IpFiltrSourceMac.3 = 00:00:00:00:00:00/00:00:00:00:00:00
WESTERMO-SW6-FIREWALL-MIB::cfgFwL2IpFiltrDestinationMac.3 =
    ↪ 00:00:00:00:00:00/00:00:00:00:00:00
WESTERMO-SW6-FIREWALL-MIB::cfgFwL2IpFiltrEthertype.3 = 0806
WESTERMO-SW6-FIREWALL-MIB::cfgFwL2IpFiltrVlan.3 = -1
WESTERMO-SW6-FIREWALL-MIB::cfgFwL2IpFiltrInputInterface.3 = any
WESTERMO-SW6-FIREWALL-MIB::cfgFwL2IpFiltrComment.3 = allow ARP lookups for the mgmt
    ↪ interface

WESTERMO-SW6-FIREWALL-MIB::cfgFwL2IpFiltrEnabled.4 = 1
WESTERMO-SW6-FIREWALL-MIB::cfgFwL2IpFiltrBridge.4 = 0
WESTERMO-SW6-FIREWALL-MIB::cfgFwL2IpFiltrAction.4 = 0
WESTERMO-SW6-FIREWALL-MIB::cfgFwL2IpFiltrPriority.4 = 2000
WESTERMO-SW6-FIREWALL-MIB::cfgFwL2IpFiltrSource.4 = 0.0.0.0/32
WESTERMO-SW6-FIREWALL-MIB::cfgFwL2IpFiltrDestination.4 = 255.255.255.255/32
WESTERMO-SW6-FIREWALL-MIB::cfgFwL2IpFiltrProtocol.4 = 17
WESTERMO-SW6-FIREWALL-MIB::cfgFwL2IpFiltrSourcePort.4 = 68
WESTERMO-SW6-FIREWALL-MIB::cfgFwL2IpFiltrDestinationPort.4 = 67
WESTERMO-SW6-FIREWALL-MIB::cfgFwL2IpFiltrSourceMac.4 = 00:00:00:00:00:00/00:00:00:00:00:00
WESTERMO-SW6-FIREWALL-MIB::cfgFwL2IpFiltrDestinationMac.4 =
    ↪ 00:00:00:00:00:00/00:00:00:00:00:00
WESTERMO-SW6-FIREWALL-MIB::cfgFwL2IpFiltrEthertype.4 = 0800
WESTERMO-SW6-FIREWALL-MIB::cfgFwL2IpFiltrVlan.4 = -1
WESTERMO-SW6-FIREWALL-MIB::cfgFwL2IpFiltrInputInterface.4 = wlan0
WESTERMO-SW6-FIREWALL-MIB::cfgFwL2IpFiltrComment.4 = allow DHCP from wlan0

WESTERMO-SW6-FIREWALL-MIB::cfgFwL2IpFiltrEnabled.5 = 1
WESTERMO-SW6-FIREWALL-MIB::cfgFwL2IpFiltrBridge.5 = 0
WESTERMO-SW6-FIREWALL-MIB::cfgFwL2IpFiltrAction.5 = 0
WESTERMO-SW6-FIREWALL-MIB::cfgFwL2IpFiltrPriority.5 = 2001
WESTERMO-SW6-FIREWALL-MIB::cfgFwL2IpFiltrSource.5 = 0.0.0.0/32
WESTERMO-SW6-FIREWALL-MIB::cfgFwL2IpFiltrDestination.5 = 255.255.255.255/32
WESTERMO-SW6-FIREWALL-MIB::cfgFwL2IpFiltrProtocol.5 = 17
WESTERMO-SW6-FIREWALL-MIB::cfgFwL2IpFiltrSourcePort.5 = 68
WESTERMO-SW6-FIREWALL-MIB::cfgFwL2IpFiltrDestinationPort.5 = 67
WESTERMO-SW6-FIREWALL-MIB::cfgFwL2IpFiltrSourceMac.5 = 00:00:00:00:00:00/00:00:00:00:00:00
WESTERMO-SW6-FIREWALL-MIB::cfgFwL2IpFiltrDestinationMac.5 =
    ↪ 00:00:00:00:00:00/00:00:00:00:00:00
WESTERMO-SW6-FIREWALL-MIB::cfgFwL2IpFiltrEthertype.5 = 0800
WESTERMO-SW6-FIREWALL-MIB::cfgFwL2IpFiltrVlan.5 = -1
WESTERMO-SW6-FIREWALL-MIB::cfgFwL2IpFiltrInputInterface.5 = wlan1
WESTERMO-SW6-FIREWALL-MIB::cfgFwL2IpFiltrComment.5 = allow DHCP from wlan1

WESTERMO-SW6-FIREWALL-MIB::cfgFwL2IpFiltrEnabled.6 = 1
WESTERMO-SW6-FIREWALL-MIB::cfgFwL2IpFiltrBridge.6 = 0
WESTERMO-SW6-FIREWALL-MIB::cfgFwL2IpFiltrAction.6 = 0
WESTERMO-SW6-FIREWALL-MIB::cfgFwL2IpFiltrPriority.6 = 3000
WESTERMO-SW6-FIREWALL-MIB::cfgFwL2IpFiltrSource.6 = 0.0.0.0/0

```

```

WESTERMO-SW6-FIREWALL-MIB::cfgFwL2IpFiltrDestination.6 = 10.0.11.1/32
WESTERMO-SW6-FIREWALL-MIB::cfgFwL2IpFiltrProtocol.6 = -1
WESTERMO-SW6-FIREWALL-MIB::cfgFwL2IpFiltrSourcePort.6 = -1
WESTERMO-SW6-FIREWALL-MIB::cfgFwL2IpFiltrDestinationPort.6 = -1
WESTERMO-SW6-FIREWALL-MIB::cfgFwL2IpFiltrSourceMac.6 = 00:00:00:00:00:00/00:00:00:00:00:00
WESTERMO-SW6-FIREWALL-MIB::cfgFwL2IpFiltrDestinationMac.6 =
    ↪ 00:00:00:00:00:00/00:00:00:00:00:00
WESTERMO-SW6-FIREWALL-MIB::cfgFwL2IpFiltrEthertype.6 = 0806
WESTERMO-SW6-FIREWALL-MIB::cfgFwL2IpFiltrVlan.6 = -1
WESTERMO-SW6-FIREWALL-MIB::cfgFwL2IpFiltrInputInterface.6 = any
WESTERMO-SW6-FIREWALL-MIB::cfgFwL2IpFiltrComment.6 = allow ARP lookups for the gateway

WESTERMO-SW6-FIREWALL-MIB::cfgFwL2IpFiltrEnabled.7 = 1
WESTERMO-SW6-FIREWALL-MIB::cfgFwL2IpFiltrBridge.7 = 0
WESTERMO-SW6-FIREWALL-MIB::cfgFwL2IpFiltrAction.7 = 0
WESTERMO-SW6-FIREWALL-MIB::cfgFwL2IpFiltrPriority.7 = 4000
WESTERMO-SW6-FIREWALL-MIB::cfgFwL2IpFiltrSource.7 = 0.0.0.0/0
WESTERMO-SW6-FIREWALL-MIB::cfgFwL2IpFiltrDestination.7 = 0.0.0.0/0
WESTERMO-SW6-FIREWALL-MIB::cfgFwL2IpFiltrProtocol.7 = -1
WESTERMO-SW6-FIREWALL-MIB::cfgFwL2IpFiltrSourcePort.7 = -1
WESTERMO-SW6-FIREWALL-MIB::cfgFwL2IpFiltrDestinationPort.7 = -1
WESTERMO-SW6-FIREWALL-MIB::cfgFwL2IpFiltrSourceMac.7 = 00:00:00:00:00:00/00:00:00:00:00:00
WESTERMO-SW6-FIREWALL-MIB::cfgFwL2IpFiltrDestinationMac.7 = 00:14:5a:02:11:95/ff:ff:ff:ff:ff:ff
WESTERMO-SW6-FIREWALL-MIB::cfgFwL2IpFiltrEthertype.7 = 0000
WESTERMO-SW6-FIREWALL-MIB::cfgFwL2IpFiltrVlan.7 = -1
WESTERMO-SW6-FIREWALL-MIB::cfgFwL2IpFiltrInputInterface.7 = any
WESTERMO-SW6-FIREWALL-MIB::cfgFwL2IpFiltrComment.7 = allow all frames to MAC of gateway

WESTERMO-SW6-FIREWALL-MIB::cfgFwL2IpFiltrEnabled.8 = 1
WESTERMO-SW6-FIREWALL-MIB::cfgFwL2IpFiltrBridge.8 = 0
WESTERMO-SW6-FIREWALL-MIB::cfgFwL2IpFiltrAction.8 = 0
WESTERMO-SW6-FIREWALL-MIB::cfgFwL2IpFiltrPriority.8 = 4001
WESTERMO-SW6-FIREWALL-MIB::cfgFwL2IpFiltrSource.8 = 0.0.0.0/0
WESTERMO-SW6-FIREWALL-MIB::cfgFwL2IpFiltrDestination.8 = 0.0.0.0/0
WESTERMO-SW6-FIREWALL-MIB::cfgFwL2IpFiltrProtocol.8 = -1
WESTERMO-SW6-FIREWALL-MIB::cfgFwL2IpFiltrSourcePort.8 = -1
WESTERMO-SW6-FIREWALL-MIB::cfgFwL2IpFiltrDestinationPort.8 = -1
WESTERMO-SW6-FIREWALL-MIB::cfgFwL2IpFiltrSourceMac.8 = 00:14:5a:02:11:95/ff:ff:ff:ff:ff:ff
WESTERMO-SW6-FIREWALL-MIB::cfgFwL2IpFiltrDestinationMac.8 =
    ↪ 00:00:00:00:00:00/00:00:00:00:00:00
WESTERMO-SW6-FIREWALL-MIB::cfgFwL2IpFiltrEthertype.8 = 0000
WESTERMO-SW6-FIREWALL-MIB::cfgFwL2IpFiltrVlan.8 = -1
WESTERMO-SW6-FIREWALL-MIB::cfgFwL2IpFiltrInputInterface.8 = any
WESTERMO-SW6-FIREWALL-MIB::cfgFwL2IpFiltrComment.8 = allow all frames from MAC of gateway

```

5 Country Codes

Product regulatory limits and operating parameters are controlled by product software driver with the country code settings. The country code limits are equally valid for Client (STA) and Access Point operation mode. Not all country codes are supported by all product variants and versions.

5.1 Configuration

The configuration values as follows are relevant for the system to compute the allowed output power at antenna port and radiated:

- set [cfgWlanDevAntennaGain](#) Set the antenna gain in dBi. The selected country code describes the range. In some regions the antenna type is fixed. The user cannot change the antenna gain.
- set [cfgWlanDevPower](#) to limit the output power to a defined EIRP level. The modem software will calculate a value which is at this maximum level or lower
- set [cfgWlanDevFrequency](#) to set the nominal operation frequency

5.2 Regions for 802.11n products

5.2.1 Country code WORLD

Country code 'WORLD' is the default country code. The country code shall be changed to the correct country by the user before installation.

	Min	Max
Frequencies 2.4 GHz	2412, 2417, 2422, 2427, 2432, 2437, 2442, 2447, 2452, 2457, 2462	
- Antenna Gain	0 dBi	12 dBi
- EIRP (one antenna)	6 dBm	18 dBm
- EIRP (two antennas)	9 dBm	18 dBm
- EIRP (three antennas)	11 dBm	18 dBm

5.2.2 Region E

- Max. EIRP Power are lower than regulatory power limits to respect the maximum limit at all conditions.
- Client (STA) and Access Point maximal EIRP differences at 5 GHz are depends on operation mode. Access Point mode is DFS master, client (STA) is DFS slave.
- DFS weather channels are disabled due to ETSI requirements.
- For DFS frequencies a modem in client (STA) mode will scan passive (no probe requests)

5.2.2.1 Country code EU

Country code 'EU' applies for Europe.

The country code can be set by:

- Set `cfgWlanGblCountry` to EU

Note 1: EIRP of 28 dBm applies for 5500 to 5700 in Access Point mode (DFS master).

	Min	Max
Frequencies 2.4 GHz	2412, 2417, 2422, 2427, 2432, 2437, 2442, 2447, 2452, 2457, 2462, 2467, 2472	
- Antenna Gain	0 dBi	12 dBi
- EIRP (one antenna)	6 dBm	18 dBm
- EIRP (two antennas)	9 dBm	18 dBm
- EIRP (three antennas)	11 dBm	18 dBm
Frequencies 5 GHz	5180, 5200, 5220, 5240, 5260, 5280, 5300, 5320, 5500, 5520, 5540, 5560, 5580, 5660, 5680, 5700	
- Antenna Gain	0 dBi	15 dBi
- EIRP (one antenna)	6 dBm	21 dBm, 28 dBm (Note 1)
- EIRP (two antennas)	9 dBm	21 dBm, 28 dBm (Note 1)
- EIRP (three antennas)	11 dBm	21 dBm, 28 dBm (Note 1)
- Indoor frequencies	5180 to 5320	
- DFS (slave/master)	5260 to 5700	
Frequencies 5.8 GHz	5745, 5765, 5785, 5805, 5825, 5845, 5865	
- Antenna Gain	0 dBi	7 dBi
- EIRP (one antenna)	6 dBm	13 dBm
- EIRP (two antennas)	9 dBm	13 dBm
- EIRP (three antennas)	11 dBm	13 dBm

5.2.2.2 Country code CHINA

Country code 'CHINA' applies for China using low gain antennas in the 2.4 GHz band.

The country code can be set by:

- Set `cfgWlanGlbCountry` to CHINA

	Min	Max
Frequencies 2.4 GHz	2412, 2417, 2422, 2427, 2432, 2437, 2442, 2447, 2452, 2457, 2462, 2467, 2472	
- Antenna Gain	0 dBi	9 dBi
- EIRP (one antenna)	6 dBm	18 dBm
- EIRP (two antennas)	9 dBm	18 dBm
- EIRP (three antennas)	11 dBm	18 dBm
Frequencies 5.8 GHz	5745, 5765, 5785, 5805, 5825	
- Antenna Gain	0 dBi	14 dBi
- EIRP (one antenna)	6 dBm	31 dBm
- EIRP (two antennas)	9 dBm	31 dBm
- EIRP (three antennas)	11 dBm	31 dBm

5.2.2.3 Country code CHINA_HIGH_GAIN

Country code 'CHINA_HIGH_GAIN' applies for China using high gain antennas in the 2.4 GHz band.

The country code can be set by:

- Set [cfgWlanGlbCountry](#) to CHINA_HIGH_GAIN

	Min	Max
Frequencies 2.4 GHz	2412, 2417, 2422, 2427, 2432, 2437, 2442, 2447, 2452, 2457, 2462, 2467, 2472	
- Antenna Gain	10 dBi	14 dBi
- EIRP (one antenna)	6 dBm	25 dBm
- EIRP (two antennas)	9 dBm	25 dBm
- EIRP (three antennas)	11 dBm	25 dBm
Frequencies 5.8 GHz	5745, 5765, 5785, 5805, 5825	
- Antenna Gain	0 dBi	14 dBi
- EIRP (one antenna)	6 dBm	31 dBm
- EIRP (two antennas)	9 dBm	31 dBm
- EIRP (three antennas)	11 dBm	31 dBm

5.2.2.4 Country code AUS_NZL

Country code 'AUS_NZL' applies for Australia and New Zealand.

The country code can be set by:

- Set `cfgWlanGlbCountry` to AUS_NZL

	Min	Max
Frequencies 2.4 GHz	2412, 2417, 2422, 2427, 2432, 2437, 2442, 2447, 2452, 2457, 2462, 2467, 2472	
- Antenna Gain	0 dBi	12 dBi
- EIRP (one antenna)	6 dBm	34 dBm
- EIRP (two antennas)	9 dBm	34 dBm
- EIRP (three antennas)	11 dBm	34 dBm
Frequencies 5 GHz	5180, 5200, 5220, 5240, 5260, 5280, 5300, 5320, 5500, 5520, 5540, 5560, 5580, 5660, 5680, 5700	
- Antenna Gain	0 dBi	15 dBi
- EIRP (one antenna)	6 dBm	28 dBm
- EIRP (two antennas)	9 dBm	28 dBm
- EIRP (three antennas)	11 dBm	28 dBm
- Indoor frequencies	5180 to 5320	
- DFS	5260 to 5700	
Frequencies 5.8 GHz	5745, 5765, 5785, 5805, 5825	
- Antenna Gain	0 dBi	7 dBi
- EIRP (one antenna)	6 dBm	34 dBm
- EIRP (two antennas)	9 dBm	34 dBm
- EIRP (three antennas)	11 dBm	34 dBm

5.2.2.5 Country code SINGAPORE

Country code 'SINGAPORE' applies for Singapore.

The country code can be set by:

- Set `cfgWlanGlbCountry` to SINGAPORE

	Min	Max
Frequencies 2.4 GHz	2412, 2417, 2422, 2427, 2432, 2437, 2442, 2447, 2452, 2457, 2462, 2467, 2472	
- Antenna Gain	0 dBi	12 dBi
- EIRP (one antenna)	6 dBm	21 dBm
- EIRP (two antennas)	9 dBm	21 dBm
- EIRP (three antennas)	11 dBm	21 dBm
Frequencies 5 GHz	5180, 5200, 5220, 5240, 5260, 5280, 5300, 5320, 5500, 5520, 5540, 5560, 5580, 5660, 5680, 5700	
- Antenna Gain	0 dBi	15 dBi
- EIRP (one antenna)	6 dBm	28 dBm
- EIRP (two antennas)	9 dBm	28 dBm
- EIRP (three antennas)	11 dBm	28 dBm
- Indoor frequencies	5180 to 5320	
- DFS	5260 to 5700	
Frequencies 5.8 GHz	5745, 5765, 5785, 5805, 5825, 5845, 5865	
- Antenna Gain	0 dBi	7 dBi
- EIRP (one antenna)	6 dBm	13 dBm
- EIRP (two antennas)	9 dBm	13 dBm
- EIRP (three antennas)	11 dBm	13 dBm

5.2.3 Region U

- For this region the antenna type is fixed. The user cannot change the antenna gain.
- Max. EIRP Power are lower than regulatory power limits to respect the maximum limit at all conditions.
- DFS weather channels are disabled due to FCC requirements.
- For DFS frequencies a modem in client (STA) mode will scan passive (no probe requests)

5.2.3.1 Country codes USA and CANADA

Country code 'USA' applies for the United States. Country code 'CANADA' applies for the Canada.

The country code can be set by:

- Set `cfgWlanGblCountry` to USA or CANADA

	Min	Max
Frequencies 2.4 GHz	2412, 2417, 2422, 2427, 2432, 2437, 2442, 2447, 2452, 2457, 2462	
- Antenna Types	Refer to Antennas	
- EIRP	Refer to Antennas	
Frequencies 5 GHz	5180, 5200, 5220, 5240, 5260, 5280, 5300, 5320, 5500, 5520, 5540, 5560, 5580, 5660, 5680, 5700	
- Antenna Types	Refer to Antennas	
- EIRP	Refer to Antennas	
- Indoor frequencies	5180 to 5240	
- DFS (slave/master)	5260 to 5720	
Frequencies 5.8 GHz	5745, 5765, 5785, 5805, 5825	
- Antenna Types	Refer to Antennas	
- EIRP	Refer to Antennas	

5.2.3.2 Antennas

Only the PROFESSIONAL user can configure the attached antenna for each product by configuring one of the antennas listed in the Table 5.1. The current antenna configuration of the product is shown on the Web Interface login page. More information about EIRP can be found in the Web Interface under *Application > Regulatory Domain Manager*.

Antenna	Description
H&S SPA 2400/75/8/0/V	1 to 3 active antenna ports allowed Gain 2.4GHz = 8 dBi, Gain 5GHz = na Max. EIRP (2412 to 2462) = 22 dBm Examples: - H&S SPA 2400/75/8/0/V
Tekfun F51-N	1 to 3 active antenna ports allowed Gain 2.4GHz = 5 dBi, Gain 5GHz = 7 dBi Max. EIRP (2412 to 2462) = 24 dBm Max. EIRP (5180 to 5240) = 19 dBm (USA) Max. EIRP (5180 to 5240) = 21 dBm (CANADA) Max. EIRP (5260 to 5320) = 28 dBm (USA) Max. EIRP (5260 to 5320) = 21 dBm (CANADA) Max. EIRP (5500 to 5700) = 28 dBm

Max. EIRP (5745 to 5825) = 28 dBm

Examples:

- Tekfun F51-N
- Westermo ICL 5GHz MIMO Antenna

H&S SPA 5600/40/14/0/V_2

1 or 2 active antenna ports allowed

Gain 2.4GHz = na, Gain 5GHz = 14 dBi

Max. EIRP (5180 to 5240) = 19 dBm (USA)

Max. EIRP (5180 to 5240) = 21 dBm (CANADA)

Max. EIRP (5260 to 5320) = 28 dBm (USA)

Max. EIRP (5260 to 5320) = 21 dBm (CANADA)

Max. EIRP (5500 to 5700) = 28 dBm

Max. EIRP (5745 to 5825) = 30 dBm

Examples:

- H&S SPA 5600/40/14/0/V_2

H&S SPA 5600/65/9/0/MIMO

1 to 3 active antenna ports allowed

Gain 2.4GHz = na, Gain 5GHz = 9 dBi

Max. EIRP (5180 to 5240) = 19 dBm (USA)

Max. EIRP (5180 to 5240) = 21 dBm (CANADA)

Max. EIRP (5260 to 5320) = 28 dBm (USA)

Max. EIRP (5260 to 5320) = 21 dBm (CANADA)

Max. EIRP (5500 to 5700) = 28 dBm

Max. EIRP (5745 to 5825) = 30 dBm

Examples:

- H&S SPA 5600/65/9/0/MIMO

H&S SPA 5600/45/12/10/V

1 or 2 active antenna ports allowed

Gain 2.4GHz = na, Gain 5GHz = 12 dBi

Max. EIRP (5180 to 5240) = 19 dBm (USA)

Max. EIRP (5180 to 5240) = 21 dBm (CANADA)

Max. EIRP (5260 to 5320) = 28 dBm (USA)

Max. EIRP (5260 to 5320) = 21 dBm (CANADA)

Max. EIRP (5500 to 5700) = 28 dBm

Max. EIRP (5745 to 5825) = 28 dBm

Examples:

- H&S SPA 5600/45/12/10/V

H&S SPA 5600/40/14/0/V_2 with Neratec 105072

1 or 2 active antenna ports allowed

Gain 2.4GHz = na, Gain 5GHz = 13 dBi

Max. EIRP (5745-5825) = 34 dBm

Examples:

Table 5.1: *Region U antennas of radio0*

5.3 Regions for 802.11ac products

5.3.1 Region E

- Max. EIRP Power are lower than regulatory power limits to respect the maximum limit at all conditions.
- Client (STA) and Access Point maximal EIRP differences at 5 GHz are depends on operation mode. Access Point mode is DFS master, client (STA) is DFS slave.
- For DFS frequencies a modem in client (STA) mode will scan passive (no probe requests)

5.3.1.1 Europe

Following country codes (<country>) are supported for Europe: 'AU', 'BE', 'BG', 'HR', 'CY', 'CZ', 'DK', 'EE', 'FI', 'FR', 'DE', 'GR', 'HU', 'IE', 'IT', 'LV', 'LT', 'LU', 'MT', 'NL', 'NO', 'PL', 'PT', 'RO', 'SK', 'SI', 'ES', 'SE', 'GB', 'IS', 'LI', 'CH', 'TR'

The country code can be set by:

- Set `cfgWlanGlblCountry` to <country>

	Min	Max
Frequencies 2.4 GHz	2412, 2417, 2422, 2427, 2432, 2437, 2442, 2447, 2452, 2457, 2462, 2467, 2472	
- Antenna Gain	0 dBi	12 dBi
- EIRP (one antenna)	6 dBm	20 dBm
- EIRP (two antennas)	9 dBm	20 dBm
- EIRP (three antennas)	11 dBm	20 dBm
- EIRP (four antennas)	12 dBm	20 dBm
Frequencies 5 GHz	5180, 5200, 5220, 5240, 5260, 5280, 5300, 5320, 5500, 5520, 5540, 5560, 5580, 5600, 5620, 5640, 5660, 5680, 5700	
- Antenna Gain	0 dBi	15 dBi
- EIRP (one antenna)	6 dBm	27 dBm
- EIRP (two antennas)	9 dBm	27 dBm
- EIRP (three antennas)	11 dBm	27 dBm
- EIRP (four antennas)	12 dBm	27 dBm
- Indoor frequencies	5180 to 5320	
- DFS (slave/master)	5260 to 5700	

5.3.2 Region U

- For this region the antenna type is fixed. The user cannot change the antenna gain.
- Max. EIRP Power are lower than regulatory power limits to respect the maximum limit at all conditions.
- DFS weather channels are disabled due to FCC requirements.
- For DFS frequencies a modem in client (STA) mode will scan passive (no probe requests)

5.3.2.1 North America

Following country codes (<country>) are supported for North America: 'US', 'CA'

The country code can be set by:

- Set `cfgWlanGlblCountry` to <country>

	Min	Max
Frequencies 2.4 GHz	2412, 2417, 2422, 2427, 2432, 2437, 2442, 2447, 2452, 2457, 2462	
- Antenna Types	Refer to Antennas	
- EIRP	Refer to Antennas	
Frequencies 5 GHz	5180, 5200, 5220, 5240, 5260, 5280, 5300, 5320, 5500, 5520, 5540, 5560, 5580, 5600, 5620, 5640, 5660, 5680, 5700, 5720	
- Antenna Types	Refer to Antennas	
- EIRP	Refer to Antennas	
- Indoor frequencies	5180 to 5240	
- DFS (slave/master)	5260 to 5720	
Frequencies 5.8 GHz	5745, 5765, 5785, 5805, 5825	
- Antenna Types	Refer to Antennas	
- EIRP	Refer to Antennas	

5.3.2.2 Antennas

Only the PROFESSIONAL user can configure the attached antenna for each the product by configuring one of the antennas listed in the Table 5.2 and 5.3, respectively. The current antenna configuration of the product is shown on the Web Interface login page. Please contact your support for more information.

Antenna	Description
5GHz 5dBi	1 to 4 active antenna ports allowed Gain 2.4GHz = na, Gain 5GHz = 5 dBi Max. EIRP (5180 to 5240) = 23 dBm (USA) Max. EIRP (5180 to 5240) = 21 dBm (CANADA) Max. EIRP (5260 to 5320) = 22 dBm (USA) Max. EIRP (5260 to 5320) = 21 dBm (CANADA) Max. EIRP (5500 to 5700) = 21 dBm Max. EIRP (5745 to 5825) = 29 dBm Examples: - H&S SENCITY®Omni-S MIMO Antenna
5GHz 4dBi	1 to 4 active antenna ports allowed Gain 2.4GHz = na, Gain 5GHz = 4 dBi Max. EIRP (5180 to 5240) = 22 dBm (USA) Max. EIRP (5180 to 5240) = 21 dBm (CANADA) Max. EIRP (5260 to 5320) = 21 dBm

Max. EIRP (5500 to 5700) = 20 dBm
 Max. EIRP (5745 to 5825) = 28 dBm

5GHz 10dBi 1 to 4 active antenna ports allowed
 Gain 2.4GHz = na, Gain 5GHz = 10 dBi
 Max. EIRP (5180 to 5240) = 23 dBm (USA)
 Max. EIRP (5180 to 5240) = 21 dBm (CANADA)
 Max. EIRP (5260 to 5320) = 27 dBm (USA)
 Max. EIRP (5260 to 5320) = 21 dBm (CANADA)
 Max. EIRP (5500 to 5700) = 26 dBm
 Max. EIRP (5745 to 5825) = 34 dBm

Examples:

- H&S SPA 5600/65/9/0/MIMO
- Antonics OmPlecs®-TOP 200 AMR MF-06 -1-
- Antonics OmPlecs®-TOP 200 AMR MF-06 -4-
- Mars MA-WO6960-DP6
- Mars MA-WO3860-MIMO

Table 5.2: Region U antennas of radio0

Antenna	Description
2.4GHz 11dBi	1 or 2 active antenna ports allowed Gain 2.4GHz = 11 dBi, Gain 5GHz = na Max. EIRP (2412 to 2462) = 29 dBm Examples: - H&S SPA 2400/75/8/0/V
5GHz 10dBi	1 or 2 active antenna ports allowed Gain 2.4GHz = na, Gain 5GHz = 10 dBi Max. EIRP (5180 to 5240) = 30 dBm (USA) Max. EIRP (5180 to 5240) = 21 dBm (CANADA) Max. EIRP (5260 to 5320) = 28 dBm (USA) Max. EIRP (5260 to 5320) = 21 dBm (CANADA) Max. EIRP (5500 to 5700) = 28 dBm Max. EIRP (5745 to 5825) = 31 dBm Examples: - H&S SPA 5600/65/9/0/MIMO
2.4GHz, 5GHz 10dBi	1 or 2 active antenna ports allowed Gain 2.4GHz = 10 dBi, Gain 5GHz = 10 dBi Max. EIRP (2412 to 2462) = 28 dBm Max. EIRP (5180 to 5240) = 30 dBm (USA) Max. EIRP (5180 to 5240) = 21 dBm (CANADA) Max. EIRP (5260 to 5320) = 28 dBm (USA)

Max. EIRP (5260 to 5320) = 21 dBm (CANADA)
 Max. EIRP (5500 to 5700) = 28 dBm
 Max. EIRP (5745 to 5825) = 31 dBm

Examples:

- Antonics OmPlecs®-TOP 200 AMR MF-06 -1-
- Antonics OmPlecs®-TOP 200 AMR MF-06 -2-
- Mars MA-WO6960-DP6
- Mars MA-WO3860-MIMO

2.4GHz, 5GHz 7dBi 1 or 2 active antenna ports allowed
 Gain 2.4GHz = 7 dBi, Gain 5GHz = 7 dBi
 Max. EIRP (2412 to 2462) = 25 dBm
 Max. EIRP (5180 to 5240) = 27 dBm (USA)
 Max. EIRP (5180 to 5240) = 21 dBm (CANADA)
 Max. EIRP (5260 to 5320) = 30 dBm (USA)
 Max. EIRP (5260 to 5320) = 21 dBm (CANADA)
 Max. EIRP (5500 to 5700) = 28 dBm
 Max. EIRP (5745 to 5825) = 28 dBm

Examples:

- Tekfun F51-N

2.4GHz, 5GHz 5dBi 1 or 2 active antenna ports allowed
 Gain 2.4GHz = 5 dBi, Gain 5GHz = 5 dBi
 Max. EIRP (2412 to 2462) = 23 dBm
 Max. EIRP (5180 to 5240) = 25 dBm (USA)
 Max. EIRP (5180 to 5240) = 21 dBm (CANADA)
 Max. EIRP (5260 to 5320) = 28 dBm (USA)
 Max. EIRP (5260 to 5320) = 21 dBm (CANADA)
 Max. EIRP (5500 to 5700) = 28 dBm
 Max. EIRP (5745 to 5825) = 26 dBm

Examples:

- H&S SENCITY®Omni-S MIMO Antenna

2.4GHz, 5GHz 2dBi 1 or 2 active antenna ports allowed
 Gain 2.4GHz = 2 dBi, Gain 5GHz = 2 dBi
 Max. EIRP (2412 to 2462) = 20 dBm
 Max. EIRP (5180 to 5240) = 22 dBm (USA)
 Max. EIRP (5180 to 5240) = 21 dBm (CANADA)
 Max. EIRP (5260 to 5320) = 25 dBm (USA)
 Max. EIRP (5260 to 5320) = 21 dBm (CANADA)
 Max. EIRP (5500 to 5700) = 25 dBm
 Max. EIRP (5745 to 5825) = 23 dBm

Table 5.3: Region U antennas of radio1

6 Security Considerations

During commissioning it is often desirable to not have the devices locked down. This helps to debug and analyze issues. However once the commissioning phase is complete the devices should be locked down and access restricted. The following chapter provides some points what to look for and which config parameters to change to reduce the chances of an incident.

6.1 Physical Interfaces

Disable ports which are not in use. This will reduce that chance that an unauthorized party gains unnoticed access to the backbone.

Configuration File Example: Disable the second port (eth1)

```
# Config.Format = raw
WESTERMO-SW6-MIB::cfgNetEthEnabled.1 = 0
```

6.2 Network Concept

6.2.1 Local Administrative Access

Contrary to disabling unused ports, it may be desirable to provide local administrative access to the device. Use cases are to allow access to the device when the backbone is lost, or someone is working physically on site of the device. In such a case keep the normally unused port enabled but on its own bridge with its own IP. This will allow local access to the administrative interface of the device, but no further access to the backbone.

Configuration File Example: Enable local administrative access on second port (eth1)

```
# Config.Format = raw
# eth0
WESTERMO-SW6-MIB::cfgNetEthEnabled.0 = 1
WESTERMO-SW6-MIB::cfgNetEthBridge.0 = 0
# eth1
```

```
WESTERMO-SW6-MIB::cfgNetEthEnabled.1 = 1
WESTERMO-SW6-MIB::cfgNetEthBridge.1 = 1
WESTERMO-SW6-MIB::cfgNetEthVlanMode.1 = 1
WESTERMO-SW6-MIB::cfgNetEthTag.1 = 0
# br1.vlan0
WESTERMO-SW6-MIB::cfgNetVlanEnabled.1 = 1
WESTERMO-SW6-MIB::cfgNetVlanBridge.1 = 1
WESTERMO-SW6-MIB::cfgNetVlanVid.1 = 0
# IP on br1.vlan0
WESTERMO-SW6-MIB::cfgNetIpEnabled.2 = 1
WESTERMO-SW6-MIB::cfgNetIpAddr.2 = 192.168.1.20/24
WESTERMO-SW6-MIB::cfgNetIpProto.2 = 0
WESTERMO-SW6-MIB::cfgNetIpInterface.2 = br1.vlan0
```

6.2.2 Remote Administrative Access

To access the administrative interface of the device from an NMS or manually it may be desirable to have an IP address on the interface towards the backbone. However this administrative access should not be provided in a way that users of the AP have access to it. A good practice is to separate user-data and administrative data via VLANs.

In the following example we configure:

- disable eth1
- eth0 in br0 as trunk-port for vlans 9 and 13
- wlan0 in br0 as access-port for vlan 9
- br0.vlan13 as access-port for vlan 13 with a local IP

Configuration File Example: Allow remote administrative access on separate VLAN

```
# Config.Format = raw
# disable second port eth1
WESTERMO-SW6-MIB::cfgNetEthEnabled.1 = 0
# eth0 in br0 as trunk with vlans 9 and 13
WESTERMO-SW6-MIB::cfgNetEthEnabled.0 = 1
WESTERMO-SW6-MIB::cfgNetEthBridge.0 = 0
WESTERMO-SW6-MIB::cfgNetEthVlanMode.0 = 0
WESTERMO-SW6-MIB::cfgNetEthTrunk.0 = 9, 13
# wlan0 in br0 as access port for vlan 9
WESTERMO-SW6-MIB::cfgNetEthEnabled.0 = 1
WESTERMO-SW6-MIB::cfgNetEthBridge.0 = 0
WESTERMO-SW6-MIB::cfgNetEthVlanMode.0 = 1
WESTERMO-SW6-MIB::cfgNetEthTag.0 = 9
```

```
# br0.vlan13
WESTERMO-SW6-MIB::cfgNetVlanEnabled.1 = 1
WESTERMO-SW6-MIB::cfgNetVlanBridge.1 = 0
WESTERMO-SW6-MIB::cfgNetVlanVid.1 = 13
# IP on br0.vlan13
WESTERMO-SW6-MIB::cfgNetIpEnabled.2 = 1
WESTERMO-SW6-MIB::cfgNetIpAddr.2 = 10.11.12.13/24
WESTERMO-SW6-MIB::cfgNetIpProto.2 = 0
WESTERMO-SW6-MIB::cfgNetIpInterface.2 = br0.vlan13
```

6.3 Service Restrictions

To reduce the attack surface on a device, all services which are not required should be disabled.

Services to disable:

- CLI - WESTERMO-SW6-MIB::cfgCliEnabled.0
- HTTP - WESTERMO-SW6-MIB::cfgHttpEnabled.0
- LLDP - WESTERMO-SW6-MIB::cfgLldpEnabled.0
- MDNS - WESTERMO-SW6-MIB::cfgMdnsEnabled.0
- DHCP-Server - WESTERMO-SW6-MIB::cfgDhcpGlobalEnabled.0

6.3.1 CLI

As a general rule, the CLI should be disabled. In case the CLI must be used for any purpose whatsoever, it is recommended to use it with an SSH connection (WESTERMO-SW6-MIB::cfgCliSshEnabled.0) rather than a Telnet connection. Note that Telnet is considered insecure and is therefore disabled by default. If the CLI is activated it is bound to 0.0.0.0:22 for SSH. This means it is listening to all addresses by default. Consider binding the CLI to a specific address. The entries WESTERMO-SW6-MIB::cfgCliSshAddress.0 and WESTERMO-SW6-MIB::cfgCliTelnetAddress.0 allow the CLI to be reached only via the addresses under which it must be accessible. Assuming that the administrative interface is accessible at 192.168.1.20, the CLI should be bound to 192.168.1.20:22 and 192.168.1.20:23 respectively. However, the configuration parameters of the CLI allow for multiple addresses and ports.

6.3.2 SNMP

The SNMP service can not be disabled since it is the main internal configuration interface of the device. It is bound by default to 0.0.0.0:161. The entry WESTERMO-SW6-MIB::[cfgSnmpdAddress.0](#) allows to bind the service only to the addresses it should be reachable on. e.g. If the administrative interface is reachable at 192.168.1.20, the SNMP server should be bound to 192.168.1.20:161. If SNMP is not used, access to it can be prevented by binding it to the localhost only. e.g. 127.0.0.1:161. The SNMP server can be bound to multiple addresses and/or ports.

6.3.3 HTTP

The HTTP server provides access to the Web Interface via http and https. By default http on port 80 redirects to https on port 443. Whenever possible, keep the http to https redirection enabled ('WESTERMO-SW6-MIB::[cfgHttpRedirectEnabled.0](#)'). The entries WESTERMO-SW6-MIB::[cfgHttpHttpAddress.0](#) and WESTERMO-SW6-MIB::[cfgHttpHttpsAddress.0](#) allow to bind the service only to the addresses it should be reachable on. e.g. If the administrative interface is reachable at 192.168.1.20, the HTTP server should be bound to 192.168.1.20:80 respectively 192.168.1.20:443. If unencrypted access to port 80 is not used, access to it can be prevented by binding WESTERMO-SW6-MIB::[cfgHttpHttpAddress.0](#) to the localhost only. e.g. 127.0.0.1:80. The HTTP server can be bound to multiple addresses and/or ports.

6.4 Passwords

All passwords should be changed from their default value. A list of the relevant config parameters:

- CLI - username: WESTERMO-SW6-MIB::[cfgCliUsername.0](#)
- CLI - password: WESTERMO-SW6-MIB::[cfgCliPassword.0](#)
- HTTP - password: WESTERMO-SW6-MIB::[cfgHttpPassword.0](#)
- SNMP - admin-community/passphrase: WESTERMO-SW6-MIB::[cfgSnmpdComAdmin.0](#)
- SNMP - maintainer-community/passphrase: WESTERMO-SW6-MIB::[cfgSnmpdComMaintainer.0](#)
- SNMP - monitor-community/passphrase: WESTERMO-SW6-MIB::[cfgSnmpdComMonitor.0](#)
- WLAN - PSK-passphrase: WESTERMO-SW6-MIB::[cfgWlanIfacePassword.X](#)

6.4.1 Strength Of PSK-Passphrase

When using a private shared key (PSK), it is strongly suggested to use a maximum sized random string of characters as PSK-passphrase. The maximum length of the passphrase is 63 characters, and should be 63 characters long. This is a countermeasure against an adversary trying to brute-force the passphrase. Ensure that the passphrase does not contain any human readable words in any language, since they are susceptible to dictionary attacks.

7 Default settings

The following table shows default value of selected configuration parameters.

Network Settings:

cfgNetIpAddr	192.168.1.20/24
cfgNetIpProto	static(0)
cfgNetIpInterface	br0.vlan0
cfgNetEthBridge	br0(0)
cfgNetWlanBridge	br0(0)
cfgNetVlanBridge	br0(0)
cfgNetVlanVid	0

WLAN physical device Settings (*RT 11n* family products):

cfgWlanGlblCountry	WORLD
cfgWlanDevModulation	ng(10)
cfgWlanDevBandwidth	ht20(0)
cfgWlanDevFrequency	2412
cfgWlanDevPower	9

WLAN logical interface Settings:

cfgWlanIfaceMode	AP(0)
cfgWlanIfaceSsid	Rmodem1
cfgWlanIfaceEncryption	wpa2(3)
cfgWlanIfacePassword	password
cfgWlanIfaceBitrates	auto(-1)
cfgWlanIfaceScanList	0

Routing settings:

cfgRouteDefGateway	0.0.0.0 (disabled)
------------------------------------	--------------------

8 WESTERMO-SW6-MIB

8.1 configuration

8.1.1 cfgSystem

8.1.1.1 cfgSysHostname

The hostname of the device

Valid characters for hostnames are ASCII(7) letters from a to z, the digits from 0 to 9, and the hyphen (-). A hostname may not start with a hyphen.

Applies to AP and STA.

<i>Type</i>	DisplayString
<i>Range</i>	1 - 255
<i>Access</i>	readwrite
<i>OID</i>	1.3.6.1.4.1.16177.1.400.1.1.1.1

8.1.1.2 cfgSysSearchdomainTable

Domain Search List

Configure the domain search list by adding entries in this table.

The domain search list, as well as the local domain name (see `cfgSysDomain`), is used by the resolver to create a fully qualified domain name from a relative name.

Applies to AP and STA.

<i>Range</i>	0 - 5
<i>OID</i>	1.3.6.1.4.1.16177.1.400.1.1.1.10

8.1.1.2.1 cfgSysSearchdomainTableEntry

Searchdomain configuration.

<i>OID</i>	1.3.6.1.4.1.16177.1.400.1.1.1.10.1
------------	------------------------------------

8.1.1.2.1.1 cfgSysSdIndex

Table Entry Index

<i>Range</i>	0 - 5
<i>Access</i>	noaccess
<i>OID</i>	1.3.6.1.4.1.16177.1.400.1.1.1.10.1.1

8.1.1.2.1.2 cfgSysSdSearch

Search Domain List Entry

This entry will be ignored when set to 'none'.

Example:

- example.com
- subdomain.otherdomain.org

Applies to AP and STA.

<i>Type</i>	DisplayString
<i>Range</i>	1 - 255
<i>Access</i>	readwrite
<i>OID</i>	1.3.6.1.4.1.16177.1.400.1.1.1.10.1.2

8.1.1.3 cfgSysNameserverTable

Nameserver configuration.

<i>Range</i>	0 - 15
<i>OID</i>	1.3.6.1.4.1.16177.1.400.1.1.1.11

8.1.1.3.1 cfgSysNameserverTableEntry

Nameserver configuration.

<i>OID</i>	1.3.6.1.4.1.16177.1.400.1.1.1.11.1
------------	------------------------------------

8.1.1.3.1.1 cfgSysNsIndex

Table Entry Index

<i>Range</i>	0 - 15
<i>Access</i>	noaccess
<i>OID</i>	1.3.6.1.4.1.16177.1.400.1.1.1.11.1.1

8.1.1.3.1.2 cfgSysNsType

Type of the Nameserver Entry

- **none(0)**: Disables this entry
- **server(1)**: Uses the address specified by `cfgSysNsServer`
- **dhcpinterface(2)**: Uses nameservers provided by a DHCP-client referenced by `cfgSysNsDhcpInterface`

Applies to AP and STA.

<i>Enumeration</i>	none (0), server (1), dhcpinterface (2)
<i>Access</i>	readwrite
<i>OID</i>	1.3.6.1.4.1.16177.1.400.1.1.1.11.1.2

8.1.1.3.1.3 cfgSysNsServer

Nameserver Address

This parameter is only used when `cfgSysNsType` is set to **server(1)**.

Applies to AP and STA.

<i>Type</i>	IpAddress
<i>Access</i>	readwrite
<i>OID</i>	1.3.6.1.4.1.16177.1.400.1.1.1.11.1.3

8.1.1.3.1.4 cfgSysNsDhcpInterface

DHCP Client Interface

This parameter is only used when `cfgSysNsType` is set to **interface(2)**.

Name of an interface on which a DHCP client is running. This may be a DHCP client defined by `cfgNetIpTable` or a wwan or ovpn interface which have their own means of handling DHCP.

Examples:

- wlan0
- ovpn0
- macvlan2
- wwan0
- br0.vlan7

Applies to AP and STA.

Type	DisplayString
Range	1 - 16
Access	readwrite
OID	1.3.6.1.4.1.16177.1.400.1.1.1.11.1.4

8.1.1.4 cfgSysTimezone

POSIX timezone string, defines the local time

For more strings also see http://wiki.openwrt.org/doc/uci/system#time_zones

Example:

- Europe/Zurich: CET-1CEST,M3.5.0,M10.5.0/3

Applies to AP and STA.

Type	DisplayString
Range	1 - 255
Access	readwrite
OID	1.3.6.1.4.1.16177.1.400.1.1.1.2

8.1.1.5 cfgSysDomain

Local Domain Name of the Device

Will be ignored when set to 'none'.

The local domain name, as well as the domain search list (as configured in `cfgSysSearchdomainTable`), is used by the resolver to create a fully qualified domain name from a relative name.

Example:

- yourdomain.org

- subdomain.yourdomain.org

Note:

It is recommended to not use the domain `local` because it collides with mDNS.

Applies to AP and STA.

<i>Type</i>	DisplayString
<i>Range</i>	1 - 255
<i>Access</i>	readwrite
<i>OID</i>	1.3.6.1.4.1.16177.1.400.1.1.1.3

8.1.2 cfgCli

8.1.2.1 cfgCliEnabled

CLI feature support configuration.

Applies to AP and STA.

<i>Enumeration</i>	disabled (0), enabled (1)
<i>Access</i>	readwrite
<i>OID</i>	1.3.6.1.4.1.16177.1.400.1.1.100.1

8.1.2.2 cfgCliUsername

CLI username.

Applies to AP and STA.

<i>Type</i>	DisplayString
<i>Range</i>	1 - 31
<i>Access</i>	readwrite
<i>OID</i>	1.3.6.1.4.1.16177.1.400.1.1.100.2

8.1.2.3 cfgCliPassword

CLI password.

For SSH, a password is mandatory. Accessing the device via telnet without using a password is done by setting the password to an empty string (zero string).

Applies to AP and STA.

Type	DisplayString
Range	0 - 63
Access	readwrite
OID	1.3.6.1.4.1.16177.1.400.1.1.100.3

8.1.2.4 cfgCliTelnetEnabled

CLI telnet support configuration.

Applies to AP and STA.

Enumeration	disabled (0), enabled (1)
Access	readwrite
OID	1.3.6.1.4.1.16177.1.400.1.1.100.4

8.1.2.5 cfgCliSshEnabled

CLI ssh support configuration.

Applies to AP and STA.

Enumeration	disabled (0), enabled (1)
Access	readwrite
OID	1.3.6.1.4.1.16177.1.400.1.1.100.6

8.1.2.6 cfgCliTelnetAddress

Address to which the telnet server for CLI binds.

The default is '0.0.0.0:23'. Multiple space and/or comma separated tuples are allowed.

Examples:

- 192.168.1.20:23
- 192.168.1.20:23, 192.168.2.20:8023, 172.16.32.32:10023
- 192.168.1.20:23 192.168.2.20:8023 172.16.32.32:10023

Applies to AP and STA.

<i>Type</i>	DisplayString
<i>Range</i>	1 - 255
<i>Access</i>	readwrite
<i>OID</i>	1.3.6.1.4.1.16177.1.400.1.1.100.8

8.1.2.7 cfgCliSshAddress

Address to which the ssh server for CLI binds.

The default is '0.0.0.0:22'. Multiple space and/or comma separated tuples are allowed.

Examples:

- 192.168.1.20:22
- 192.168.1.20:22, 192.168.2.20:8022, 172.16.32.32:10022
- 192.168.1.20:22 192.168.2.20:8022 172.16.32.32:10022

Applies to AP and STA.

<i>Type</i>	DisplayString
<i>Range</i>	1 - 255
<i>Access</i>	readwrite
<i>OID</i>	1.3.6.1.4.1.16177.1.400.1.1.100.9

8.1.3 cfgVpn

8.1.3.1 cfgVpnOpenvpn

8.1.3.1.1 cfgVpnOpenvpnTable

OpenVPN Instance Table

Each entry in this table represents one instance of the OpenVPN service and is in a one-to-one relation with the OpenVPN Interface whose index is identical. Also see `cfgNetOpenvpnTable`.

<i>Range</i>	0 - 15
<i>OID</i>	1.3.6.1.4.1.16177.1.400.1.1.1003.1.1

8.1.3.1.1.1 cfgVpnOpenvpnTableEntry

OpenVPN Instance Table Entry

<i>OID</i>	1.3.6.1.4.1.16177.1.400.1.1.1003.1.1.1
------------	--

cfgVpnOpenvpnIndex

Table Entry Index

<i>Range</i>	0 - 15
<i>Access</i>	noaccess
<i>OID</i>	1.3.6.1.4.1.16177.1.400.1.1.1003.1.1.1.1

cfgVpnOpenvpnDevType

Interface Type

OpenVPN is designed to work with virtual network interface either of type tunnel or tap. The interface types on both sides of an OpenVPN connection must match.

Interface types are:

- **tun(0)**: to encapsulate IPv4 or IPv6 (OSI Layer 3), or
- **tap(1)**: to encapsulate Ethernet 802.3 (OSI Layer 2).

To be able to bridge an OpenVPN interface with `cfgNetOpenvpnBridge` its type must be **tap(1)**.

<i>Enumeration</i>	tun (0), tap (1)
<i>Access</i>	readwrite
<i>OID</i>	1.3.6.1.4.1.16177.1.400.1.1.1003.1.1.1.10

cfgVpnOpenvpnCustomOptions

Custom OpenVPN Options

These options are appended to the OpenVPN configuration. This allows to set options not available via other configuration items. Set to `none` when no additional options shall be added.

When setting multiple options, separate them with a semicolon ;.

The full list of all available options is at: <https://openvpn.net/community-resources/reference-manual-for-openvpn-2-4/>

Prohibited options are:

- `ipchange`

- route-up
- route-pre-down
- up
- down
- script-security
- cd
- chroot
- log
- client-connect
- client-disconnect
- learn-address
- auth-user-pass-verify
- auth-user-pass
- tls-verify
- ca
- cert
- key
- secret
- askpass
- tls-auth

Essentially everything which calls a script, changes files, or otherwise allows to adjust sensitive setting on the system.

When a custom option is set which is configurable via an existing parameter, the custom option will take precedence. A message is shown in syslog/TRAP notifying about the duplicate entry.

Type	DisplayString
Range	4 - 4095
Access	readwrite
OID	1.3.6.1.4.1.16177.1.400.1.1.1003.1.1.1.1000

cfgVpnOpenvpnKeepaliveInterval

Keep-Alive Interval

Send keep-alive packets to the remote OpenVPN instance if no packets have been sent for at least the given number of seconds.

This parameter has two intended uses:

- Compatibility with stateful firewalls
- To provide a basis for the remote OpenVPN instance to detect the existence of its peer

Note: If OpenVPN is in client mode, this parameter may be overridden by the server.

Type	Integer32
Access	readwrite
OID	1.3.6.1.4.1.16177.1.400.1.1.1003.1.1.1.11

cfgVpnOpenvpnKeepaliveTimeout

Keep-Alive Timeout

This parameter specifies the number of seconds that trigger a restart of the OpenVPN instance if no keep-alive or other packet has been received from the remote side, see `cfgVpnOpenvpnKeepaliveInterval`.

Note: If OpenVPN is in client mode, this parameter may be overridden by the server.

Type	Integer32
Access	readwrite
OID	1.3.6.1.4.1.16177.1.400.1.1.1003.1.1.1.12

cfgVpnOpenvpnConnectRetry

Connect Retry Interval

Number of seconds to wait between connection attempts. Repeated reconnection attempts are slowed down after 5 retries per remote by doubling the wait time after each unsuccessful attempt.

The parameter `cfgVpnOpenvpnConnectRetryLimit` specifies the maximum value of wait time in seconds at which it gets capped.

Range	1 - 65535
Access	readwrite
OID	1.3.6.1.4.1.16177.1.400.1.1.1003.1.1.1.13

cfgVpnOpenvpnConnectRetryLimit

Connect Retry Interval Limit

The maximum value of wait time in seconds (see `cfgVpnOpenvpnConnectRetry`) at which it gets capped.

Range	1 - 65535
Access	readwrite
OID	1.3.6.1.4.1.16177.1.400.1.1.1003.1.1.1.14

cfgVpnOpenvpnCompress

Compression Algorithm

Specify a compression algorithm:

- **disabled(0)**: Data compression is turned off.
- **allowPush(1)**: Data compression is turned off, but may be enabled by the server later.
- **lzo(2)**: Lempel-Ziv-Oberhumer (LZO) algorithm
- **lz4(3)**: LZ4 algorithm (faster than LZO)
- **lz4v2(4)** OpenVPN optimised version of the LZ4 algorithm

The LZO algorithm provides a slightly better compression ratio than the LZ4 compression. However, it is considerably slower and should not be used unless for backward compatibility.

<i>Enumeration</i>	disabled (0), allowPush (1), lzo (2), lz4 (3), lz4v2 (4)
<i>Access</i>	readwrite
<i>OID</i>	1.3.6.1.4.1.16177.1.400.1.1.1003.1.1.1.15

cfgVpnOpenvpnVerb

Log Verbosity

Each log verbosity level shows all messages from the previous levels. Level 3 is recommended for a good summary of what is happening.

- **0**: No output except fatal errors
- **1 - 4**: Normal usage range
- **5**: Output R and W characters to the console for each packet read and write operation, uppercase is used for TCP/UDP packets and lowercase is used for TUN/TAP packets.
- **6 - 11**: Levels for debugging purposes

<i>Type</i>	Integer32
<i>Access</i>	readwrite
<i>OID</i>	1.3.6.1.4.1.16177.1.400.1.1.1003.1.1.1.16

cfgVpnOpenvpnKeyPassword

Password to Unlock Private Key

This is only required if the key is encrypted.

<i>Type</i>	DisplayString
<i>Access</i>	readwrite
<i>OID</i>	1.3.6.1.4.1.16177.1.400.1.1.1003.1.1.1.17

cfgVpnOpenvpnKeyType

Key Type

- **asymmetric(0)**: Use an asymmetric encryption with certificate, private and public keys (TLS).
- **symmetric(1)**: Use encryption with a static key.
- **combined(2)**: Use asymmetric encryption and encrypt the control channel with a static key.

Note: The key material for the asymmetric encryption is managed by the Certificate Store, see `setCrtFileSelector`.

<i>Enumeration</i>	asymmetric (0), symmetric (1), combined (2)
<i>Access</i>	readwrite
<i>OID</i>	1.3.6.1.4.1.16177.1.400.1.1.1003.1.1.1.18

cfgVpnOpenvpnKeyDirection

Key Direction

This option is active when `cfgVpnOpenvpnKeyType` is set to either **symmetric(1)** or **combined(2)**.

- **omitted(-1)**: No direction is defined
- **zero(0)**: Use direction 0
- **one(1)**: Use direction 1

When the direction parameter is omitted, 2 keys are used bidirectionally: One for HMAC and the other for encryption/decryption.

With a direction specified, 4 keys are used: One per direction for HMAC and encryption.

Note: The direction parameter should always be complementary on either side of the connection, i.e. one side should use '0' and the other should use '1', or both sides should omit it altogether.

<i>Enumeration</i>	omitted (-1), zero (0), one (1)
<i>Access</i>	readwrite
<i>OID</i>	1.3.6.1.4.1.16177.1.400.1.1.1003.1.1.1.19

cfgVpnOpenvpnMode

Mode

- **client(0)**: OpenVPN instance is a client and connects to a server
- **server(1)**: **NOT IMPLEMENTED YET**

<i>Enumeration</i>	client (0), server (1)
<i>Access</i>	readwrite
<i>OID</i>	1.3.6.1.4.1.16177.1.400.1.1.1003.1.1.1.2

cfgVpnOpenvpnRemoteCertTls

Remote Certificate Verification

Verify if the Extended Key Usage field in the certificate of the remote host has the correct type.

- **disabled(0)**: Do not verify the remote certificate.
- **client(1)**: Check for client type.
- **server(2)**: Check for server type.

<i>Enumeration</i>	disabled (0), client (1), server (2)
<i>Access</i>	readwrite
<i>OID</i>	1.3.6.1.4.1.16177.1.400.1.1.1003.1.1.1.20

cfgVpnOpenvpnVerifyX509Name

X.509 Certificate Verification Method

The following verification methods are available:

- **disabled(0)**: No verification is done
- **name(1)**: Match the exact CN (Common Name)
- **prefix(2)**: Match the prefix of the CN
- **subject(3)**: Match the complete subject DN

The `cfgVpnOpenvpnVerifyX509String` parameter defines the string to be matched.

<i>Type</i>	'nodetype': 'namednumber', 'number': '1'
<i>Enumeration</i>	disabled (0), name (1), prefix (2), subject (3)
<i>Access</i>	readwrite
<i>OID</i>	1.3.6.1.4.1.16177.1.400.1.1.1003.1.1.1.21

cfgVpnOpenvpnVerifyX509String

X.509 Certificate Verification String

If a X.509 certificate verification method is enabled (see `cfgVpnOpenvpnVerifyX509Name`), this parameter defines the string to be compared by the verification method.

Type	DisplayString
Access	readwrite
OID	1.3.6.1.4.1.16177.1.400.1.1.1003.1.1.1.22

cfgVpnOpenvpnUsername

Username

Authenticate with the server using the given username.

It is disabled when set to **none**.

Type	DisplayString
Range	1 - 255
Access	readwrite
OID	1.3.6.1.4.1.16177.1.400.1.1.1003.1.1.1.23

cfgVpnOpenvpnPassword

Password

Authenticate with the server using the given password.

It is disabled when set to **none**.

Type	DisplayString
Range	1 - 255
Access	readwrite
OID	1.3.6.1.4.1.16177.1.400.1.1.1003.1.1.1.24

cfgVpnOpenvpnLocal

Local IP Address

The OpenVPN instance binds to the given IP address only. Address 0.0.0.0 binds the OpenVPN instance to all interfaces.

Type	DisplayString
Range	7 - 15
Access	readwrite
OID	1.3.6.1.4.1.16177.1.400.1.1.1003.1.1.1.3

cfgVpnOpenvpnLPort

Local TCP/UDP Port

Specifies the TCP/UDP port number for bind. If the local port number is set to 0, OpenVPN uses a random port number.

<i>Range</i>	0 - 65535
<i>Access</i>	readwrite
<i>OID</i>	1.3.6.1.4.1.16177.1.400.1.1.1003.1.1.1.4

cfgVpnOpenvpnRemote

Remote Host Name or IP Address

The OpenVPN client tries to connect to a server at the given remote host name or IP address.

The remote option will be omitted from the OpenVPN config file when set to 'none'. This allows to specify your own remote entries via the custom options (see `cfgVpnOpenvpnCustomOptions`).

<i>Type</i>	DisplayString
<i>Range</i>	6 - 255
<i>Access</i>	readwrite
<i>OID</i>	1.3.6.1.4.1.16177.1.400.1.1.1003.1.1.1.5

cfgVpnOpenvpnRPort

Remote TCP/UDP Port

Specifies the TCP/UDP port to which the connection is created. This is the port on which the OpenVPN server is listening.

<i>Range</i>	1 - 65535
<i>Access</i>	readwrite
<i>OID</i>	1.3.6.1.4.1.16177.1.400.1.1.1003.1.1.1.6

cfgVpnOpenvpnProto

Transport Protocol

The following transport protocols are available:

- **UDP(0)**: User Datagram Protocol
- **TCP(1)**: Transmission Control Protocol

<i>Enumeration</i>	udp (0), tcp (1)
<i>Access</i>	readwrite
<i>OID</i>	1.3.6.1.4.1.16177.1.400.1.1.1003.1.1.1.7

cfgVpnOpenvpnAuth

Packet Authentication

Authenticate packets with a Hash-based Message Authentication Code HMAC using the given message digest algorithm.

In static-key encryption mode, the HMAC key is included in the key file. In TLS mode, the HMAC key is dynamically generated and shared between peers via the TLS control channel.

Examples:

- **SHA256**
- **SHA3-512**
- **none**: to disable HMAC packet authentication

For a full list of supported algorithms please consult the user manual or execute **openvpn --show-digests** on a device.

Type	DisplayString
Access	readwrite
OID	1.3.6.1.4.1.16177.1.400.1.1.1003.1.1.1.8

cfgVpnOpenvpnCipher

Packet Encryption

Encrypt packets with the given cipher algorithm.

Examples:

- **AES-256-CBC**
- **AES-256-GCM**
- **none**: to disable packet encryption

For a full list of supported algorithms please consult the user manual.

Type	DisplayString
Access	readwrite
OID	1.3.6.1.4.1.16177.1.400.1.1.1003.1.1.1.9

8.1.3.2 cfgVpnIpsec

8.1.3.2.1 cfgVpnIpsecTable

IPsec Table

The used IPsec implementation is strongSwan. For more detailed configuration explanations and examples see <https://wiki.strongswan.org/projects/strongswan/wiki/IpsecConf>

<i>Range</i>	0 - 15
<i>OID</i>	1.3.6.1.4.1.16177.1.400.1.1.1003.2.1

8.1.3.2.1.1 cfgVpnIpsecTableEntry

IPsec Table Entry

<i>OID</i>	1.3.6.1.4.1.16177.1.400.1.1.1003.2.1.1
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cfgVpnIpsecIndex

Table Entry Index

<i>Range</i>	0 - 15
<i>Access</i>	noaccess
<i>OID</i>	1.3.6.1.4.1.16177.1.400.1.1.1003.2.1.1.1

cfgVpnIpsecType

IPsec Type

The type of the connection; currently the accepted values are:

- **tunnel(0)**: signifying a host-to-host, host-to-subnet, or subnet-to-subnet tunnel
- **transport(1)**: signifying host-to-host transport mode
- **transportProxy(2)**: signifying the special Mobile IPv6 transport proxy mode
- **passthrough(3)**: signifying that no IPsec processing should be done at all
- **drop(4)**: signifying that packets should be discarded.

<i>Enumeration</i>	tunnel (0), transport (1), transportProxy (2), passthrough (3), drop (4)
<i>Access</i>	readwrite
<i>OID</i>	1.3.6.1.4.1.16177.1.400.1.1.1003.2.1.1.100

cfgVpnIpsecCustomOptions

Custom IPsec Options

These options are appended to the IPsec configuration. This allows to set options not available via other configuration items. Set to `none` when no additional options shall be added.

When setting multiple options, separate them with a semicolon ;.

The full list of all available options is at: <https://wiki.strongswan.org/projects/strongswan/wiki/ConnSection>

Prohibited options are:

- left
- leftid
- leftsubnet
- leftauth
- right
- rightid
- rightauth
- rightsubnet
- keyexchange
- mobike
- ikelifetime
- lifetime
- keyingtries
- type
- ike
- esp
- dpddelay
- dpdtimeout

Essentially everything which is possible to set by other SNMP commands. Applies to AP and STA.

Type	DisplayString
Range	4 - 4095
Access	readwrite
OID	1.3.6.1.4.1.16177.1.400.1.1.1003.2.1.1.1000

cfgVpnIpsecAuto

IPsec Auto Startup Operation

What operation, if any, should be done automatically at IPsec startup.

- **ignore(0)**: Ignores the connection. This is equal to deleting a connection from the config file.
- **add(1)**: Loads a connection without starting it.
- **route(2)**: Loads a connection and installs kernel traps. If traffic is detected between leftsubnet (cfgVpnIpsecLeftSubnet) and rightsubnet (cfgVpnIpsecRightSubnet), a connection is established.
- **start(3)**: loads a connection and brings it up immediately.

Relevant only locally, other end need not agree on it.

<i>Enumeration</i>	ignore (0), add (1), route (2), start (3)
<i>Access</i>	readwrite
<i>OID</i>	1.3.6.1.4.1.16177.1.400.1.1.1003.2.1.1.101

cfgVpnIpsecKeyExchange

Key Exchange Method

Which protocol should be used to initialize the connection.

<i>Enumeration</i>	ikev1 (1), ikev2 (2)
<i>Access</i>	readwrite
<i>OID</i>	1.3.6.1.4.1.16177.1.400.1.1.1003.2.1.1.102

cfgVpnIpsecMobike

Disable or Enable IKEv2 MOBIKE Protocol

Enables the IKEv2 MOBIKE protocol defined by RFC 4555. If set to no, the charon daemon will not actively propose MOBIKE as initiator and ignore the MOBIKE_SUPPORTED notify as responder.

<i>Enumeration</i>	no (0), yes (1)
<i>Access</i>	readwrite
<i>OID</i>	1.3.6.1.4.1.16177.1.400.1.1.1003.2.1.1.103

cfgVpnIpsecIke

IKE/ISAKMP SA Encryption/Authentication Algorithms

Comma-separated list of IKE/ISAKMP SA encryption/authentication algorithms to be used, e.g. aes128-sha256-modp3072. The notation is encryption-integrity[-prf]-dhgroup. In IKEv2, multiple algorithms and proposals may be included, such as aes128-aes256-sha1-modp3072-modp2048,3des-sha1-md5-modp1024.

It is possible to configure a PRF algorithm different to that defined for integrity protection. If no PRF is configured, the algorithms defined for integrity are proposed as PRF. The prf keywords are the same as the integrity algorithms, but have a prf prefix (such as prfsha1, prfsha256 or prfaesxcbc).

Defaults to aes128-sha256-modp3072 (aes128-sha1-modp2048,3des-sha1-modp1536 before 5.4.0) for IKEv1. The daemon adds its extensive default proposal to this default or the configured value. To restrict it to the configured proposal an exclamation mark (!) can be added at the end.

Refer to IKEv1CipherSuites and IKEv2CipherSuites for a list of valid keywords.

Note: As a responder both daemons accept the first supported proposal received from the peer. In order to restrict a responder to only accept specific cipher suites, the strict flag (!, exclamation mark) can be used, e.g: aes256-sha512-modp4096!

Type	DisplayString
Access	readwrite
OID	1.3.6.1.4.1.16177.1.400.1.1.1003.2.1.1.104

cfgVpnIpsecEsp

ESP Encryption/Authentication Algorithms

Comma-separated list of ESP encryption/authentication algorithms to be used for the connection, e.g. aes128-sha256. The notation is encryption-integrity[-dhgroup][-esnmodes]. For IKEv2, multiple algorithms (separated by -) of the same type can be included in a single proposal. IKEv1 only includes the first algorithm in a proposal. Only either the ah or the esp keyword may be used, AH+ESP bundles are not supported.

Defaults to aes128-sha256. The daemon adds its extensive default proposal to this default or the configured value. To restrict it to the configured proposal an exclamation mark (!) can be added at the end.

Note: As a responder, the daemon defaults to selecting the first configured proposal that's also supported by the peer. By disabling charon.prefer_configured_proposals in strongswan.conf this may be changed to selecting the first acceptable proposal sent by the peer instead. In order to restrict a responder to only accept specific cipher suites, the strict flag (!, exclamation mark) can be used, e.g: aes256-sha512-modp4096!

If dh-group is specified, CHILD_SA rekeying and initial negotiation include a separate Diffie-Hellman exchange (this also applies to IKEv1 Quick Mode). However, for IKEv2, the keys of the CHILD_SA created implicitly with the IKE_SA will always be derived from the IKE_SA's key material. So any DH group specified here will only apply when the CHILD_SA is later rekeyed or is created with a separate CREATE_CHILD_SA exchange. Therefore, a proposal mismatch might not immediately be noticed when the SA is established, but may later cause rekeying to fail.

Valid values for esnmodes are esn and noesn. Specifying both negotiates extended sequence number support with the peer, the default is noesn.

Refer to IKEv1CipherSuites and IKEv2CipherSuites for a list of valid keywords.

Type	DisplayString
Access	readwrite
OID	1.3.6.1.4.1.16177.1.400.1.1.1003.2.1.1.105

cfgVpnIpsecIkeLifetime

IKE Lifetime

How long the keying channel of a connection (ISAKMP or IKE SA) should last before being renegotiated.

Type	Integer32
Access	readwrite
OID	1.3.6.1.4.1.16177.1.400.1.1.1003.2.1.1.106

cfgVpnIpsecLifetime

Connection Instance Lifetime

How long a particular instance of a connection (a set of encryption/authentication keys for user packets) should last, from successful negotiation to expiry; acceptable values are an integer optionally followed by s (a time in seconds) or a decimal number followed by m, h, or d (a time in minutes, hours, or days respectively) (default 1h, maximum 24h). Normally, the connection is renegotiated (via the keying channel) before it expires (see margintime). The two ends need not exactly agree on lifetime, although if they do not, there will be some clutter of superseded connections on the end which thinks the lifetime is longer.

Applies to AP and STA.

Type	Integer32
Access	readwrite
OID	1.3.6.1.4.1.16177.1.400.1.1.1003.2.1.1.107

cfgVpnIpsecKeyingTries

Keying Tries

When set to -1 means %forever, otherwise what is set.

Examples:

- -1: try forever
- 3

Type	Integer32
Access	readwrite
OID	1.3.6.1.4.1.16177.1.400.1.1.1003.2.1.1.108

cfgVpnIpsecDpdAction

Dead Peer Detection Protocol Usage

Controls the use of the Dead Peer Detection protocol (DPD, RFC 3706) where R_U_THERE notification messages (IKEv1) or empty INFORMATIONAL messages (IKEv2) are periodically sent in order to check the liveness of the IPsec peer. The values clear, hold, and restart all activate DPD and determine the action to perform on a timeout. With **clear(1)** the connection is closed with no further actions taken. **hold(2)** installs a trap policy, which will catch matching traffic and tries to re-negotiate the connection on demand. **restart(3)** will immediately trigger an attempt to re-negotiate the connection.

The default is **none(0)** which disables the active sending of DPD messages.

Applies to AP and STA.

<i>Enumeration</i>	none (0), clear (1), hold (2), restart (3)
<i>Access</i>	readwrite
<i>OID</i>	1.3.6.1.4.1.16177.1.400.1.1.1003.2.1.1.109

cfgVpnIpsecDpdDelay

Dead Peer Detection Delay

Defines the period time interval (in seconds) with which R_U_THERE messages/INFORMATIONAL exchanges are sent to the peer. These are only sent if no other traffic is received. In IKEv2, a value of 0 sends no additional INFORMATIONAL messages and uses only standard messages (such as those to rekey) to detect dead peers.

<i>Type</i>	Integer32
<i>Access</i>	readwrite
<i>OID</i>	1.3.6.1.4.1.16177.1.400.1.1.1003.2.1.1.110

cfgVpnIpsecDpdTimeout

DPD Timeout

Defines the timeout interval in seconds, after which all connections to a peer are deleted in case of inactivity. This only applies to IKEv1, in IKEv2 the default retransmission timeout applies, as every exchange is used to detect dead peers.

<i>Type</i>	Integer32
<i>Access</i>	readwrite
<i>OID</i>	1.3.6.1.4.1.16177.1.400.1.1.1003.2.1.1.111

cfgVpnIpsecKeyPassword

Password to Unlock Private Key

This is only required if the key is encrypted.

Type	DisplayString
Access	readwrite
OID	1.3.6.1.4.1.16177.1.400.1.1.1003.2.1.1.112

cfgVpnIpsecPassword

IPsec Password

When `cfgVpnIpsecRightAuth` is set to **psk**.

A preshared secret is most conveniently represented as a sequence of characters. The sequence cannot contain newline or double-quote characters. Alternatively, preshared secrets can be represented as hexadecimal or Base64 encoded binary values. A character sequence beginning with 0x is interpreted as sequence hexadecimal digits. Similarly, a character sequence beginning with 0s is interpreted as Base64 encoded binary data.

Type	DisplayString
Access	readwrite
OID	1.3.6.1.4.1.16177.1.400.1.1.1003.2.1.1.113

cfgVpnIpsecCloseAction

Action When Link is Closed by Remote Peer

Defines the action to take if the remote peer unexpectedly closes a CHILD_SA. This may happen when the remote site is reconfigured, or goes down for maintenance. A closeaction should not be used if the peer uses reauthentication (see `cfgVpnIpsecReauth`) or uniqueids checking, as these events might trigger the defined action when not desired.

Available actions are:

- **none(0)** No action is taken. This disables the Close Action.
- **clear(1)** The connection is closed with no further actions taken.
- **hold(2)** Installs a trap policy, which will catch matching traffic and tries to re-negotiate the connection on demand.
- **restart(3)** Will immediately trigger an attempt to re-negotiate the connection.

Enumeration	none (0), clear (1), hold (2), restart (3)
Access	readwrite
OID	1.3.6.1.4.1.16177.1.400.1.1.1003.2.1.1.114

cfgVpnIpsecReauth

Reauthentication of Peer During Rekeying

Whether rekeying of an IKE_SA should also reauthenticate the peer. In IKEv1, reauthentication is always done.

In IKEv2, a value of:

- **disabled(0)** Rekeys without uninstalling the IPsec SAs
- **enabled(1)** Creates a new IKE_SA from scratch and tries to recreate all IPsec SAs

<i>Enumeration</i>	disabled (0), enabled (1)
<i>Access</i>	readwrite
<i>OID</i>	1.3.6.1.4.1.16177.1.400.1.1.1003.2.1.1.115

cfgVpnIpsecLeft

Left or Local

left = ip address | fqdn | %any | %any4 | %any6 | range | subnet

The IP address of the participant's public-network interface or one of several magic values. The value %any for the local endpoint signifies an address to be filled in (by automatic keying) during negotiation. If the local peer initiates the connection setup the routing table will be queried to determine the correct local IP address. In case the local peer is responding to a connection setup then any IP address that is assigned to a local interface will be accepted. The value %any4 restricts address selection to IPv4 addresses, the value %any6 restricts address selection to IPv6 addresses.

The prefix % in front of a fully-qualified domain name or an IP address will implicitly set leftallowany=yes.

leftallowany is a modifier for left, making it behave as %any although a concrete IP address has been assigned. Recommended for dynamic IP addresses that can be resolved by DynDNS at IPsec startup or update time.

If %any is used for the remote endpoint it literally means any IP address.

If an FQDN is assigned it is resolved every time a configuration lookup is done. If DNS resolution times out, the lookup is delayed for that time.

Connections can be limited to a specific range of hosts. To do so a range (10.1.0.0-10.2.255.255) or a subnet (10.1.0.0/16) can be specified, and multiple addresses, ranges and subnets can be separated by commas. While one can freely combine these items, to initiate the connection at least one non-range/subnet is required.

Please note that with the usage of wildcards multiple connection descriptions might match a given incoming connection attempt. The most specific description is used in that case.

Type	DisplayString
Range	1 - 255
Access	readwrite
OID	1.3.6.1.4.1.16177.1.400.1.1.1003.2.1.1.2

cfgVpnIpsecRight

Right or Remote

right = ip address | fqdn | %any | %any4 | %any6 | range | subnet

The IP address of the participant's public-network interface or one of several magic values. The value %any for the local endpoint signifies an address to be filled in (by automatic keying) during negotiation. If the local peer initiates the connection setup the routing table will be queried to determine the correct local IP address. In case the local peer is responding to a connection setup then any IP address that is assigned to a local interface will be accepted. The value %any4 restricts address selection to IPv4 addresses, the value %any6 restricts address selection to IPv6 addresses.

The prefix % in front of a fully-qualified domain name or an IP address will implicitly set rightallowany=yes.

rightallowany is a modifier for right, making it behave as %any although a concrete IP address has been assigned. Recommended for dynamic IP addresses that can be resolved by DynDNS at IPsec startup or update time.

If %any is used for the remote endpoint it literally means any IP address.

If an FQDN is assigned it is resolved every time a configuration lookup is done. If DNS resolution times out, the lookup is delayed for that time.

Connections can be limited to a specific range of hosts. To do so a range (10.1.0.0-10.2.255.255) or a subnet (10.1.0.0/16) can be specified, and multiple addresses, ranges and subnets can be separated by commas. While one can freely combine these items, to initiate the connection at least one non-range/subnet is required.

Please note that with the usage of wildcards multiple connection descriptions might match a given incoming connection attempt. The most specific description is used in that case.

Type	DisplayString
Range	1 - 255
Access	readwrite
OID	1.3.6.1.4.1.16177.1.400.1.1.1003.2.1.1.3

cfgVpnIpsecLeftSubnet

IPsec Left Subnet

leftsubnet = ip subnet[[proto/port]][,...]

Private subnet behind the left/local participant, expressed as network/netmask. The configured subnets of the peers may differ, the protocol narrows it to the greatest common subnet.

This is also done for IKEv1, but as this may lead to problems with other implementations, make sure to configure identical subnets in such configurations. IKEv2 supports multiple subnets separated by commas, IKEv1 only interprets the first subnet of such a definition. This is due to a limitation of the IKEv1 protocol, which only allows a single pair of subnets per CHILD_SA. So to tunnel several subnets, a conn entry has to be defined and brought up for each pair of subnets.

The optional part after each subnet enclosed in square brackets specifies a protocol/port to restrict the selector for that subnet.

Examples:

- 10.0.0.1[tcp/http],10.0.0.2[6/80]
- fec1::1[udp],10.0.0.0/16[53]

Instead of omitting either value %any can be used to the same effect, e.g. fec1::1[udp/%any],10.0.0.0/16[%any]

If the protocol is icmp or ipv6-icmp the port is interpreted as ICMP message type if it is less than 256, or as type and code if it greater or equal to 256, with the type in the most significant 8 bits and the code in the least significant 8 bits.

The port value can alternatively take the value %opaque for RFC 4301 OPAQUE selectors, or a numerical range in the form 1024-65535. None of the kernel backends currently supports opaque or port ranges and uses %any for policy installation instead.

Instead of specifying a subnet, %dynamic can be used to replace it with the IKE address, having the same effect as omitting left|rightsubnet completely. Using %dynamic can be used to define multiple dynamic selectors, each having a potentially different protocol/port definition.

Type	DisplayString
Range	1 - 255
Access	readwrite
OID	1.3.6.1.4.1.16177.1.400.1.1.1003.2.1.1.4

cfgVpnIpsecRightSubnet

IPsec Right Subnet

rightsubnet = ip subnet[[proto/port]][,...]

Private subnet behind the right/remote participant, expressed as network/netmask. The configured subnets of the peers may differ, the protocol narrows it to the greatest common subnet.

This is also done for IKEv1, but as this may lead to problems with other implementations, make sure to configure identical subnets in such configurations. IKEv2 supports multiple subnets separated by commas, IKEv1 only interprets the first subnet of such a definition. This is due to a limitation of the IKEv1 protocol, which only allows a single pair of subnets per CHILD_SA. So to tunnel several subnets a conn entry has to be defined and brought up for each pair of subnets.

The optional part after each subnet enclosed in square brackets specifies a protocol/port to restrict the selector for that subnet.

Examples:

- 10.0.0.1[tcp/http],10.0.0.2[6/80]
- fec1::1[udp],10.0.0.0/16[/53]

Instead of omitting either value %any can be used to the same effect, e.g. fec1::1[udp/%any],10.0.0.0/16[%any]

If the protocol is icmp or ipv6-icmp the port is interpreted as ICMP message type if it is less than 256, or as type and code if it greater or equal to 256, with the type in the most significant 8 bits and the code in the least significant 8 bits.

The port value can alternatively take the value %opaque for RFC 4301 OPAQUE selectors, or a numerical range in the form 1024-65535. None of the kernel backends currently supports opaque or port ranges and uses %any for policy installation instead.

Instead of specifying a subnet, %dynamic can be used to replace it with the IKE address, having the same effect as omitting left/rightsubnet completely. Using %dynamic can be used to define multiple dynamic selectors, each having a potentially different protocol/port definition.

Type	DisplayString
Range	1 - 255
Access	readwrite
OID	1.3.6.1.4.1.16177.1.400.1.1.1003.2.1.1.5

cfgVpnIpsecLeftId

IPsec Left ID

leftid = id

How the left/local participant should be identified for authentication; defaults to left or the subject of the certificate configured with leftcert. If leftcert is configured, the identity has to be confirmed by the certificate, that is, it has to match the full subject DN or one of the subjectAltName extensions contained in the certificate.

Can be an IP address, a fully-qualified domain name, an email address or a Distinguished Name for which the ID type is determined automatically and the string is converted to the appropriate encoding.

The rules for this conversion are described on IdentityParsing (see <https://wiki.strongswan.org/projects/strongswan/>

In certain special situations the identity parsing above might be inadequate or produce the wrong result. Examples are the need to encode a FQDN as KEY_ID or the string parser being unable to produce the correct binary ASN.1 encoding of a certificate's DN. For these situations it is possible to enforce a specific identity type and to provide the binary encoding of the identity. To do this a prefix may be used, followed by a colon (:). If the number sign (#) follows the colon, the remaining data is interpreted as hex encoding, otherwise the string is used as is as the identification data. Note: The latter implies that no conversion is performed for non-string identities. For example, ipv4:10.0.0.1 does not create a valid ID_IPV4_ADDR IKE identity, as it does not get converted to binary 0x0a000001. Instead, one could use ipv4:#0a000001 to get a valid identity, but just using the implicit type with automatic conversion is usually simpler. The same applies to the ASN.1 encoded types.

The following prefixes are known: ipv4, ipv6, rfc822, email, userfqdn, fqdn, dns, asn1dn, asn1gn and keyid.

Custom type prefixes may be specified by surrounding the numerical type value with curly brackets.

rightid for IKEv2 connections optionally takes a % as prefix in front of the identity. If given it prevents the daemon from sending IDr in its IKE_AUTH request and will allow it to verify the configured identity against the subject and subjectAltNames contained in the responder's certificate (otherwise, it is only compared with the IDr returned by the responder). The IDr sent by the initiator might otherwise prevent the responder from finding a config if it has configured a different value for leftid.

Type	DisplayString
Range	1 - 255
Access	readwrite
OID	1.3.6.1.4.1.16177.1.400.1.1.1003.2.1.1.6

cfgVpnIpsecRightId

IPsec Right ID

rightid = id

How the right/remote participant should be identified for authentication; defaults to right or the subject of the certificate configured with rightcert. If rightcert is configured, the identity has to be confirmed by the certificate, that is, it has to match the full subject DN or one of the subjectAltName extensions contained in the certificate.

Can be an IP address, a fully-qualified domain name, an email address or a Distinguished Name for which the ID type is determined automatically and the string is converted to the appropriate encoding. The rules for this conversion are described on IdentityParsing (see <https://wiki.strongswan.org/projects/strongswan/>

In certain special situations the identity parsing above might be inadequate or produce the wrong result. Examples are the need to encode a FQDN as KEY_ID or the string parser being unable to

produce the correct binary ASN.1 encoding of a certificate's DN. For these situations it is possible to enforce a specific identity type and to provide the binary encoding of the identity. To do this a prefix may be used, followed by a colon (:). If the number sign (#) follows the colon, the remaining data is interpreted as hex encoding, otherwise the string is used as is as the identification data. Note: The latter implies that no conversion is performed for non-string identities. For example, ipv4:10.0.0.1 does not create a valid ID_IPV4_ADDR IKE identity, as it does not get converted to binary 0x0a000001. Instead, one could use ipv4:#0a000001 to get a valid identity, but just using the implicit type with automatic conversion is usually simpler. The same applies to the ASN.1 encoded types.

The following prefixes are known: ipv4, ipv6, rfc822, email, userfqdn, fqdn, dns, asn1dn, asn1gn and keyid.

Custom type prefixes may be specified by surrounding the numerical type value with curly brackets.

rightid for IKEv2 connections optionally takes a % as prefix in front of the identity. If given it prevents the daemon from sending IDr in its IKE_AUTH request and will allow it to verify the configured identity against the subject and subjectAltNames contained in the responder's certificate (otherwise, it is only compared with the IDr returned by the responder). The IDr sent by the initiator might otherwise prevent the responder from finding a config if it has configured a different value for leftid.

Type	DisplayString
Range	1 - 255
Access	readwrite
OID	1.3.6.1.4.1.16177.1.400.1.1.1003.2.1.1.7

cfgVpnIpsecLeftAuth

IPsec Left Authentication

leftauth = auth method

Examples:

- psk
- pubkey
- pubkey-sha256-sha512
- ecdsa-384
- bliss-sha512

Authentication method to use locally (left). Acceptable values are pubkey for public key encryption (RSA/ECDSA/BLISS), psk for pre-shared key authentication, eap to use the Extensible Authentication Protocol, and xauth for IKEv1 eXtended Authentication.

To require a trustchain public key strength for the remote side, specify the key type followed by the minimum strength in bits (for example ecdsa-384 or rsa-2048-ecdsa-256). To limit the acceptable set of hashing algorithms for trustchain validation, append hash algorithms to pubkey or a key

strength definition (for example pubkey-sha256-sha512, rsa-2048-sha256-sha384-sha512, or rsa-2048-sha256-ecdsa-256-sha256-sha384).

Unless explicit IKEv2 signature constraints are configured (see below), such key types and hash algorithms are also applied as constraints against IKEv2 signature authentication schemes used by the remote side.

If both peers support RFC 7427 (Signature Authentication in IKEv2) specific hash algorithms to be used during IKEv2 authentication may be configured. The syntax is the same as above, but with ike: prefix. For example, with ike:pubkey-sha384-sha256 a public key signature scheme with either SHA-384 or SHA-256 would get used for authentication, in that order and depending on the hash algorithms supported by the peer. If no specific hash algorithms are configured, the default is to prefer an algorithm that matches or exceeds the strength of the signature key. If no constraints with ike: prefix are configured any signature scheme constraint (without ike: prefix) will also apply to IKEv2 authentication.

RSASSA-PSS signatures are supported. To use or require them, configure rsa/pss instead of rsa as in e.g. ike:rsa/pss-sha256. If pubkey or rsa constraints are configured, RSASSA-PSS signatures will only be used/accepted if enabled in strongswan.conf.

In the case of eap, an optional EAP method can be appended. Currently defined methods are eap-aka, eap-gtc, eap-md5, eap-mschapv2, eap-peap, eap-sim, eap-tls, eap-ttls, eap-dynamic, and eap-radius. Alternatively, IANA assigned EAP method numbers are accepted. Vendor specific EAP methods are defined in the form eap-type-vendor (e.g. eap-7-12345).

Signature and trust chain constraints for EAP-(T)TLS may be defined. To do so, append a colon to the EAP method, followed by the key type/size and hash algorithm as discussed above. For xauth, an XAuth authentication backend can be specified, such as xauth-generic or xauth-eap. If XAuth is used in leftauth, Hybrid authentication is used. For traditional XAuth authentication, define XAuth in leftauth2.

Type	DisplayString
Range	1 - 255
Access	readwrite
OID	1.3.6.1.4.1.16177.1.400.1.1.1003.2.1.1.8

cfgVpnIpsecRightAuth

IPsec Right Authentication

rightauth = auth method

Examples:

- psk
- pubkey

- pubkey-sha256-sha512
- ecdsa-384
- bliss-sha512

Authentication method to require from the remote (right) side. Acceptable values are pubkey for public key encryption (RSA/ECDSA/BLISS), psk for pre-shared key authentication, eap to require the use of the Extensible Authentication Protocol, and xauth for IKEv1 eXtended Authentication.

To require a trustchain public key strength for the remote side, specify the key type followed by the minimum strength in bits (for example ecdsa-384 or rsa-2048-ecdsa-256). To limit the acceptable set of hashing algorithms for trustchain validation, append hash algorithms to pubkey or a key strength definition (for example pubkey-sha256-sha512, rsa-2048-sha256-sha384-sha512, or rsa-2048-sha256-ecdsa-256-sha256-sha384).

Unless explicit IKEv2 signature constraints are configured (see below), such key types and hash algorithms are also applied as constraints against IKEv2 signature authentication schemes used by the remote side.

If both peers support RFC 7427 (Signature Authentication in IKEv2) specific hash algorithms to be used during IKEv2 authentication may be configured. The syntax is the same as above, but with ike: prefix. For example, with ike:pubkey-sha384-sha256 a public key signature scheme with either SHA-384 or SHA-256 would get used for authentication, in that order and depending on the hash algorithms supported by the peer. If no specific hash algorithms are configured, the default is to prefer an algorithm that matches or exceeds the strength of the signature key. If no constraints with ike: prefix are configured any signature scheme constraint (without ike: prefix) will also apply to IKEv2 authentication.

RSASSA-PSS signatures are supported. To use or require them, configure rsa/pss instead of rsa as in e.g. ike:rsa/pss-sha256. If pubkey or rsa constraints are configured, RSASSA-PSS signatures will only be used/accepted if enabled in strongswan.conf.

In the case of eap, an optional EAP method can be appended. Currently defined methods are eap-aka, eap-gtc, eap-md5, eap-mschapv2, eap-peap, eap-sim, eap-tls, eap-ttls, eap-dynamic, and eap-radius. Alternatively, IANA assigned EAP method numbers are accepted. Vendor specific EAP methods are defined in the form eap-type-vendor (e.g. eap-7-12345).

Signature and trust chain constraints for EAP-(T)TLS may be defined. To do so, append a colon to the EAP method, followed by the key type/size and hash algorithm as discussed above. For xauth, an XAuth authentication backend can be specified, such as xauth-generic or xauth-eap. If XAuth is used in leftauth, Hybrid authentication is used. For traditional XAuth authentication, define XAuth in leftauth2.

Type	DisplayString
Range	1 - 255
Access	readwrite
OID	1.3.6.1.4.1.16177.1.400.1.1.1003.2.1.1.9

8.1.3.2.2 cfgVpnIpsecGlobal

8.1.3.2.2.1 cfgVpnIpsecGlbVirtualTunnelInterface

Virtual Tunnel Interfaces Disabled or Enabled

Whether IPsec should use virtual tunnel interfaces (vti) or not.

Generally IPsec processing is based on policies. After regular route lookups are done, the OS kernel consults its SPD for a matching policy and if one is found that is associated with an IPsec SA, the packet is processed (e.g. encrypted and sent as ESP packet).

It is also possible to configure route-based VPNs. Here IPsec processing does not (only) depend on negotiated policies but may e.g. be controlled by routing packets to a specific interface.

Most of these approaches also allow easy capture of plaintext traffic, which, depending on the operating system, might not be that straight-forward with policy-based VPNs. Another advantage this approach is that the MTU can be specified for the tunneling devices allowing to fragment packets before tunneling them in case PMTUD does not work properly.

VTI devices act like a wrapper around existing IPsec policies. This means you can't just route arbitrary packets to a VTI device to get them tunneled, the established IPsec policies have to match too. However, you can negotiate 0.0.0.0/0 traffic selectors on both ends to allow tunneling anything that's routed via VTI device.

It's important to note that VTI tunnel devices are a local feature, no additional encapsulation (like with GRE) is added, so the other end does not have to be aware that VTI devices are used in addition to regular IPsec policies.

<i>Enumeration</i>	disabled (0), enabled (1)
<i>Access</i>	readwrite
<i>OID</i>	1.3.6.1.4.1.16177.1.400.1.1.1003.2.2.1

8.1.3.2.2.2 cfgVpnIpsecGlbCustomOptions

Custom Global IPsec Options

Set to none when no additional options shall be added.

When setting multiple options, separate them with a semicolon ;.

Example to enable aggressive mode for PSK with IKEv1:

- `charon.i_dont_care_about_security_and_use_aggressive_mode_psk=yes`

For a full list of all available options, please see: <https://wiki.strongswan.org/projects/strongswan/wiki/Strongswanconf>

Type	DisplayString
Access	readwrite
OID	1.3.6.1.4.1.16177.1.400.1.1.1003.2.2.2

8.1.3.2.3 cfgVpnIpsecDebug

8.1.3.2.3.1 cfgVpnIpsecDbgAsn

asn: Low-level encoding/decoding (ASN.1, X.509 etc.)

- **silent(-1):** Absolutely silent
- **basic(0):** Very basic auditing logs (e.g. SA up/SA down)
- **generic(1):** Generic control flow with errors, a good default to see whats going on
- **detailed(2):** More detailed debugging control flow
- **raw(3):** Including RAW data dumps in hex
- **sensitive(4):** Also include sensitive material in dumps, e.g. keys

Enumeration	silent (-1), basic (0), generic (1), detailed (2), raw (3), sensitive (4)
Access	readwrite
OID	1.3.6.1.4.1.16177.1.400.1.1.1003.2.3.1

8.1.3.2.3.2 cfgVpnIpsecDbgJob

job: Jobs queuing/processing and thread pool management

- **silent(-1):** Absolutely silent
- **basic(0):** Very basic auditing logs (e.g. SA up/SA down)
- **generic(1):** Generic control flow with errors, a good default to see whats going on
- **detailed(2):** More detailed debugging control flow
- **raw(3):** Including RAW data dumps in hex
- **sensitive(4):** Also include sensitive material in dumps, e.g. keys

Enumeration	silent (-1), basic (0), generic (1), detailed (2), raw (3), sensitive (4)
Access	readwrite
OID	1.3.6.1.4.1.16177.1.400.1.1.1003.2.3.10

8.1.3.2.3.3 cfgVpnIpsecDbgKnl

knl: IPsec/Networking kernel interface

- **silent(-1)**: Absolutely silent
- **basic(0)**: Very basic auditing logs (e.g. SA up/SA down)
- **generic(1)**: Generic control flow with errors, a good default to see whats going on
- **detailed(2)**: More detailed debugging control flow
- **raw(3)**: Including RAW data dumps in hex
- **sensitive(4)**: Also include sensitive material in dumps, e.g. keys

<i>Enumeration</i>	silent (-1), basic (0), generic (1), detailed (2), raw (3), sensitive (4)
<i>Access</i>	readwrite
<i>OID</i>	1.3.6.1.4.1.16177.1.400.1.1.1003.2.3.11

8.1.3.2.3.4 cfgVpnIpsecDbgLib

lib: libstrongwan library messages

- **silent(-1)**: Absolutely silent
- **basic(0)**: Very basic auditing logs (e.g. SA up/SA down)
- **generic(1)**: Generic control flow with errors, a good default to see whats going on
- **detailed(2)**: More detailed debugging control flow
- **raw(3)**: Including RAW data dumps in hex
- **sensitive(4)**: Also include sensitive material in dumps, e.g. keys

<i>Enumeration</i>	silent (-1), basic (0), generic (1), detailed (2), raw (3), sensitive (4)
<i>Access</i>	readwrite
<i>OID</i>	1.3.6.1.4.1.16177.1.400.1.1.1003.2.3.12

8.1.3.2.3.5 cfgVpnIpsecDbgMgr

mgr: IKE_SA manager, handling synchronization for IKE_SA access

- **silent(-1)**: Absolutely silent
- **basic(0)**: Very basic auditing logs (e.g. SA up/SA down)
- **generic(1)**: Generic control flow with errors, a good default to see whats going on
- **detailed(2)**: More detailed debugging control flow
- **raw(3)**: Including RAW data dumps in hex
- **sensitive(4)**: Also include sensitive material in dumps, e.g. keys

<i>Enumeration</i>	silent (-1), basic (0), generic (1), detailed (2), raw (3), sensitive (4)
<i>Access</i>	readwrite
<i>OID</i>	1.3.6.1.4.1.16177.1.400.1.1.1003.2.3.13

8.1.3.2.3.6 cfgVpnIpsecDbgNet

net: IKE network communication

- **silent(-1)**: Absolutely silent
- **basic(0)**: Very basic auditing logs (e.g. SA up/SA down)
- **generic(1)**: Generic control flow with errors, a good default to see whats going on
- **detailed(2)**: More detailed debugging control flow
- **raw(3)**: Including RAW data dumps in hex
- **sensitive(4)**: Also include sensitive material in dumps, e.g. keys

<i>Enumeration</i>	silent (-1), basic (0), generic (1), detailed (2), raw (3), sensitive (4)
<i>Access</i>	readwrite
<i>OID</i>	1.3.6.1.4.1.16177.1.400.1.1.1003.2.3.14

8.1.3.2.3.7 cfgVpnlpsecDbgPts

pts: Platform Trust Service

- **silent(-1)**: Absolutely silent
- **basic(0)**: Very basic auditing logs (e.g. SA up/SA down)
- **generic(1)**: Generic control flow with errors, a good default to see whats going on
- **detailed(2)**: More detailed debugging control flow
- **raw(3)**: Including RAW data dumps in hex
- **sensitive(4)**: Also include sensitive material in dumps, e.g. keys

<i>Enumeration</i>	silent (-1), basic (0), generic (1), detailed (2), raw (3), sensitive (4)
<i>Access</i>	readwrite
<i>OID</i>	1.3.6.1.4.1.16177.1.400.1.1.1003.2.3.15

8.1.3.2.3.8 cfgVpnlpsecDbgTls

tls: libtls library messages

- **silent(-1)**: Absolutely silent
- **basic(0)**: Very basic auditing logs (e.g. SA up/SA down)
- **generic(1)**: Generic control flow with errors, a good default to see whats going on
- **detailed(2)**: More detailed debugging control flow
- **raw(3)**: Including RAW data dumps in hex
- **sensitive(4)**: Also include sensitive material in dumps, e.g. keys

<i>Enumeration</i>	silent (-1), basic (0), generic (1), detailed (2), raw (3), sensitive (4)
<i>Access</i>	readwrite
<i>OID</i>	1.3.6.1.4.1.16177.1.400.1.1.1003.2.3.16

8.1.3.2.3.9 cfgVpnlpsecDbgTnc

tnc: Trusted Network Connect

- **silent(-1)**: Absolutely silent
- **basic(0)**: Very basic auditing logs (e.g. SA up/SA down)
- **generic(1)**: Generic control flow with errors, a good default to see whats going on
- **detailed(2)**: More detailed debugging control flow
- **raw(3)**: Including RAW data dumps in hex
- **sensitive(4)**: Also include sensitive material in dumps, e.g. keys

<i>Enumeration</i>	silent (-1), basic (0), generic (1), detailed (2), raw (3), sensitive (4)
<i>Access</i>	readwrite
<i>OID</i>	1.3.6.1.4.1.16177.1.400.1.1.1003.2.3.17

8.1.3.2.3.10 cfgVpnlpsecDbgCfg

cfg: Configuration management and plugins

- **silent(-1)**: Absolutely silent
- **basic(0)**: Very basic auditing logs (e.g. SA up/SA down)
- **generic(1)**: Generic control flow with errors, a good default to see whats going on
- **detailed(2)**: More detailed debugging control flow
- **raw(3)**: Including RAW data dumps in hex
- **sensitive(4)**: Also include sensitive material in dumps, e.g. keys

<i>Enumeration</i>	silent (-1), basic (0), generic (1), detailed (2), raw (3), sensitive (4)
<i>Access</i>	readwrite
<i>OID</i>	1.3.6.1.4.1.16177.1.400.1.1.1003.2.3.2

8.1.3.2.3.11 cfgVpnlpsecDbgChd

chd: CHILD_SA/IPsec SA

- **silent(-1)**: Absolutely silent
- **basic(0)**: Very basic auditing logs (e.g. SA up/SA down)
- **generic(1)**: Generic control flow with errors, a good default to see whats going on
- **detailed(2)**: More detailed debugging control flow
- **raw(3)**: Including RAW data dumps in hex
- **sensitive(4)**: Also include sensitive material in dumps, e.g. keys

<i>Enumeration</i>	silent (-1), basic (0), generic (1), detailed (2), raw (3), sensitive (4)
<i>Access</i>	readwrite
<i>OID</i>	1.3.6.1.4.1.16177.1.400.1.1.1003.2.3.3

8.1.3.2.3.12 cfgVpnlpsecDbgDmn

dmn: Main daemon setup/cleanup/signal handling

- **silent(-1)**: Absolutely silent
- **basic(0)**: Very basic auditing logs (e.g. SA up/SA down)
- **generic(1)**: Generic control flow with errors, a good default to see whats going on
- **detailed(2)**: More detailed debugging control flow
- **raw(3)**: Including RAW data dumps in hex
- **sensitive(4)**: Also include sensitive material in dumps, e.g. keys

<i>Enumeration</i>	silent (-1), basic (0), generic (1), detailed (2), raw (3), sensitive (4)
<i>Access</i>	readwrite
<i>OID</i>	1.3.6.1.4.1.16177.1.400.1.1.1003.2.3.4

8.1.3.2.3.13 cfgVpnlpsecDbgEnc

enc: Packet encoding/decoding encryption/decryption operations

- **silent(-1)**: Absolutely silent
- **basic(0)**: Very basic auditing logs (e.g. SA up/SA down)
- **generic(1)**: Generic control flow with errors, a good default to see whats going on
- **detailed(2)**: More detailed debugging control flow
- **raw(3)**: Including RAW data dumps in hex
- **sensitive(4)**: Also include sensitive material in dumps, e.g. keys

<i>Enumeration</i>	silent (-1), basic (0), generic (1), detailed (2), raw (3), sensitive (4)
<i>Access</i>	readwrite
<i>OID</i>	1.3.6.1.4.1.16177.1.400.1.1.1003.2.3.5

8.1.3.2.3.14 cfgVpnlpsecDbgEsp

esp: libipsec library messages

- **silent(-1)**: Absolutely silent
- **basic(0)**: Very basic auditing logs (e.g. SA up/SA down)
- **generic(1)**: Generic control flow with errors, a good default to see whats going on
- **detailed(2)**: More detailed debugging control flow
- **raw(3)**: Including RAW data dumps in hex
- **sensitive(4)**: Also include sensitive material in dumps, e.g. keys

<i>Enumeration</i>	silent (-1), basic (0), generic (1), detailed (2), raw (3), sensitive (4)
<i>Access</i>	readwrite
<i>OID</i>	1.3.6.1.4.1.16177.1.400.1.1.1003.2.3.6

8.1.3.2.3.15 cfgVpnlpsecDbgIke

ike: IKE_SA/ISAKMP SA

- **silent(-1)**: Absolutely silent
- **basic(0)**: Very basic auditing logs (e.g. SA up/SA down)
- **generic(1)**: Generic control flow with errors, a good default to see whats going on
- **detailed(2)**: More detailed debugging control flow
- **raw(3)**: Including RAW data dumps in hex
- **sensitive(4)**: Also include sensitive material in dumps, e.g. keys

<i>Enumeration</i>	silent (-1), basic (0), generic (1), detailed (2), raw (3), sensitive (4)
<i>Access</i>	readwrite
<i>OID</i>	1.3.6.1.4.1.16177.1.400.1.1.1003.2.3.7

8.1.3.2.3.16 cfgVpnlpsecDbgImc

imc: Integrity Measurement Collector

- **silent(-1)**: Absolutely silent
- **basic(0)**: Very basic auditing logs (e.g. SA up/SA down)
- **generic(1)**: Generic control flow with errors, a good default to see whats going on
- **detailed(2)**: More detailed debugging control flow
- **raw(3)**: Including RAW data dumps in hex
- **sensitive(4)**: Also include sensitive material in dumps, e.g. keys

<i>Enumeration</i>	silent (-1), basic (0), generic (1), detailed (2), raw (3), sensitive (4)
<i>Access</i>	readwrite
<i>OID</i>	1.3.6.1.4.1.16177.1.400.1.1.1003.2.3.8

8.1.3.2.3.17 cfgVpnlpsecDbgImv

imv: Integrity Measurement Verifier

- **silent(-1)**: Absolutely silent
- **basic(0)**: Very basic auditing logs (e.g. SA up/SA down)
- **generic(1)**: Generic control flow with errors, a good default to see whats going on
- **detailed(2)**: More detailed debugging control flow
- **raw(3)**: Including RAW data dumps in hex
- **sensitive(4)**: Also include sensitive material in dumps, e.g. keys

<i>Enumeration</i>	silent (-1), basic (0), generic (1), detailed (2), raw (3), sensitive (4)
<i>Access</i>	readwrite
<i>OID</i>	1.3.6.1.4.1.16177.1.400.1.1.1003.2.3.9

8.1.3.3 cfgVpnWireguard

8.1.3.3.1 cfgVpnWireguardTable

Wireguard Table

<i>Range</i>	0 - 15
<i>OID</i>	1.3.6.1.4.1.16177.1.400.1.1.1003.5.1

8.1.3.3.1.1 cfgVpnWireguardTableEntry

Wireguard Table Entry

<i>OID</i>	1.3.6.1.4.1.16177.1.400.1.1.1003.5.1.1
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cfgVpnWgIndex

Table Entry Index

<i>Range</i>	0 - 15
<i>Access</i>	noaccess
<i>OID</i>	1.3.6.1.4.1.16177.1.400.1.1.1003.5.1.1.1

cfgVpnWgName

Name of the Wireguard Interface

<i>Type</i>	DisplayString
<i>Range</i>	1 - 255
<i>Access</i>	readonly
<i>OID</i>	1.3.6.1.4.1.16177.1.400.1.1.1003.5.1.1.2

cfgVpnWgListenPort

Wireguard Listen Port

Specify the port which is used by this wireguard instance.

Set this to a fixed value when expecting inbound connections. The official wireguard port is 51820.

A random port is used when set to 0.

<i>Range</i>	0 - 65535
<i>Access</i>	readwrite
<i>OID</i>	1.3.6.1.4.1.16177.1.400.1.1.1003.5.1.1.3

cfgVpnWgPrivateKey

Wireguard Private Key

Base 64 encoded private key used by this wireguard instance. The public key in `cfgVpnWgPublicKey` is derived from this private key.

Will automatically generate a new private/public key-pair when set to `generate` or `not_yet_generated`.

May be generated on the CLI with the command `wg genkey`.

Examples:

- `generate`
- `not_yet_generated`
- `4CwLw8p8UGdv6baTN1dMxhxVq+m779vI2IjDpSFecW8=`

Type	DisplayString
Range	8 - 44
Access	readwrite
OID	1.3.6.1.4.1.16177.1.400.1.1.1003.5.1.1.4

cfgVpnWgPublicKey

Wireguard Public Key

Base 64 encoded public key provided by this wireguard instance. This key is derived from what is set in `cfgVpnWgPrivateKey`.

When the private key changes or is regenerated, the content of this field is updated during the apply.

Configure this public key on the remote peer(s).

Type	DisplayString
Access	readonly
OID	1.3.6.1.4.1.16177.1.400.1.1.1003.5.1.1.5

8.1.3.3.2 cfgVpnWireguardPeersTable

Wireguard Peers Table

Range	0 - 255
OID	1.3.6.1.4.1.16177.1.400.1.1.1003.5.2

8.1.3.3.2.1 cfgVpnWireguardPeersTableEntry

Wireguard Peers Table Entry

<i>OID</i>	1.3.6.1.4.1.16177.1.400.1.1.1003.5.2.1
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cfgVpnWgPIndex

Table Entry Index

<i>Range</i>	0 - 255
<i>Access</i>	noaccess
<i>OID</i>	1.3.6.1.4.1.16177.1.400.1.1.1003.5.2.1.1

cfgVpnWgPEnabled

Wireguard Peer Disabled or Enabled

Disable or enabled this peer instance.

<i>Enumeration</i>	disabled (0), enabled (1)
<i>Access</i>	readwrite
<i>OID</i>	1.3.6.1.4.1.16177.1.400.1.1.1003.5.2.1.2

cfgVpnWgPInstance

Wireguard Peer Instance

Specify a wireguard instance defined in `cfgVpnWireguardTable`. All peers with a matching instance are set up for the referenced instance.

<i>Range</i>	0 - 15
<i>Access</i>	readwrite
<i>OID</i>	1.3.6.1.4.1.16177.1.400.1.1.1003.5.2.1.3

cfgVpnWgPPeer

Wireguard Peer Public Key

Specifys the remote peer by its public key. This Peer is considered disabled when set to `none`.

Example:

*Su05SN6Wlje1PFIHMO8C2GmCzPp1R85ciwY Ao6yvlhA=

Type	DisplayString
Range	4 - 44
Access	readwrite
OID	1.3.6.1.4.1.16177.1.400.1.1.1003.5.2.1.4

cfgVpnWgPEndpoint

Wireguard Peer Endpoint

Specifies the remote end by IP and port.

Set to 0.0.0.0:0, when the local peer accepts connections, but does not initiate by itself.

Examples:

- 192.168.1.20:51820
- 0.0.0.0:0

Type	DisplayString
Access	readwrite
OID	1.3.6.1.4.1.16177.1.400.1.1.1003.5.2.1.5

cfgVpnWgPAllowedIps

Wireguard Peer Allowed IPs

The Allowed IPs refers to the addresses inside the tunnel. It has two meanings:

In TX direction it acts as routing table. Frames with a destination matching the Allowed IPs are encrypted. All other frames that are routed to the interface but don't match the Allowed IPs are dropped.

In RX direction it acts as ACL. Only frames where the source matches the Allowed IPs are accepted. Everything else is dropped.

Multiple space and/or comma separated networks in CIDR notation may be specified.

When set to none, no frames will be sent nor received.

Examples:

- none
- 0.0.0.0/0
- 192.168.0.0/16, 172.16.0.0/12, 10.0.0.0/8

Type	DisplayString
Access	readwrite
OID	1.3.6.1.4.1.16177.1.400.1.1.1003.5.2.1.6

cfgVpnWgPPsk

Wireguard Peer Private Shared Key

Base 64 encoded privately shared key. Provides an additional layer of cryptography with a symmetric key for post-quantum resistance. Has negligible impact on performance. A separate key should be used for every peer.

Set to none when not used.

May be generated on the CLI with the command `wg genpsk`.

Examples:

- none
- F6wPakowlilChk4FRcHrAP+/jO5jdsQ7xphXi8UzG6Y=

Type	DisplayString
Range	4 - 44
Access	readwrite
OID	1.3.6.1.4.1.16177.1.400.1.1.1003.5.2.1.7

cfgVpnWgPPersistentKeepalive

Wireguard Peer Persistent Keepalive

Interval in seconds to send a keepalive message to the peer.

Only set this, when connecting through NAT or a firewall blocking inbound connections. When not set to 0, a sane value is 25.

Range	0 - 65535
Access	readwrite
OID	1.3.6.1.4.1.16177.1.400.1.1.1003.5.2.1.8

8.1.3.4 cfgVpnTunnelEndPoint

8.1.3.4.1 cfgVpnTunnelEndPointTable

Tunnel Endpoint Table

<i>Range</i>	0 - 15
<i>OID</i>	1.3.6.1.4.1.16177.1.400.1.1.1003.6.1

8.1.3.4.1.1 cfgVpnTunnelEndPointTableEntry

Tunnel Endpoint Table Entry

<i>OID</i>	1.3.6.1.4.1.16177.1.400.1.1.1003.6.1.1
------------	--

cfgVpnTepIndex

Table Entry Index

<i>Range</i>	0 - 15
<i>Access</i>	noaccess
<i>OID</i>	1.3.6.1.4.1.16177.1.400.1.1.1003.6.1.1.1

cfgVpnTepName

Name of the Tunnel Endpoint Interface

<i>Type</i>	DisplayString
<i>Range</i>	1 - 255
<i>Access</i>	readonly
<i>OID</i>	1.3.6.1.4.1.16177.1.400.1.1.1003.6.1.1.2

cfgVpnTepTunnelType

Tunnel Endpoint Type

- **gre(0)**: Allows to tunnel L3 frames. Can not be bridged.
- **gretap(1)**: Allows to tunnel L2 frames. Can be bridged.

<i>Enumeration</i>	gre (0), gretap (1)
<i>Access</i>	readwrite
<i>OID</i>	1.3.6.1.4.1.16177.1.400.1.1.1003.6.1.1.3

cfgVpnTepSource

Source Address of the Tunnel

Can be set to 0.0.0.0 to let the system select the appropriate address based on the routing table.

Type	DisplayString
Range	7 - 15
Access	readwrite
OID	1.3.6.1.4.1.16177.1.400.1.1.1003.6.1.1.4

cfgVpnTepDestination

Destination Address of the Tunnel

Encapsulated frames are sent to this address.

Type	DisplayString
Range	7 - 15
Access	readwrite
OID	1.3.6.1.4.1.16177.1.400.1.1.1003.6.1.1.5

cfgVpnTepTos

Outer TOS Field of the Tunnel

The outer IP header will have it's TOS field set to the value specified here.

To force the outer TOS header to an explicit value, set the field to a value between '00' and 'ff'.

The outer TOS header may inherit its TOS field from the inner IP header. To achieve this set to 'inherit'. Non-IP frames will have the value 00.

Usually, only the upper most 3 bits have a significant impact on prioritisation, thus it is recommended to preferably use the values: 00, 20, 40, 60, 80, a0, c0, e0

Examples:

- inherit
- 00
- a0
- ff

Type	DisplayString
Range	2 - 7
Access	readwrite
OID	1.3.6.1.4.1.16177.1.400.1.1.1003.6.1.1.6

cfgVpnTepRxKeyId

RX Tunnel Key ID

The key allows to run multiple tunnels between peers in parallel. The RX Key specifies which value is expected in the header of frames during reception.

This is a 32 bit number which may be specified directly (a number between 0 and 4294967295) or as an IP address-like dotted quad: '123.123.0.255'.

The value configured in this field should match the `cfgVpnTepTxKeyId` on the remote side.

Set to -1 to not expect a key.

Examples:

- -1
- 0
- 4000
- 100.0.0.1

Type	DisplayString
Range	1 - 15
Access	readwrite
OID	1.3.6.1.4.1.16177.1.400.1.1.1003.6.1.1.7

cfgVpnTepTxKeyId

TX Tunnel Key ID

The key allows to run multiple tunnels between peers in parallel. The TX Key specifies which value is set in the header of frames during transmission.

This is a 32 bit number which may be specified directly (a number between 0 and 4294967295) or as an IP address-like dotted quad: '123.123.0.255'.

The value configured in this field should match the `cfgVpnTepRxKeyId` on the remote side.

Set to -1 to not set a key.

Examples:

- -1
- 0
- 4000
- 100.0.0.1

Type	DisplayString
Range	1 - 15
Access	readwrite
OID	1.3.6.1.4.1.16177.1.400.1.1.1003.6.1.1.8

8.1.4 cfgCellular

8.1.4.1 cfgCellSimTable

SIM Parameter Table

Applies to cellular products only.

Range	0 - 0
OID	1.3.6.1.4.1.16177.1.400.1.1.101.1

8.1.4.1.1 cfgCellSimTableEntry

SIM Parameter Table Entry

Applies to cellular products only.

OID	1.3.6.1.4.1.16177.1.400.1.1.101.1.1
-----	-------------------------------------

8.1.4.1.1.1 cfgCellSimIndex

Table Entry Index

Range	0 - 0
Access	noaccess
OID	1.3.6.1.4.1.16177.1.400.1.1.101.1.1.1

8.1.4.1.1.2 cfgCellSimSlot1

SIM Slot 1

This value references the index of the desired SIM profile in the `cfgCellSimProfileTable` for the SIM card in slot 1. A value of -1 disables slot 1.

Applies to cellular products only.

Type	Integer32
Access	readwrite
OID	1.3.6.1.4.1.16177.1.400.1.1.101.1.1.2

8.1.4.1.1.3 cfgCellSimSlot2

SIM Slot 2

This value references the index of the desired SIM profile in the `cfgCellSimProfileTable` for the SIM card in slot 2. A value of -1 disables slot 2.

Applies to cellular products only.

Type	Integer32
Access	readwrite
OID	1.3.6.1.4.1.16177.1.400.1.1.101.1.1.3

8.1.4.1.1.4 cfgCellSimPrimarySlot

Primary SIM Slot

First connection attempts are always done with the SIM card in the primary slot. Choose either

- **slot1(1)**
- **slot2(2)**

as primary SIM slot.

Note: If only one slot is activated, this parameter has no effect.

Applies to cellular products only.

Enumeration	slot1 (1), slot2 (2)
Access	readwrite
OID	1.3.6.1.4.1.16177.1.400.1.1.101.1.1.4

8.1.4.1.1.5 cfgCellSimUnlockTimeout

SIM Unlock Timeout

The time in milliseconds how long the unlocking process waits until the SIM card is ready to be unlocked. Increase the timeout if older SIMs cannot be unlocked. A high timeout affects the performance of SIM rotation.

Note: Changes become effective after restarting the device.

Applies to cellular products only.

<i>Range</i>	300 - 30000
<i>Access</i>	readwrite
<i>OID</i>	1.3.6.1.4.1.16177.1.400.1.1.101.1.1.5

8.1.4.2 cfgCellSimProfileTable

SIM Profiles

Applies to cellular products only.

<i>Range</i>	0 - 9
<i>OID</i>	1.3.6.1.4.1.16177.1.400.1.1.101.2

8.1.4.2.1 cfgCellSimProfileTableEntry

SIM Profile

Applies to cellular products only.

<i>OID</i>	1.3.6.1.4.1.16177.1.400.1.1.101.2.1
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8.1.4.2.1.1 cfgCellSimProfileIndex

Table Entry Index

<i>Range</i>	0 - 9
<i>Access</i>	noaccess
<i>OID</i>	1.3.6.1.4.1.16177.1.400.1.1.101.2.1.1

8.1.4.2.1.2 cfgCellSimProfileApn

Access Point Name

If a specific Access Point Name (APN) is required by the service provider, it can be defined using this entry, otherwise it can be left at its default value **auto**.

Note: If this entry is set to **auto** the authentication entries `cfgCellSimProfileUsername`, `cfgCellSimProfilePassword` and `cfgCellSimProfileAuthType` are ignored.

Applies to cellular products only.

Type	DisplayString
Range	1 - 255
Access	readwrite
OID	1.3.6.1.4.1.16177.1.400.1.1.101.2.1.2

8.1.4.2.1.3 cfgCellSimProfileUsername

Username

If the service provider requires authentication, the username can be specified by this entry. If no authentication type is selected, see `cfgCellSimProfileAuthType`, this entry is ignored.

Applies to cellular products only.

Type	DisplayString
Range	1 - 255
Access	readwrite
OID	1.3.6.1.4.1.16177.1.400.1.1.101.2.1.3

8.1.4.2.1.4 cfgCellSimProfilePassword

Password

Set the password for the user defined with `cfgCellSimProfileUsername`.

Applies to cellular products only.

Type	DisplayString
Range	1 - 255
Access	readwrite
OID	1.3.6.1.4.1.16177.1.400.1.1.101.2.1.4

8.1.4.2.1.5 cfgCellSimProfilePinEnabled

PIN Authentication Disabled or Enabled

Set to **enabled(1)** if PIN authentication is required for the SIM card in the corresponding slot.

Applies to cellular products only.

Enumeration	disabled (0), enabled (1)
Access	readwrite
OID	1.3.6.1.4.1.16177.1.400.1.1.101.2.1.5

8.1.4.2.1.6 cfgCellSimProfilePin

PIN of the SIM Card

The PIN is ignored, when PIN authentication is disabled, see `cfgCellSimProfilePinEnabled`.

Applies to cellular products only.

<i>Type</i>	DisplayString
<i>Range</i>	4 - 4
<i>Access</i>	readwrite
<i>OID</i>	1.3.6.1.4.1.16177.1.400.1.1.101.2.1.6

8.1.4.2.1.7 cfgCellSimProfileAuthType

Authentication Type

Select the authentication type, which is required by the service provider. If no authentication is required set this entry to its default value **none(0)**, otherwise choose one of the supported authentication types:

- **pap(1)**,
- **chap(2)**, or
- **both(3)**.

Applies to cellular products only.

<i>Enumeration</i>	none (0), pap (1), chap (2), both (3)
<i>Access</i>	readwrite
<i>OID</i>	1.3.6.1.4.1.16177.1.400.1.1.101.2.1.7

8.1.4.2.1.8 cfgCellSimProfileRoaming

Roaming Disabled or Enabled

Set to **enabled(1)** to roam to other available cellular networks outside the range of the home network. The SIM card and network provider must support roaming.

Note: Activated roaming may incur additional costs!

Applies to cellular products only.

<i>Enumeration</i>	disabled (0), enabled (1)
<i>Access</i>	readwrite
<i>OID</i>	1.3.6.1.4.1.16177.1.400.1.1.101.2.1.8

8.1.4.3 cfgCellConnectionManagement

8.1.4.3.1 cfgCellConnMgmtSimRotationEnabled

SIM Rotation Disabled or Enabled

Set to **enabled(1)** for rotating between primary and secondary SIM after a connection loss.

Applies to cellular products only.

<i>Enumeration</i>	disabled (0), enabled (1)
<i>Access</i>	readwrite
<i>OID</i>	1.3.6.1.4.1.16177.1.400.1.1.101.3.1

8.1.4.3.2 cfgCellConnMgmtMonMode

Monitor Mode

The monitor mode defines which algorithm the connection management is using to evaluate whether a cellular connection is up. The following monitor modes are supported:

- **signal(0)**: Monitor the cellular signal (e.g. RSSI level)
- **remoteHosts(1)**: Monitor the cellular network (e.g. the availability of remote network hosts).

Adjusting the monitor algorithm can be achieved by setting the `cfgCellConnMgmtMonPeriod` and `cfgCellConnMgmtMonPeriod`.

Using the **remoteHosts(1)** mode requires at least one active remote host in the `cfgCellConnMgmtMonRemoteTab`.

Applies to cellular products only.

<i>Enumeration</i>	signal (0), remoteHosts (1)
<i>Access</i>	readwrite
<i>OID</i>	1.3.6.1.4.1.16177.1.400.1.1.101.3.2

8.1.4.3.3 cfgCellConnMgmtMonPeriod

Monitor Period

The monitor period defines the time in seconds between two consecutive evaluations of the cellular connection.

Applies to cellular products only.

<i>Range</i>	1 - 86400
<i>Access</i>	readwrite
<i>OID</i>	1.3.6.1.4.1.16177.1.400.1.1.101.3.3

8.1.4.3.4 cfgCellConnMgmtMonCount

Monitor Count

The monitor count defines the needed amount of consecutive failed connection tests before a connection is considered down.

Applies to cellular products only.

<i>Range</i>	1 - 10
<i>Access</i>	readwrite
<i>OID</i>	1.3.6.1.4.1.16177.1.400.1.1.101.3.4

8.1.4.3.5 cfgCellConnMgmtMonRemoteTable

Monitored Remotes

All hosts listed in this table are used by the **network(1)** mode, defined in `cfgCellConnMgmtMonMode`. If the connection management uses this algorithm to evaluate the status of the connection, it tests all hosts one after the other. As soon as one host is available the cellular connection is considered up.

Applies to cellular products only.

<i>Range</i>	0 - 3
<i>OID</i>	1.3.6.1.4.1.16177.1.400.1.1.101.3.5

8.1.4.3.5.1 cfgCellConnMgmtMonRemoteTableEntry

Monitored Remote

Applies to cellular products only.

<i>OID</i>	1.3.6.1.4.1.16177.1.400.1.1.101.3.5.1
------------	---------------------------------------

cfgCellConnMgmtMonRemoteIndex

Table Entry Index

Applies to cellular products only.

<i>Range</i>	0 - 3
<i>Access</i>	noaccess
<i>OID</i>	1.3.6.1.4.1.16177.1.400.1.1.101.3.5.1.1

cfgCellConnMgmtMonRemoteType

Monitoring Type

The following methods are provided to check the availability of a remote host via the cellular network:

- **disabled(0)**: Ignore this host.
- **icmp(1)**: Use ICMP to ping the host.
- **tcp(2)**: Probe a TCP/IP port of the host.

Applies to cellular products only.

<i>Enumeration</i>	disabled (0), icmp (1), tcp (2)
<i>Access</i>	readwrite
<i>OID</i>	1.3.6.1.4.1.16177.1.400.1.1.101.3.5.1.2

cfgCellConnMgmtMonRemoteAddress

Remote Address

The address of a remote host can either be defined by an IP address or a host name. If `cfgCellConnMgmtMonRemoteType` is set to **tcp(2)**, an additional port number must be specified. The format of a valid remote address is:

<IP address|host name>[:port]

Examples:

- **icmp(1)**: 8.8.8.8
- **tcp(2)**: www.example.com:80

Applies to cellular products only.

<i>Type</i>	DisplayString
<i>Range</i>	1 - 255
<i>Access</i>	readwrite
<i>OID</i>	1.3.6.1.4.1.16177.1.400.1.1.101.3.5.1.3

8.1.4.4 cfgCellDbgTable

Cellular Debug Parameters

<i>Range</i>	0 - 0
<i>OID</i>	1.3.6.1.4.1.16177.1.400.1.1.101.4

8.1.4.4.1 cfgCellDbgTableEntry

Cellular Debug Parameters Entry

<i>OID</i>	1.3.6.1.4.1.16177.1.400.1.1.101.4.1
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8.1.4.4.1.1 cfgCellDbgIndex

Table Entry Index

<i>Range</i>	0 - 0
<i>Access</i>	noaccess
<i>OID</i>	1.3.6.1.4.1.16177.1.400.1.1.101.4.1.1

8.1.4.4.1.2 cfgCellDbgSignal

TECHPREVIEW: Signal Level Trap Disable or Enable

Persistent default value to enable or disable the signal level trap. These log messages are subject to change. DO NOT PARSE!

Applies to cellular products only.

<i>Enumeration</i>	disabled (0), enabled (1)
<i>Access</i>	readwrite
<i>OID</i>	1.3.6.1.4.1.16177.1.400.1.1.101.4.1.2

8.1.4.4.1.3 cfgCellDbgConnection

TECHPREVIEW: Connection State Trap Disable or Enable

Persistent default value to enable or disable the connection state trap. These log messages are subject to change. DO NOT PARSE!

Applies to cellular products only.

<i>Enumeration</i>	disabled (0), enabled (1)
<i>Access</i>	readwrite
<i>OID</i>	1.3.6.1.4.1.16177.1.400.1.1.101.4.1.3

8.1.5 cfgLogging

8.1.5.1 cfgLogRemote

8.1.5.1.1 cfgLogRemoteTable

List of Syslog Destinations

<i>Range</i>	0 - 3
<i>OID</i>	1.3.6.1.4.1.16177.1.400.1.1.11.2.1

8.1.5.1.1.1 cfgLogRemoteTableEntry

List of Syslog Destinations

<i>OID</i>	1.3.6.1.4.1.16177.1.400.1.1.11.2.1.1
------------	--------------------------------------

cfgLogRemoteIndex

Table Entry Index

<i>Range</i>	0 - 3
<i>Access</i>	noaccess
<i>OID</i>	1.3.6.1.4.1.16177.1.400.1.1.11.2.1.1.1

cfgLogRemoteEnabled

Log Syslog messages to a remote server in accordance to RFC 5424.

Applies to AP and STA.

<i>Enumeration</i>	disabled (0), enabled (1)
<i>Access</i>	readwrite
<i>OID</i>	1.3.6.1.4.1.16177.1.400.1.1.11.2.1.1.2

cfgLogRemoteLevel

Log only messages with equal or higher priority than prio N (0-7).

Applies to AP and STA.

<i>Enumeration</i>	emergency (0), alert (1), critical (2), error (3), warning (4), notice (5), info (6), debug (7)
<i>Access</i>	readonly
<i>OID</i>	1.3.6.1.4.1.16177.1.400.1.1.11.2.1.1.3

cfgLogRemoteProtocol

Protocol to Send Log Messages

The udp(0) protocol complies with the standard syslog protocol.

Applies to AP and STA.

<i>Enumeration</i>	udp (0), tcp (1)
<i>Access</i>	readwrite
<i>OID</i>	1.3.6.1.4.1.16177.1.400.1.1.11.2.1.1.4

cfgLogRemotelp

Remote IP address.

Applies to AP and STA.

<i>Type</i>	IpAddress
<i>Access</i>	readwrite
<i>OID</i>	1.3.6.1.4.1.16177.1.400.1.1.11.2.1.1.5

cfgLogRemotePort

Remote Port

Applies to AP and STA.

<i>Type</i>	Integer32
<i>Access</i>	readwrite
<i>OID</i>	1.3.6.1.4.1.16177.1.400.1.1.11.2.1.1.6

cfgLogRemoteType

TECHPREVIEW: Bitfield to Control Remote Syslog Type

- **0x00** - no remote syslog

- **0x01** - standard syslog
- **0x02** - security syslog
- **0x04** - commissioning syslog

Applies to AP and STA.

<i>Range</i>	0 - 255
<i>Access</i>	readwrite
<i>OID</i>	1.3.6.1.4.1.16177.1.400.1.1.11.2.1.1.7

8.1.6 cfgSnmpp

8.1.6.1 cfgSnmppd

8.1.6.1.1 cfgSnmppdLocation

SNMP system location.

Applies to AP and STA.

<i>Type</i>	DisplayString
<i>Range</i>	1 - 255
<i>Access</i>	readwrite
<i>OID</i>	1.3.6.1.4.1.16177.1.400.1.1.12.1.1

8.1.6.1.2 cfgSnmppdCommunity

8.1.6.1.2.1 cfgSnmppdComAdmin

Password for the administrator.

This is the community or the passphrase for the user administrator depending on the cfgSnmppdVersion:

- **v2c:** community string for administrator
- **v3usm:** passphrase for authentication and privacy for user admin

Applies to AP and STA.

Type	DisplayString
Range	1 - 255
Access	readwrite
OID	1.3.6.1.4.1.16177.1.400.1.1.12.1.100.1

8.1.6.1.2.2 cfgSnmpdComMaintainer

Password for the maintainer.

This is the community or the passphrase for the user maintainer depending on the cfgSnmpdVersion:

- **v2c:** community string for maintainer
- **v3usm:** passphrase for authentication and privacy for user maintainer

Applies to AP and STA.

Type	DisplayString
Range	1 - 255
Access	readwrite
OID	1.3.6.1.4.1.16177.1.400.1.1.12.1.100.2

8.1.6.1.2.3 cfgSnmpdComMonitor

Password for the monitor.

This is the community or the passphrase for the user monitor depending on the cfgSnmpdVersion:

- **v2c:** community string for monitor
- **v3usm:** passphrase for authentication and privacy for user monitor

Applies to AP and STA.

Type	DisplayString
Range	1 - 255
Access	readwrite
OID	1.3.6.1.4.1.16177.1.400.1.1.12.1.100.3

8.1.6.1.3 cfgSnmpdContact

SNMP contact.

Applies to AP and STA.

Type	DisplayString
Range	1 - 255
Access	readwrite
OID	1.3.6.1.4.1.16177.1.400.1.1.12.1.2

8.1.6.1.4 cfgSnmpdVersion

SNMP version, either v2c(0) or v3usm(1), for the users** admin, maintainer and monitor.**

Please refer to the user guide for more information about the access rights of the three predefined users.

The User-based Security Model (USM), which is the default Security Module for SNMPv3, with the authentication type cfgSnmpdAuthType and the privacy protocol defined in cfgSnmpdPrivType is implemented and can be chosen by setting this value to **v3usm(1)**.

Setting **v3usm(1)** disables access to the device via SNMPv2.

SNMPv3 USM configuration:

- **User:** admin, maintainer or monitor
- **Authentication Protocol:** cfgSnmpdAuthType
- **Privacy Protocol:** cfgSnmpdPrivType

Applies to AP and STA.

Enumeration	v2c (0), v3usm (1)
Access	readwrite
OID	1.3.6.1.4.1.16177.1.400.1.1.12.1.3

8.1.6.1.5 cfgSnmpdName

SNMP node name often used in NMS.

By convention, this is the node's fully-qualified domain name.

Applies to AP and STA.

Type	DisplayString
Range	1 - 255
Access	readwrite
OID	1.3.6.1.4.1.16177.1.400.1.1.12.1.4

8.1.6.1.6 cfgSnmpdEnabled

Enable or disable the SNMP Agent.

Applies to AP and STA.

<i>Enumeration</i>	disabled (0), enabled (1)
<i>Access</i>	readwrite
<i>OID</i>	1.3.6.1.4.1.16177.1.400.1.1.12.1.5

8.1.6.1.7 cfgSnmpdAddress

Address to which the SNMP Agent binds.

The default is 'udp:0.0.0.0:161'. When defining a configuration, default arguments may be omitted. Multiple space and/or comma separated tuples are allowed.

Supported transport protocols are:

- UDP

Examples:

- udp:192.168.1.20:161
- 192.168.1.20, 10.0.0.1:10161, udp:172.16.32.32:30161
- 192.168.1.20 10.0.0.1:10161 udp:172.16.32.32:30161

More at <http://www.net-snmp.org/docs/man/snmpd.examples.html#lBAE>

Applies to AP and STA.

<i>Type</i>	DisplayString
<i>Range</i>	1 - 255
<i>Access</i>	readwrite
<i>OID</i>	1.3.6.1.4.1.16177.1.400.1.1.12.1.6

8.1.6.1.8 cfgSnmpdAuthType

TECHPREVIEW: Authentication Type for SNMPv3

- **sha1(2)**: SHA-1
- **sha384(5)**: SHA-384
- **sha512(6)**: SHA-512

Applies to AP and STA.

<i>Enumeration</i>	sha1 (2), sha384 (5), sha512 (6)
<i>Access</i>	readwrite
<i>OID</i>	1.3.6.1.4.1.16177.1.400.1.1.12.1.7

8.1.6.1.9 cfgSnmpdPrivType

TECHPREVIEW: Encryption Type for SNMPv3

- **aes128(2):** AES-128
- **aes256(4):** AES-256

Applies to AP and STA.

<i>Enumeration</i>	aes128 (2), aes256 (4)
<i>Access</i>	readwrite
<i>OID</i>	1.3.6.1.4.1.16177.1.400.1.1.12.1.8

8.1.6.2 cfgSnmpTrap

8.1.6.2.1 cfgSnmpTrapEnabled

Enable SNMP Notifications.

Applies to AP and STA.

<i>Enumeration</i>	disabled (0), enabled (1)
<i>Access</i>	readwrite
<i>OID</i>	1.3.6.1.4.1.16177.1.400.1.1.12.10.1

8.1.6.2.2 cfgSnmpTrapAuthProtocol

TECHPREVIEW: SNMP Authentication Protocol

Set the authentication protocol used for authenticated SNMPv3 messages.

Note: Only used if `cfgSnmpTrapVersion` is **usm(2)**.

Applies to AP and STA.

<i>Enumeration</i>	none (0), sha1 (2), sha384 (5), sha512 (6)
<i>Access</i>	readwrite
<i>OID</i>	1.3.6.1.4.1.16177.1.400.1.1.12.10.10

8.1.6.2.3 cfgSnmpTrapAuthPassword

TECHPREVIEW: Authentication Password

Set the authentication password used for authenticated SNMPv3 messages.

Note: Only used if `cfgSnmpTrapVersion` is **usm(2)**.

Applies to AP and STA.

<i>Type</i>	DisplayString
<i>Range</i>	0 - 255
<i>Access</i>	readwrite
<i>OID</i>	1.3.6.1.4.1.16177.1.400.1.1.12.10.11

8.1.6.2.4 cfgSnmpTrapPrivProtocol

TECHPREVIEW: SNMP Privacy Protocol

Note: Only used if `cfgSnmpTrapVersion` is **usm(2)**.

Applies to AP and STA.

<i>Enumeration</i>	none (0), aes128 (2), aes256 (4)
<i>Access</i>	readwrite
<i>OID</i>	1.3.6.1.4.1.16177.1.400.1.1.12.10.12

8.1.6.2.5 cfgSnmpTrapPrivPassword

TECHPREVIEW: Privacy Password

Set the privacy password used for encrypted SNMPv3 messages.

Note: Only used if `cfgSnmpTrapVersion` is **usm(2)**.

Applies to AP and STA.

<i>Type</i>	DisplayString
<i>Range</i>	0 - 255
<i>Access</i>	readwrite
<i>OID</i>	1.3.6.1.4.1.16177.1.400.1.1.12.10.13

8.1.6.2.6 cfgSnmpTrapTimeout

TECHPREVIEW: SNMP Trap Timeout

Specifies the timeout in seconds between retries.

Applies to AP and STA.

<i>Range</i>	1 - 2147483647
<i>Access</i>	readwrite
<i>OID</i>	1.3.6.1.4.1.16177.1.400.1.1.12.10.14

8.1.6.2.7 cfgSnmpTrapRetries

TECHPREVIEW: SNMP Trap Retries

Specifies the number of retries to be used for INFORM notifications.

Applies to AP and STA.

<i>Range</i>	0 - 2147483647
<i>Access</i>	readwrite
<i>OID</i>	1.3.6.1.4.1.16177.1.400.1.1.12.10.15

8.1.6.2.8 cfgSnmpTrapVersion

SNMP Version

SNMP notifications can be sent in the following versions:

- **v1(0):** Version 1 (RFCs 1155-1157), this version is obsolete and will default to **v2c(1)**.
- **v2c(1):** Version 2c (RFCs 1901-1908)
- **usm(2):** Version 3 (RFCs 2571-2574)

Applies to AP and STA.

<i>Enumeration</i>	v1 (0), v2c (1), usm (2)
<i>Access</i>	readwrite
<i>OID</i>	1.3.6.1.4.1.16177.1.400.1.1.12.10.2

8.1.6.2.9 cfgSnmpTrapCommunity

SNMP Community.

Note: Only used for SNMP versions v1(0) and v2c(1).

Applies to AP and STA.

Type	DisplayString
Range	1 - 255
Access	readwrite
OID	1.3.6.1.4.1.16177.1.400.1.1.12.10.3

8.1.6.2.10 cfgSnmptTrapDest

IP address of the trap receiver.

Applies to AP and STA.

Type	IpAddress
Access	readwrite
OID	1.3.6.1.4.1.16177.1.400.1.1.12.10.4

8.1.6.2.11 cfgSnmptTrapType

TECHPREVIEW: SNMP Notification Type

- **trap(1):** Send all SNMP notifications as TRAPv1 or TRAPv2 depending on the SNMP version **v1(0)** or **v2c(1)**, respectively.
- **inform(2):** Send all SNMP notifications as INFORM. Note that INFORM PDUs are not supported for SNMP version **v1(0)**.

Applies to AP and STA.

Enumeration	trap (1), inform (2)
Access	readwrite
OID	1.3.6.1.4.1.16177.1.400.1.1.12.10.5

8.1.6.2.12 cfgSnmptTrapEngineId

TECHPREVIEW: SNMP Authoritative Engine ID

Set the authoritative (security) engine ID used for SNMPv3 messages, given as a hexadecimal string (optionally prefixed by '0x'). The value must be between 5 and 32 octets long. This engine ID will be discovered automatically if this parameter is set to 'auto'.

Note: Only used if `cfgSnmptTrapVersion` is **usm(2)**.

Applies to AP and STA.

Type	DisplayString
Range	0 - 255
Access	readwrite
OID	1.3.6.1.4.1.16177.1.400.1.1.12.10.6

8.1.6.2.13 cfgSnmptapUser

TECHPREVIEW: SNMP User / Security Name

Set the user name used for authenticated SNMPv3 messages.

Note: Only used if cfgSnmptapVersion is **usm(2)**.

Applies to AP and STA.

Type	DisplayString
Range	0 - 255
Access	readwrite
OID	1.3.6.1.4.1.16177.1.400.1.1.12.10.9

8.1.7 cfgDhcp

8.1.7.1 cfgDhcpGlobal

8.1.7.1.1 cfgDhcpGlobalEnabled

Enable DNS/DHCP Server Functionality

Enumeration	disabled (0), enabled (1)
Access	readwrite
OID	1.3.6.1.4.1.16177.1.400.1.1.13.1.1

8.1.7.2 cfgDhcpDnsmasqTable

DNS/DHCP Server Instances

Dnsmasq is used as DNS and DHCP server. A single instance may serve multiple interfaces and scopes. It may run as DNS server only, or as DHCP server only.

Range	0 - 9
OID	1.3.6.1.4.1.16177.1.400.1.1.13.2

8.1.7.2.1 cfgDhcpDnsmasqTableEntry

DNS/DHCP Server Instances

Dnsmasq is used as DNS and DHCP server. A single instance may serve multiple interfaces and scopes. It may run as DNS server only, or as DHCP server only.

<i>OID</i>	1.3.6.1.4.1.16177.1.400.1.1.13.2.1
------------	------------------------------------

8.1.7.2.1.1 cfgDhcpDnsmasqIndex

Table Entry Index

<i>Range</i>	0 - 9
<i>Access</i>	noaccess
<i>OID</i>	1.3.6.1.4.1.16177.1.400.1.1.13.2.1.1

8.1.7.2.1.2 cfgDhcpDnsmasqDnsDomainOverrideParameter

Reference to Domain Override ID

All entries in the `cfgDhcpDomainOverrideTable` with a `cfgDhcpDmnOvrId` that matches the value set here will be used for this DNS server instance.

<i>Range</i>	0 - 255
<i>Access</i>	readwrite
<i>OID</i>	1.3.6.1.4.1.16177.1.400.1.1.13.2.1.10

8.1.7.2.1.3 cfgDhcpDnsmasqDnsHostOverrideParameter

Reference to Host Override ID

All entries in the `cfgDhcpHostOverrideTable` with a `cfgDhcpHstOvrId` that matches the value set here will be used for this DNS server instance.

<i>Range</i>	0 - 255
<i>Access</i>	readwrite
<i>OID</i>	1.3.6.1.4.1.16177.1.400.1.1.13.2.1.11

8.1.7.2.1.4 cfgDhcpDnsmasqScopeParameter

Parameter to Set Which Scope ID to use for the DHCP Server

This is used in conjunction with the scope ID parameter `cfgDhcpScopeId`.

<i>Range</i>	0 - 8
<i>Access</i>	readwrite
<i>OID</i>	1.3.6.1.4.1.16177.1.400.1.1.13.2.1.2

8.1.7.2.1.5 cfgDhcpDnsmasqDnsPort

DNS Server Port

The UDP and TCP port on which the DNS service is running.

<i>Range</i>	1 - 65535
<i>Access</i>	readwrite
<i>OID</i>	1.3.6.1.4.1.16177.1.400.1.1.13.2.1.3

8.1.7.2.1.6 cfgDhcpDnsmasqDnsListenAddress

DNS Server Listen Address

Multiple space and/or comma separated addresses are allowed.

When set to auto, this instance binds to all addresses on interfaces specified by `cfgDhcpScopeInterface` in the referenced `cfgDhcpDnsmasqScopeParameter`. When no interfaces are referenced, this instance binds to any address in the system except localhost (127.0.0.1).

When set to wildcard, this instance binds to 0.0.0.0:<port>.

Examples:

- auto
- wildcard
- 127.0.0.1, 192.168.1.20
- 127.0.0.1 192.168.1.20
- 192.168.1.20,10.0.88.1,10.0.99.1

Note: When a dnsmasq instance binds a port to wildcard, no other instance may bind the same port.

<i>Type</i>	DisplayString
<i>Range</i>	1 - 255
<i>Access</i>	readwrite
<i>OID</i>	1.3.6.1.4.1.16177.1.400.1.1.13.2.1.4

8.1.7.2.1.7 cfgDhcpDnsmasqDnsResolveOrder

Upstream DNS Server Resolve Order

Defines in what order the upstream DNS servers shall be queried.

- **any(0):** Send queries to any of the upstream servers and try to favour servers that are known to be up.
- **strictorder(1):** Try each query with each server strictly in the order they are configured in the `cfgSysNameserverTable`.
- **all(2):** Send all queries to all available servers. The reply from the server which answers first will be returned to the original requester.

<i>Enumeration</i>	any (0), strictorder (1), all (2)
<i>Access</i>	readwrite
<i>OID</i>	1.3.6.1.4.1.16177.1.400.1.1.13.2.1.5

8.1.7.2.1.8 cfgDhcpDnsmasqDnsEnabled

DNS Server Disabled or Enabled

Enable the DNS Server functionality of this instance.

<i>Enumeration</i>	disabled (0), enabled (1)
<i>Access</i>	readwrite
<i>OID</i>	1.3.6.1.4.1.16177.1.400.1.1.13.2.1.6

8.1.7.2.1.9 cfgDhcpDnsmasqDhcpEnabled

DHCP Server Disabled or Enabled

Enable the DHCP Server functionality of this instance.

<i>Enumeration</i>	disabled (0), enabled (1)
<i>Access</i>	readwrite
<i>OID</i>	1.3.6.1.4.1.16177.1.400.1.1.13.2.1.7

8.1.7.2.1.10 cfgDhcpDnsmasqDnsStopDnsRebind

DNS Rebind Protection

Reject and log to syslog addresses from upstream nameservers which are in the private ranges (RFC1918). This blocks an attack where a browser behind a firewall is used to probe machines on the local network.

When using split-DNS, use `cfgDhcpDnsmasqDnsRebindDomainOk` to specify domains that are allowed to resolve to private addresses.

<i>Enumeration</i>	disabled (0), enabled (1)
<i>Access</i>	readwrite
<i>OID</i>	1.3.6.1.4.1.16177.1.400.1.1.13.2.1.8

8.1.7.2.1.11 cfgDhcpDnsmasqDnsRebindDomainOk

Allowed Domains for DNS Rebind Protection

This entry is active when `cfgDhcpDnsmasqDnsStopDnsRebind` is enabled.

Enter a space and/or comma separated list of domains which are allowed to resolve to private addresses (RFC1918).

No Domains are excepted when set to `none`.

Examples:

- example.com, example.net, example.org
- yourdomain.com anotherdomain.com

Note: Subdomains are included when excepting domains. e.g when `domain.net` is set, then `subdomain1.domain.net` and `subdomain2.domain.net` are excepted as well.

<i>Type</i>	DisplayString
<i>Access</i>	readwrite
<i>OID</i>	1.3.6.1.4.1.16177.1.400.1.1.13.2.1.9

8.1.7.3 cfgDhcpScopeTable

DHCP Instance Configs

<i>Range</i>	0 - 7
<i>OID</i>	1.3.6.1.4.1.16177.1.400.1.1.13.3

8.1.7.3.1 cfgDhcpScopeTableEntry

DHCP Instance Configs

<i>OID</i>	1.3.6.1.4.1.16177.1.400.1.1.13.3.1
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8.1.7.3.1.1 cfgDhcpScopeIndex

Table Entry Index

<i>Range</i>	0 - 7
<i>Access</i>	noaccess
<i>OID</i>	1.3.6.1.4.1.16177.1.400.1.1.13.3.1.1

8.1.7.3.1.2 cfgDhcpScopeDhcpOptions

DHCP Scope Custom Options

This config option allows to specify arbitrary DHCP options which will be sent to DHCP clients. Set to `none` when no additional options are required. Enter the DHCP option number followed by a comma followed by the arguments of the DHCP option.

Multiple DHCP options may be specified, separated by a space.

Examples:

- 15, domain.example.com
- 119, search-domain.example.com, another-search.example.com
- 121, 10.0.32.0/24, 10.0.8.6, 10.0.33.0/24, 10.0.8.7
- 15, domain.example.com 119, search-domain.example.com

For a full list of available DHCP options see: <https://www.iana.org/assignments/bootp-dhcp-parameters/bootp-dhcp-parameters.xhtml>

Note: To be able to set DHCP Option 3 and 6 via this configuration item, you first need to disable the respective direct configuration item. Set 0.0.0.0 to `cfgDhcpScopeGateway` for option 3. Set 0.0.0.0 to `cfgDhcpScopeDnsServer1` for option 6.

Applies to AP and STA.

<i>Type</i>	DisplayString
<i>Range</i>	1 - 1024
<i>Access</i>	readwrite
<i>OID</i>	1.3.6.1.4.1.16177.1.400.1.1.13.3.1.14

8.1.7.3.1.3 cfgDhcpScopeAutoGateway

DHCP Scope Auto Gateway

When no default gateway is configured in `cfgDhcpScopeGateway` or `cfgDhcpScopeDhcpOptions`, then this options will automatically assign the IP address of this DHCP server as default gateway.

Applies to AP and STA.

<i>Enumeration</i>	disabled (0), enabled (1)
<i>Access</i>	readwrite
<i>OID</i>	1.3.6.1.4.1.16177.1.400.1.1.13.3.1.15

8.1.7.3.1.4 cfgDhcpScopeAutoDns

DHCP Scope Auto DNS

When no DNS servers are configured in `cfgDhcpScopeDnsServer1`, `cfgDhcpScopeDnsServer2` and `cfgDhcpScopeDhcpOptions`, then this options will automatically assign the IP address of this DHCP server as DNS server.

This option has no effect, when `cfgDhcpDnsmasqDnsEnabled` of the `dnsmasq` instance referencing this scope is disabled.

Applies to AP and STA.

<i>Enumeration</i>	disabled (0), enabled (1)
<i>Access</i>	readwrite
<i>OID</i>	1.3.6.1.4.1.16177.1.400.1.1.13.3.1.16

8.1.7.3.1.5 cfgDhcpScopeld

Scope ID

This is used in conjunction with the DHCP parameter `cfgDhcpDnsmasqScopeParameter`.

Applies to AP and STA.

<i>Range</i>	0 - 8
<i>Access</i>	readwrite
<i>OID</i>	1.3.6.1.4.1.16177.1.400.1.1.13.3.1.2

8.1.7.3.1.6 cfgDhcpScopeInterface

DHCP Server Listening Interface

Network interface on which the DHCP server listen for DHCP requests. The interface on which the server runs must have an address configured. The DHCP server offers lease addresses based on the assigned address, the `cfgDhcpScopeStart` offset and the `cfgDhcpScopeLimit`. If an interface has multiple addresses, then the first address in the order specified in the `cfgNetIpTable` is used.

Examples:

- eth1

- br0.vlan0
- macvlan0

Applies to AP and STA.

Type	DisplayString
Range	1 - 255
Access	readwrite
OID	1.3.6.1.4.1.16177.1.400.1.1.13.3.1.3

8.1.7.3.1.7 cfgDhcpScopeStart

DHCP Scope Start Offset

Specifies the first address of the scope as an offset from the network address and can be calculated as:

network address + cfgDhcpScopeStart

The network address is derived from the first configured IP address on the interface specified by cfgDhcpScopeInterface.

Examples:

- br0.vlan0 has 192.168.1.20/24. The network address of this CIDR block is 192.168.1.0. With cfgDhcpScopeStart set to 100, the lowest address which will be assigned is 192.168.1.100.
- br0.vlan99 has 172.29.101.7/23. The network address of this CIDR block is 172.29.100.0. With cfgDhcpScopeStart set to 306, the lowest address which will be assigned is 172.29.101.50.
- br0.vlan1000 has 10.0.8.140/26. The network address of this CIDR block is 10.0.8.128. With cfgDhcpScopeStart set to 22, the lowest address which will be assigned is 10.0.8.150.

Applies to AP and STA.

Type	Integer32
Access	readwrite
OID	1.3.6.1.4.1.16177.1.400.1.1.13.3.1.4

8.1.7.3.1.8 cfgDhcpScopeLimit

DHCP Scope Limit

Limits the number of addresses which are assigned to clients. Thus this parameter defines how many addresses are in the scope. The highest address assigned to a client can be calculated as:

network address + cfgDhcpScopeStart + cfgDhcpScopeLimit - 1

Examples:

- br0.vlan0 has 192.168.1.20/24. The network address of this CIDR block is 192.168.1.0. With `cfgDhcpScopeStart` set to 100 and `cfgDhcpScopeLimit` set to 150, the highest address which will be assigned is 192.168.1.249.
- br0.vlan99 has 172.29.101.7/23. The network address of this CIDR block is 172.29.100.0. With `cfgDhcpScopeStart` set to 306 and `cfgDhcpScopeLimit` set to 100, the highest address which will be assigned is 172.29.101.149.
- br0.vlan1000 has 10.0.8.140/26. The network address of this CIDR block is 10.0.8.128. With `cfgDhcpScopeStart` set to 22 and `cfgDhcpScopeLimit` set to 30 the highest address which will be assigned is 10.0.8.179.

Note: When `cfgDhcpScopeLimit` is set to a value greater than the remaining size of the CIDR block, the highest address is set to the last address in the block. Essentially the size of the scope is reduced until it fits the block.

Applies to AP and STA.

Type	Integer32
Access	readwrite
OID	1.3.6.1.4.1.16177.1.400.1.1.13.3.1.5

8.1.7.3.1.9 `cfgDhcpScopeLeasetime`

DHCP Scope Lease Time

Specifies the lease time of addresses handed out to clients.

Examples:

- 12h
- 30m
- 180s

Applies to AP and STA.

Type	DisplayString
Range	1 - 255
Access	readwrite
OID	1.3.6.1.4.1.16177.1.400.1.1.13.3.1.6

8.1.7.3.1.10 `cfgDhcpScopeGateway`

DHCP Scope Default Gateway (DHCP Option 3)

Specifies the default gateway address handed out to clients. A value of 0.0.0.0 means not used. IPv4 only.

Applies to AP and STA.

<i>Type</i>	IpAddress
<i>Access</i>	readwrite
<i>OID</i>	1.3.6.1.4.1.16177.1.400.1.1.13.3.1.7

8.1.7.3.1.11 cfgDhcpScopeDnsServer1

DHCP Scope Primary DNS Server (DHCP Option 6)

Specifies the primary DNS server address handed out to clients. A value of 0.0.0.0 means not used. IPv4 only.

Applies to AP and STA.

<i>Type</i>	IpAddress
<i>Access</i>	readwrite
<i>OID</i>	1.3.6.1.4.1.16177.1.400.1.1.13.3.1.8

8.1.7.3.1.12 cfgDhcpScopeDnsServer2

DHCP Scope Secondary DNS Server (DHCP Option 6)

Specifies the secondary DNS server address handed out to clients. If the primary DNS server is not configured, this entry will also be ignored. A value of 0.0.0.0 means not used. IPv4 only.

Applies to AP and STA.

<i>Type</i>	IpAddress
<i>Access</i>	readwrite
<i>OID</i>	1.3.6.1.4.1.16177.1.400.1.1.13.3.1.9

8.1.7.4 cfgDhcpDomainOverrideTable

DNS Server Host Override Table

<i>Range</i>	0 - 255
<i>OID</i>	1.3.6.1.4.1.16177.1.400.1.1.13.4

8.1.7.4.1 cfgDhcpDomainOverrideTableEntry

DNS Server Domain Override Table

<i>OID</i>	1.3.6.1.4.1.16177.1.400.1.1.13.4.1
------------	------------------------------------

8.1.7.4.1.1 cfgDhcpDmnOvrIndex

Table Entry Index

<i>Range</i>	0 - 255
<i>Access</i>	noaccess
<i>OID</i>	1.3.6.1.4.1.16177.1.400.1.1.13.4.1.1

8.1.7.4.1.2 cfgDhcpDmnOvrId

Domain Override Entry ID

This is used in conjunction with `cfgDhcpDnsmasqDnsDomainOverrideParameter`.

All entries set to the value referenced are used by the DNS server.

<i>Range</i>	0 - 255
<i>Access</i>	readwrite
<i>OID</i>	1.3.6.1.4.1.16177.1.400.1.1.13.4.1.2

8.1.7.4.1.3 cfgDhcpDmnOvrEnabled

Domain Override Entry Disabled or Enabled

<i>Enumeration</i>	disabled (0), enabled (1)
<i>Access</i>	readwrite
<i>OID</i>	1.3.6.1.4.1.16177.1.400.1.1.13.4.1.3

8.1.7.4.1.4 cfgDhcpDmnOvrDomain

Domain Override Entry Domain

Domain to match. Queries for the matched domain are sent to the server specified in `cfgDhcpHstOvrServer`. This entry is considered disabled when set to `none`.

All subdomains of a specified domain are included. Multiple servers may be specified for the same domain. These servers are queried in the order defined in `cfgDhcpDnsmasqDnsResolveOrder`.

When set to *, this server is queried for all domains, and the servers specified in `cfgSysNameserverTable` are ignored. Multiple servers may be specified for the same domain. These servers are queried in the order defined in `cfgDhcpDnsmasqDnsResolveOrder`.

Examples:

- none
- example.com
- mydomain.net
- *

To exclude a subdomain from another override, set the corresponding `cfgDhcpDmnOvrServer` to #.

Example:

- specify example.com via 192.168.1.20
- specify blocked.example.com via #
- example.com is queried via 192.168.1.20
- subdomain.example.com is queried via 192.168.1.20
- blocked.example.com is queried via `cfgSysNameserverTable`

<i>Type</i>	DisplayString
<i>Range</i>	1 - 255
<i>Access</i>	readwrite
<i>OID</i>	1.3.6.1.4.1.16177.1.400.1.1.13.4.1.4

8.1.7.4.1.5 cfgDhcpDmnOvrServer

Domain Override Entry Server

The server which is queried for the domain and its subdomains specified in `cfgDhcpDmnOvrDomain`.

The format of the server is: IPv4-Address [#Destination-Port] [@Source-Address | @Source-Interface] [

- The IPv4-Address is the destination to which requests are sent.
- The Destination-Port specifies on which port the server is running. When omitted, the default port 53 is used.
- The Source-Address specifies from which address the requests are sent. Instead of a Source-Address, a Source-Interface may be specified. This is used in combination with dynamic addresses, for example CARP. When omitted, the routing table provides the source to be used.
- The Source-Port may be used to specify from which port requests are sent. When omitted, a random port is used.

This entry is considered disabled when set to none.

Set to # to exempt a subdomain, (see `cfgDhcpDmnOvrDomain`).

Examples:

- none
- #
- 192.168.1.20
- 192.168.1.20#10053
- 192.168.1.20@192.168.1.2
- 192.168.1.20#10053@192.168.1.2
- 192.168.1.20@192.168.1.2#12345
- 192.168.1.20#10053@192.168.1.2#12345
- 192.168.1.20@br0.vlan0
- 192.168.1.20@br0.vlan0#12345
- 192.168.1.20#10053@br0.vlan0#12345

<i>Type</i>	DisplayString
<i>Range</i>	1 - 255
<i>Access</i>	readwrite
<i>OID</i>	1.3.6.1.4.1.16177.1.400.1.1.13.4.1.5

8.1.7.5 `cfgDhcpHostOverrideTable`

DNS Server Host Override Table

<i>Range</i>	0 - 255
<i>OID</i>	1.3.6.1.4.1.16177.1.400.1.1.13.5

8.1.7.5.1 `cfgDhcpHostOverrideTableEntry`

DNS Server Host Override Table

<i>OID</i>	1.3.6.1.4.1.16177.1.400.1.1.13.5.1
------------	------------------------------------

8.1.7.5.1.1 `cfgDhcpHstOvrIndex`

Table Entry Index

<i>Range</i>	0 - 255
<i>Access</i>	noaccess
<i>OID</i>	1.3.6.1.4.1.16177.1.400.1.1.13.5.1.1

8.1.7.5.1.2 cfgDhcpHstOvrId

Host Override Entry ID

This is used in conjunction with `cfgDhcpDnsmasqDnsHostOverrideParameter`.

All entries set to the value referenced are used by the DNS server.

<i>Range</i>	0 - 255
<i>Access</i>	readwrite
<i>OID</i>	1.3.6.1.4.1.16177.1.400.1.1.13.5.1.2

8.1.7.5.1.3 cfgDhcpHstOvrEnabled

Host Override Entry Disabled or Enabled

<i>Enumeration</i>	disabled (0), enabled (1)
<i>Access</i>	readwrite
<i>OID</i>	1.3.6.1.4.1.16177.1.400.1.1.13.5.1.3

8.1.7.5.1.4 cfgDhcpHstOvrHost

Host Override Entry Host

FQDN or name to match. Is considered disabled when set to `none`.

Examples:

- none
- host.example.com
- somenamewithouttld

<i>Type</i>	DisplayString
<i>Range</i>	1 - 255
<i>Access</i>	readwrite
<i>OID</i>	1.3.6.1.4.1.16177.1.400.1.1.13.5.1.4

8.1.7.5.1.5 cfgDhcpHstOvrAddress

Host Override Entry Address

IP Address to return.

This is the address which is returned when a query is received for the FQDN or name configured in `cfgDhcpHstOvrDomain`.

Type	DisplayString
Range	1 - 255
Access	readwrite
OID	1.3.6.1.4.1.16177.1.400.1.1.13.5.1.5

8.1.8 cfgNtp

8.1.8.1 cfgNtpEnabled

Synchronize the system time with given server.

Applies to AP and STA.

Enumeration	disabled (0), enabled (1)
Access	readwrite
OID	1.3.6.1.4.1.16177.1.400.1.1.14.1

8.1.8.2 cfgNtpServer1

NTP server 1

If the IP is set to 0.0.0.0 the NTP client will only listen to broadcast packages.

Applies to AP and STA.

Type	IpAddress
Access	readwrite
OID	1.3.6.1.4.1.16177.1.400.1.1.14.2

8.1.8.3 cfgNtpServer2

NTP server 2

Used as fallback if server 1 cannot be reached.

Applies to AP and STA.

Type	IpAddress
Access	readwrite
OID	1.3.6.1.4.1.16177.1.400.1.1.14.3

8.1.9 cfgHttp

8.1.9.1 cfgHttpUser

Web administrator username.

Applies to AP and STA.

<i>Type</i>	DisplayString
<i>Range</i>	1 - 255
<i>Access</i>	readonly
<i>OID</i>	1.3.6.1.4.1.16177.1.400.1.1.15.1

8.1.9.2 cfgHttpPassword

Web administrator password.

Applies to AP and STA.

<i>Type</i>	DisplayString
<i>Range</i>	5 - 126
<i>Access</i>	readwrite
<i>OID</i>	1.3.6.1.4.1.16177.1.400.1.1.15.2

8.1.9.3 cfgHttpEnabled

Configure if the webserver shall be started.

When disabling the webserver, make sure you still have another way to access the device, e.g by CLI or via SNMP.

Applies to AP and STA.

<i>Enumeration</i>	disabled (0), enabled (1)
<i>Access</i>	readwrite
<i>OID</i>	1.3.6.1.4.1.16177.1.400.1.1.15.3

8.1.9.4 cfgHttpRedirectEnabled

If enabled, all access to `cfgHttpHttpAddress` shall be redirected to `cfgHttpHttpsAddress`. This does not disable http.

Applies to AP and STA.

<i>Enumeration</i>	disabled (0), enabled (1)
<i>Access</i>	readwrite
<i>OID</i>	1.3.6.1.4.1.16177.1.400.1.1.15.4

8.1.9.5 cfgHttpHttpAddress

Address to which the http server binds.

The default is '0.0.0.0:80'. Multiple space and/or comma separated tuples are allowed.

Examples:

- 192.168.1.20:80
- 192.168.1.20:80, 192.168.2.20:8080, 172.16.32.32:10080
- 192.168.1.20:80 192.168.2.20:8080 172.16.32.32:10080

Applies to AP and STA.

<i>Type</i>	DisplayString
<i>Range</i>	1 - 255
<i>Access</i>	readwrite
<i>OID</i>	1.3.6.1.4.1.16177.1.400.1.1.15.5

8.1.9.6 cfgHttpHttpsAddress

Address to which the https server binds.

The default is '0.0.0.0:443'. Multiple space and/or comma separated tuples are allowed.

Examples:

- 192.168.1.20:443
- 192.168.1.20:443, 192.168.2.20:8443, 172.16.32.32:10443
- 192.168.1.20:443 192.168.2.20:8443 172.16.32.32:10443

Applies to AP and STA.

<i>Type</i>	DisplayString
<i>Range</i>	1 - 255
<i>Access</i>	readwrite
<i>OID</i>	1.3.6.1.4.1.16177.1.400.1.1.15.6

8.1.10 cfgLldp

8.1.10.1 cfgLldpEnabled

Enable LLDP.

Applies to AP and STA.

<i>Enumeration</i>	disabled (0), enabled (1)
<i>Access</i>	readwrite
<i>OID</i>	1.3.6.1.4.1.16177.1.400.1.1.16.1

8.1.10.2 cfgLldpDescription

LLDP Description

Applies to AP and STA.

<i>Type</i>	DisplayString
<i>Range</i>	0 - 255
<i>Access</i>	readwrite
<i>OID</i>	1.3.6.1.4.1.16177.1.400.1.1.16.2

8.1.11 cfgMdns

8.1.11.1 cfgMdnsEnabled

Disable or Enable mDNS

<i>Enumeration</i>	disabled (0), enabled (1)
<i>Access</i>	readwrite
<i>OID</i>	1.3.6.1.4.1.16177.1.400.1.1.17.1

8.1.11.2 cfgMdnsNetwork

mDNS Aware Network Interfaces

Multiple interfaces may be specified as a space and/or comma separated list.

Examples: * br0.vlan0 * br0.vlan0, br0.vlan7, br0.vlan99 * br0.vlan0 br0.vlan66

Type	DisplayString
Range	1 - 255
Access	readwrite
OID	1.3.6.1.4.1.16177.1.400.1.1.17.2

8.1.12 cfgNetwork

8.1.12.1 cfgNetEthernetTable

Ethernet Network Interfaces

Range	0 - 9
OID	1.3.6.1.4.1.16177.1.400.1.1.2.1

8.1.12.1.1 cfgNetEthernetTableEntry

Ethernet Network Interfaces

OID	1.3.6.1.4.1.16177.1.400.1.1.2.1.1
-----	-----------------------------------

8.1.12.1.1.1 cfgNetEthIndex

Table Entry Index

Range	0 - 9
Access	noaccess
OID	1.3.6.1.4.1.16177.1.400.1.1.2.1.1.1

8.1.12.1.1.2 cfgNetEthTrunk

Eth Trunk

This entry is active when cfgNetEthVlanMode is set to **trunk(0)** or **nativeuntagged(3)**.

It specifies which 802.1q VLANs are accepted ingress and egress on the respective port. All unspecified VLANs are dropped. Set this entry to -1 to allow all VLANs. Untagged traffic is considered as VLAN 0.

The format of this entry is a space or comma separated list. To describe ranges the character '-' can be used.

Examples:

- '0,12,24,69'
- '7 56 127'
- '0, 84, 99, 2000'
- '0, 12-17, 3000-4000'
- '0-99 101-199 201-299, 301-4094'

Applies to AP and STA.

Type	DisplayString
Range	1 - 255
Access	readwrite
OID	1.3.6.1.4.1.16177.1.400.1.1.2.1.1.10

8.1.12.1.1.3 cfgNetEthTag

Eth Tag

This entry is active when `cfgNetEthVlanMode` is set to **access(1)** or **nativeuntagged(3)**. It specifies which 802.1q VLAN should be used for untagged ingress and egress traffic. Set this entry to -1 to disable it.

Applies to AP and STA.

Range	-1 - 4094
Access	readwrite
OID	1.3.6.1.4.1.16177.1.400.1.1.2.1.1.11

8.1.12.1.1.4 cfgNetEthVlanMode

Eth VLAN Mode

- **trunk(0):** A trunk port carries packets on one or more specified VLANs specified in the `cfgNetEthTrunk` entry. A packet that ingresses on a trunk port is in the VLAN specified in its 802.1q header, or VLAN 0 if the packet has no 802.1q header (untagged frame). A packet that egresses through a trunk port will have an 802.1q header if it has a nonzero VLAN ID. Frames

egressing on VLAN 0 have their tag stripped (egress untagged). Any packet that ingresses on a trunk port tagged with a VLAN that the port does not trunk is dropped.

- **access(1):** An access port carries packets on exactly one VLAN specified in `cfgNetEthTag`. Packets egressing on an access port have no 802.1q header (egress untagged). Any packet with an 802.1q header with a nonzero VLAN ID that ingresses on an access port is dropped, regardless of whether the VLAN ID in the header is the access port's VLAN ID or not.
- **nativeuntagged(3):** A native-untagged port resembles a trunk port, with the exception that a packet without an 802.1q header (ingress untagged) is automatically in the native VLAN specified in `cfgNetEthTag`. Frames egressing in the native VLAN are automatically untagged (egress untagged).

Applies to AP and STA.

<i>Enumeration</i>	trunk (0), access (1), nativeuntagged (3)
<i>Access</i>	readwrite
<i>OID</i>	1.3.6.1.4.1.16177.1.400.1.1.2.1.1.12

8.1.12.1.1.5 cfgNetEthProtected

Eth Port Protection

This feature only applies to bridged interfaces.

The protected port feature allows bridged ports to be designated as protected. Traffic between protected ports is blocked. Protected ports can send traffic to unprotected ports. Unprotected ports can send traffic to any port.

<i>Enumeration</i>	disabled (0), enabled (1)
<i>Access</i>	readwrite
<i>OID</i>	1.3.6.1.4.1.16177.1.400.1.1.2.1.1.13

8.1.12.1.1.6 cfgNetEthLldpEnabled

When `cfgLldpEnabled` is enabled, this parameter controls if the interface takes part in LLDP operation.

Applies to AP and STA.

<i>Enumeration</i>	disabled (0), enabled (1)
<i>Access</i>	readwrite
<i>OID</i>	1.3.6.1.4.1.16177.1.400.1.1.2.1.1.15

8.1.12.1.1.7 cfgNetEthMtu

The MTU of the Ethernet Interface

The minimum value is 68. The maximum value is 9000. The default value is -1, which means that the system default value (usually 1500) will be used.

When this interface is part of a bridge, the here configured MTU affects the MTU of the bridge. The bridge will have the smallest MTU of all its bridge members.

Example:

- br0 contains the interfaces eth0, eth1 and wlan0
- eth0 is set to 1400
- eth1 is set to 1200
- wlan0 is set to -1 (default 1500)
- This will result in an MTU of 1200 for br0

Applies to AP and STA.

<i>Range</i>	-1 - 9000
<i>Access</i>	readwrite
<i>OID</i>	1.3.6.1.4.1.16177.1.400.1.1.2.1.1.16

8.1.12.1.1.8 cfgNetEthName

Name of the ethernet interface.

Applies to AP and STA.

<i>Type</i>	DisplayString
<i>Range</i>	1 - 255
<i>Access</i>	readonly
<i>OID</i>	1.3.6.1.4.1.16177.1.400.1.1.2.1.1.2

8.1.12.1.1.9 cfgNetEthEnabled

Ethernet interface disabled or enabled.

Applies to AP and STA.

<i>Enumeration</i>	disabled (0), enabled (1)
<i>Access</i>	readwrite
<i>OID</i>	1.3.6.1.4.1.16177.1.400.1.1.2.1.1.3

8.1.12.1.1.10 cfgNetEthBridge

Ethernet Interface Bridge Membership

If value ≥ 0 then interface is part of bridge.

- -1: none
- 0: br0
- 1: br1
- X: brX

Bridges with an index ≥ 100 are special bridges which forward link local traffic. This can be used for wireless links in 4addr mode which should act as a cable-replacement.

Note: Such a bridge may only contain 2 interfaces!

Example:

- wlan0 and eth0 in br100, with eth1 as management interface

Applies to AP and STA.

Type	Integer32
Access	readwrite
OID	1.3.6.1.4.1.16177.1.400.1.1.2.1.1.7

8.1.12.1.1.11 cfgNetEthAutoneg

Enable or Disable Auto Negotiation of the PHY

- **forced(0):** Forces the speed and duplex defined by `cfgNetEthSpeed`. Only 10Mbit and 100Mbit rates are allowed in forced mode. 1000Mbit requires the mode to be auto.
- **auto(1):** Advertises the supported auto negotiation defined by `cfgNetEthSpeed`.

Applies to AP and STA.

Enumeration	forced (0), auto (1)
Access	readwrite
OID	1.3.6.1.4.1.16177.1.400.1.1.2.1.1.8

8.1.12.1.1.12 cfgNetEthSpeed

Defines a bitmask containing the possible speed/duplex combinations.

- 0x01 (1) = 10Mbit/Half
- 0x02 (2) = 10Mbit/Full
- 0x04 (4) = 100Mbit/Half
- 0x08 (8) = 100Mbit/Full

- 0x20 (32) = 1000Mbit/Full

When `cfgNetEthSpeed` is **forced(0)** only a single bit may be active. Only 10Mbit and 100Mbit rates are allowed in forced mode. 1000Mbit requires the mode to be auto.

Examples:

- **1:** Force 10Mbit half duplex
- **8:** Force 100Mbit full duplex

When `cfgNetEthSpeed` is **auto(1)** multiple bits may be set which are used to advertise the supported speed/duplex. 1000Mbit/Half is not supported.

Examples:

- **12:** Advertise 100 Mbit, half/full duplex (4 + 8)
- **15:** Advertise 10/100 Mbit, half/full duplex (1 + 2 + 4 + 8)
- **32:** Advertise only 1000 Mbit full duplex
- **47:** Advertise all speeds (1 + 2 + 4 + 8 + 32)

Applies to AP and STA.

Type	Integer32
Access	readwrite
OID	1.3.6.1.4.1.16177.1.400.1.1.2.1.1.9

8.1.12.2 cfgNetWwanTable

Cellular Network Interfaces

Applies to cellular products only.

Range	0 - 0
OID	1.3.6.1.4.1.16177.1.400.1.1.2.11

8.1.12.2.1 cfgNetWwanTableEntry

Cellular Network Interface

Applies to cellular products only.

OID	1.3.6.1.4.1.16177.1.400.1.1.2.11.1
-----	------------------------------------

8.1.12.2.1.1 cfgNetWwanIndex

Table Entry Index

Applies to cellular products only.

<i>Range</i>	0 - 0
<i>Access</i>	noaccess
<i>OID</i>	1.3.6.1.4.1.16177.1.400.1.1.2.11.1.1

8.1.12.2.1.2 cfgNetWwanMtu

The MTU of the Cellular Network Interface

The default value is -1, which means that the system default value (usually 1500) will be used. This value may be provided by the service provider and may change depending on the provider.

It is not recommended to set a value, unless you know what you do.

Applies to cellular products only.

<i>Range</i>	-1 - 65535
<i>Access</i>	readwrite
<i>OID</i>	1.3.6.1.4.1.16177.1.400.1.1.2.11.1.16

8.1.12.2.1.3 cfgNetWwanName

Name of the Cellular Network Interface

The name of cellular network interfaces at system level is typically derived from the term Wireless Wide Area Network (WWAN), e.g. wwan0.

Applies to cellular products only.

<i>Type</i>	DisplayString
<i>Range</i>	1 - 255
<i>Access</i>	readonly
<i>OID</i>	1.3.6.1.4.1.16177.1.400.1.1.2.11.1.2

8.1.12.2.1.4 cfgNetWwanEnabled

Cellular Network Interface Disabled or Enabled

Applies to cellular products only.

<i>Enumeration</i>	disabled (0), enabled (1)
<i>Access</i>	readwrite
<i>OID</i>	1.3.6.1.4.1.16177.1.400.1.1.2.11.1.3

8.1.12.3 cfgNetOpenvpnTable

OpenVPN Interface Table

This table is in a one-to-one relation with the `cfgVpnOpenvpnTable`, where both indices match.

<i>Range</i>	0 - 15
<i>OID</i>	1.3.6.1.4.1.16177.1.400.1.1.2.12

8.1.12.3.1 cfgNetOpenvpnTableEntry

OpenVPN Interface Entry

<i>OID</i>	1.3.6.1.4.1.16177.1.400.1.1.2.12.1
------------	------------------------------------

8.1.12.3.1.1 cfgNetOpenvpnIndex

Table Entry Index

<i>Range</i>	0 - 15
<i>Access</i>	noaccess
<i>OID</i>	1.3.6.1.4.1.16177.1.400.1.1.2.12.1.1

8.1.12.3.1.2 cfgNetOpenvpnTrunk

OpenVPN Interface Trunk

This entry is active when `cfgNetOpenvpnVlanMode` is set to **trunk(0)** or **nativeuntagged(3)**.

It specifies which 802.1q VLANs are accepted ingress and egress on the respective port. All unspecified VLANs are dropped. Set this entry to -1 to allow all VLANs. Untagged traffic is considered as VLAN 0.

The format of this entry is a space or comma separated list. To describe ranges the character '-' can be used.

Examples:

- '0,12,24,69'

- '7 56 127'
- '0, 84, 99, 2000'
- '0, 12-17, 3000-4000'
- '0-99 101-199 201-299, 301-4094'

Applies to AP and STA.

Type	DisplayString
Range	1 - 255
Access	readwrite
OID	1.3.6.1.4.1.16177.1.400.1.1.2.12.1.10

8.1.12.3.1.3 cfgNetOpenvpnTag

OpenVPN Interface Tag

This entry is active when `cfgNetOpenvpnVlanMode` is set to: **access(1)** or **native-untagged(3)**. It specifies which 802.1q VLAN should be used for untagged ingress and egress traffic. Set this entry to -1 to disable it.

Applies to AP and STA.

Range	-1 - 4094
Access	readwrite
OID	1.3.6.1.4.1.16177.1.400.1.1.2.12.1.11

8.1.12.3.1.4 cfgNetOpenvpnVlanMode

OpenVPN Interface VLAN Mode

This entry specifies how the port should behave:

- **trunk(0)**: A trunk port carries packets on one or more specified VLANs specified in the `cfgNetOpenvpnTrunk` entry. A packet that ingresses on a trunk port is in the VLAN specified in its 802.1q header, or VLAN 0 if the packet has no 802.1q header (untagged frame). A packet that egresses through a trunk port will have an 802.1q header if it has a nonzero VLAN id. Frames egressing on VLAN 0 have their tag stripped (egress untagged). Any packet that ingresses on a trunk port tagged with a VLAN that the port does not trunk is dropped.
- **access(1)**: An access port carries packets on exactly one VLAN specified in the `cfgNetOpenvpnTag`. Packets egressing on an access port have no 802.1q header (egress untagged). Any packet with an 802.1q header with a nonzero VLAN id that ingresses on an access port is dropped, regardless of whether the VLAN id in the header is the access port's VLAN id.
- **nativeuntagged(3)**: A native-untagged port resembles a trunk port, with the exception that a packet without an 802.1q header (ingress untagged) is automatically in the native-vlan specified

in `cfgNetOpenvpnTag`. Frames egressing in the native-vlan are automatically untagged (egress untagged).

Applies to AP and STA.

<i>Enumeration</i>	trunk (0), access (1), nativeuntagged (3)
<i>Access</i>	readwrite
<i>OID</i>	1.3.6.1.4.1.16177.1.400.1.1.2.12.1.12

8.1.12.3.1.5 `cfgNetOpenvpnProtected`

OpenVPN Port Protection

This feature only applies to bridged interfaces.

The protected port feature allows bridged ports to be designated as protected. Traffic between protected ports is blocked. Protected ports can send traffic to unprotected ports. Unprotected ports can send traffic to any port.

<i>Enumeration</i>	disabled (0), enabled (1)
<i>Access</i>	readwrite
<i>OID</i>	1.3.6.1.4.1.16177.1.400.1.1.2.12.1.13

8.1.12.3.1.6 `cfgNetOpenvpnLldpEnabled`

LLDP Port Operation

When `cfgLldpEnabled` is enabled, this parameter controls if the interface takes part in LLDP operation.

Applies to AP and STA.

<i>Enumeration</i>	disabled (0), enabled (1)
<i>Access</i>	readwrite
<i>OID</i>	1.3.6.1.4.1.16177.1.400.1.1.2.12.1.15

8.1.12.3.1.7 `cfgNetOpenvpnMtu`

The MTU of the OpenVPN Interface

The minimum allowed valid value is 68. The maximum allowed valid value is 65535. The default value is -1, which does not change what is set by the system (usually 1500).

Applies to AP and STA.

<i>Range</i>	-1 - 65535
<i>Access</i>	readwrite
<i>OID</i>	1.3.6.1.4.1.16177.1.400.1.1.2.12.1.16

8.1.12.3.1.8 cfgNetOpenvpnName

Name of the OpenVPN Interface

Applies to AP and STA.

<i>Type</i>	DisplayString
<i>Range</i>	1 - 255
<i>Access</i>	readonly
<i>OID</i>	1.3.6.1.4.1.16177.1.400.1.1.2.12.1.2

8.1.12.3.1.9 cfgNetOpenvpnEnabled

OpenVPN Interface Disabled or Enabled

Applies to AP and STA.

<i>Enumeration</i>	disabled (0), enabled (1)
<i>Access</i>	readwrite
<i>OID</i>	1.3.6.1.4.1.16177.1.400.1.1.2.12.1.3

8.1.12.3.1.10 cfgNetOpenvpnBridge

OpenVPN Interface Bridge Membership

If value ≥ 0 then interface is part of bridge.

- -1: none
- 0: br0
- 1: br1
- X: brX

Bridges with an index ≥ 100 are special bridges which forward link local traffic. This can be used for tunnels which act as a cable-replacement.

Note: Such a bridge may only contain 2 interfaces!

Example:

- ovpn0 and eth0 in br100, with eth1 as management interface

Note: Interfaces may only be configured in a bridge when `cfgVpnOpenvpnDevType` is set to 1 (tap).

Applies to AP and STA.

<i>Type</i>	Integer32
<i>Access</i>	readwrite
<i>OID</i>	1.3.6.1.4.1.16177.1.400.1.1.2.12.1.7

8.1.12.4 cfgNetIpsecTable

IPsec Network Interfaces.

<i>Range</i>	0 - 15
<i>OID</i>	1.3.6.1.4.1.16177.1.400.1.1.2.13

8.1.12.4.1 cfgNetIpsecTableEntry

IPsec Network Interfaces

<i>OID</i>	1.3.6.1.4.1.16177.1.400.1.1.2.13.1
------------	------------------------------------

8.1.12.4.1.1 cfgNetIpsecIndex

Table Entry Index

<i>Range</i>	0 - 15
<i>Access</i>	noaccess
<i>OID</i>	1.3.6.1.4.1.16177.1.400.1.1.2.13.1.1

8.1.12.4.1.2 cfgNetIpsecMtu

IPsec Tunnel MTU

When VTI is enabled (see `cfgVpnIpsecGlblVirtualTunnelInterface`), this allows to manually set the MTU of the IPsec tunnel.

Applies to AP and STA.

<i>Range</i>	-1 - 65515
<i>Access</i>	readwrite
<i>OID</i>	1.3.6.1.4.1.16177.1.400.1.1.2.13.1.16

8.1.12.4.1.3 cfgNetIpssecName

Name of the IPsec interface

Applies to AP and STA.

Type	DisplayString
Range	1 - 255
Access	readonly
OID	1.3.6.1.4.1.16177.1.400.1.1.2.13.1.2

8.1.12.4.1.4 cfgNetIpssecEnabled

IPsec Interface Disabled or Enabled

Applies to AP and STA.

Enumeration	disabled (0), enabled (1)
Access	readwrite
OID	1.3.6.1.4.1.16177.1.400.1.1.2.13.1.3

8.1.12.5 cfgNetFlowControllerTable

FlowController Interfaces.

Range	0 - 15
OID	1.3.6.1.4.1.16177.1.400.1.1.2.14

8.1.12.5.1 cfgNetFlowControllerTableEntry

FlowController Network Interfaces.

OID	1.3.6.1.4.1.16177.1.400.1.1.2.14.1
-----	------------------------------------

8.1.12.5.1.1 cfgNetFcIndex

Table Entry Index

Range	0 - 15
Access	noaccess
OID	1.3.6.1.4.1.16177.1.400.1.1.2.14.1.1

8.1.12.5.1.2 cfgNetFcTrunk

TECHPREVIEW: Flow Controller Trunk

This entry is active when `cfgNetFcVlanMode` is set to **trunk(0)** or **nativeuntagged(3)**.

It specifies which 802.1q VLANs are accepted ingress and egress on the respective port. All unspecified VLANs are dropped. Set this entry to -1 to allow all VLANs. Untagged traffic is considered as VLAN 0.

The format of this entry is a space or comma separated list. To describe ranges the character '-' can be used.

Examples:

- '0,12,24,69'
- '7 56 127'
- '0, 84, 99, 2000'
- '0, 12-17, 3000-4000'
- '0-99 101-199 201-299, 301-4094'

Applies to AP and STA.

Type	DisplayString
Range	1 - 255
Access	readwrite
OID	1.3.6.1.4.1.16177.1.400.1.1.2.14.1.10

8.1.12.5.1.3 cfgNetFcTag

TECHPREVIEW: Flow Controller Tag

This entry is active when `cfgNetFcVlanMode` is set to: **access(1)** or **native-untagged(3)**.

It specifies which 802.1q VLAN should be used for untagged ingress and egress traffic. Set this entry to -1 to disable it.

Applies to AP and STA.

Range	-1 - 4094
Access	readwrite
OID	1.3.6.1.4.1.16177.1.400.1.1.2.14.1.11

8.1.12.5.1.4 cfgNetFcVlanMode

TECHPREVIEW: Flow Controller VLAN Mode

- **trunk(0):** A trunk port carries packets on one or more specified VLANs specified in the `cfgNetFcTrunk` entry. A packet that ingresses on a trunk port is in the VLAN specified in its 802.1q header, or VLAN 0 if the packet has no 802.1q header (untagged frame). A packet that egresses through a trunk port will have an 802.1q header if it has a nonzero VLAN id. Frames egressing on VLAN 0 have their tag stripped (egress untagged). Any packet that ingresses on a trunk port tagged with a VLAN that the port does not trunk is dropped.
- **access(1):** An access port carries packets on exactly one VLAN specified in the `cfgNetFcTag`. Packets egressing on an access port have no 802.1q header (egress untagged). Any packet with an 802.1q header with a nonzero VLAN id that ingresses on an access port is dropped, regardless of whether the VLAN id in the header is the access port's VLAN id.
- **nativeuntagged(3):** A native-untagged port resembles a trunk port, with the exception that a packet without an 802.1q header (ingress untagged) is automatically in the native-vlan specified in `cfgNetFcTag`. Frames egressing in the native-vlan are automatically untagged (egress untagged).

Applies to AP and STA.

<i>Enumeration</i>	trunk (0), access (1), nativeuntagged (3)
<i>Access</i>	readwrite
<i>OID</i>	1.3.6.1.4.1.16177.1.400.1.1.2.14.1.12

8.1.12.5.1.5 `cfgNetFcProtected`

TECHPREVIEW: Flow Controller Port Protection

This feature only applies to bridged interfaces.

The protected port feature allows bridged ports to be designated as protected. Traffic between protected ports is blocked. Protected ports can send traffic to unprotected ports. Unprotected ports can send traffic to any port.

<i>Enumeration</i>	disabled (0), enabled (1)
<i>Access</i>	readwrite
<i>OID</i>	1.3.6.1.4.1.16177.1.400.1.1.2.14.1.13

8.1.12.5.1.6 `cfgNetFcLldpEnabled`

TECHPREVIEW: LLDP Disabled or Enabled

When `cfgLldpEnabled` is enabled, this parameter controls if the interface takes part in LLDP operation.

Applies to AP and STA.

<i>Enumeration</i>	disabled (0), enabled (1)
<i>Access</i>	readwrite
<i>OID</i>	1.3.6.1.4.1.16177.1.400.1.1.2.14.1.15

8.1.12.5.1.7 cfgNetFcFlowMode

TECHPREVIEW: Flow Controller Mode

- **normal(0):** Behaves like a normal bridge
- **fullduplex(1):** Transmits frames on the bridge-master, and receives frames on all other members of the bridge. The bridge-master is the first interface which is added to a bride. The order in which interfaces are added is ascending in interface number: eth, wlan, ovpn, fc.

Applies to AP and STA.

<i>Enumeration</i>	normal (0), fullduplex (1)
<i>Access</i>	readwrite
<i>OID</i>	1.3.6.1.4.1.16177.1.400.1.1.2.14.1.17

8.1.12.5.1.8 cfgNetFcName

TECHPREVIEW: Name of the Flow Controller Bridge

Applies to AP and STA.

<i>Type</i>	DisplayString
<i>Range</i>	1 - 255
<i>Access</i>	readonly
<i>OID</i>	1.3.6.1.4.1.16177.1.400.1.1.2.14.1.2

8.1.12.5.1.9 cfgNetFcEnabled

TECHPREVIEW: Flow Controller Bridge Disabled or Enabled

Applies to AP and STA.

<i>Enumeration</i>	disabled (0), enabled (1)
<i>Access</i>	readwrite
<i>OID</i>	1.3.6.1.4.1.16177.1.400.1.1.2.14.1.3

8.1.12.5.1.10 cfgNetFcBridge

TECHPREVIEW: Flow Controller Bridge Membership

If value ≥ 0 then interface is part of bridge.

- -1: none
- 0: br0
- 1: br1
- X: brX

Bridges with an index ≥ 100 are special bridges which forward link local traffic. This can be used for wireless links in 4addr mode which should act as a cable-replacement.

Note: Such a bridge may only contain 2 interfaces!

Example:

- wlan0 and eth0 in br100, with eth1 as management interface

Applies to AP and STA.

Type	Integer32
Access	readwrite
OID	1.3.6.1.4.1.16177.1.400.1.1.2.14.1.7

8.1.12.6 cfgNetTunnelEndPointTable

Tunnel Endpoint Interfaces

Range	0 - 15
OID	1.3.6.1.4.1.16177.1.400.1.1.2.15

8.1.12.6.1 cfgNetTunnelEndPointTableEntry

Tunnel Endpoint Interfaces

OID	1.3.6.1.4.1.16177.1.400.1.1.2.15.1
-----	------------------------------------

8.1.12.6.1.1 cfgNetTepIndex

Table Entry Index

Range	0 - 15
Access	noaccess
OID	1.3.6.1.4.1.16177.1.400.1.1.2.15.1.1

8.1.12.6.1.2 cfgNetTepTrunk

Tunnel Endpoint Trunk

This entry is active when `cfgNetTepVlanMode` is set to **trunk(0)** or **nativeuntagged(3)**.

It specifies which 802.1q VLANs are accepted ingress and egress on the respective port. All unspecified VLANs are dropped. Set this entry to -1 to allow all VLANs. Untagged traffic is considered as VLAN 0.

The format of this entry is a space or comma separated list. To describe ranges the character '-' can be used.

Examples:

- '0,12,24,69'
- '7 56 127'
- '0, 84, 99, 2000'
- '0, 12-17, 3000-4000'
- '0-99 101-199 201-299, 301-4094'

Applies to AP and STA.

Type	DisplayString
Range	1 - 255
Access	readwrite
OID	1.3.6.1.4.1.16177.1.400.1.1.2.15.1.10

8.1.12.6.1.3 cfgNetTepTag

Tunnel Endpoint Tag

This entry is active when `cfgNetTepVlanMode` is set to: **access(1):** or **nativeuntagged(3)**. It specifies which 802.1q VLAN should be used for untagged ingress and egress traffic. Set this entry to -1 to disable it.

Applies to AP and STA.

Range	-1 - 4094
Access	readwrite
OID	1.3.6.1.4.1.16177.1.400.1.1.2.15.1.11

8.1.12.6.1.4 cfgNetTepVlanMode

Tunnel Endpoint VLAN Mode

- **trunk(0):** A trunk port carries packets on one or more specified VLANs specified in the `cfgNetTepTrunk` entry. A packet that ingresses on a trunk port is in the VLAN specified in its 802.1q header, or VLAN 0 if the packet has no 802.1q header (untagged frame). A packet that egresses through a trunk port will have an 802.1q header if it has a nonzero VLAN id. Frames egressing on VLAN 0 have their tag stripped (egress untagged). Any packet that ingresses on a trunk port tagged with a VLAN that the port does not trunk is dropped.
- **access(1):** An access port carries packets on exactly one VLAN specified in the `cfgNetTepTag`. Packets egressing on an access port have no 802.1q header (egress untagged). Any packet with an 802.1q header with a nonzero VLAN id that ingresses on an access port is dropped, regardless of whether the VLAN id in the header is the access port's VLAN id.
- **nativeuntagged(3):** A native-untagged port resembles a trunk port, with the exception that a packet without an 802.1q header (ingress untagged) is automatically in the native-vlan specified in `cfgNetTepTag`. Frames egressing in the native-vlan are automatically untagged (egress untagged).

Applies to AP and STA.

<i>Enumeration</i>	trunk (0), access (1), nativeuntagged (3)
<i>Access</i>	readwrite
<i>OID</i>	1.3.6.1.4.1.16177.1.400.1.1.2.15.1.12

8.1.12.6.1.5 cfgNetTepProtected

Tunnel Endpoint Port Protection

This feature only applies to bridged interfaces.

The protected port feature allows bridged ports to be designated as protected. Traffic between protected ports is blocked. Protected ports can send traffic to unprotected ports. Unprotected ports can send traffic to any port.

<i>Enumeration</i>	disabled (0), enabled (1)
<i>Access</i>	readwrite
<i>OID</i>	1.3.6.1.4.1.16177.1.400.1.1.2.15.1.13

8.1.12.6.1.6 cfgNetTepLldpEnabled

Tunnel Endpoint LLDP Operation on Port

When `cfgLldpEnabled` is enabled, this parameter controls if the interface takes part in LLDP operation.

Applies to AP and STA.

<i>Enumeration</i>	disabled (0), enabled (1)
<i>Access</i>	readwrite
<i>OID</i>	1.3.6.1.4.1.16177.1.400.1.1.2.15.1.15

8.1.12.6.1.7 cfgNetTepMtu

MTU of the Tunnel Endpoint Interface

The minimum allowed valid value is 68. The maximum allowed valid value depends on the used tunnel type specified in `cfgVpnTepTunnelType`. The default value is -1, which doesn't change what is set by the system. The default value as well depends on the tunnel type specified in `cfgVpnTepTunnelType`.

Type	Usual Default	Maximum
gre	1476 (1500-20-4)	65504
gretap	1462 (1500-20-4-14)	65490

The outer IP header is 20 bytes. The GRE header is 4 bytes. For gretap there is an additional ethernet header of 14 bytes.

The default value is derived from the MTU of the interface over which the tunnel will transmit frames based on the routing table lookup based on the content of `cfgVpnTepDestination`.

When this interface is part of a bridge, the here configured MTU affects the MTU of the bridge. The bridge will have the smallest MTU of all its bridge members.

Example:

- br0 contains the interfaces eth0, eth1 and tep0
- eth0 is set to -1 (default 1500)
- eth1 is set to 9000
- tep0 is set to 1300
- This will result in an MTU of 1300 for br0

Applies to AP and STA.

<i>Range</i>	-1 - 65504
<i>Access</i>	readwrite
<i>OID</i>	1.3.6.1.4.1.16177.1.400.1.1.2.15.1.16

8.1.12.6.1.8 cfgNetTepName

Name of the Tunnel Endpoint

Applies to AP and STA.

Type	DisplayString
Range	1 - 255
Access	readonly
OID	1.3.6.1.4.1.16177.1.400.1.1.2.15.1.2

8.1.12.6.1.9 cfgNetTepEnabled

Tunnel Endpoint Disabled or Enabled

Applies to AP and STA.

Enumeration	disabled (0), enabled (1)
Access	readwrite
OID	1.3.6.1.4.1.16177.1.400.1.1.2.15.1.3

8.1.12.6.1.10 cfgNetTepBridge

Tunnel Endpoint Interface Bridge Membership

If value ≥ 0 then interface is part of bridge.

- -1: none
- 0: br0
- 1: br1
- X: brX

Bridges with an index ≥ 100 are special bridges which forward link local traffic. This can be used for wireless links in 4addr mode which should act as a cable-replacement.

Note: Such a bridge may only contain 2 interfaces!

Example:

- tep0 and wlan0 in br100, with eth0/1 as management interface

Not all types of tunnels configured in `cfgVpnTepTunnelType` are able to be bridged. Currently supported types to be bridged are:

- **gretap(1):**

Applies to AP and STA.

Type	Integer32
Access	readwrite
OID	1.3.6.1.4.1.16177.1.400.1.1.2.15.1.7

8.1.12.7 cfgNetWireguardTable

Wireguard Interfaces

Range	0 - 15
OID	1.3.6.1.4.1.16177.1.400.1.1.2.16

8.1.12.7.1 cfgNetWireguardTableEntry

Wireguard Interfaces

OID	1.3.6.1.4.1.16177.1.400.1.1.2.16.1
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8.1.12.7.1.1 cfgNetWgIndex

Table Entry Index

Range	0 - 15
Access	noaccess
OID	1.3.6.1.4.1.16177.1.400.1.1.2.16.1.1

8.1.12.7.1.2 cfgNetWgMtu

The MTU of the Wireguard Interface

The default value is -1, which does not change what is set by the system (usually 1420).

Range	-1 - 65504
Access	readwrite
OID	1.3.6.1.4.1.16177.1.400.1.1.2.16.1.16

8.1.12.7.1.3 cfgNetWgName

Name of the Wireguard Interface

Type	DisplayString
Range	1 - 255
Access	readonly
OID	1.3.6.1.4.1.16177.1.400.1.1.2.16.1.2

8.1.12.7.1.4 cfgNetWgEnabled

Wireguard Interface Disabled or Enabled

<i>Enumeration</i>	disabled (0), enabled (1)
<i>Access</i>	readwrite
<i>OID</i>	1.3.6.1.4.1.16177.1.400.1.1.2.16.1.3

8.1.12.8 cfgNetWlanTable

WLAN Network Interfaces

<i>Range</i>	0 - 63
<i>OID</i>	1.3.6.1.4.1.16177.1.400.1.1.2.2

8.1.12.8.1 cfgNetWlanTableEntry

WLAN Network Interfaces

<i>OID</i>	1.3.6.1.4.1.16177.1.400.1.1.2.2.1
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8.1.12.8.1.1 cfgNetWlanIndex

Table Entry Index

<i>Range</i>	0 - 63
<i>Access</i>	noaccess
<i>OID</i>	1.3.6.1.4.1.16177.1.400.1.1.2.2.1.1

8.1.12.8.1.2 cfgNetWlanTrunk

Wlan Trunk

This entry is active when `cfgNetWlanVlanMode` is set to **trunk(0)** or **nativeuntagged(3)**.

It specifies which 802.1q VLANs are accepted ingress and egress on the respective port. All unspecified VLANs are dropped. Set this entry to -1 to allow all VLANs. Untagged traffic is considered as VLAN 0.

The format of this entry is a space or comma separated list. To describe ranges the character '-' can be used.

Examples:

- '0,12,24,69'
- '7 56 127'
- '0, 84, 99, 2000'
- '0, 12-17, 3000-4000'
- '0-99 101-199 201-299, 301-4094'

Applies to AP and STA.

Type	DisplayString
Range	1 - 255
Access	readwrite
OID	1.3.6.1.4.1.16177.1.400.1.1.2.2.1.10

8.1.12.8.1.3 cfgNetWlanTag

Wlan Tag

This entry is active when `cfgNetWlanVlanMode` is set to **access(1)** or **nativeuntagged(3)**. It specifies which 802.1q VLAN should be used for untagged ingress and egress traffic. Set this entry to -1 to disable it.

Applies to AP and STA.

Range	-1 - 4094
Access	readwrite
OID	1.3.6.1.4.1.16177.1.400.1.1.2.2.1.11

8.1.12.8.1.4 cfgNetWlanVlanMode

WLAN VLAN Mode

- **trunk(0):** A trunk port carries packets on one or more specified VLANs specified in the `cfgNetWlanTrunk` entry. A packet that ingresses on a trunk port is in the VLAN specified in its 802.1q header, or VLAN 0 if the packet has no 802.1q header (untagged frame). A packet that egresses through a trunk port will have an 802.1q header if it has a nonzero VLAN ID. Frames egressing on VLAN 0 have their tag stripped (egress untagged). Any packet that ingresses on a trunk port tagged with a VLAN that the port does not trunk is dropped.
- **access(1):** An access port carries packets on exactly one VLAN specified in the `cfgNetWlanTag`. Packets egressing on an access port have no 802.1q header (egress untagged). Any packet with an 802.1q header with a nonzero VLAN ID that ingresses on an access port is dropped, regardless of whether the VLAN ID in the header is the access port's VLAN ID.
- **nativeuntagged(3):** A native-untagged port resembles a trunk port, with the exception that a packet without an 802.1q header (ingress untagged) is automatically in the native VLAN specified in `cfgNetWlanTag`. Frames egressing in the native VLAN are automatically untagged (egress untagged).

Applies to AP and STA.

<i>Enumeration</i>	trunk (0), access (1), nativeuntagged (3)
<i>Access</i>	readwrite
<i>OID</i>	1.3.6.1.4.1.16177.1.400.1.1.2.2.1.12

8.1.12.8.1.5 cfgNetWlanProtected

WLAN Port Protection

This feature only applies to bridged interfaces.

The protected port feature allows bridged ports to be designated as protected. Traffic between protected ports is blocked. Protected ports can send traffic to unprotected ports. Unprotected ports can send traffic to any port.

<i>Enumeration</i>	disabled (0), enabled (1)
<i>Access</i>	readwrite
<i>OID</i>	1.3.6.1.4.1.16177.1.400.1.1.2.2.1.13

8.1.12.8.1.6 cfgNetWlanLldpEnabled

When `cfgLldpEnabled` **is enabled, this parameter controls if** the interface takes part in LLDP operation.

Applies to AP and STA.

<i>Enumeration</i>	disabled (0), enabled (1)
<i>Access</i>	readwrite
<i>OID</i>	1.3.6.1.4.1.16177.1.400.1.1.2.2.1.15

8.1.12.8.1.7 cfgNetWlanMtu

The MTU of the Wireless Interface

The minimum value is 256. The maximum value is 2304. The default value is -1, which means that the system default value (usually 1500) will be used.

When this interface is part of a bridge, the here configured MTU affects the MTU of the bridge. The bridge will have the smallest MTU of all its bridge members.

Example:

- br0 contains the interfaces eth0, eth1 and wlan0
- eth0 is set to -1 (default 1500)

- eth1 is set to 9000
- wlan0 is set to 2000
- This will result in an MTU of 1500 for br0

Applies to AP and STA.

<i>Range</i>	-1 - 2304
<i>Access</i>	readwrite
<i>OID</i>	1.3.6.1.4.1.16177.1.400.1.1.2.2.1.16

8.1.12.8.1.8 cfgNetWlanName

Name of the wireless interface

Applies to AP and STA.

<i>Type</i>	DisplayString
<i>Range</i>	1 - 255
<i>Access</i>	readonly
<i>OID</i>	1.3.6.1.4.1.16177.1.400.1.1.2.2.1.2

8.1.12.8.1.9 cfgNetWlanEnabled

Wireless interface disabled or enabled

Applies to AP and STA.

<i>Enumeration</i>	disabled (0), enabled (1)
<i>Access</i>	readwrite
<i>OID</i>	1.3.6.1.4.1.16177.1.400.1.1.2.2.1.3

8.1.12.8.1.10 cfgNetWlanBridge

Wireless Interface Bridge Membership

If value ≥ 0 then interface is part of bridge.

- -1: none
- 0: br0
- 1: br1
- X: brX

Bridges with an index ≥ 100 are special bridges which forward link local traffic. This can be used for wireless links in 4addr mode which should act as a cable-replacement.

Note: Such a bridge may only contain 2 interfaces!

Example:

- wlan0 and eth0 in br100, with eth1 as management interface

Applies to AP and STA.

Type	Integer32
Access	readwrite
OID	1.3.6.1.4.1.16177.1.400.1.1.2.2.1.7

8.1.12.9 cfgNetVlanTable

VLAN Network Interfaces

Range	0 - 127
OID	1.3.6.1.4.1.16177.1.400.1.1.2.3

8.1.12.9.1 cfgNetVlanTableEntry

VLAN Network Interfaces

OID	1.3.6.1.4.1.16177.1.400.1.1.2.3.1
-----	-----------------------------------

8.1.12.9.1.1 cfgNetVlanIndex

Table Entry Index

Range	0 - 127
Access	noaccess
OID	1.3.6.1.4.1.16177.1.400.1.1.2.3.1.1

8.1.12.9.1.2 cfgNetVlanProtected

VLAN Port Protection

This feature only applies to bridged interfaces.

The protected port feature allows bridged ports to be designated as protected. Traffic between protected ports is blocked. Protected ports can send traffic to unprotected ports. Unprotected ports can send traffic to any port.

<i>Enumeration</i>	disabled (0), enabled (1)
<i>Access</i>	readwrite
<i>OID</i>	1.3.6.1.4.1.16177.1.400.1.1.2.3.1.13

8.1.12.9.1.3 cfgNetVlanMtu

The MTU of the VLAN Interface

The minimum value is 68. The maximum value is 9000. The default value is -1, which means that the MTU is inherited from the bridge MTU.

VLAN interfaces are internal ports of a bridge. Based on the MTU of the bridge, these internal ports inherit the MTU of the bridge. The Bridge MTU is the minimum MTU of its member-interfaces. VLAN interfaces are not considered member-interfaces when deriving the bridge MTU. When setting the VLAN MTU, it is not possible to set a higher MTU, than the value inherited from the bridge. Setting a value higher than the bridge MTU, will not result in an error, but instead set the MTU to the value inherited from the bridge.

Applies to AP and STA.

<i>Range</i>	-1 - 9000
<i>Access</i>	readwrite
<i>OID</i>	1.3.6.1.4.1.16177.1.400.1.1.2.3.1.16

8.1.12.9.1.4 cfgNetVlanPriority

The VLAN Priority (PCP) of the VLAN Interface

Sets the PCP of the 802.1Q header to the value specified.

When set to -1, will not set anything. However the PCP may be inherited from the priority of a frame received via another interface, e.g. wlan (TID), when L2 prioritisation is enabled (see cfgQosL3PrioEnabled).

<i>Range</i>	-1 - 7
<i>Access</i>	readwrite
<i>OID</i>	1.3.6.1.4.1.16177.1.400.1.1.2.3.1.18

8.1.12.9.1.5 cfgNetVlanName

Name of the VLAN interface.

Applies to AP and STA.

Type	DisplayString
Range	1 - 255
Access	readonly
OID	1.3.6.1.4.1.16177.1.400.1.1.2.3.1.2

8.1.12.9.1.6 cfgNetVlanEnabled

VLAN interface disabled or enabled.

Applies to AP and STA.

Enumeration	disabled (0), enabled (1)
Access	readwrite
OID	1.3.6.1.4.1.16177.1.400.1.1.2.3.1.3

8.1.12.9.1.7 cfgNetVlanBridge

VLAN Interface Bridge Membership

If value ≥ 0 then interface is part of bridge.

- -1: none
- 0: br0
- 1: br1
- X: brX

VLAN interfaces are always of type access.

When set to -1, the VLAN interface will be created on the parent interface defined by `cfgNetVlanParent`.

Applies to AP and STA.

Type	Integer32
Access	readwrite
OID	1.3.6.1.4.1.16177.1.400.1.1.2.3.1.7

8.1.12.9.1.8 cfgNetVlanParent

VLAN Parent

Name of the physical parent interface on which the VLAN resides.

This entry is only active when the VLAN interface is not part of a bridge (`cfgNetVlanBridge = -1`).

Applies to AP and STA.

Type	DisplayString
Range	1 - 255
Access	readwrite
OID	1.3.6.1.4.1.16177.1.400.1.1.2.3.1.8

8.1.12.9.1.9 cfgNetVlanVid

VLAN ID

Applies to AP and STA.

Range	0 - 4094
Access	readwrite
OID	1.3.6.1.4.1.16177.1.400.1.1.2.3.1.9

8.1.12.10 cfgNetIpTable

IP address configuration

Range	0 - 127
OID	1.3.6.1.4.1.16177.1.400.1.1.2.6

8.1.12.10.1 cfgNetIpTableEntry

IP Address Configuration

OID	1.3.6.1.4.1.16177.1.400.1.1.2.6.1
-----	-----------------------------------

8.1.12.10.1.1 cfgNetIpIndex

Table Entry Index

Range	0 - 127
Access	noaccess
OID	1.3.6.1.4.1.16177.1.400.1.1.2.6.1.1

8.1.12.10.1.2 cfgNetIpCarpId

Reference to CARP-Instance ID

The referenced ID is reflected by `cfgNetCarpIndex`. This parameter is only active when `cfgNetIpProto` is set to **carp(5)**. Indicates that the referenced CARP instance processes this IP.

<i>Range</i>	0 - 15
<i>Access</i>	readwrite
<i>OID</i>	1.3.6.1.4.1.16177.1.400.1.1.2.6.1.10

8.1.12.10.1.3 cfgNetIpComment

User Comment

This parameter has no operational function. It allows to store a comment about the use of this IP address.

<i>Type</i>	DisplayString
<i>Access</i>	readwrite
<i>OID</i>	1.3.6.1.4.1.16177.1.400.1.1.2.6.1.11

8.1.12.10.1.4 cfgNetIpEnabled

IP Address Disabled or Enabled

<i>Enumeration</i>	disabled (0), enabled (1)
<i>Access</i>	readwrite
<i>OID</i>	1.3.6.1.4.1.16177.1.400.1.1.2.6.1.3

8.1.12.10.1.5 cfgNetIpAddr

IP Address

The IPv4 address (using CIDR notation) of the interface specified in `cfgNetIpInterface`.

<i>Type</i>	DisplayString
<i>Range</i>	5 - 50
<i>Access</i>	readwrite
<i>OID</i>	1.3.6.1.4.1.16177.1.400.1.1.2.6.1.4

8.1.12.10.1.6 cfgNetIpProto

Protocol

This parameter defines which protocol is used to configure the IPv4 address for this interface.

- **static(0):** Indicates that the address is manually configured to a specified address given by the IPv4 address parameter of this interface configuration.
- **dhcp(1):** Indicates that an IPv4 address will be obtained by the DHCP client. In case the DHCP client is unable to get a valid IPv4 address the static IP address will be used as a fallback.

- **linkLocal(4):** Indicates that an IPv4 link local address (an address in the range of 169.254.0.1 to 169.254.255.254, randomly chosen by the system) will be used on this interface. `cfgNetIpAddr` is then not used for the interface. **Note:** Only one interface of the device can use a linkLocal protocol.
- **carp(5):** Indicates that an IPv4 address will be set by the CARP instance specified in `cfgNetIpCarpId`, when the CARP instance has the state MASTER.

Note: Only one interface of the device can use a link local protocol.

For wireless interfaces, the following additional modes are available:

- **dhcpForceRenew(2):** Indicates that an IPv4 address will be obtained by the DHCP client. In case the DHCP client is unable to get a valid IPv4 address the static IP address will be used as a fallback. On a STA this mode will perform a DHCP RENEW after every connection to an AP. This is useful if the device is roaming between different DHCP servers.
- **dhcpForceRelease(3):** Indicates that an IPv4 address will be obtained by the DHCP client. In case the DHCP client is unable to get a valid IPv4 address, the static IP address will be used as a fallback. On a STA this mode will perform a DHCP RELEASE followed by a DHCP DISCOVER after every connection to an AP. This is useful if the device is roaming between different DHCP servers which don't send NAK to an unknown device sending RENEW.

<i>Enumeration</i>	static (0), dhcp (1), dhcpForceRenew (2), dhcpForceRelease (3), linkLocal (4), carp (5)
<i>Access</i>	readwrite
<i>OID</i>	1.3.6.1.4.1.16177.1.400.1.1.2.6.1.6

8.1.12.10.1.7 cfgNetIpInterface

Interface Name

The name of the network interface on which the specified IP address is created.

<i>Type</i>	DisplayString
<i>Range</i>	1 - 255
<i>Access</i>	readwrite
<i>OID</i>	1.3.6.1.4.1.16177.1.400.1.1.2.6.1.8

8.1.12.11 cfgNetCarpTable

Redundant IP Addresses with the Common Address Redundancy Protocol (CARP)

<i>Range</i>	0 - 15
<i>OID</i>	1.3.6.1.4.1.16177.1.400.1.1.2.7

8.1.12.11.1 cfgNetCarpTableEntry

Redundant IP Addresses with the Common Address Redundancy Protocol (CARP)

<i>OID</i>	1.3.6.1.4.1.16177.1.400.1.1.2.7.1
------------	-----------------------------------

8.1.12.11.1.1 cfgNetCarpIndex

Table Entry Index

<i>Range</i>	0 - 15
<i>Access</i>	noaccess
<i>OID</i>	1.3.6.1.4.1.16177.1.400.1.1.2.7.1.1

8.1.12.11.1.2 cfgNetCarpVhid

The Virtual Host ID

This is a unique number that is used to identify the redundancy group to other nodes in the group, and to distinguish between groups on the same network.

This must be the same on all members of the group.

Applies to AP and STA.

<i>Range</i>	1 - 255
<i>Access</i>	readwrite
<i>OID</i>	1.3.6.1.4.1.16177.1.400.1.1.2.7.1.10

8.1.12.11.1.3 cfgNetCarpPassword

The CARP authentication password

This is the password which is used to encrypt the CARP frames when talking to other CARP-enabled hosts in this redundancy group.

This must be the same on all members of the group.

Applies to AP and STA.

<i>Type</i>	DisplayString
<i>Range</i>	1 - 255
<i>Access</i>	readwrite
<i>OID</i>	1.3.6.1.4.1.16177.1.400.1.1.2.7.1.11

8.1.12.11.1.4 cfgNetCarpAdvbase

Advertisement base in seconds

This parameter specifies how often to transmit advertisement frames that we're a member of the redundancy group.

This time is divided with `cfgNetCarpAdvdivider` to decrease the failovertime of the redundancy group.

Applies to AP and STA.

<i>Range</i>	1 - 255
<i>Access</i>	readwrite
<i>OID</i>	1.3.6.1.4.1.16177.1.400.1.1.2.7.1.12

8.1.12.11.1.5 cfgNetCarpAdvskew

Advertisement base in 1/255th of a second

This parameter specifies how much to skew the advbase when sending CARP advertisements.

By manipulating advskew, the master of a CARP group can be chosen. The higher the number, the less often frames are transmitted and the less preferred the host will be when determining the master.

This time is divided with `cfgNetCarpAdvdivider` to decrease the failovertime of the redundancy group.

Applies to AP and STA.

<i>Range</i>	0 - 254
<i>Access</i>	readwrite
<i>OID</i>	1.3.6.1.4.1.16177.1.400.1.1.2.7.1.13

8.1.12.11.1.6 cfgNetCarpAdvdivider

Divider for `cfgNetCarpAdvbase` and `cfgNetCarpAdvskew`

This parameter specifies how much the Advbase and Advskew are divided to speed the algorithm up. With a factor of 1, the Advbase and Advskew are unchanged.

Increasing the Advdivider to 10 speeds the algorithm up to allow a minimum advertise time of 100ms instead of 1s.

This must be the same on all members of the group.

Applies to AP and STA.

<i>Range</i>	1 - 100
<i>Access</i>	readwrite
<i>OID</i>	1.3.6.1.4.1.16177.1.400.1.1.2.7.1.14

8.1.12.11.1.7 cfgNetCarpRatio

The dead ratio after which an existing master is considered dead

A slave device will wait:

$\text{cfgNetCarpRatio} * (\text{cfgNetCarpAdvbase} + \text{cfgNetCarpAdvskew})$

before considering the current master as dead and attempt to become the new master. Since The locally configured values for Advbase and Advskew are used, if there is a better suited master (lower Advbase/Advskew), it will start advertising before the local device.

Applies to AP and STA.

<i>Range</i>	1 - 100
<i>Access</i>	readwrite
<i>OID</i>	1.3.6.1.4.1.16177.1.400.1.1.2.7.1.15

8.1.12.11.1.8 cfgNetCarpPreempt

Preempt other masters when the local device is better

Allow hosts within a redundancy group that have a better Advbase and Advskew to preempt the current master.

Applies to AP and STA.

<i>Enumeration</i>	disabled (0), enabled (1)
<i>Access</i>	readwrite
<i>OID</i>	1.3.6.1.4.1.16177.1.400.1.1.2.7.1.16

8.1.12.11.1.9 cfgNetCarpPreemptdemote

Preempt other master when they demote themselves

Allow hosts within a redundancy group that have a better Advbase and Advskew to preempt the current master when it demotes itself (skew >= 240).

Preemptdemote only works when cfgNetCarpPreempt is used.

Preemptdemote also controls if a master demotes itself when another CARP-instance in the same `cfgNetCarpLocalInterfaceGroup` goes down or is demoted.

Applies to AP and STA.

<i>Enumeration</i>	disabled (0), enabled (1)
<i>Access</i>	readwrite
<i>OID</i>	1.3.6.1.4.1.16177.1.400.1.1.2.7.1.17

8.1.12.11.1.10 `cfgNetCarpLocalInterfaceGroup`

The local carp interface group the carp interface belongs to.

When `cfgNetCarpPreempt` or `cfgNetCarpPreemptdemote` is set to enabled, the interface will demote itself (skew = 240) when another interface within the same local group goes down or is demoted.

Applies to AP and STA.

<i>Range</i>	0 - 255
<i>Access</i>	readwrite
<i>OID</i>	1.3.6.1.4.1.16177.1.400.1.1.2.7.1.18

8.1.12.11.1.11 `cfgNetCarpSyncInterface`

Name of the CARP control interface.

This interface is used to transmit and receive CARP advertisements.

This interface is required to have its own unique IP address.

Applies to AP and STA.

<i>Type</i>	DisplayString
<i>Range</i>	1 - 255
<i>Access</i>	readwrite
<i>OID</i>	1.3.6.1.4.1.16177.1.400.1.1.2.7.1.19

8.1.12.11.1.12 `cfgNetCarpMcastIp`

Multicast address on which CARP advertisements are transmitted.

The default multicast address for CARP is 224.0.0.18.

Use this parameter to specify a custom multicast destination.

Applies to AP and STA.

Type	IpAddress
Access	readwrite
OID	1.3.6.1.4.1.16177.1.400.1.1.2.7.1.21

8.1.12.11.1.13 cfgNetCarpEnabled

CARP interface disabled or enabled

Applies to AP and STA.

Enumeration	disabled (0), enabled (1)
Access	readwrite
OID	1.3.6.1.4.1.16177.1.400.1.1.2.7.1.3

8.1.12.12 cfgNetMacVlanTable

MACVLAN Network Interfaces

Range	0 - 63
OID	1.3.6.1.4.1.16177.1.400.1.1.2.9

8.1.12.12.1 cfgNetMacVlanTableEntry

MACVLAN Network Interfaces

OID	1.3.6.1.4.1.16177.1.400.1.1.2.9.1
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8.1.12.12.1.1 cfgNetMacVlanIndex

Table Entry Index

Range	0 - 63
Access	noaccess
OID	1.3.6.1.4.1.16177.1.400.1.1.2.9.1.1

8.1.12.12.1.2 cfgNetMacVlanMac

MAC address of the MACVLAN

Set 00:00:00:00:00:00 to use a random MAC address.

Format: 00:14:5a:02:10:42

Applies to AP and STA.

<i>Type</i>	DisplayString
<i>Range</i>	17 - 17
<i>Access</i>	readwrite
<i>OID</i>	1.3.6.1.4.1.16177.1.400.1.1.2.9.1.15

8.1.12.12.1.3 cfgNetMacVlanMtu

The MTU of the MACVLAN Interface

The minimum value is 68. The maximum value is 9000. The default value is -1, which means that the MTU is inherited from the parent interface.

MACVLAN interfaces are virtual interfaces which reside on another interface as parent. Based on the MTU of the parent, these virtual interfaces inherit the MTU of the parent. Setting the MACVLAN MTU to a value higher than the MTU inherited from the parent, will result in an error.

Applies to AP and STA.

<i>Range</i>	-1 - 9000
<i>Access</i>	readwrite
<i>OID</i>	1.3.6.1.4.1.16177.1.400.1.1.2.9.1.16

8.1.12.12.1.4 cfgNetMacVlanName

Name of the MACVLAN interface.

Applies to AP and STA.

<i>Type</i>	DisplayString
<i>Range</i>	1 - 255
<i>Access</i>	readonly
<i>OID</i>	1.3.6.1.4.1.16177.1.400.1.1.2.9.1.2

8.1.12.12.1.5 cfgNetMacVlanEnabled

MACVLAN interface disabled or enabled.

Applies to AP and STA.

<i>Enumeration</i>	disabled (0), enabled (1)
<i>Access</i>	readwrite
<i>OID</i>	1.3.6.1.4.1.16177.1.400.1.1.2.9.1.3

8.1.12.12.1.6 cfgNetMacVlanParent

Name of the parent interface on which the MACVLAN resides.

Applies to AP and STA.

<i>Type</i>	DisplayString
<i>Range</i>	1 - 255
<i>Access</i>	readwrite
<i>OID</i>	1.3.6.1.4.1.16177.1.400.1.1.2.9.1.8

8.1.13 cfgWireless

8.1.13.1 cfgWlanDeviceTable

Wireless Hardware Modules

<i>Range</i>	0 - 1
<i>OID</i>	1.3.6.1.4.1.16177.1.400.1.1.3.1

8.1.13.1.1 cfgWlanDeviceTableEntry

Wireless Hardware Modules

<i>OID</i>	1.3.6.1.4.1.16177.1.400.1.1.3.1.1
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8.1.13.1.1.1 cfgWlanDevIndex

Table Entry Index

<i>Range</i>	0 - 1
<i>Access</i>	noaccess
<i>OID</i>	1.3.6.1.4.1.16177.1.400.1.1.3.1.1.1

8.1.13.1.1.2 cfgWlanDevDistance

Maximum distance in meters a client can be apart from the access point. Even though the distance is set in meters, the slot time settings change in 450m steps.

Applies to AP and STA. 802.11n products only.

<i>Range</i>	0 - 114750
<i>Access</i>	readwrite
<i>OID</i>	1.3.6.1.4.1.16177.1.400.1.1.3.1.1.10

8.1.13.1.1.3 cfgWlanDevRts

RTS/CTS Threshold

Frames equal or longer in bytes than this value require a RTS/CTS handshake.

RTS/CTS is used in hidden node situations. In 11bg and b mode, these frames are sent in DSSS modulation at 11b data rates. Otherwise (pure-g and a) OFDM rates are used.

The following settings are special:

- **-1** disable value, RTS/CTS is disabled.
- **0** minimum value, RTS/CTS is always used.
- **2346** maximum value legacy-rates, RTS/CTS is enabled for maximum sized frames.
- **65535** maximum value n-rates, RTS/CTS is enabled for maximal aggregate sized frames.

Note: It is not recommended to use RTS/CTS in AP mode.

Applies to AP and STA.

<i>Range</i>	-1 - 65535
<i>Access</i>	readwrite
<i>OID</i>	1.3.6.1.4.1.16177.1.400.1.1.3.1.1.11

8.1.13.1.1.4 cfgWlanDevFragments

Fragmentation Threshold

Frames longer in bytes than this threshold will be fragmented.

Fragmentation can be used to reduce the number of retransmissions. The following settings are special

- **-1** disable value, fragmentation is disabled
- **256** minimum value, frames above 256 are fragmented.
- **2346** maximum value, essentially the same as disabled.

Applies to AP and STA. 802.11n products only.

<i>Range</i>	-1 - 2346
<i>Access</i>	readwrite
<i>OID</i>	1.3.6.1.4.1.16177.1.400.1.1.3.1.1.12

8.1.13.1.1.5 cfgWlanDevShortRetry

Number of times the transmission of the RTS frame will be retried if there is no CTS received from the AP.

Applies to AP and STA. 802.11n products only.

<i>Range</i>	1 - 10
<i>Access</i>	readwrite
<i>OID</i>	1.3.6.1.4.1.16177.1.400.1.1.3.1.1.13

8.1.13.1.1.6 cfgWlanDevLongRetry

Number of times the unicast data frames will be retried if there is no ACK from the receiver.

Applies to AP and STA. 802.11n products only.

<i>Range</i>	1 - 10
<i>Access</i>	readwrite
<i>OID</i>	1.3.6.1.4.1.16177.1.400.1.1.3.1.1.14

8.1.13.1.1.7 cfgWlanDevAntennaGain

Antenna gain in dBi.

If multiple antennas with different gains are connected, the value of the antenna with the highest gain shall be configured.

Applies to AP and STA.

<i>Type</i>	Integer32
<i>Access</i>	readwrite
<i>OID</i>	1.3.6.1.4.1.16177.1.400.1.1.3.1.1.15

8.1.13.1.1.8 cfgWlanDevTxAntenna

Number of Wireless Transmitter Antenna Ports

This is a bitmask to enable/disable the chains.

Examples:

- 1(0001) = A1 (chain 0) enabled
- 3(0011) = A1 and A2 (chain 0 and 1) enabled
- 7(0111) = A1, A2 and A3 (chain 0, 1 and 2) enabled
- 15(1111) = A1, A2, A3 and A4 (chain 0, 1, 2 and 3) enabled

Note: Number of available antennas depends on the product. Please check the data-sheet of your product.

Applies to AP and STA.

Type	Integer32
Access	readwrite
OID	1.3.6.1.4.1.16177.1.400.1.1.3.1.1.16

8.1.13.1.1.9 cfgWlanDevRxAntenna

Number of Wireless Receiver Antenna Ports

This is a bitmask to enable/disable the chains.

Examples:

- 1(0001) = A1 (chain 0) enabled
- 3(0011) = A1 and A2 chain 0 and 1) enabled
- 7(0111) = A1, A2 and A3 (chain 0, 1 and 2) enabled
- 15(1111) = A1, A2, A3 and A4 (chain 0, 1, 2 and 3) enabled

Note: Number of available antennas depends on the product. Please check the data-sheet of your product.

Applies to AP and STA.

Type	Integer32
Access	readwrite
OID	1.3.6.1.4.1.16177.1.400.1.1.3.1.1.17

8.1.13.1.1.10 cfgWlanDevPhy

The map between physical device and radio.

Applies to AP and STA.

Type	DisplayString
Range	1 - 255
Access	readonly
OID	1.3.6.1.4.1.16177.1.400.1.1.3.1.1.18

8.1.13.1.1.11 cfgWlanDevName

Name of the Wireless Device.

Applies to AP and STA.

Type	DisplayString
Range	1 - 255
Access	readonly
OID	1.3.6.1.4.1.16177.1.400.1.1.3.1.1.2

8.1.13.1.1.12 cfgWlanDevHtCapabilities

HT capability flags:

- [LDPC] = 1 Enable support for LDPC coding
- [SHORT-GI-20] = 32 Allow short GI for 20 MHz
- [SHORT-GI-40] = 64 Allow short GI for 40 MHz
- [TX-STBC] = 128 Enable support for TX-STBC
- [RX-STBC1] = 256 Enable support for RX-STBC1
- [DSSS_CCK-40] = 4096 Enable support for DSSS/CCK Mode in 40 MHz
- [40-INTOLERANT] = 16384 Advertise 40 MHz intolerance

Applies to AP. 802.11n products only.

Range	0 - 65535
Access	readwrite
OID	1.3.6.1.4.1.16177.1.400.1.1.3.1.1.24

8.1.13.1.1.13 cfgWlanDevQmrrString

A list of rate controller quadruples per queue.

Each quadruple consist of (mcs-rate [0-31], tries [0-15], rts_cts [0-1], sgi [0-1]) with 4 entries per queue.

16 values together are for a single queue.

The values are in the form: rate1, try1, rts_cts1, sgi1, rate2, . . . , rate4, try4, rts_cts4, sgi4.

The order of the queues is VO, VI, BE, BK.

QMRR override for a specific queue is disabled when its respective try1 value is 0.

When QMRR override is disabled, the normal minstrel or other configured overrides, are used.

Frames in the VO queue are never aggregated.

All characters other than numbers are ignored

Example:

[(7 1 0 0) (4 2 0 0) (2 3 0 0) (0 4 1 0)] [(7 1 0 0) (4 1 0 1) (2 1 0 1) (0 1 1 1)] [(7 1 0 0) (4 1 0 0) (0 0 0 0) (0 0 0 0)] [(7 1 0 0) (0 0 0 0) (0 0 0 0) (0 0 0 0)]

Applies to AP and STA. 802.11n products only.

Type	DisplayString
Access	readwrite
OID	1.3.6.1.4.1.16177.1.400.1.1.3.1.1.26

8.1.13.1.1.14 cfgWlanDevAtfSchedulingAlgorithm

ATF Scheduling Algorithm

The following airtime scheduling algorithms are available:

- **fair(0)**: try to redistribute unused airtime.
- **strict(1)**: strictly follow the given airtime.

Applies to 802.11ac products in AP mode.

Enumeration	fair (0), strict (1)
Access	readwrite
OID	1.3.6.1.4.1.16177.1.400.1.1.3.1.1.27

8.1.13.1.1.15 cfgWlanDevModulation

Physical Wireless Device Modulation Mode

For an AP the following modulation modes are configurable:

- **g(2)**: This modulation mode uses OFDM data rates up to 54 MBit/s in the frequency band between 2.4 and 2.4835 GHz. It supports the 802.11g standard.

- **bg(3)**: This modulation mode uses data rates up to 54 MBit/s in the frequency band between 2.4 and 2.4835 GHz. It supports the 802.11bg standard. The modulation is either DSSS for the slower rates or OFDM for the faster ones.
- **a(4)**: Mode supports data rates up to 54 MBit/s in the 5GHz frequency band and only OFDM modulation.
- **ng(10)**: For 2.4GHz
- **na(12)**: For 5GHz.
- **ac(28)**: 11ac mode for 5GHz.

Note: Some products do not support all modulations. Please check the data-sheet of your product.

Note:: The STA scans both frequency bands (2.4 GHz and 5GHz) regardless of the modulation. The STA will adapt to the modulation of the Access Point during connection.

Note:: For a STA the modulation can be used to explicitly disable n-rates even though n-rates would be supported by the Access Point. The n-rates data rates support up to 300 MBit/s in the 2.4GHz and 5GHz frequency band and only OFDM modulation.

Setting modulation to **10(ng)** or **12(na)** will enable n-rates if n-rates are supported by the Access Point, while setting modulation to **g(2)**, **bg(3)** or **a(4)** will disable n-rates. This feature applies to 802.11n products only.

Applies to AP and STA.

<i>Enumeration</i>	g (2), bg (3), a (4), ng (10), na (12), ac (28)
<i>Access</i>	readwrite
<i>OID</i>	1.3.6.1.4.1.16177.1.400.1.1.3.1.1.4

8.1.13.1.1.16 cfgWlanDevBandwidth

Wireless Bandwidth Mode specifies the Bandwidth of the Channel.

- **bw20(0)**: for 20MHz wide channel.
- **bw40Plus(1)**: for 40MHz wide channel with the side channel on the top.
- **bw40Minus(2)**: for 40MHz wide channel with the side channel on the bottom.
- **bwQuarter(3)**: for 5MHz wide channel (quarter rate).
- **bwHalf(4)**: for 10MHz wide channel (half rate).
- **bw80(5)**: 80MHz wide channel (only in 11ac mode)
- **bw160(6)**: 160MHz wide channel (only in 11ac mode)
- **bw8080(7)**: 80+80MHz wide channel (only in 11ac mode)
- **bwAuto(8)**: automatic channel width (only in 11ac mode)

Note: **bw40Plus(1)** and **bw40Minus(2)** may not be usable on all channels.

Examples:

The following table shows examples of which channels may be used. The full list can be found in IEEE 802.11n Annex J. Depending on the country, not all frequencies may be available.

Band	bw40Plus(1)	bw40Minus(2)
2.4 GHz	2412 to 2452	2432 to 2472
5 GHz	5180, 5220, 5260, etc.	5200, 5240, 5280, etc.

Note: Some products do not support all bandwidths. Please check the data-sheet of your product.

Applies to AP and STA.

<i>Enumeration</i>	bw20 (0), bw40Plus (1), bw40Minus (2), bwQuarter (3), bwHalf (4), bw80 (5), bw160 (6), bw8080 (7), bwAuto (8)
<i>Access</i>	readwrite
<i>OID</i>	1.3.6.1.4.1.16177.1.400.1.1.3.1.1.5

8.1.13.1.1.17 cfgWlanDevFrequency

Wireless Frequency in MHz.

In AP mode, setting the wireless frequency to zero(0) enables the automatic channel selection (ACS) feature. This forces the AP to choose the best channel for operation.

In STA mode, the `cfgWlanIfaceScanList` is used to configure which frequencies are to be scanned.

Applies to AP and STA.

<i>Type</i>	Integer32
<i>Access</i>	readwrite
<i>OID</i>	1.3.6.1.4.1.16177.1.400.1.1.3.1.1.6

8.1.13.1.1.18 cfgWlanDevPower

TX Power Limit (EIRP)

Max. limit for wireless output power as effective isotropic radiated power (EIRP) in dBm including array gain and antenna gain.

Note: This parameter only limits the maximum output power. The effective output power might be lower (regulatory limits, rate depended limits).

EIRP (dBm) = antenna port power (dBm) + array gain (dB) + antenna gain (dBi)

- The antenna port power in dBm defines the power transmitted per antenna port (chain).
- The array gain in dB defines the gain which is achieved by the use of multiple antenna ports (chains). The number of active antenna ports (chains) is defined by `cfgWlanDevTxAntenna`. The array gain depends on number of active antenna ports (chains) as following:
 - One antenna port (chain) = 0 dB
 - Two antenna ports (chains) = 3 dB
 - Three antenna ports (chains) = 5 dB
 - Four antenna ports (chains) = 6 dB
- The antenna gain in dBi defines the gain which is achieved by the antenna. The antenna gain is configured by `cfgWlanDevAntennaGain`.

Applies to AP and STA.

<i>Range</i>	6 - 50
<i>Access</i>	readwrite
<i>OID</i>	1.3.6.1.4.1.16177.1.400.1.1.3.1.1.8

8.1.13.2 `cfgWlan802dot1xTable`

Wireless 802dot1x

<i>Range</i>	0 - 63
<i>OID</i>	1.3.6.1.4.1.16177.1.400.1.1.3.10

8.1.13.2.1 `cfgWlan802dot1xTableEntry`

Wireless 802dot1x entry

<i>OID</i>	1.3.6.1.4.1.16177.1.400.1.1.3.10.1
------------	------------------------------------

8.1.13.2.1.1 `cfgWlan802dot1xIndex`

Table Entry Index

<i>Range</i>	0 - 63
<i>Access</i>	noaccess
<i>OID</i>	1.3.6.1.4.1.16177.1.400.1.1.3.10.1.1

8.1.13.2.1.2 `cfgWlan802dot1xIdentity`

The identity string for EAP

Applies to STA.

Type	DisplayString
Range	1 - 255
Access	readwrite
OID	1.3.6.1.4.1.16177.1.400.1.1.3.10.1.10

8.1.13.2.1.3 cfgWlan802dot1xClientKeyPassword

The password to unlock the private key

This is only required if the key is encrypted.

Applies to STA.

Type	DisplayString
Access	readwrite
OID	1.3.6.1.4.1.16177.1.400.1.1.3.10.1.17

8.1.13.2.1.4 cfgWlan802dot1xTlsControlParams

Bitfield to control TLS behaviour

- **0x1** ignore certificate validity time
- **0x2** ignore ca certificate

Applies to STA.

Range	0 - 3
Access	readwrite
OID	1.3.6.1.4.1.16177.1.400.1.1.3.10.1.18

8.1.13.2.1.5 cfgWlan802dot1xName

Name of the virtual wireless interface

Applies to AP and STA.

Type	DisplayString
Range	1 - 255
Access	readonly
OID	1.3.6.1.4.1.16177.1.400.1.1.3.10.1.2

8.1.13.2.1.6 cfgWlan802dot1xCiphers

OpenSSL Cipher String for the client

This is an OpenSSL specific configuration option for configuring the default cipher.

Please read the documentation for a list of all available ciphers and used syntax.

Used only if `cfgWlanIfaceEncryption = eap(6)`.

Examples:

- ECDHE-RSA-AES128-GCM-SHA256
- DHE-RSA-AES128-GCM-SHA256
- ECDHE-RSA-AES256-GCM-SHA384:ECDHE-ECDSA-AES256-GCM-SHA384:ECDHE-RSA-AES256-SHA384

Applies to STA.

Type	DisplayString
Access	readwrite
OID	1.3.6.1.4.1.16177.1.400.1.1.3.10.1.21

8.1.13.2.1.7 cfgWlan802dot1xOwnIpAddr

The own IP address of the authenticator

This field is used as NAS-IP-Address RADIUS attribute. Set this to the IP address with which the authenticator will communicate with the RADIUS server.

Applies to AP.

Type	IpAddress
Access	readwrite
OID	1.3.6.1.4.1.16177.1.400.1.1.3.10.1.3

8.1.13.2.1.8 cfgWlan802dot1xAuthServerParameter

Reference ID to the radius auth server table

Uses all parameters in the `cfgWlan802dot1xAuthServerTable` which have as `cfgWlan802dot1xAuthSrvId` the value set here.

Applies to AP.

Type	Integer32
Access	readwrite
OID	1.3.6.1.4.1.16177.1.400.1.1.3.10.1.4

8.1.13.2.1.9 cfgWlan802dot1xAcctServerParameter

Reference ID to the radius acct server table

Uses all parameters in the `cfgWlan802dot1xAcctServerTable` which have as `cfgWlan802dot1xAcctSrvId` the value set here.

Applies to AP.

Type	Integer32
Access	readwrite
OID	1.3.6.1.4.1.16177.1.400.1.1.3.10.1.5

8.1.13.2.1.10 cfgWlan802dot1xRetryPrimaryInterval

Retry interval to return to the primary RADIUS server in seconds

RADIUS client code will automatically try to use the next server when the current server is not replying to requests. If this interval is set, primary server will be retried after configured amount of time even if the currently used secondary server is still working. Set to 0 to disable.

Applies to AP.

Range	0 - 86400
Access	readwrite
OID	1.3.6.1.4.1.16177.1.400.1.1.3.10.1.6

8.1.13.2.1.11 cfgWlan802dot1xInterimAccountingInterval

Interim accounting update interval in seconds

If this is set (larger than 0) and `acct_server` is configured, `hostapd` will send interim accounting updates every N seconds. Set to 0 to disable.

Note: If set, this overrides possible `Acct-Interim-Interval` attribute in `Access-Accept` message. Thus, this value should not be configured in `hostapd.conf` if RADIUS server is used to control the interim interval. This value should not be less 600 (10 minutes) and must not be less than 60 (1 minute).

Applies to AP.

<i>Range</i>	0 - 86400
<i>Access</i>	readwrite
<i>OID</i>	1.3.6.1.4.1.16177.1.400.1.1.3.10.1.7

8.1.13.2.1.12 cfgWlan802dot1xNasId

Optional NAS Identifier string for RADIUS messages

When used, this should be unique to the NAS within the scope of the RADIUS server. For example, a fully qualified domain name can be used here. When using IEEE 802.11r, nas_identifier must be set and must be between 1 and 48 octets long.

Applies to AP.

<i>Type</i>	DisplayString
<i>Range</i>	1 - 48
<i>Access</i>	readwrite
<i>OID</i>	1.3.6.1.4.1.16177.1.400.1.1.3.10.1.8

8.1.13.2.1.13 cfgWlan802dot1xEapType

Specify the EAP type

Applies to STA.

<i>Enumeration</i>	tls (0), peap (1), ttls (2)
<i>Access</i>	readwrite
<i>OID</i>	1.3.6.1.4.1.16177.1.400.1.1.3.10.1.9

8.1.13.3 cfgWlan802dot1xAuthServerTable

Wireless 802dot1x AuthServer

<i>Range</i>	0 - 255
<i>OID</i>	1.3.6.1.4.1.16177.1.400.1.1.3.11

8.1.13.3.1 cfgWlan802dot1xAuthServerTableEntry

Wireless 802dot1x AuthServer entry

<i>OID</i>	1.3.6.1.4.1.16177.1.400.1.1.3.11.1
------------	------------------------------------

8.1.13.3.1.1 cfgWlan802dot1xAuthSrvIndex

Table Entry Index

<i>Range</i>	0 - 255
<i>Access</i>	noaccess
<i>OID</i>	1.3.6.1.4.1.16177.1.400.1.1.3.11.1.1

8.1.13.3.1.2 cfgWlan802dot1xAuthSrvEnabled

Enable this entry in the auth server list

Applies to AP.

<i>Enumeration</i>	disabled (0), enabled (1)
<i>Access</i>	readwrite
<i>OID</i>	1.3.6.1.4.1.16177.1.400.1.1.3.11.1.2

8.1.13.3.1.3 cfgWlan802dot1xAuthSrvId

ID of the authorisation server table

The configuration item `cfgWlan802dot1xAuthServerParameter` references to this ID.

Applies to AP.

<i>Type</i>	Integer32
<i>Access</i>	readwrite
<i>OID</i>	1.3.6.1.4.1.16177.1.400.1.1.3.11.1.3

8.1.13.3.1.4 cfgWlan802dot1xAuthSrvIpAddr

IP of the RADIUS server against which will be authenticated

Applies to AP.

<i>Type</i>	IpAddress
<i>Access</i>	readwrite
<i>OID</i>	1.3.6.1.4.1.16177.1.400.1.1.3.11.1.4

8.1.13.3.1.5 cfgWlan802dot1xAuthSrvPort

Port of the RADIUS server against which will be authenticated

Applies to AP.

Type	Integer32
Access	readwrite
OID	1.3.6.1.4.1.16177.1.400.1.1.3.11.1.5

8.1.13.3.1.6 cfgWlan802dot1xAuthSrvSharedSecret

Password to connect to the specified RADIUS server

Applies to AP.

Type	DisplayString
Range	1 - 255
Access	readwrite
OID	1.3.6.1.4.1.16177.1.400.1.1.3.11.1.6

8.1.13.4 cfgWlan802dot1xAcctServerTable

Wireless 802dot1x AcctServer

Range	0 - 255
OID	1.3.6.1.4.1.16177.1.400.1.1.3.12

8.1.13.4.1 cfgWlan802dot1xAcctServerTableEntry

Wireless 802dot1x AcctServer entry

OID	1.3.6.1.4.1.16177.1.400.1.1.3.12.1
-----	------------------------------------

8.1.13.4.1.1 cfgWlan802dot1xAcctSrvIndex

Table Entry Index

Range	0 - 255
Access	noaccess
OID	1.3.6.1.4.1.16177.1.400.1.1.3.12.1.1

8.1.13.4.1.2 cfgWlan802dot1xAcctSrvEnabled

Enable this entry in the acct server list.

Applies to AP.

<i>Enumeration</i>	disabled (0), enabled (1)
<i>Access</i>	readwrite
<i>OID</i>	1.3.6.1.4.1.16177.1.400.1.1.3.12.1.2

8.1.13.4.1.3 cfgWlan802dot1xAcctSrvId

ID of the accounting server table

The configuration item `cfgWlan802dot1xAcctServerParameter` refers to this ID.

Applies to AP.

<i>Type</i>	Integer32
<i>Access</i>	readwrite
<i>OID</i>	1.3.6.1.4.1.16177.1.400.1.1.3.12.1.3

8.1.13.4.1.4 cfgWlan802dot1xAcctSrvIpAddr

IP of the RADIUS accounting server Set to 0.0.0.0 to disable.

Applies to AP.

<i>Type</i>	IpAddress
<i>Access</i>	readwrite
<i>OID</i>	1.3.6.1.4.1.16177.1.400.1.1.3.12.1.4

8.1.13.4.1.5 cfgWlan802dot1xAcctSrvPort

Port of the RADIUS accounting server

Applies to AP.

<i>Type</i>	Integer32
<i>Access</i>	readwrite
<i>OID</i>	1.3.6.1.4.1.16177.1.400.1.1.3.12.1.5

8.1.13.4.1.6 cfgWlan802dot1xAcctSrvSharedSecret

Password to connect to the specified RADIUS accounting server

Applies to AP.

Type	DisplayString
Range	1 - 255
Access	readwrite
OID	1.3.6.1.4.1.16177.1.400.1.1.3.12.1.6

8.1.13.5 cfgWlan802dot11rTable

Wireless 802dot11r

Range	0 - 63
OID	1.3.6.1.4.1.16177.1.400.1.1.3.13

8.1.13.5.1 cfgWlan802dot11rTableEntry

Wireless 802dot11r Entry

OID	1.3.6.1.4.1.16177.1.400.1.1.3.13.1
-----	------------------------------------

8.1.13.5.1.1 cfgWlan802dot11rIndex

Table Entry Index

Range	0 - 63
Access	noaccess
OID	1.3.6.1.4.1.16177.1.400.1.1.3.13.1.1

8.1.13.5.1.2 cfgWlan802dot11rR0KHParameter

Reference ID to the R0KH parameter table

Uses all parameters in the `cfgWlan802dot11rR0KHTable` which have as `cfgWlan802dot11rR0KHTblId` the value set here.

Applies to AP. 802.11n products only.

Type	Integer32
Access	readwrite
OID	1.3.6.1.4.1.16177.1.400.1.1.3.13.1.10

8.1.13.5.1.3 cfgWlan802dot11rR1KHParameter

Reference ID to the R1KH parameter table

Uses all parameters in the `cfgWlan802dot11r1KHTb1Id` which have as `cfgWlan802dot11r1KHTb1Id` the value set here.

Applies to AP. 802.11n products only.

Type	Integer32
Access	readwrite
OID	1.3.6.1.4.1.16177.1.400.1.1.3.13.1.11

8.1.13.5.1.4 cfgWlan802dot11rExpirationEnabled

Enable/disable PMK-R0/-R1 expiration forcing

If set to **enabled(1)**, AP forces PMK-R0s and PMK-R1s expiration once a day (see `cfgWlan802dot11rExpirationTime`).

Applies to AP. 802.11n products only.

Enumeration	disabled (0), enabled (1)
Access	readwrite
OID	1.3.6.1.4.1.16177.1.400.1.1.3.13.1.12

8.1.13.5.1.5 cfgWlan802dot11rExpirationTime

Daily PMK-R0/-R1 expiration time

Define time (hour:minute) at which PMK-R0s and PMK-R1s expiration is daily forced (if `cfgWlan802dot11rExpirationEnabled` is **enabled(1)**).

The time is referenced to the local time as define in `cfgSysTimezone`

Examples:

- 00:00 - force expiration each day at midnight
- 01:00 - force expiration each day at 01:00
- 23:05 - force expiration each day at 23:05

Applies to AP. 802.11n products only.

Type	DisplayString
Range	5 - 5
Access	readwrite
OID	1.3.6.1.4.1.16177.1.400.1.1.3.13.1.13

8.1.13.5.1.6 cfgWlan802dot11rVlan

The 802.1q VLAN tag of 802.11r backbone traffic

Defines the VLAN tag with which all 802.11r management frames (ethertype 0x88b7) are transmitted on the backbone.

Setting a value of 0 disables the VLAN header and transmits the frames untagged.

The priority of all tagged frames is set to 0x7.

When this value is set to a value other than 0, make sure that the VLAN mode of the wireless interface `cfgNetWlanVlanMode` allows tagged frames. Also ensure that `cfgNetWlanTrunk` includes all VLANs or this specific VLAN.

Applies to AP. 802.11n products only.

<i>Range</i>	0 - 4094
<i>Access</i>	readwrite
<i>OID</i>	1.3.6.1.4.1.16177.1.400.1.1.3.13.1.14

8.1.13.5.1.7 `cfgWlan802dot11rName`

Name of the virtual wireless interface

Applies to AP and STA. 802.11n products only.

<i>Type</i>	DisplayString
<i>Range</i>	1 - 255
<i>Access</i>	readonly
<i>OID</i>	1.3.6.1.4.1.16177.1.400.1.1.3.13.1.2

8.1.13.5.1.8 `cfgWlan802dot11rEnabled`

Enable usage of 802.11r on this device

Applies to AP and STA. 802.11n products only.

<i>Enumeration</i>	disabled (0), enabled (1)
<i>Access</i>	readwrite
<i>OID</i>	1.3.6.1.4.1.16177.1.400.1.1.3.13.1.3

8.1.13.5.1.9 `cfgWlan802dot11rMobilityDomain`

Mobility Domain identifier (dot11FTMobilityDomainID, MDID)

MDID is used to indicate a group of APs (within an ESS, i.e., sharing the same SSID) between which a STA can use Fast BSS Transition. 2-octet identifier as a hex string.

Examples:

- a1b2
- faba
- 5678

Applies to AP. 802.11n products only.

Type	DisplayString
Range	4 - 4
Access	readwrite
OID	1.3.6.1.4.1.16177.1.400.1.1.3.13.1.4

8.1.13.5.1.10 cfgWlan802dot11rPmkR0KeyHolderIdentifier

PMK-R0 Key Holder identifier (dot11FTR0KeyHolderID)

Configure this in the field `cfgWlan802dot1xNasId`.

Applies to AP. 802.11n products only.

Type	DisplayString
Range	1 - 48
Access	readonly
OID	1.3.6.1.4.1.16177.1.400.1.1.3.13.1.5

8.1.13.5.1.11 cfgWlan802dot11rPmkR0Lifetime

PMK-RO lifetime in seconds (dot11FTR0KeyLifetime)

When Session-Timeout attribute is provided by RADIUS server, then $\min(\text{Session-Timeout}, \text{cfgWlan802dot11rPmkR0Lifetime})$ is used.

Ranges

- 0 - Infinite lifetime (disabled)
- 1..59 - Reserved, do not use
- 60..2147483647 - Allowed lifetime in seconds

Applies to AP. 802.11n products only.

<i>Range</i>	0 - 2147483647
<i>Access</i>	readwrite
<i>OID</i>	1.3.6.1.4.1.16177.1.400.1.1.3.13.1.6

8.1.13.5.1.12 cfgWlan802dot11rPmkR1KeyHolderIdentifier

PMK-R1 Key Holder identifier (dot11FTR0KeyHolderID)

6-octet identifier as a hex string. This may be the same as the local MAC address. Default magic number 000102030405 means use own mac address (bssid).

Format: 020102030405

Applies to AP. 802.11n products only.

<i>Type</i>	DisplayString
<i>Range</i>	12 - 12
<i>Access</i>	readwrite
<i>OID</i>	1.3.6.1.4.1.16177.1.400.1.1.3.13.1.7

8.1.13.5.1.13 cfgWlan802dot11rPmkR1Push

Whether PMK-R1 push is enabled at R0KH

- **0** do not push PMK-R1 to all configured R1KHs (default).
- **1** push PMK-R1 to all configured R1KHs whenever a new PMK-R0 is derived.

Applies to AP. 802.11n products only.

<i>Enumeration</i>	donotpush (0), push (1)
<i>Access</i>	readwrite
<i>OID</i>	1.3.6.1.4.1.16177.1.400.1.1.3.13.1.9

8.1.13.6 cfgWlan802dot11rR0KHTable

Wireless 802dot11r R0KH

List of R0KHs in the same Mobility Domain. This list is used to map R0KH-ID (NAS Identifier) to a destination MAC address when requesting PMK-R1 key from the R0KH that the STA used during the Initial Mobility Domain Association.

Format: <MAC address> <NAS Identifier> <128-bit key as hex string>

Examples:

r0kh=02:01:02:03:04:05 r0kh-1.example.com 000102030405060708090a0b0c0d0e0f
r0kh=02:01:02:03:04:06 r0kh-2.example.com 00112233445566778899aabbccddeeff

This may also contain a wildcard entry to transmit a request to the broadcast address instead of a unicast. This has the advantage that not all potential R0KH have to be configured. The provided key has to match the configured wildcard key in the `cfgWlan802dot11rR1KHTable`

Wildcard entry

r1kh=ff:ff:ff:ff:ff:ff * 0123456789abcdef0123456789abcdef

Applies to AP. 802.11n products only.

<i>Range</i>	0 - 511
<i>OID</i>	1.3.6.1.4.1.16177.1.400.1.1.3.14

8.1.13.6.1 cfgWlan802dot11rR0KHTableEntry

Wireless 802dot11r R0KH Entry

<i>OID</i>	1.3.6.1.4.1.16177.1.400.1.1.3.14.1
------------	------------------------------------

8.1.13.6.1.1 cfgWlan802dot11rR0KHIndex

Table Entry Index

<i>Range</i>	0 - 511
<i>Access</i>	noaccess
<i>OID</i>	1.3.6.1.4.1.16177.1.400.1.1.3.14.1.1

8.1.13.6.1.2 cfgWlan802dot11rR0KHId

ID of the R0KH table

The configuration item `cfgWlan802dot11rR0KHParameter` references to this ID.

Applies to AP. 802.11n products only.

<i>Type</i>	Integer32
<i>Access</i>	readwrite
<i>OID</i>	1.3.6.1.4.1.16177.1.400.1.1.3.14.1.2

8.1.13.6.1.3 cfgWlan802dot11rR0KHEnabled

Enable this entry in the R0KH list

Applies to AP. 802.11n products only.

<i>Enumeration</i>	disabled (0), enabled (1)
<i>Access</i>	readwrite
<i>OID</i>	1.3.6.1.4.1.16177.1.400.1.1.3.14.1.3

8.1.13.6.1.4 cfgWlan802dot11rR0KHDestinationMac

MAC addresses of possible R0KHs from which PMK-R1 can be requested

Format: 02:01:02:03:04:05

Applies to AP. 802.11n products only.

<i>Type</i>	DisplayString
<i>Range</i>	17 - 17
<i>Access</i>	readwrite
<i>OID</i>	1.3.6.1.4.1.16177.1.400.1.1.3.14.1.4

8.1.13.6.1.5 cfgWlan802dot11rR0KHHID

NAS Identifier of all R0KHs to map to the MAC address

See the field: `cfgWlan802dot11rPmkR0KeyHolderIdentifier`.

Applies to AP. 802.11n products only.

<i>Type</i>	DisplayString
<i>Range</i>	1 - 48
<i>Access</i>	readwrite
<i>OID</i>	1.3.6.1.4.1.16177.1.400.1.1.3.14.1.5

8.1.13.6.1.6 cfgWlan802dot11rR0KHKey

Static Key of the R0KH

Connecting R1KHs need to have this key configured.

Format: 000102030405060708090a0b0c0d0e0f

Applies to AP. 802.11n products only.

Type	DisplayString
Range	32 - 32
Access	readwrite
OID	1.3.6.1.4.1.16177.1.400.1.1.3.14.1.6

8.1.13.7 cfgWlan802dot11rR1KHTable

Wireless 802dot11r R1KH

List of R1KHs in the same Mobility Domain This list is used to map R1KH-ID to a destination MAC address when sending PMK-R1 key from the R0KH. This is also the list of authorized R1KHs in the MD that can request PMK-R1 keys.

Format: <MAC address> <R1KH-ID> <128-bit key as hex string>

Examples:

```
r1kh=02:01:02:03:04:05 02:11:22:33:44:55 000102030405060708090a0b0c0d0e0f
r1kh=02:01:02:03:04:06 02:11:22:33:44:66 00112233445566778899aabbccddeeff
```

This may also contain a wildcard entry allowing everyone to request a PMK-R1 from this R0KH. The provided key has to match the configured wildcard key in the cfgWlan802dot11rR0KHTable

Wildcard entry

```
r1kh=00:00:00:00:00:00 00:00:00:00:00:00 0123456789abcdef0123456789abcdef
```

Applies to AP. 802.11n products only.

Range	0 - 511
OID	1.3.6.1.4.1.16177.1.400.1.1.3.15

8.1.13.7.1 cfgWlan802dot11rR1KHTableEntry

Wireless 802dot11r R1KH Entry

OID	1.3.6.1.4.1.16177.1.400.1.1.3.15.1
-----	------------------------------------

8.1.13.7.1.1 cfgWlan802dot11rR1KHIndex

Table Entry Index

<i>Range</i>	0 - 511
<i>Access</i>	noaccess
<i>OID</i>	1.3.6.1.4.1.16177.1.400.1.1.3.15.1.1

8.1.13.7.1.2 cfgWlan802dot11rR1KHId

ID of the R1KH table

The configuration item `cfgWlan802dot11rR1KHParameter` references to this ID.

Applies to AP.

<i>Type</i>	Integer32
<i>Access</i>	readwrite
<i>OID</i>	1.3.6.1.4.1.16177.1.400.1.1.3.15.1.2

8.1.13.7.1.3 cfgWlan802dot11rR1KHEnabled

Enable this entry in the R1KH list

Applies to AP. 802.11n products only.

<i>Enumeration</i>	disabled (0), enabled (1)
<i>Access</i>	readwrite
<i>OID</i>	1.3.6.1.4.1.16177.1.400.1.1.3.15.1.3

8.1.13.7.1.4 cfgWlan802dot11rR1KHDestinationMac

MAC addresses of R1KHs which can request PMK-R1 from the local R0KH

Format: 02:01:02:03:04:05

Applies to AP.

<i>Type</i>	DisplayString
<i>Range</i>	17 - 17
<i>Access</i>	readwrite
<i>OID</i>	1.3.6.1.4.1.16177.1.400.1.1.3.15.1.4

8.1.13.7.1.5 cfgWlan802dot11rR1KHHID

PMK-R1 Key Holder identifier (dot11FTR1KeyHolderID)

6-octet identifier as a hex string to map to the MAC. This may be the same as the destination MAC. See the field `cfgWlan802dot11rPmkR1KeyHolderIdentifier`.

Format: 02:01:02:03:04:05

Applies to AP. 802.11n products only.

Type	DisplayString
Range	17 - 17
Access	readwrite
OID	1.3.6.1.4.1.16177.1.400.1.1.3.15.1.5

8.1.13.7.1.6 cfgWlan802dot11rR1KHKey

Static Key of the R0KH

These keys are used wenn sending updates to R1KHs from the local R0KH. The respective key has to match the respective entry on the target in the field `cfgWlan802dot11rR0KHKey`.

Format: 000102030405060708090a0b0c0d0e0f

Applies to AP. 802.11n products only.

Type	DisplayString
Range	32 - 32
Access	readwrite
OID	1.3.6.1.4.1.16177.1.400.1.1.3.15.1.6

8.1.13.8 cfgWlanNeighbourTable

Hostapd Neighbour Table

Range	0 - 511
OID	1.3.6.1.4.1.16177.1.400.1.1.3.16

8.1.13.8.1 cfgWlanNeighbourTableEntry

Hostapd Neighbour Table Entry

OID	1.3.6.1.4.1.16177.1.400.1.1.3.16.1
-----	------------------------------------

8.1.13.8.1.1 cfgWlanNeighbourIndex

Table Entry Index

<i>Range</i>	0 - 511
<i>Access</i>	noaccess
<i>OID</i>	1.3.6.1.4.1.16177.1.400.1.1.3.16.1.1

8.1.13.8.1.2 cfgWlanNeighbourId

ID of the neighbour table

The configuration item `cfgWlanIfaceNeighbourParameter` references to this ID.

Applies to AP. 802.11n products only.

<i>Type</i>	Integer32
<i>Access</i>	readwrite
<i>OID</i>	1.3.6.1.4.1.16177.1.400.1.1.3.16.1.2

8.1.13.8.1.3 cfgWlanNeighbourEnabled

Enable this entry in the list

Applies to AP. 802.11n products only.

<i>Enumeration</i>	disabled (0), enabled (1)
<i>Access</i>	readwrite
<i>OID</i>	1.3.6.1.4.1.16177.1.400.1.1.3.16.1.3

8.1.13.8.1.4 cfgWlanNeighbourBSSID

BSSID (MAC address) of the neighbour

Format: 00:14:5a:02:10:42

Applies to AP.

<i>Type</i>	DisplayString
<i>Range</i>	17 - 17
<i>Access</i>	readwrite
<i>OID</i>	1.3.6.1.4.1.16177.1.400.1.1.3.16.1.4

8.1.13.8.1.5 cfgWlanNeighbourFrequency

Frequency in MHz of the neighbour

Applies to AP. 802.11n products only.

<i>Range</i>	1 - 6000
<i>Access</i>	readwrite
<i>OID</i>	1.3.6.1.4.1.16177.1.400.1.1.3.16.1.5

8.1.13.9 cfgWlanInterfaceTable

Wireless Virtual Interfaces

<i>Range</i>	0 - 63
<i>OID</i>	1.3.6.1.4.1.16177.1.400.1.1.3.2

8.1.13.9.1 cfgWlanInterfaceTableEntry

Wireless Virtual Interface Entry

<i>OID</i>	1.3.6.1.4.1.16177.1.400.1.1.3.2.1
------------	-----------------------------------

8.1.13.9.1.1 cfgWlanInterfaceIndex

Table Entry Index

<i>Range</i>	0 - 63
<i>Access</i>	noaccess
<i>OID</i>	1.3.6.1.4.1.16177.1.400.1.1.3.2.1.1

8.1.13.9.1.2 cfgWlanInterfaceDtim

Number of beacons between transmission of DTIM element

This attribute specifies the number of beacon intervals that shall elapse between transmission of beacons frames containing a TIM element whose DTIM count field is 0. This value is transmitted in the DTIM Period field of Beacon frames.

The DTIM counter is used to signal to power saving sleeping clients how long they can sleep between wakeups to get data from the AP.

Applies to AP. 802.11n products only.

<i>Range</i>	1 - 255
<i>Access</i>	readwrite
<i>OID</i>	1.3.6.1.4.1.16177.1.400.1.1.3.2.1.10

8.1.13.9.1.3 cfgWlanIfaceAplsolate

Enable AP clients isolation

If enabled, clients connected to this AP can't communicate to each other.

Applies to AP.

<i>Enumeration</i>	disabled (0), enabled (1)
<i>Access</i>	readwrite
<i>OID</i>	1.3.6.1.4.1.16177.1.400.1.1.3.2.1.100

8.1.13.9.1.4 cfgWlanIfaceBitrates

Fixed MCS Index For 802.11n Rates

Set to -1 to not force an MCS index (auto-rate).

Allows multiple space and/or comma separated indices which are then used in auto rate.

This entry is only active when `cfgWlanDevModulation` is set to **ng(10)** or **na(12)**

Examples:

- -1
- 0 1 2 3 4 5 6 7
- 0, 4, 7, 8, 12, 15

Applies to AP and STA. 802.11n products only.

<i>Type</i>	DisplayString
<i>Access</i>	readwrite
<i>OID</i>	1.3.6.1.4.1.16177.1.400.1.1.3.2.1.11

8.1.13.9.1.5 cfgWlanIfaceBeaconInterval

Time in kus (1.024 ms) between the sending of beacon frames

Applies to AP. 802.11n products only.

<i>Range</i>	15 - 65535
<i>Access</i>	readwrite
<i>OID</i>	1.3.6.1.4.1.16177.1.400.1.1.3.2.1.12

8.1.13.9.1.6 cfgWlanIfaceLlcBroadcastVlan

VLAN to send broadcast LLC frame after Handoff

A space and/or comma separated list of VLANs. This list defines to which VLANs the broadcast LLC frame is sent. The broadcast LLC frames is sent when a STA connects to an AP. This broadcast LLC frame is used to update the FDB of all switches on the backbone which are involved on the path on which data flows.

The value 0 specifies that the frame is sent untagged.

When this value is set to a value other than 0, make sure that the VLAN mode of the wireless interface `cfgNetWlanVlanMode` allows tagged frames. Also ensure that `cfgNetWlanTrunk` includes the specified VLAN(s).

Examples:

- 69
- 12, 24, 69
- 0 12 24 69

Applies to AP.

<i>Type</i>	DisplayString
<i>Range</i>	1 - 255
<i>Access</i>	readwrite
<i>OID</i>	1.3.6.1.4.1.16177.1.400.1.1.3.2.1.120

8.1.13.9.1.7 cfgWlanIfaceAtfSsidEnabled

ATF SSID Disabled or Enabled

When this parameter is set to **disabled(0)** no airtime is reserved for the SSID.

Applies to 802.11ac products in AP mode.

<i>Enumeration</i>	disabled (0), enabled (1)
<i>Access</i>	readwrite
<i>OID</i>	1.3.6.1.4.1.16177.1.400.1.1.3.2.1.121

8.1.13.9.1.8 cfgWlanIfaceAtfSsidAirtime

ATF SSID Airtime Percentage

Specify the percentage of each second of airtime assigned to the SSID on this virtual interface. The sum of all assigned airtime shares on the same physical device must not exceed 100%.

Applies to 802.11ac products in AP mode.

<i>Range</i>	0 - 100
<i>Access</i>	readwrite
<i>OID</i>	1.3.6.1.4.1.16177.1.400.1.1.3.2.1.122

8.1.13.9.1.9 cfgWlanIfaceWmeParameter

Reference ID to the WME parameter table

Uses all parameters in the `cfgWlanWmeTable` which have as `cfgWlanWmeId` the value set here.

Applies to AP and STA. 802.11n products only.

<i>Type</i>	Integer32
<i>Access</i>	readwrite
<i>OID</i>	1.3.6.1.4.1.16177.1.400.1.1.3.2.1.13

8.1.13.9.1.10 cfgWlanIfaceWmeEnabled

Enables usage of the WME parameter table

When using legacy rates (a-rates and g-rates) this is optional. When using n-rates this has to be enabled at all times.

Applies to AP. 802.11n products only.

<i>Enumeration</i>	disabled (0), enabled (1)
<i>Access</i>	readwrite
<i>OID</i>	1.3.6.1.4.1.16177.1.400.1.1.3.2.1.14

8.1.13.9.1.11 cfgWlanIfaceScanList

Index to specify a frequency list to be scanned

Is only active when `cfgWlanIfaceMode` is set to `sta(1)`. Set to **-1**, to scan the frequency defined in `cfgWlanDevFrequency`. Set to **-2**, to scan all frequencies allowed by the country code.

Applies to STA. 802.11n products only.

<i>Range</i>	-2 - 23
<i>Access</i>	readwrite
<i>OID</i>	1.3.6.1.4.1.16177.1.400.1.1.3.2.1.15

8.1.13.9.1.12 cfgWlanIfacelgnoreBroadcastSsid

Hide SSID

Send empty SSID in beacons and ignore probe request frames that do not specify the full SSID, i.e., require stations to know SSID.

Applies to AP.

<i>Enumeration</i>	disabled (0), enabled (1)
<i>Access</i>	readwrite
<i>OID</i>	1.3.6.1.4.1.16177.1.400.1.1.3.2.1.16

8.1.13.9.1.13 cfgWlanIfaceMacaddrAcl

Mode of the MAC Access Control List

- **acceptunlessdeny(0):** Accept unless deny filter. Accept every MAC unless it is on the list defined in `cfgWlanAclBlackTable`.
- **denyunlessaccept(1):** Deny unless accept filter. Deny every MAC unless it is on the list defined in `cfgWlanAclWhiteTable`.
- **radius(2):** Use RADIUS to accept/deny clients. The local Accept and Deny list are searched first and take priority.

Applies to AP. 802.11n products only.

<i>Enumeration</i>	acceptunlessdeny (0), denyunlessaccept (1), radius (2)
<i>Access</i>	readwrite
<i>OID</i>	1.3.6.1.4.1.16177.1.400.1.1.3.2.1.17

8.1.13.9.1.14 cfgWlanIfaceMaxNumSta

Maximum Number of Clients

802.11ac products have an upper limit of 512 clients.

Applies to AP.

<i>Range</i>	1 - 2007
<i>Access</i>	readwrite
<i>OID</i>	1.3.6.1.4.1.16177.1.400.1.1.3.2.1.18

8.1.13.9.1.15 cfgWlanIfaceBssid

BSSID of AP

Set 00:00:00:00:00:00 to use the MAC address stored in the flash of the wireless card itself. If this is the second or more virtual AP on this card it will automatically set the locally assigned bit and add an increasing counter in the leading 0 range.

When the device is operating in STA mode the MAC address of the wireless interface can be configured.

Format: 00:14:5a:02:10:42

Applies to AP and STA.

Type	DisplayString
Range	17 - 17
Access	readwrite
OID	1.3.6.1.4.1.16177.1.400.1.1.3.2.1.19

8.1.13.9.1.16 cfgWlanIfaceName

Name of the virtual wireless interface

Applies to AP and STA.

Type	DisplayString
Range	1 - 255
Access	readonly
OID	1.3.6.1.4.1.16177.1.400.1.1.3.2.1.2

8.1.13.9.1.17 cfgWlanIfaceLegacyRates

Wireless legacy data rates

- **11b:** 1, 2, 5.5, 11 Mbps
- **11a/g:** 6, 9, 12, 18, 24, 36, 48, 54 Mbps

The values are interpreted as flags:

- ****auto(0)***
- **1Mbps(1)**
- **2Mbps(2)**
- **5.5Mbps(4)**
- **6Mbps(8)**
- **9Mbps(16)**

- 11Mbps(32)
- 12Mbps(64)
- 18Mbps(128)
- 24Mbps(256)
- 36Mbps(512)
- 48Mbps(1024)
- **54Mbps(2048)

When `cfgWlanDevBandwidth` is equal 3 (quarter rates) the rate is quarter i.e. 36Mbps becomes 9MBps. When `cfgWlanDevBandwidth` is equal 4 (half rates) the rate is halved i.e. 36Mbps becomes 18MBps.

This entry only has an effect on clients which only can use g-rates or a-rates. For clients which support MCS-rates the entry `cfgWlanIfaceBitrates` can be used to allow specific rates.

Applies to AP and STA. 802.11n products only.

<i>Range</i>	0 - 2048
<i>Access</i>	readwrite
<i>OID</i>	1.3.6.1.4.1.16177.1.400.1.1.3.2.1.20

8.1.13.9.1.18 `cfgWlanIface4addr`

This option allows to bridge the STA side

When used on the STA, the corresponding AP has to enable this feature as well. This option may not be enabled together with `cfgWlanIfaceL2nat`.

Applies to AP and STA.

<i>Enumeration</i>	disabled (0), enabled (1)
<i>Access</i>	readwrite
<i>OID</i>	1.3.6.1.4.1.16177.1.400.1.1.3.2.1.21

8.1.13.9.1.19 `cfgWlanIfaceInactivityTimeout`

Allowed idle time before station is removed

If a station does not send anything in `ap_max_inactivity` seconds, an empty data frame is sent to it in order to verify whether it is still in range. If this frame is not ACKed, the station will be disassociated and then deauthenticated. This feature is used to clear the station table of old entries when the STAs move out of range.

Applies to AP. 802.11n products only.

<i>Range</i>	15 - 65535
<i>Access</i>	readwrite
<i>OID</i>	1.3.6.1.4.1.16177.1.400.1.1.3.2.1.23

8.1.13.9.1.20 cfgWlanIfaceUseVendorSsid

Enable Vendor Element containing the SSID

When `cfgWlanIfaceIgnoreBroadcastSsid` is enabled, a passively scanning STA (forced or because of DFS) has no way of detecting the AP it tries to find. On an AP this options adds the hidden SSID as vendor element. On a STA this options allows it to use the vendor element in the beacon.

Applies to AP and STA. 802.11n products only.

<i>Enumeration</i>	disabled (0), enabled (1)
<i>Access</i>	readwrite
<i>OID</i>	1.3.6.1.4.1.16177.1.400.1.1.3.2.1.26

8.1.13.9.1.21 cfgWlanIfaceDevice

Maps the virtual wireless interface to the radio device

Applies to AP and STA.

<i>Enumeration</i>	radio0 (0), radio1 (1)
<i>Access</i>	readwrite
<i>OID</i>	1.3.6.1.4.1.16177.1.400.1.1.3.2.1.3

8.1.13.9.1.22 cfgWlanIfaceleeee80211w

Controls usage of 802.11w Management Frame Protection (MFP)

On AP, if set to **optional(1)**, MFP will be used only for clients which have it also enabled (either **optional(1)** or **required(2)**). If set to **required(2)**, only MFP enabled clients will be able to connect.

On STA, if set to **optional(1)**, MFP will be used only for Access Points which have it also enabled (either **optional(1)** or **required(2)**). If set to **required(2)**, client will connect only to MFP enabled Access Points.

Applies to AP and STA.

<i>Enumeration</i>	disabled (0), optional (1), required (2)
<i>Access</i>	readwrite
<i>OID</i>	1.3.6.1.4.1.16177.1.400.1.1.3.2.1.30

8.1.13.9.1.23 cfgWlanIfaceleeee80211wMaxTimeout

802.11w Management Frame Protection (MFP) timeout

Association SA query maximum timeout (in TU = 1.024 ms; for MFP) (maximum time to wait for a SA query response).

Applies to AP.

<i>Range</i>	1 - 4000
<i>Access</i>	readwrite
<i>OID</i>	1.3.6.1.4.1.16177.1.400.1.1.3.2.1.31

8.1.13.9.1.24 cfgWlanIfaceleeee80211wRetryTimeout

802.11w Management Frame Protection (MFP) retry timeout

Association SA query retry timeout (in TU = 1.024 ms; for MFP) (time between two subsequent SA query requests).

Applies to AP.

<i>Range</i>	1 - 4000
<i>Access</i>	readwrite
<i>OID</i>	1.3.6.1.4.1.16177.1.400.1.1.3.2.1.32

8.1.13.9.1.25 cfgWlanIfaceMode

Wireless Operation Mode

Allowed modes are:

- **ap(0)**: defines the interface as Access Point (AP)
- **sta(1)**: defines the interface as Station (STA)
- **monitor(2)**: defines the interface as Monitor (MON)
- **mesh(3)**: defines the interface as Mesh (MESH)

Note: Some products do not support all modes. Please check the data-sheet of your product.

Applies to AP and STA.

<i>Enumeration</i>	ap (0), sta (1), monitor (2), mesh (3)
<i>Access</i>	readwrite
<i>OID</i>	1.3.6.1.4.1.16177.1.400.1.1.3.2.1.4

8.1.13.9.1.26 cfgWlanIfaceAcsList

Index to specify a frequency list from `cfgWlanFreqTable`

Used for Automated Channel Selection (ACS) and for channel switch on Radar event to specify a list of frequencies to be chosen from.

To disable set to **-1**.

Applies to AP.

<i>Type</i>	Integer32
<i>Access</i>	readwrite
<i>OID</i>	1.3.6.1.4.1.16177.1.400.1.1.3.2.1.40

8.1.13.9.1.27 cfgWlanIfaceSsid

The Service Set Identifier (SSID) of the wireless interface

This is the arbitrary name of the wireless network this interface is part of.

Applies to AP and STA.

<i>Type</i>	DisplayString
<i>Range</i>	1 - 32
<i>Access</i>	readwrite
<i>OID</i>	1.3.6.1.4.1.16177.1.400.1.1.3.2.1.5

8.1.13.9.1.28 cfgWlanIfaceEncryption

Wireless Encryption Mode

Supported encryption modes:

- **open(0)** means open network without encryption.
- **psk(3)** means WPA-PSK security standard (IEEE 802.11i, AES-CCMP)
- **eap(6)** WPA-EAP (EAP-TLS/TTLS/PEAP)
- **sae(7)** enables SAE (WPA3-Personal)
- **owe(8)** Opportunistic Wireless Encryption (WPA3 Enhanced Open)
- **saepsk(9)** enables WPA3-Personal Transition mode (SAE + PSK)

WPA-EAP enables 802.1X support.

Applies to AP and STA.

<i>Enumeration</i>	open (0), psk (3), eap (6), sae (7), owe (8), saepsk (9)
<i>Access</i>	readwrite
<i>OID</i>	1.3.6.1.4.1.16177.1.400.1.1.3.2.1.6

8.1.13.9.1.29 cfgWlanIfaceNeighbourReport

Enable/disable neighbour reporting

A STA can request the neighbour table from an AP and use this information to improve its handoff decision.

Applies to AP. 802.11n products only.

<i>Enumeration</i>	disabled (0), enabled (1)
<i>Access</i>	readwrite
<i>OID</i>	1.3.6.1.4.1.16177.1.400.1.1.3.2.1.60

8.1.13.9.1.30 cfgWlanIfaceNeighbourParameter

Reference ID to the neighbour table

Uses all neighbours in the `cfgWlanNeighbourTable` which have as `cfgWlanNeighbourId` the value set here.

Applies to AP. 802.11n products only.

<i>Type</i>	Integer32
<i>Access</i>	readwrite
<i>OID</i>	1.3.6.1.4.1.16177.1.400.1.1.3.2.1.61

8.1.13.9.1.31 cfgWlanIfacePassword

Wireless password if an encryption is in use

Each character in the pass-phrase must have an encoding in the range of 32 to 126 (decimal), inclusive. (IEEE Std. 802.11i-2004, Annex H.4.1) The space character is included in this range.

Applies to AP and STA.

<i>Type</i>	DisplayString
<i>Range</i>	8 - 63
<i>Access</i>	readwrite
<i>OID</i>	1.3.6.1.4.1.16177.1.400.1.1.3.2.1.7

8.1.13.9.1.32 cfgWlanIfacePassiveScanning

Wireless Scanning Mode

If the scanning mode is set to active(0) the station will send a probe request to detect available access points if it's allowed by the country code. If the country code restricts active scanning it is automatically set to passive.

If the scanning mode is set to passive(1) the station will always perform passive scanning to detect available access points. Since the listening time is related to the beacon period used at the AP, the passive scan mode `cfgWlanHoPassiveChanTime` (STA) must be at least as long as the maximum beacon period `cfgWlanIfaceBeaconInterval` (AP).

Applies to STA. 802.11n products only.

<i>Enumeration</i>	active (0), passive (1)
<i>Access</i>	readwrite
<i>OID</i>	1.3.6.1.4.1.16177.1.400.1.1.3.2.1.8

8.1.13.9.1.33 cfgWlanIfaceL2nat

This option allows to bridge the STA side

It is intended to be used in setups where the STA is doing handoff between multiple APs. There is no configuration on the APs required. This is an alternative to `cfgWlanIface4addr`. This option may not be enabled together with `cfgWlanIface4addr`.

Applies to STA.

<i>Enumeration</i>	disabled (0), enabled (1)
<i>Access</i>	readwrite
<i>OID</i>	1.3.6.1.4.1.16177.1.400.1.1.3.2.1.80

8.1.13.9.1.34 cfgWlanIfaceL2natLearningMode

Type of frames from which the L2nat IP/MAC table may be learned

When `cfgWlanIfaceL2nat` is enabled a table of the association between IP-addresses and MAC-addresses is kept. This options specifies from which type of frames this association may be learned:

- **both(0):** MAC/IP association is learned from ARP frames and from IP frames.
- **arp(1):** MAC/IP association is learned only from ARP frames.

Applies to STA.

<i>Enumeration</i>	both (0), arp (1)
<i>Access</i>	readwrite
<i>OID</i>	1.3.6.1.4.1.16177.1.400.1.1.3.2.1.81

8.1.13.9.1.35 cfgWlanIfaceL2natDefaultDestination

Default destination for L2nat

This option defines the default MAC address to send frames to. Whenever a unicast frame is received for which no learned entry exists, or which isn't a L3 frame (e.g a custom L2 protocol), the frame will be sent to the address specified here.

This address is by default the broadcast address.

Format: ff:ff:ff:ff:ff:ff

When the STA is directly connected to a router, the only possible destination is the MAC of the attached router. However it is cumbersome to manually configure the MAC address for the router. When this field is set to 00:00:00:00:00:00, it is in router-auto-learn-mode. Depending on `cfgWlanIfaceL2natLearningMode` it will automatically learn the default destination to the MAC of the attached router from the flowing frames. In this mode `cfgWlanIfaceL2natLearningMode` should be set to **arp(1)**.

Applies to STA.

<i>Type</i>	DisplayString
<i>Range</i>	17 - 17
<i>Access</i>	readwrite
<i>OID</i>	1.3.6.1.4.1.16177.1.400.1.1.3.2.1.82

8.1.13.9.1.36 cfgWlanIfaceBeaconMiss

Number of consecutive beacons misses before the station disconnects

Applies to STA. 802.11n products only.

<i>Type</i>	Integer32
<i>Access</i>	readwrite
<i>OID</i>	1.3.6.1.4.1.16177.1.400.1.1.3.2.1.9

8.1.13.9.1.37 cfgWlanIfaceTimeAdvertisement

Enable AP to include local time in association response frames

This provides a mean for APs to distribute its local system time to STAs in setups where devices have no RTC and start up with an invalid system time.

If configured, AP embeds its current system time as IE in association response frames, which STAs can use to update to before they get access to trusted time sources like NTP.

Note that enabling this feature increases the size of response frames, therefore this feature shall only be activated where required and defaults to disabled.

Applies to AP. 802.11n products only.

<i>Enumeration</i>	disabled (0), enabled (1)
<i>Access</i>	readwrite
<i>OID</i>	1.3.6.1.4.1.16177.1.400.1.1.3.2.1.90

8.1.13.10 cfgWlanHandoffTable

Wireless handoff parameters

<i>Range</i>	0 - 63
<i>OID</i>	1.3.6.1.4.1.16177.1.400.1.1.3.3

8.1.13.10.1 cfgWlanHandoffTableEntry

Wireless Handoff Parameters Entry

<i>OID</i>	1.3.6.1.4.1.16177.1.400.1.1.3.3.1
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8.1.13.10.1.1 cfgWlanHoIndex

Table Entry Index

<i>Range</i>	0 - 63
<i>Access</i>	noaccess
<i>OID</i>	1.3.6.1.4.1.16177.1.400.1.1.3.3.1.1

8.1.13.10.1.2 cfgWlanHoFilterLongX

IIR Low-Pass Filter Parameter X for RSSI measurements.

Parameter X is the weighting for RSSI current value. It must be greater than or equal to cfgWlanHoFilterLongY.

Applies to STA. 802.11n products only.

Type	Integer32
Access	readwrite
OID	1.3.6.1.4.1.16177.1.400.1.1.3.3.1.10

8.1.13.10.1.3 cfgWlanHoFilterLongY

IIR Low-Pass Filter Parameter Y for RSSI measurements.

The parameter Y is the weighting for the previous average value. It must be smaller than or equal to cfgWlanHoFilterLongX.

Applies to STA. 802.11n products only.

Type	Integer32
Access	readwrite
OID	1.3.6.1.4.1.16177.1.400.1.1.3.3.1.11

8.1.13.10.1.4 cfgWlanHoScanRateLimitTime

Scan Rate Limit Time

Time in milliseconds (4ms steps) in which a number of attempts to connect to an AP can be tried. The option only affects t2gv1 (see cfgWlanHoProfile).

Applies to STA. 802.11n products only.

Range	4 - 65535
Access	readwrite
OID	1.3.6.1.4.1.16177.1.400.1.1.3.3.1.12

8.1.13.10.1.5 cfgWlanHoScanRateLimitTries

Scan Rate Limit Tries

Number of attempts to connect to an AP before the AP is blacklisted and ignored. The AP is removed from the blacklist after cfgWlanHoScanRateLimitTime since the first attempt. This option only affects t2gv1 (see cfgWlanHoProfile).

Applies to STA. 802.11n products only.

Range	1 - 255
Access	readwrite
OID	1.3.6.1.4.1.16177.1.400.1.1.3.3.1.13

8.1.13.10.1.6 cfgWlanHoPassiveChanTime

Time in milliseconds (4ms steps) we stay on a channel during passive scanning and wait for beacons. Must be at least as long as the maximum beacon period `cfgWlanIfaceBeaconInterval` used at the AP.

Applies to STA. 802.11n products only.

<i>Range</i>	1 - 65535
<i>Access</i>	readwrite
<i>OID</i>	1.3.6.1.4.1.16177.1.400.1.1.3.3.1.16

8.1.13.10.1.7 cfgWlanHoLevelLow

Scanning level low in RSSI

When the RSSI level to the current connected AP is below this value, perform a handoff to the next AP.

This value is only active when `cfgWlanHoProfile` is set to T2Gv3. When configured on an AP, this value is advertised for clients to use. When configured on a STA, this is the default value which is used when the currently connected AP doesn't provide a different value.

Applies to AP and STA. 802.11n products only.

<i>Type</i>	Integer32
<i>Access</i>	readwrite
<i>OID</i>	1.3.6.1.4.1.16177.1.400.1.1.3.3.1.18

8.1.13.10.1.8 cfgWlanHoLevelHigh

Scanning level high in RSSI

When the RSSI level to the current connected AP is higher or equal this value, perform a handoff to the next AP.

This value is only active when `cfgWlanHoProfile` is set to T2Gv3. When configured on an AP, this value is advertised for clients to use. When configured on a STA, this is the default value which is used when the currently connected AP doesn't provide a different value.

Applies to AP and STA. 802.11n products only.

<i>Type</i>	Integer32
<i>Access</i>	readwrite
<i>OID</i>	1.3.6.1.4.1.16177.1.400.1.1.3.3.1.19

8.1.13.10.1.9 cfgWlanHofaceName

Name of the virtual wireless interface

Applies to STA. 802.11n products only.

Type	DisplayString
Range	1 - 255
Access	readonly
OID	1.3.6.1.4.1.16177.1.400.1.1.3.3.1.2

8.1.13.10.1.10 cfgWlanHoDistanceNear

Distance level near

When the measured distance (units) to the current connected AP is lower or equal this value, perform a handoff to the next AP.

This value is only active when `cfgWlanHoProfile` is set to T2Gv3. When configured on an AP, this value is advertised for clients to use. When configured on a STA, this is the default value which is used when the currently connected AP doesn't provide a different value.

Note: Distance value is not in meters.

Applies to AP and STA. 802.11n products only.

Range	0 - 114750
Access	readwrite
OID	1.3.6.1.4.1.16177.1.400.1.1.3.3.1.20

8.1.13.10.1.11 cfgWlanHoDistanceFar

Distance level far

When the measured distance (units) to the current connected AP is greater or equal this value, perform a handoff to the next AP.

This value is only active when `cfgWlanHoProfile` is set to T2Gv3. When configured on an AP, this value is advertised for clients to use. When configured on a STA, this is the default value which is used when the currently connected AP doesn't provide a different value.

Note: Distance value is not in meters.

Applies to AP and STA. 802.11n products only.

<i>Range</i>	0 - 114750
<i>Access</i>	readwrite
<i>OID</i>	1.3.6.1.4.1.16177.1.400.1.1.3.3.1.21

8.1.13.10.1.12 cfgWlanHoDistanceMeasurementPeriod

Distance ranging measurement period in milliseconds.

Setting zero disables distance handoff. This value is only active when `cfgWlanHoProfile` is set to T2Gv3.

Typical ranging measurement period values are in the range from 200 ms to 1000 ms. Values lower than 100 ms are not recommended.

Applies to STA. 802.11n products only.

<i>Range</i>	0 - 100000
<i>Access</i>	readwrite
<i>OID</i>	1.3.6.1.4.1.16177.1.400.1.1.3.3.1.22

8.1.13.10.1.13 cfgWlanHoDistanceFilterX

IIR Low-Pass Filter Parameter X for distance measurements.

Parameter X is the weighting for the previous average distance value. It must be smaller than or equal to `cfgWlanHoDistanceFilterY`.

This value is only active when `cfgWlanHoProfile` is set to T2Gv3.

Applies to STA. 802.11n products only.

<i>Type</i>	Integer32
<i>Access</i>	readwrite
<i>OID</i>	1.3.6.1.4.1.16177.1.400.1.1.3.3.1.23

8.1.13.10.1.14 cfgWlanHoDistanceFilterY

IIR Low-Pass Filter Parameter Y for distance measurements.

Parameter Y is the weighting for the current distance value. It must be greater than or equal to `cfgWlanHoDistanceFilterX`.

This value is only active when `cfgWlanHoProfile` is set to T2Gv3.

Applies to STA. 802.11n products only.

Type	Integer32
Access	readwrite
OID	1.3.6.1.4.1.16177.1.400.1.1.3.3.1.24

8.1.13.10.1.15 cfgWlanHoProfile

Handoff Profile

A description of available handoff profiles in terms of use case, functionality and associated parameters.

t2gv1(1) Train to Ground v1 (legacy)

Legacy profile for DT30 compatibility. It does not provide advanced security features. The handoff decision is based on low RSSI measurements. An AP is blacklisted after a defined number of attempts and recovers after a defined timeout.

- Encryption: psk/eap (WPA2)
- Scanning: Foreground
- Neighbour reporting: None
- Associated Parameters: `cfgWlanHoScanningLevel`, `cfgWlanHoScanRateLimitTime`, `cfgWlanHoScanR`

t2gv2(2) Train to Ground v2 (background scan)

Performs a background scan that optimizes operation in ISO-frequency systems. If neighbor reporting is enabled on the AP `cfgWlanIfaceNeighbourReport`, the STA will scan directly. Intended for operation with third-party products.

- Encryption: psk/eap/sae (WPA2/WPA3)
- Scanning: Background
- Neighbour reporting: Frequency
- Associated Parameters: `cfgWlanHoScanningLevel`

t2gv2fg(3) Train to Ground v2 (foreground scan)

Performs a foreground scan that optimizes operation in multi-frequency systems. If neighbor reporting is enabled on the AP `cfgWlanIfaceNeighbourReport`, the STA will scan directly. Intended for operation with third-party products.

- Encryption: psk/eap/sae (WPA2/WPA3)
- Scanning: Foreground
- Neighbour reporting: Frequency
- Associated Parameters: `cfgWlanHoScanningLevel`

t2gv3(4) Train to Ground v3 (RSSI low/high, Distance near/far)

If neighbor reporting is enabled on the AP `cfgWlanIfaceNeighbourReport`, the STA will scan directly and set the handoff parameters according to the received AP values. Otherwise, or if a neighbour report is lost, the handoff values for RSSI and Distance defined on the STA are used as backup. To enable distance handoff, the measurement period must be set.

- Encryption: psk/eap/sae (WPA2/WPA3)
- Scanning: Foreground
- Neighbour reporting: Frequency, RSSI low/high, Distance near/far
- Associated Parameters: `cfgWlanHoLevelLow`, `cfgWlanHoLevelHigh`, `cfgWlanHoDistanceNear`, `cfgWlanHoDistanceFar`, `cfgWlanHoDistanceMeasurementPeriod`

Applies to STA. 802.11n products only.

<i>Enumeration</i>	t2gv1 (1), t2gv2 (2), t2gv2fg (3), t2gv3 (4)
<i>Access</i>	readwrite
<i>OID</i>	1.3.6.1.4.1.16177.1.400.1.1.3.3.1.3

8.1.13.10.1.16 `cfgWlanHoScanningLevel`

Scanning level in RSSI

When the RSSI level of the currently connected access point drops below the value configured in this parameter, the STA will scan for better access points on all frequencies specified by the scan list configured in `cfgWlanIfaceScanList` and `cfgWlanFreqTable`.

Note: This parameter is inactive when using `cfgWlanHoProfile` t2gv3(4). Define the thresholds in `cfgWlanHoLevelLow` and `cfgWlanHoLevelHigh` instead.

Applies to STA. 802.11n products only.

<i>Range</i>	0 - 95
<i>Access</i>	readwrite
<i>OID</i>	1.3.6.1.4.1.16177.1.400.1.1.3.3.1.5

8.1.13.10.1.17 `cfgWlanHoBeacons`

Number of beacons which have to be received from an AP before a decision about handoff is allowed. Essentially forces the STA to stay on a given AP for `cfgWlanHoBeacons` * `cfgWlanIfaceBeaconInterval` before doing another handoff.

Applies to STA. 802.11n products only.

<i>Range</i>	4 - 20
<i>Access</i>	readwrite
<i>OID</i>	1.3.6.1.4.1.16177.1.400.1.1.3.3.1.6

8.1.13.10.1.18 cfgWlanHoRecovery

Recovery time in milliseconds after a successful handoff

During this time no further handoff will be executed.

Applies to STA. 802.11n products only.

<i>Range</i>	0 - 2000
<i>Access</i>	readwrite
<i>OID</i>	1.3.6.1.4.1.16177.1.400.1.1.3.3.1.7

8.1.13.11 cfgWlanFreqTable

Frequency list entry

<i>Range</i>	0 - 23
<i>OID</i>	1.3.6.1.4.1.16177.1.400.1.1.3.4

8.1.13.11.1 cfgWlanFreqTableEntry

Frequency list entry

<i>OID</i>	1.3.6.1.4.1.16177.1.400.1.1.3.4.1
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8.1.13.11.1.1 cfgWlanFIndex

Table Entry Index

<i>Range</i>	0 - 23
<i>Access</i>	noaccess
<i>OID</i>	1.3.6.1.4.1.16177.1.400.1.1.3.4.1.1

8.1.13.11.1.2 cfgWlanFFreq8

Frequency in MHz, 0 is interpreted as empty

Applies to AP and STA.

<i>Type</i>	Integer32
<i>Access</i>	readwrite
<i>OID</i>	1.3.6.1.4.1.16177.1.400.1.1.3.4.1.10

8.1.13.11.1.3 cfgWlanFFreq9

Frequency in MHz, 0 is interpreted as empty

Applies to AP and STA.

Type	Integer32
Access	readwrite
OID	1.3.6.1.4.1.16177.1.400.1.1.3.4.1.11

8.1.13.11.1.4 cfgWlanFFreq10

Frequency in MHz, 0 is interpreted as empty

Applies to AP and STA.

Type	Integer32
Access	readwrite
OID	1.3.6.1.4.1.16177.1.400.1.1.3.4.1.12

8.1.13.11.1.5 cfgWlanFFreq11

Frequency in MHz, 0 is interpreted as empty

Applies to AP and STA.

Type	Integer32
Access	readwrite
OID	1.3.6.1.4.1.16177.1.400.1.1.3.4.1.13

8.1.13.11.1.6 cfgWlanFFreq12

Frequency in MHz, 0 is interpreted as empty

Applies to AP and STA.

Type	Integer32
Access	readwrite
OID	1.3.6.1.4.1.16177.1.400.1.1.3.4.1.14

8.1.13.11.1.7 cfgWlanFFreq13

Frequency in MHz, 0 is interpreted as empty

Applies to AP and STA.

Type	Integer32
Access	readwrite
OID	1.3.6.1.4.1.16177.1.400.1.1.3.4.1.15

8.1.13.11.1.8 cfgWlanFFreq14

Frequency in MHz, 0 is interpreted as empty

Applies to AP and STA.

Type	Integer32
Access	readwrite
OID	1.3.6.1.4.1.16177.1.400.1.1.3.4.1.16

8.1.13.11.1.9 cfgWlanFFreq15

Frequency in MHz, 0 is interpreted as empty

Applies to AP and STA.

Type	Integer32
Access	readwrite
OID	1.3.6.1.4.1.16177.1.400.1.1.3.4.1.17

8.1.13.11.1.10 cfgWlanFFreq16

Frequency in MHz, 0 is interpreted as empty

Applies to AP and STA.

Type	Integer32
Access	readwrite
OID	1.3.6.1.4.1.16177.1.400.1.1.3.4.1.18

8.1.13.11.1.11 cfgWlanFFreq17

Frequency in MHz, 0 is interpreted as empty

Applies to AP and STA.

Type	Integer32
Access	readwrite
OID	1.3.6.1.4.1.16177.1.400.1.1.3.4.1.19

8.1.13.11.1.12 cfgWlanFFreq0

Frequency in MHz, 0 is interpreted as empty

Applies to AP and STA.

Type	Integer32
Access	readwrite
OID	1.3.6.1.4.1.16177.1.400.1.1.3.4.1.2

8.1.13.11.1.13 cfgWlanFFreq18

Frequency in MHz, 0 is interpreted as empty

Applies to AP and STA.

Type	Integer32
Access	readwrite
OID	1.3.6.1.4.1.16177.1.400.1.1.3.4.1.20

8.1.13.11.1.14 cfgWlanFFreq19

Frequency in MHz, 0 is interpreted as empty

Applies to AP and STA.

Type	Integer32
Access	readwrite
OID	1.3.6.1.4.1.16177.1.400.1.1.3.4.1.21

8.1.13.11.1.15 cfgWlanFFreq20

Frequency in MHz, 0 is interpreted as empty

Applies to AP and STA.

Type	Integer32
Access	readwrite
OID	1.3.6.1.4.1.16177.1.400.1.1.3.4.1.22

8.1.13.11.1.16 cfgWlanFFreq21

Frequency in MHz, 0 is interpreted as empty

Applies to AP and STA.

Type	Integer32
Access	readwrite
OID	1.3.6.1.4.1.16177.1.400.1.1.3.4.1.23

8.1.13.11.1.17 cfgWlanFFreq22

Frequency in MHz, 0 is interpreted as empty

Applies to AP and STA.

Type	Integer32
Access	readwrite
OID	1.3.6.1.4.1.16177.1.400.1.1.3.4.1.24

8.1.13.11.1.18 cfgWlanFFreq23

Frequency in MHz, 0 is interpreted as empty

Applies to AP and STA.

Type	Integer32
Access	readwrite
OID	1.3.6.1.4.1.16177.1.400.1.1.3.4.1.25

8.1.13.11.1.19 cfgWlanFFreq1

Frequency in MHz, 0 is interpreted as empty

Applies to AP and STA.

Type	Integer32
Access	readwrite
OID	1.3.6.1.4.1.16177.1.400.1.1.3.4.1.3

8.1.13.11.1.20 cfgWlanFFreq2

Frequency in MHz, 0 is interpreted as empty

Applies to AP and STA.

Type	Integer32
Access	readwrite
OID	1.3.6.1.4.1.16177.1.400.1.1.3.4.1.4

8.1.13.11.1.21 cfgWlanFFreq3

Frequency in MHz, 0 is interpreted as empty

Applies to AP and STA.

Type	Integer32
Access	readwrite
OID	1.3.6.1.4.1.16177.1.400.1.1.3.4.1.5

8.1.13.11.1.22 cfgWlanFFreq4

Frequency in MHz, 0 is interpreted as empty

Applies to AP and STA.

Type	Integer32
Access	readwrite
OID	1.3.6.1.4.1.16177.1.400.1.1.3.4.1.6

8.1.13.11.1.23 cfgWlanFFreq5

Frequency in MHz, 0 is interpreted as empty

Applies to AP and STA.

Type	Integer32
Access	readwrite
OID	1.3.6.1.4.1.16177.1.400.1.1.3.4.1.7

8.1.13.11.1.24 cfgWlanFFreq6

Frequency in MHz, 0 is interpreted as empty

Applies to AP and STA.

Type	Integer32
Access	readwrite
OID	1.3.6.1.4.1.16177.1.400.1.1.3.4.1.8

8.1.13.11.1.25 cfgWlanFFreq7

Frequency in MHz, 0 is interpreted as empty

Applies to AP and STA.

Type	Integer32
Access	readwrite
OID	1.3.6.1.4.1.16177.1.400.1.1.3.4.1.9

8.1.13.12 cfgWlanWmeTable

Wireless Multimedia Extensions Table

Wireless Multimedia Extensions (WME) based on the IEEE 802.11e standard. It provides basic Quality of Service (QoS) features to IEEE 802.11 networks.

The levels of priority in EDCA are called access categories (ACs). The contention window (CW) can be set according to the traffic expected for each access category, with a wider window needed for categories with heavier traffic.

Applies to AP. 802.11n products only.

Range	0 - 31
OID	1.3.6.1.4.1.16177.1.400.1.1.3.5

8.1.13.12.1 cfgWlanWmeTableEntry

WME table entry

OID	1.3.6.1.4.1.16177.1.400.1.1.3.5.1
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8.1.13.12.1.1 cfgWlanWmeIndex

Table Entry Index

Range	0 - 31
Access	noaccess
OID	1.3.6.1.4.1.16177.1.400.1.1.3.5.1.1

8.1.13.12.1.2 cfgWlanWmeApAifs

Arbitration inter-frame space (AIFS)

Is used on the AP itself.

Applies to AP.

<i>Range</i>	0 - 255
<i>Access</i>	readwrite
<i>OID</i>	1.3.6.1.4.1.16177.1.400.1.1.3.5.1.10

8.1.13.12.1.3 cfgWlanWmeApBurst

Maximum length for bursting (equivalent to TxOpLimit)

This value is in units of 32us.

Is used on the AP itself.

Applies to AP. 802.11n products only.

<i>Range</i>	0 - 255
<i>Access</i>	readwrite
<i>OID</i>	1.3.6.1.4.1.16177.1.400.1.1.3.5.1.11

8.1.13.12.1.4 cfgWlanWmeld

ID of the WME parameter table

The virtual wireless interface references to this ID, specified by `cfgWlanIfaceWmeParameter`.

Applies to AP. 802.11n products only.

<i>Type</i>	Integer32
<i>Access</i>	readwrite
<i>OID</i>	1.3.6.1.4.1.16177.1.400.1.1.3.5.1.2

8.1.13.12.1.5 cfgWlanWmeAc

WME Access Category

The following categories are available:

- **none(0)** use driver default value of queue
- **BK - background(1)**
- **BE - besteffort(2)**
- **VI - video(3)**
- **VO - voice(4)**

Frames in the VO queue are never aggregated.

Applies to AP. 802.11n products only.

<i>Enumeration</i>	none (0), background (1), besteffort (2), video (3), voice (4)
<i>Access</i>	readwrite
<i>OID</i>	1.3.6.1.4.1.16177.1.400.1.1.3.5.1.3

8.1.13.12.1.6 cfgWlanWmeCwMin

Contention window minimum in exponential form

Is used on STAs connected to this AP: Real value = $(2^n) - 1$

Applies to AP. 802.11n products only.

<i>Range</i>	0 - 12
<i>Access</i>	readwrite
<i>OID</i>	1.3.6.1.4.1.16177.1.400.1.1.3.5.1.4

8.1.13.12.1.7 cfgWlanWmeCwMax

Contention window maximum in exponential form

Is used on STAs connected to this AP: Real value = $(2^n) - 1$

Applies to AP. 802.11n products only.

<i>Range</i>	0 - 12
<i>Access</i>	readwrite
<i>OID</i>	1.3.6.1.4.1.16177.1.400.1.1.3.5.1.5

8.1.13.12.1.8 cfgWlanWmeAifs

Arbitration inter-frame space (AIFS)

Is used on STAs connected to this AP.

Applies to AP. 802.11n products only.

<i>Range</i>	1 - 255
<i>Access</i>	readwrite
<i>OID</i>	1.3.6.1.4.1.16177.1.400.1.1.3.5.1.6

8.1.13.12.1.9 cfgWlanWmeTxOpMax

Transmit Opportunity

A Transmit Opportunity (TXOP) is a bound time interval during which a station can send as many frames as possible (as long as the duration of the transmissions does not exceed the maximum duration of the TXOP). A value of 0 indicates that a single MSDU or MMPDU in addition to a possible RTS/CTS or CTS to itself may be transmitted at any PHY rate for each TXOP. This value is in units of 32 us.

Is used on STAs connected to this AP.

Applies to AP. 802.11n products only.

<i>Range</i>	0 - 255
<i>Access</i>	readwrite
<i>OID</i>	1.3.6.1.4.1.16177.1.400.1.1.3.5.1.7

8.1.13.12.1.10 cfgWlanWmeApCwMin

Contention window minimum

Allowed values: 1, 3, 7, 15, 31, 63, 127, 255, 511, 1023.

Is used on the AP itself.

Applies to AP. 802.11n products only.

<i>Range</i>	1 - 1023
<i>Access</i>	readwrite
<i>OID</i>	1.3.6.1.4.1.16177.1.400.1.1.3.5.1.8

8.1.13.12.1.11 cfgWlanWmeApCwMax

Contention window maximum

Allowed values: 1, 3, 7, 15, 31, 63, 127, 255, 511, 1023

cwMax has to be greater or equal cwMin.

Is used on the AP itself.

Applies to AP. 802.11n products only.

<i>Range</i>	1 - 1023
<i>Access</i>	readwrite
<i>OID</i>	1.3.6.1.4.1.16177.1.400.1.1.3.5.1.9

8.1.13.13 cfgWlanDbgTable

Wireless Handoff Debug Parameters

<i>Range</i>	0 - 63
<i>OID</i>	1.3.6.1.4.1.16177.1.400.1.1.3.6

8.1.13.13.1 cfgWlanDbgTableEntry

Wireless Handoff Debug Parameters Entry

<i>OID</i>	1.3.6.1.4.1.16177.1.400.1.1.3.6.1
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8.1.13.13.1.1 cfgWlanDbgIndex

Table Entry Index

<i>Range</i>	0 - 63
<i>Access</i>	noaccess
<i>OID</i>	1.3.6.1.4.1.16177.1.400.1.1.3.6.1.1

8.1.13.13.1.2 cfgWlanDbgRatelimit

Persistent default value to enable/disable the rate limiter messages in standard syslog.

These log messages are subject to change. DO NOT PARSE!

Applies to STA. 802.11n products only.

<i>Enumeration</i>	disabled (0), enabled (1)
<i>Access</i>	readwrite
<i>OID</i>	1.3.6.1.4.1.16177.1.400.1.1.3.6.1.10

8.1.13.13.1.3 cfgWlanDbgLinkmonitor

Periodically sends a trap containing link information of all connected devices on this interface.

Applies to both AP and STA. 802.11n products only.

<i>Enumeration</i>	disabled (0), enabled (1)
<i>Access</i>	readwrite
<i>OID</i>	1.3.6.1.4.1.16177.1.400.1.1.3.6.1.11

8.1.13.13.1.4 cfgWlanDbgBeacontsf

Persistent default value to enable/disable the Beacon RSSI messages in standard syslog. The TS field contains the internal TSF (mactime) instead of the system uptime.

These log messages are subject to change. DO NOT PARSE!

Applies to STA. 802.11n products only.

<i>Enumeration</i>	disabled (0), enabled (1)
<i>Access</i>	readwrite
<i>OID</i>	1.3.6.1.4.1.16177.1.400.1.1.3.6.1.12

8.1.13.13.1.5 cfgWlanDbgRange

Persistent default value to enable/disable the distance measurement messages in standard syslog.

These log messages are subject to change. DO NOT PARSE!

Note: Distance value is not in meters.

Applies to STA. 802.11n products only.

<i>Enumeration</i>	disabled (0), enabled (1)
<i>Access</i>	readwrite
<i>OID</i>	1.3.6.1.4.1.16177.1.400.1.1.3.6.1.13

8.1.13.13.1.6 cfgWlanDbgReports

Persistent default value to enable/disable the periodical WLAN debug data reporting in standard syslog.

These log messages are subject to change. DO NOT PARSE!

Applies to STA. 802.11n products only.

<i>Enumeration</i>	disabled (0), enabled (1)
<i>Access</i>	readwrite
<i>OID</i>	1.3.6.1.4.1.16177.1.400.1.1.3.6.1.14

8.1.13.13.1.7 cfgWlanDbglfaceName

Name of the virtual wireless interface.

Applies to STA. 802.11n products only.

Type	DisplayString
Range	1 - 255
Access	readonly
OID	1.3.6.1.4.1.16177.1.400.1.1.3.6.1.2

8.1.13.13.1.8 cfgWlanDbgHandoff

Persistent default value to enable/disable the handoff trap.

These log messages are subject to change. DO NOT PARSE!

Applies to STA. 802.11n products only.

Enumeration	disabled (0), enabled (1)
Access	readwrite
OID	1.3.6.1.4.1.16177.1.400.1.1.3.6.1.3

8.1.13.13.1.9 cfgWlanDbgScan

Persistent default value to enable/disable the scan messages in standard syslog.

These log messages are subject to change. DO NOT PARSE!

Applies to STA. 802.11n products only.

Enumeration	disabled (0), enabled (1)
Access	readwrite
OID	1.3.6.1.4.1.16177.1.400.1.1.3.6.1.4

8.1.13.13.1.10 cfgWlanDbgMlme

Persistent default value to enable/disable the MLME messages in standard syslog.

These log messages are subject to change. DO NOT PARSE!

Applies to STA. 802.11n products only.

Enumeration	disabled (0), enabled (1)
Access	readwrite
OID	1.3.6.1.4.1.16177.1.400.1.1.3.6.1.5

8.1.13.13.1.11 cfgWlanDbgEvents

Persistent default value to enable/disable the events messages in standard syslog.

These log messages are subject to change. DO NOT PARSE!

Applies to STA. 802.11n products only.

<i>Enumeration</i>	disabled (0), enabled (1)
<i>Access</i>	readwrite
<i>OID</i>	1.3.6.1.4.1.16177.1.400.1.1.3.6.1.6

8.1.13.13.12 cfgWlanDbgBeaconrssi

Persistent default value to enable/disable the Beacon RSSI messages in commissioning syslog.

These log messages are subject to change. DO NOT PARSE!

Applies to STA. 802.11n products only.

<i>Enumeration</i>	disabled (0), enabled (1)
<i>Access</i>	readwrite
<i>OID</i>	1.3.6.1.4.1.16177.1.400.1.1.3.6.1.7

8.1.13.13.13 cfgWlanDbgAckrssi

Persistent default value to enable/disable the ACK RSSI messages in standard syslog.

These log messages are subject to change. DO NOT PARSE!

Applies to STA. 802.11n products only.

<i>Enumeration</i>	disabled (0), enabled (1)
<i>Access</i>	readwrite
<i>OID</i>	1.3.6.1.4.1.16177.1.400.1.1.3.6.1.8

8.1.13.13.14 cfgWlanDbgBeaconfiltered

Persistent default value to enable/disable the beacon filtered RSSI messages in commissioning syslog.

These log messages are subject to change. DO NOT PARSE!

Applies to STA. 802.11n products only.

<i>Enumeration</i>	disabled (0), enabled (1)
<i>Access</i>	readwrite
<i>OID</i>	1.3.6.1.4.1.16177.1.400.1.1.3.6.1.9

8.1.13.14 cfgWlanAclWhiteTable

Wireless MAC Access Control Whitelist

<i>Range</i>	0 - 63
<i>OID</i>	1.3.6.1.4.1.16177.1.400.1.1.3.7

8.1.13.14.1 cfgWlanAclWhiteTableEntry

Wireless MAC Access Control Whitelist Entry

<i>OID</i>	1.3.6.1.4.1.16177.1.400.1.1.3.7.1
------------	-----------------------------------

8.1.13.14.1.1 cfgWlanAclWhiteIndex

Table Entry Index

<i>Range</i>	0 - 63
<i>Access</i>	noaccess
<i>OID</i>	1.3.6.1.4.1.16177.1.400.1.1.3.7.1.1

8.1.13.14.1.2 cfgWlanAclWhiteEnabled

Disable or Enable this Entry in the ACL

Applies to AP. 802.11n products only.

<i>Enumeration</i>	disabled (0), enabled (1)
<i>Access</i>	readwrite
<i>OID</i>	1.3.6.1.4.1.16177.1.400.1.1.3.7.1.2

8.1.13.14.1.3 cfgWlanAclWhiteInterface

Name of the Wireless Interface

The ACL entry is on this specified wireless interface active.

Applies to AP. 802.11n products only.

<i>Type</i>	DisplayString
<i>Range</i>	1 - 255
<i>Access</i>	readwrite
<i>OID</i>	1.3.6.1.4.1.16177.1.400.1.1.3.7.1.3

8.1.13.14.1.4 cfgWlanAclWhiteAddr

MAC Address in the ACL

Examples:

- 00:14:5a:02:10:42
- 00:07:7c:00:00:00

Applies to AP. 802.11n products only.

Type	DisplayString
Range	17 - 17
Access	readwrite
OID	1.3.6.1.4.1.16177.1.400.1.1.3.7.1.4

8.1.13.14.1.5 cfgWlanAclWhiteMask

Mask of the MAC Address

Allows the use of ranges of MAC addresses. To be used like CIDR notation of IP addresses.

Examples:

- To match a single MAC address, specify a mask of 48.
- To match a vendor OUI, specify a mask of 24.

Applies to AP. 802.11n products only.

Range	1 - 48
Access	readwrite
OID	1.3.6.1.4.1.16177.1.400.1.1.3.7.1.5

8.1.13.15 cfgWlanAclBlackTable

Wireless MAC Access Control Blacklist

Range	0 - 63
OID	1.3.6.1.4.1.16177.1.400.1.1.3.8

8.1.13.15.1 cfgWlanAclBlackTableEntry

Wireless MAC Access Control Blacklist Entry

<i>OID</i>	1.3.6.1.4.1.16177.1.400.1.1.3.8.1
------------	-----------------------------------

8.1.13.15.1.1 cfgWlanAcIBlackIndex

Table Entry Index

<i>Range</i>	0 - 63
<i>Access</i>	noaccess
<i>OID</i>	1.3.6.1.4.1.16177.1.400.1.1.3.8.1.1

8.1.13.15.1.2 cfgWlanAcIBlackEnabled

Disable or Enable This Entry in the ACL

Applies to AP. 802.11n products only.

<i>Enumeration</i>	disabled (0), enabled (1)
<i>Access</i>	readwrite
<i>OID</i>	1.3.6.1.4.1.16177.1.400.1.1.3.8.1.2

8.1.13.15.1.3 cfgWlanAcIBlackInterface

Name of the Wireless Interface

The ACL entry is on this specified wireless interface active.

Applies to AP. 802.11n products only.

<i>Type</i>	DisplayString
<i>Range</i>	1 - 255
<i>Access</i>	readwrite
<i>OID</i>	1.3.6.1.4.1.16177.1.400.1.1.3.8.1.3

8.1.13.15.1.4 cfgWlanAcIBlackAddr

MAC Address in the ACL

Examples:

- 00:14:5a:02:10:42
- 00:07:7c:00:00:00

Applies to AP. 802.11n products only.

Type	DisplayString
Range	17 - 17
Access	readwrite
OID	1.3.6.1.4.1.16177.1.400.1.1.3.8.1.4

8.1.13.15.1.5 cfgWlanAclBlackMask

Mask of the MAC Address

Allows the use of ranges of MAC addresses. To be used like CIDR notation of IP addresses.

Examples:

- To match a single MAC address, specify a mask of 48.
- To match a vendor OUI, specify a mask of 24.

Applies to AP. 802.11n products only.

Range	1 - 48
Access	readwrite
OID	1.3.6.1.4.1.16177.1.400.1.1.3.8.1.5

8.1.13.16 cfgWlanGlobal

8.1.13.16.1 cfgWlanGlbCountry

Wireless country code

Note: Refer to documentation which countries are supported for your device.

Applies to AP and STA.

Type	DisplayString
Range	1 - 255
Access	readwrite
OID	1.3.6.1.4.1.16177.1.400.1.1.3.9.1

8.1.13.16.2 cfgWlanGlbLinkmonitorInterval

LinkMonitor interval in milliseconds

A new trap is sent at each interval.

Note: A short interval and/or numerous connections may affect system performance negatively.

Applies to AP and STA. 802.11n products only.

<i>Range</i>	200 - 60000
<i>Access</i>	readwrite
<i>OID</i>	1.3.6.1.4.1.16177.1.400.1.1.3.9.2

8.1.13.16.3 cfgWlanGlbLinkmonitorQmrrlogging

LinkMonitor QMRR logging

If enabled, the QMRR statistics collected for phy0 are periodically printed to syslog at the interval set in `cfgWlanGlbLinkmonitorInterval`.

Applies to AP and STA. 802.11n products only.

<i>Enumeration</i>	disabled (0), enabled (1)
<i>Access</i>	readwrite
<i>OID</i>	1.3.6.1.4.1.16177.1.400.1.1.3.9.3

8.1.13.16.4 cfgWlanGlbConnectionStatusWlanInterface

Persistent default of the volatile setting `swDrvConStatWlanIf`

The value set here is used to initialize `swDrvConStatWlanIf`. Initialisation happens on startup or configuration change.

Applies to AP and STA.

<i>Type</i>	DisplayString
<i>Range</i>	4 - 17
<i>Access</i>	readwrite
<i>OID</i>	1.3.6.1.4.1.16177.1.400.1.1.3.9.4

8.1.13.16.5 cfgWlanGlbAclRejectLog

ACL reject logging

If enabled, hostapd will log stations rejected based on MAC ACL.

Applies to AP. 802.11n products only.

<i>Enumeration</i>	disabled (0), enabled (1)
<i>Access</i>	readwrite
<i>OID</i>	1.3.6.1.4.1.16177.1.400.1.1.3.9.5

8.1.14 cfgRouting

8.1.14.1 cfgRouteDefault

8.1.14.1.1 cfgRouteDefGateway

Default Gateway

The default gateway defines the node on an IP network that serves as a router for any other network which is not defined in the routing table.

The default gateway specified here is always configured on routing table 254 with a metric of 0.

Type	IpAddress
Access	readwrite
OID	1.3.6.1.4.1.16177.1.400.1.1.4.1.1

8.1.14.1.2 cfgRouteDefGwOverride

Override the default gateway

Override a default gateway previously received via DHCP with the value in cfgRouteDefGateway. If this is disabled and a default gateway already exists it will not be changed.

When this flag is set, all existing default gateway(s) on table 254 are deleted before the default gateway in cfgRouteDefGateway is configured.

Enumeration	disabled (0), enabled (1)
Access	readwrite
OID	1.3.6.1.4.1.16177.1.400.1.1.4.1.2

8.1.14.2 cfgRouteTable

Static Routes

Range	0 - 265
OID	1.3.6.1.4.1.16177.1.400.1.1.4.2

8.1.14.2.1 cfgRouteTableEntry

Static Route

OID	1.3.6.1.4.1.16177.1.400.1.1.4.2.1
-----	-----------------------------------

8.1.14.2.1.1 cfgRouteTableIndex

Table Entry Index

<i>Range</i>	0 - 265
<i>Access</i>	noaccess
<i>OID</i>	1.3.6.1.4.1.16177.1.400.1.1.4.2.1.1

8.1.14.2.1.2 cfgRouteTableMetric

TECHPREVIEW: Metric of the Route

The metric allows to create multiple routes to the same destination. The higher a metric, the less priority the route has. Set to -1 to not set a metric. This implies a metric of 0.

<i>Range</i>	-1 - 2147483647
<i>Access</i>	readwrite
<i>OID</i>	1.3.6.1.4.1.16177.1.400.1.1.4.2.1.10

8.1.14.2.1.3 cfgRouteTableLookupTable

TECHPREVIEW: Routing Table on Which Route is Created

This is a space and/or comma separated list of tables on which the route will be created.

Specify 254 to create routes on the `main` table.

Valid values are > 0 and < 2000000000 . However in this range there are reserved values that may not be used:

- 128: prelocal
- 253: default
- 255: local

Use `cfgRouteRuleTable` to create policies which use the table specified here.

Examples:

- 5000
- 254, 7000
- 100 254, 8000

<i>Type</i>	DisplayString
<i>Access</i>	readwrite
<i>OID</i>	1.3.6.1.4.1.16177.1.400.1.1.4.2.1.11

8.1.14.2.1.4 cfgRouteTableMonitor

TECHPREVIEW: NLM Monitor Handling This Route

The referenced monitor has to be of type route (see `cfgNlmMonType`)

<i>Range</i>	-1 - 255
<i>Access</i>	readwrite
<i>OID</i>	1.3.6.1.4.1.16177.1.400.1.1.4.2.1.12

8.1.14.2.1.5 cfgRouteTableComment

User Comment

This parameter has no operational function. It allows to store a comment about the use of this route.

<i>Type</i>	DisplayString
<i>Access</i>	readwrite
<i>OID</i>	1.3.6.1.4.1.16177.1.400.1.1.4.2.1.13

8.1.14.2.1.6 cfgRouteTableEnabled

Disable or Enable Route Entry

<i>Enumeration</i>	disabled (0), enabled (1)
<i>Access</i>	readwrite
<i>OID</i>	1.3.6.1.4.1.16177.1.400.1.1.4.2.1.2

8.1.14.2.1.7 cfgRouteTableDestinationNetwork

Destination Network in CIDR Notation

<i>Type</i>	DisplayString
<i>Access</i>	readwrite
<i>OID</i>	1.3.6.1.4.1.16177.1.400.1.1.4.2.1.3

8.1.14.2.1.8 cfgRouteTableGateway

Gateway to Destination Network

Specify the IP address of the gateway over which the destination specified in `cfgRouteTableDestinationNetwork` is reachable.

Alternatively options are:

- **unreachable:** Create a route that blocks traffic and responds with ICMP code 1 (Host unreachable)
- **prohibit:** Create a route that blocks traffic and responds with ICMP code 13 (Communication administratively filtered)
- **blackhole:** Create a route that silently drops all traffic

<i>Type</i>	DisplayString
<i>Range</i>	5 - 50
<i>Access</i>	readwrite
<i>OID</i>	1.3.6.1.4.1.16177.1.400.1.1.4.2.1.5

8.1.14.2.1.9 cfgRouteTableSource

Source for Traffic to Destination Network

Optional, use only if you have multiple possible sources.

<i>Type</i>	DisplayString
<i>Range</i>	5 - 50
<i>Access</i>	readwrite
<i>OID</i>	1.3.6.1.4.1.16177.1.400.1.1.4.2.1.6

8.1.14.2.1.10 cfgRouteTableCarpld

The CARP Instance Which Brings the Route Up

Has to be set to -1 when this is a normal route and should not be handled by a CARP instance. All routes which have a value 0..15 are brought up by the respective CARP instance when it becomes a master for an IP. This allows to create routes which are routed over a CARP-Address.

<i>Range</i>	-1 - 15
<i>Access</i>	readwrite
<i>OID</i>	1.3.6.1.4.1.16177.1.400.1.1.4.2.1.8

8.1.14.2.1.11 cfgRouteTableInterface

TECHPREVIEW: Interface of Static Route

The interface on which the static route is created. Usually this is not required, and the system determines automatically to which interface a route applies to.

Leave this on none unless you need an explicit interface for a route. This is required when using IPsec with VTI (see `cfgVpnIpsecGlb1VirtualTunnelInterface`).

Type	DisplayString
Access	readwrite
OID	1.3.6.1.4.1.16177.1.400.1.1.4.2.1.9

8.1.14.3 cfgMRouteTable

Static Multicast Routes

Range	0 - 255
OID	1.3.6.1.4.1.16177.1.400.1.1.4.3

8.1.14.3.1 cfgMRouteTableEntry

Static Multicast Routes

OID	1.3.6.1.4.1.16177.1.400.1.1.4.3.1
-----	-----------------------------------

8.1.14.3.1.1 cfgMRouteTableIndex

Table Entry Index

Range	0 - 255
Access	noaccess
OID	1.3.6.1.4.1.16177.1.400.1.1.4.3.1.1

8.1.14.3.1.2 cfgMRouteTableEnabled

Enable/Disable this multicast route entry

Applies to AP and STA.

Enumeration	disabled (0), enabled (1)
Access	readwrite
OID	1.3.6.1.4.1.16177.1.400.1.1.4.3.1.2

8.1.14.3.1.3 cfgMRouteTableInput

Input Interface

Applies to AP and STA.

Type	DisplayString
Range	1 - 255
Access	readwrite
OID	1.3.6.1.4.1.16177.1.400.1.1.4.3.1.3

8.1.14.3.1.4 cfgMRouteTableSource

Unicast source address to listen to

If it is set to 0.0.0.0 all multicast traffic will be forwarded.

Applies to AP and STA.

Type	IpAddress
Access	readwrite
OID	1.3.6.1.4.1.16177.1.400.1.1.4.3.1.4

8.1.14.3.1.5 cfgMRouteTableGroup

Multicast group to forward

Applies to AP and STA.

Type	IpAddress
Access	readwrite
OID	1.3.6.1.4.1.16177.1.400.1.1.4.3.1.5

8.1.14.3.1.6 cfgMRouteTableOutput

Output interface(s)

Can be a list of interface names separated by spaces.

Applies to AP and STA.

Type	DisplayString
Range	1 - 255
Access	readwrite
OID	1.3.6.1.4.1.16177.1.400.1.1.4.3.1.6

8.1.14.4 cfgRouteRuleTable

TECHPREVIEW: Routing Rules

Also called policy routing.

These rules may be used to match frames based on:

- Source (cfgRouteRuleFrom)
- Destination (cfgRouteRuleTo)
- Input Interface (cfgRouteRuleInputInterface)
- Output Interface (cfgRouteRuleOutputInterface)
- TOS (Type of Service, QoS) value (cfgRouteRuleTos)

These frames are then processed with the specified routing table in cfgRouteRuleLookupTable.

Each rule should have its own unique priority set via cfgRouteRulePreference. Thus multiple overlapping rules may be created.

<i>Range</i>	0 - 255
<i>OID</i>	1.3.6.1.4.1.16177.1.400.1.1.4.4

8.1.14.4.1 cfgRouteRuleTableEntry

Routing Rule

<i>OID</i>	1.3.6.1.4.1.16177.1.400.1.1.4.4.1
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8.1.14.4.1.1 cfgRouteRuleIndex

Table Entry Index

<i>Range</i>	0 - 255
<i>Access</i>	noaccess
<i>OID</i>	1.3.6.1.4.1.16177.1.400.1.1.4.4.1.1

8.1.14.4.1.2 cfgRouteRuleLookupTable

TECHPREVIEW: Routing Table to Lookup

Frames which match the configured criteria are processed by the routing table specified here.

The default routing table is table 254 - the main table. All traffic not explicitly directed to a different table is processed by the main table.

Reserved routing table numbers may not be used and are:

- 128: prelocal

- 253: default
- 255: local

<i>Range</i>	1 - 1999999999
<i>Access</i>	readwrite
<i>OID</i>	1.3.6.1.4.1.16177.1.400.1.1.4.4.1.12

8.1.14.4.1.3 cfgRouteRuleEnabled

TECHPREVIEW: Disable or Enable Rule Entry

<i>Enumeration</i>	disabled (0), enabled (1)
<i>Access</i>	readwrite
<i>OID</i>	1.3.6.1.4.1.16177.1.400.1.1.4.4.1.2

8.1.14.4.1.4 cfgRouteRulePreference

TECHPREVIEW: Priority of the Rule

The Priority defines in what order rules are evaluated. This enables multiple overlapping rules. The lower the value, the higher the priority.

Each match should have it's own unique priority.

<i>Range</i>	10000 - 1999999999
<i>Access</i>	readwrite
<i>OID</i>	1.3.6.1.4.1.16177.1.400.1.1.4.4.1.3

8.1.14.4.1.5 cfgRouteRuleFrom

TECHPREVIEW: Source Prefix

This is an address or network in CIDR notation.

Set to any to ignore this field.

<i>Type</i>	DisplayString
<i>Access</i>	readwrite
<i>OID</i>	1.3.6.1.4.1.16177.1.400.1.1.4.4.1.4

8.1.14.4.1.6 cfgRouteRuleTo

TECHPREVIEW: Destination Prefix



This is an address or network in CIDR notation.

Set to any to ignore this field.

Type	DisplayString
Access	readwrite
OID	1.3.6.1.4.1.16177.1.400.1.1.4.4.1.5

8.1.14.4.1.7 cfgRouteRuleInputInterface

TECHPREVIEW: Input Interface

Match an interface on which traffic is received. If the interface is the loopback (lo), only traffic originating on the device itself is matched. Thus different rules for local traffic and forwarding traffic can be created.

Set to any to ignore this field.

Type	DisplayString
Access	readwrite
OID	1.3.6.1.4.1.16177.1.400.1.1.4.4.1.6

8.1.14.4.1.8 cfgRouteRuleOutputInterface

TECHPREVIEW: Output Interface

Select the outgoing device to match. The outgoing interface is only available for packets originating from local sockets that are bound to an interface.

Set to `any` to ignore this field.

<i>Type</i>	DisplayString
<i>Access</i>	readwrite
<i>OID</i>	1.3.6.1.4.1.16177.1.400.1.1.4.4.1.7

8.1.14.4.1.9 cfgRouteRuleTos

TECHPREVIEW: Type Of Service

Match frames based on the TOS field in their IP header.

0	1	2	3	4	5	6	7	+	—	+	—	+	—	+	—	+	—	+	—																	
										PRECEDENCE									TOS					MBZ												

Thus valid values are:

- 0

- 2
- 4
- 6
- 8
- a
- c
- e
- 10
- 12
- 14
- 16
- 18
- 1a
- 1c
- 1e

Set to any to ignore this field.

Type	DisplayString
Access	readwrite
OID	1.3.6.1.4.1.16177.1.400.1.1.4.4.1.8

8.1.14.5 cfgRouteDhcpTable

TECHPREVIEW: DHCP Received Routes

This table may be used to specify the metric and table of routes received via DHCP.

- Interface (cfgRouteDhcpInterface)
- Metric (cfgRouteDhcpMetric)
- Lookup Table (cfgRouteDhcpLookupTable)
- Monitor (cfgRouteDhcpMonitor)

Range	0 - 255
OID	1.3.6.1.4.1.16177.1.400.1.1.4.5

8.1.14.5.1 cfgRouteDhcpTableEntry

DHCP Received Routes

OID	1.3.6.1.4.1.16177.1.400.1.1.4.5.1
-----	-----------------------------------

8.1.14.5.1.1 cfgRouteDhcpIndex

Table Entry Index

<i>Range</i>	0 - 255
<i>Access</i>	noaccess
<i>OID</i>	1.3.6.1.4.1.16177.1.400.1.1.4.5.1.1

8.1.14.5.1.2 cfgRouteDhcpEnabled

TECHPREVIEW: Disable or Enable Entry

<i>Enumeration</i>	disabled (0), enabled (1)
<i>Access</i>	readwrite
<i>OID</i>	1.3.6.1.4.1.16177.1.400.1.1.4.5.1.2

8.1.14.5.1.3 cfgRouteDhcpInterface

TECHPREVIEW: Interface of DHCP Client

<i>Type</i>	DisplayString
<i>Access</i>	readwrite
<i>OID</i>	1.3.6.1.4.1.16177.1.400.1.1.4.5.1.3

8.1.14.5.1.4 cfgRouteDhcpMetric

TECHPREVIEW: Metric of Route(s) Received

The metric allows to create multiple routes to the same destination. The higher a metric, the less priority the route has.

Set to -1 to not explicitly set a metric. Depending on the DHCP client a different metric is implied:

- **wwan**: 600
- **eth**: 400
- **wlan**: 400
- **ovpn**: 200

<i>Range</i>	-1 - 2147483647
<i>Access</i>	readwrite
<i>OID</i>	1.3.6.1.4.1.16177.1.400.1.1.4.5.1.4

8.1.14.5.1.5 cfgRouteDhcpLookupTable

TECHPREVIEW: Routing Table on Which Route is Created

This is a space and/or comma separated list of tables on which the route will be created.

Specify 254 to create routes on the `main` table.

Valid values are > 0 and < 2000000000 . However in this range there are reserved values that may not be used:

- 128: prelocal
- 253: default
- 255: local

Use `cfgRouteRuleTable` to create policies which use the table specified here.

Examples:

- 5000
- 254, 7000
- 100 254, 8000

Type	DisplayString
Access	readwrite
OID	1.3.6.1.4.1.16177.1.400.1.1.4.5.1.5

8.1.14.5.1.6 `cfgRouteDhcpMonitor`

TECHPREVIEW: NLM Monitor That Monitors This Route

This is a reference to an NLM instance (`cfgNlmMonIndex`). If the routes received should not be monitored, set to `-1`.

Range	-1 - 255
Access	readwrite
OID	1.3.6.1.4.1.16177.1.400.1.1.4.5.1.6

8.1.15 `cfgNlm`

8.1.15.1 `cfgNlmGlobal`

8.1.15.1.1 `cfgNlmGlbEnabled`

Enable the Network Link Monitor.

Applies to AP and STA.

<i>Enumeration</i>	disabled (0), enabled (1)
<i>Access</i>	readwrite
<i>OID</i>	1.3.6.1.4.1.16177.1.400.1.1.40.1.1

8.1.15.2 cfgNlmMonitorTable

NLM Monitor Table

<i>Range</i>	0 - 255
<i>OID</i>	1.3.6.1.4.1.16177.1.400.1.1.40.2

8.1.15.2.1 cfgNlmMonitorTableEntry

NLM Monitor Table Entry

<i>OID</i>	1.3.6.1.4.1.16177.1.400.1.1.40.2.1
------------	------------------------------------

8.1.15.2.1.1 cfgNlmMonIndex

Table Entry Index

<i>Range</i>	0 - 255
<i>Access</i>	noaccess
<i>OID</i>	1.3.6.1.4.1.16177.1.400.1.1.40.2.1.1

8.1.15.2.1.2 cfgNlmMonUpAction

NLM 'UP' State Monitor Action

The action is executed on a monitor-state transition to 'up'.

Set to 0 to disable (i.e. no action).

Supported actions:

1xxx (offset: 1000, x: CARP group from 0 to 255) Un-demote CARP group defined by `cfgNetCarpLocalInterfa`

20xx (offset: 2000, x: wlan interface from 0 to 15) Enable Access Point operation on the wireless interface. Note: Currently all wireless interfaces on radio0 are enabled.

8000: Bring up all routes which reference to this NLM instance. References are defined via `cfgRouteTableMonitor` and `cfgRouteDhcpMonitor`.

8001: Bring down all routes which reference to this NLM instance. References are defined via `cfgRouteTableMonitor` and `cfgRouteDhcpMonitor`.

<i>Range</i>	0 - 8001
<i>Access</i>	readwrite
<i>OID</i>	1.3.6.1.4.1.16177.1.400.1.1.40.2.1.10

8.1.15.2.1.3 cfgNlmMonDownAction

NLM 'DOWN' State Monitor Action

The action is executed on a monitor-state transition to 'down'.

Set to 0 to disable (i.e. no action).

Supported actions:

1xxx (offset: 1000, x: CARP group from 1 to 255) Demote CARP group defined by `cfgNetCarpLocalInterfaceG`

20xx (offset: 2000, x: wlan interface from 0 to 15) Disable Access Point operation on the wireless interface. Note: Currently all wireless interfaces on radio0 are disabled.

8000: Bring down all routes which reference to this NLM instance. References are defined via `cfgRouteTableMonitor` and `cfgRouteDhcpMonitor`.

8001: Bring up all routes which reference to this NLM instance. References are defined via `cfgRouteTableMonitor` and `cfgRouteDhcpMonitor`.

<i>Range</i>	0 - 8001
<i>Access</i>	readwrite
<i>OID</i>	1.3.6.1.4.1.16177.1.400.1.1.40.2.1.11

8.1.15.2.1.4 cfgNlmMonScanLoopInterval

Scan Loop Debounce Interval in Milliseconds

If set to non-zero it will mark interface 'down' after receiving scan loop trap 415 and mark it 'up' after scan loop interval if no other 415 events have been received.

Applies to STA. 802.11n products only.

<i>Range</i>	0 - 2147483647
<i>Access</i>	readwrite
<i>OID</i>	1.3.6.1.4.1.16177.1.400.1.1.40.2.1.12

8.1.15.2.1.5 cfgNlmMonEnabled

Disable or Enable this Entry in the Monitor List

<i>Enumeration</i>	disabled (0), enabled (1)
<i>Access</i>	readwrite
<i>OID</i>	1.3.6.1.4.1.16177.1.400.1.1.40.2.1.2

8.1.15.2.1.6 cfgNlmMonInterval

Execution Interval of this Monitor in Milliseconds

The minimum interval is 100ms.

In case of an event driven monitor (e.g. `cfgNlmMonType` set to **wlan(2)**), this defines a timeout.

<i>Range</i>	0 - 2147483647
<i>Access</i>	readwrite
<i>OID</i>	1.3.6.1.4.1.16177.1.400.1.1.40.2.1.3

8.1.15.2.1.7 cfgNlmMonCount

NLM Monitor Count

The number of times the measured criteria has to be down, until the monitor is reported as down.

<i>Range</i>	0 - 65535
<i>Access</i>	readwrite
<i>OID</i>	1.3.6.1.4.1.16177.1.400.1.1.40.2.1.4

8.1.15.2.1.8 cfgNlmMonType

Objects that can be Monitored

- **phy(0)** monitor checks periodically the link status of the ethernet interfaces defined by `cfgNlmMonInterfaces`. If at least one interface in the list specified is up the monitor is considered up.
- **icmp(1)** monitor pings periodically the destination defined by `cfgNlmMonDestination`. If the destination does reply to the ECHO request within the `cfgNlmMonInterval` the monitor is considered up.
- **wlan(2)** monitor listens to link status events of the wireless interface defined by `cfgNlmMonInterfaces`.

The wlan monitor consists of 3 components:

Long Handoff Detector - Triggers when after disassociation no authorization event is detected within the configured time in `cfgNlmMonInterval`.

Scan Loop Detector - Triggers immediately on trap 415. This happens if there is no AP or only a single AP which stays below/above the Handoff thresholds. This trap is only generated when `cfgWlanHoProfile` is set to 2 or higher.

Handoff Loop Detector - Triggers if there have been `cfgNlmMonCount` number of Handoff events within `cfgNlmMonCount * (cfgNlmMonInterval + cfgNlmMonScanLoopInterval)`.

The wlan monitor recovers:

Long Handoff Detector - Immediately after the next successful authorization.

Scan Loop Detector - After the time of the last 415 trap event + the configured `cfgNlmMonScanLoopInterval`. When down, this is checked regularly in `cfgNlmMonScanLoopInterval` intervals.

Handoff Loop Detector - When there are less than `cfgNlmMonCount` Handoff events within the time-window `cfgNlmMonCount * (cfgNlmMonInterval + cfgNlmMonScanLoopInterval)`. When down, this is checked regularly in `cfgNlmMonScanLoopInterval` intervals.

- **route(3)** monitor is the same as `icmp(1)`, but it binds the source interface statically. It can only be used with actions 8000 and 8001 to bring up and tear down routes dynamically. When `cfgNlmMonDestination` is set to 0.0.0.0, this monitor is dynamically set to the received default gateway by the DHCP client. When a specific IP is set, this IP will be checked instead. This is usually only necessary when the gateway does not respond to ICMP requests.

Note: WLAN monitor is supported for 802.11n products only.

<i>Enumeration</i>	phy (0), icmp (1), wlan (2), route (3)
<i>Access</i>	readwrite
<i>OID</i>	1.3.6.1.4.1.16177.1.400.1.1.40.2.1.5

8.1.15.2.1.9 cfgNlmMonInterfaces

NLM Monitor Interface(s)

This parameter specifies the name of the interface(s) which are monitored.

Depending on the content of `cfgNlmMonType`:

- **phy(0):** List of space and/or comma separated interfaces
- **wlan(2):** A single wlan interface (802.11n products only)

All other modes ignore this parameter.

Examples:

- eth0, eth1

- wlan0

<i>Type</i>	DisplayString
<i>Range</i>	1 - 255
<i>Access</i>	readwrite
<i>OID</i>	1.3.6.1.4.1.16177.1.400.1.1.40.2.1.6

8.1.15.2.1.10 cfgNlmMonDestination

NLM Monitor Destination

This parameter specifies the IPv4 address which is to be monitored.

It will only be used when `cfgNlmMonType` is set to **icmp(1)**.

<i>Type</i>	IpAddress
<i>Access</i>	readwrite
<i>OID</i>	1.3.6.1.4.1.16177.1.400.1.1.40.2.1.7

8.1.16 cfgIpTables

8.1.17 cfgQos

8.1.17.1 cfgQosL3PrioEnabled

Layer 3 Prioritization

Controls IP Precedence based priority assignments.

Actual prioritization on the wireless link only occurs if `cfgQosWmeEnabled` is enabled as well.

Applies to AP and STA. 802.11n products only.

<i>Enumeration</i>	disabled (0), enabled (1)
<i>Access</i>	readwrite
<i>OID</i>	1.3.6.1.4.1.16177.1.400.1.1.6.1

8.1.17.2 cfgQosDscpToTidMapTable

DSCP To TID Map

Mapping table from DSCP class selector (IP TOS) to wireless priority (TID).

Applies to AP and STA. 802.11n products only.

<i>Range</i>	0 - 7
<i>OID</i>	1.3.6.1.4.1.16177.1.400.1.1.6.2

8.1.17.2.1 cfgQosDscpToTidMapTableEntry

QoS DSCP To TID Mapping Table Entry

<i>OID</i>	1.3.6.1.4.1.16177.1.400.1.1.6.2.1
------------	-----------------------------------

8.1.17.2.1.1 cfgQosDscpToTidMapTableIndex

QoS DSCP To TID Mapping Table Index

<i>Range</i>	0 - 7
<i>Access</i>	noaccess
<i>OID</i>	1.3.6.1.4.1.16177.1.400.1.1.6.2.1.1

8.1.17.2.1.2 cfgQosDscpToTidMapValue

Layer 2 Priorities For IP Precedence Values 0-7

Applies to AP and STA. 802.11n products only.

<i>Range</i>	0 - 7
<i>Access</i>	readwrite
<i>OID</i>	1.3.6.1.4.1.16177.1.400.1.1.6.2.1.2

8.1.17.3 cfgQosVlanToTidMapTable

802.1p To TID Map

Mapping table from layer 2 priorities (802.1p) to wireless priority (TID).

<i>Range</i>	0 - 7
<i>OID</i>	1.3.6.1.4.1.16177.1.400.1.1.6.3

8.1.17.3.1 cfgQosVlanToTidMapTableEntry

QoS VLAN To TID Priority Table Entry

<i>OID</i>	1.3.6.1.4.1.16177.1.400.1.1.6.3.1
------------	-----------------------------------

8.1.17.3.1.1 cfgQosVlanToTidMapTableIndex

Table Entry Index

<i>Range</i>	0 - 7
<i>Access</i>	noaccess
<i>OID</i>	1.3.6.1.4.1.16177.1.400.1.1.6.3.1.1

8.1.17.3.1.2 cfgQosVlanToTidMapValue

Layer 2 Priorities For VLAN Priorities 0-7.

Applies to AP and STA. 802.11n products only.

<i>Range</i>	0 - 7
<i>Access</i>	readwrite
<i>OID</i>	1.3.6.1.4.1.16177.1.400.1.1.6.3.1.2

8.1.17.4 cfgQosIpToTidMapTable

IP Header To TID Map

Mapping table from IP header (Source, Destination, Protocol, Port), to wireless priority (TID).

<i>Range</i>	0 - 127
<i>OID</i>	1.3.6.1.4.1.16177.1.400.1.1.6.4

8.1.17.4.1 cfgQosIpToTidMapTableEntry

IP To TID Priority Table Entry

<i>OID</i>	1.3.6.1.4.1.16177.1.400.1.1.6.4.1
------------	-----------------------------------

8.1.17.4.1.1 cfgQosIpToTidMapTableIndex

IP To TID Priority Table Index

<i>Range</i>	0 - 127
<i>Access</i>	noaccess
<i>OID</i>	1.3.6.1.4.1.16177.1.400.1.1.6.4.1.1

8.1.17.4.1.2 cfgQoslpToTidMapSrcNet

Source Network For IP Prioritization Rule

In CIDR format.

Applies to AP and STA. 802.11n products only.

<i>Type</i>	OctetString
<i>Access</i>	readwrite
<i>OID</i>	1.3.6.1.4.1.16177.1.400.1.1.6.4.1.2

8.1.17.4.1.3 cfgQoslpToTidMapDestNet

Destination Network For IP Prioritization Rule

In CIDR format.

Applies to AP and STA. 802.11n products only.

<i>Type</i>	OctetString
<i>Access</i>	readwrite
<i>OID</i>	1.3.6.1.4.1.16177.1.400.1.1.6.4.1.3

8.1.17.4.1.4 cfgQoslpToTidMapProto

Protocol For IP Prioritization Rule

Allowed protocols are:

- **any(0)**
- **udp(1)**
- **tcp(2)**

Applies to AP and STA. 802.11n products only.

<i>Enumeration</i>	any (0), udp (1), tcp (2)
<i>Access</i>	readwrite
<i>OID</i>	1.3.6.1.4.1.16177.1.400.1.1.6.4.1.4

8.1.17.4.1.5 cfgQosIpToTidMapSrcPort

Source Port For IP Prioritization Rule

Use port -1 to match any port. This setting can only be used if the protocol is set to udp or tcp.

Applies to AP and STA. 802.11n products only.

<i>Range</i>	-1 - 65536
<i>Access</i>	readwrite
<i>OID</i>	1.3.6.1.4.1.16177.1.400.1.1.6.4.1.5

8.1.17.4.1.6 cfgQosIpToTidMapDestPort

Destination Port For IP Prioritization Rule

Use port -1 to match any port. This setting can only be used if the protocol is set to udp or tcp.

Applies to AP and STA. 802.11n products only.

<i>Range</i>	-1 - 65536
<i>Access</i>	readwrite
<i>OID</i>	1.3.6.1.4.1.16177.1.400.1.1.6.4.1.6

8.1.17.4.1.7 cfgQosIpToTidMapPrecedence

Precedence To Set For IP Prioritization Rule

Applies to AP and STA. 802.11n products only.

<i>Range</i>	0 - 7
<i>Access</i>	readwrite
<i>OID</i>	1.3.6.1.4.1.16177.1.400.1.1.6.4.1.7

8.1.17.4.1.8 cfgQosIpToTidMapEnabled

Disable Or Enable IP Prioritisation Rule

Applies to AP and STA. 802.11n products only.

<i>Enumeration</i>	disabled (0), enabled (1)
<i>Access</i>	readwrite
<i>OID</i>	1.3.6.1.4.1.16177.1.400.1.1.6.4.1.8

8.1.17.5 cfgQosDefaultTid

Default TID For Frames That Do Not Match Any Other Rule

The default TID is set to to frames to be transmitted that do not match the mode to which `cfgQosL3PrioEnabled` is set and there is no matching entry in `cfgQosEthertypeToL2Table`. This means when `cfgQosL3PrioEnabled` is set to **disabled(0)**, that the frame is not a VLAN frame. And when `cfgQosL3PrioEnabled` is set to **enabled(1)**, that the frame is not an IP frame.

Applies to AP and STA. 802.11n products only.

<i>Range</i>	0 - 7
<i>Access</i>	readwrite
<i>OID</i>	1.3.6.1.4.1.16177.1.400.1.1.6.5

8.1.17.6 cfgQosEthertypeToL2Table

Set TID For Specific Ethertype

The ethertype rules are applied, when the frame to be transmitted does not match the mode to which `cfgQosL3PrioEnabled` is set. This means when `cfgQosL3PrioEnabled` is set to **disabled(0)**, that the frame is not a VLAN frame. And when `cfgQosL3PrioEnabled` is set to **enabled(1)**, that the frame is not an IP frame.

Applies to AP and STA. 802.11n products only.

<i>Range</i>	0 - 127
<i>OID</i>	1.3.6.1.4.1.16177.1.400.1.1.6.6

8.1.17.6.1 cfgQosEthertypeToL2TableEntry

Ethertype To TID Table Entry

<i>OID</i>	1.3.6.1.4.1.16177.1.400.1.1.6.6.1
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8.1.17.6.1.1 cfgQosEthertypeToL2TableIndex

IP To TID Priority Table Index

<i>Range</i>	0 - 127
<i>Access</i>	noaccess
<i>OID</i>	1.3.6.1.4.1.16177.1.400.1.1.6.6.1.1

8.1.17.6.1.2 cfgQosEthertypeToL2Enabled

Disable Or Enable Ethertype Rule

Applies to AP and STA. 802.11n products only.

<i>Enumeration</i>	disabled (0), enabled (1)
<i>Access</i>	readwrite
<i>OID</i>	1.3.6.1.4.1.16177.1.400.1.1.6.6.1.2

8.1.17.6.1.3 cfgQosEthertypeToL2Ethertype

Ethertype To Match

Some popular Ethernets:

- 0800: IPv4
- 0806: ARP
- 0835: RARP
- 8100: VLAN
- 86DD: IPv6
- 8847: MPLS unicast
- 8848: MPLS multicast
- 8892: Profinet
- 9100: stacked VLAN

Applies to AP and STA. 802.11n products only.

<i>Range</i>	4 - 4
<i>Access</i>	readwrite
<i>OID</i>	1.3.6.1.4.1.16177.1.400.1.1.6.6.1.3

8.1.17.6.1.4 cfgQosEthertypeToL2Tid

TID To Set For Ethertype

Applies to AP and STA. 802.11n products only.

<i>Range</i>	0 - 7
<i>Access</i>	readwrite
<i>OID</i>	1.3.6.1.4.1.16177.1.400.1.1.6.6.1.4

8.2 rpc

8.2.1 rpcConfiguration

8.2.1.1 rpcCfgRevert

In case there are any changes in the configuration section, which are not applied yet, they can be all reverted by writing **all(1)** to this parameter.

Reading this parameter will show the status of the last RPC. A value less than 0 means an error occurred. A value of 0 is returned if the revert process was successful.

Applies to AP and STA.

<i>Enumeration</i>	allError (-1), nop (0), all (1)
<i>Access</i>	readwrite
<i>OID</i>	1.3.6.1.4.1.16177.1.400.1.3.1.1

8.2.1.2 rpcCfgApply

All changes to any parameter in the configuration section have to be applied before they come into operation. To apply all new parameters to the device, set this parameter to **all(1)**.

Reading this parameter will show the status of the apply process. A value less than 0 indicates that an error occurred during the last apply process, **nop(0)** means no operation and indicates that no apply process is in operation and no error has occurred. The return value all(1) means the apply process is still running.

Applies to AP and STA.

<i>Enumeration</i>	allError (-1), nop (0), all (1)
<i>Access</i>	readwrite
<i>OID</i>	1.3.6.1.4.1.16177.1.400.1.3.1.2

8.2.1.3 rpcCfgFile

Export or import a configuration to or from a file respectively.

Please refer to setCfgFileUrl for more information on how to set the configuration file.

Reading this parameter will show the status of the process. A value less than 0 indicates the occurrence of an error during the last process, **nop(0)** means no operation and indicates that no process is in operation and no error has occurred. A return value greater than 0 means the process is still running.

Applies to AP and STA.

<i>Enumeration</i>	errorImport (-2), errorExport (-1), nop (0), export (1), import (2)
<i>Access</i>	readwrite
<i>OID</i>	1.3.6.1.4.1.16177.1.400.1.3.1.4

8.2.2 rpcFirmware

8.2.2.1 rpcFwFlash

Start Download/Flash of a New Firmware

To flash a new firmware to the device, define a valid URL accessible by the device. Change the firmware URL parameter `setFwFileUrl` in the settings section, if needed.

- Writing **flash(2)** to this parameter will download and validate the new firmware file. if the downloaded file is recognized as a valid firmware for this device, it will be flashed to the file system of the device.
- Writing **flashWithConfig(3)** to this parameter will download the firmware and the custom config defined with `setCfgFileUrl` and validate the new firmware file. If the download is recognized as a valid firmware for this device, it will be written to the file system of the device. The supplied custom config will be applied after the upgrade.

Reading this parameter will return the status of the firmware flash process. A value of **flashError(-2)** indicates that the flash process failed during writing. A return value of **downloadError(-1)** indicates the occurrence of an error during download or validation of the firmware/config. A value of **flash(2)** indicates that the device is currently writing the firmware to the file system.

Applies to AP and STA.

<i>Enumeration</i>	flashError (-2), downloadError (-1), nop (0), flash (2), flashWithConfig (3)
<i>Access</i>	readwrite
<i>OID</i>	1.3.6.1.4.1.16177.1.400.1.3.2.1

8.2.3 rpcSystem

8.2.3.1 rpcSysReboot

Reboot system after n seconds.

Applies to AP and STA.

<i>Type</i>	Integer32
<i>Access</i>	readwrite
<i>OID</i>	1.3.6.1.4.1.16177.1.400.1.3.3.1

8.2.3.2 rpcSysFactoryReset

Factory Reset

Perform a factory reset and reboot the device. This will reset device configuration, including administrator password and certificates, to its default state.

Note: You will not be able to communicate with the device until the factory reset has finished and the device has been rebooted.

Applies to AP and STA.

<i>Enumeration</i>	nop (0), reset (1)
<i>Access</i>	readwrite
<i>OID</i>	1.3.6.1.4.1.16177.1.400.1.3.3.2

8.2.3.3 rpcSysErrorReset

Error Reset

Writing **reset(1)** to this parameter will reset all logged warning and errors of the system. The device LEDs will indicate normal operating state after result.

Applies to AP and STA.

<i>Enumeration</i>	nop (0), reset (1)
<i>Access</i>	readwrite
<i>OID</i>	1.3.6.1.4.1.16177.1.400.1.3.3.3

8.2.3.4 rpcSysKernelLogReset

Reset Kernel Logs

- Writing **reset(1)** to this parameter will clear all kernel logs.
- Reading this parameter will show the status of the process. A **nop(0)** means no operation and points out that there is no process in operation. In case the return value is greater than 0 the process is still running.

Applies to AP and STA.

<i>Enumeration</i>	nop (0), reset (1)
<i>Access</i>	readwrite
<i>OID</i>	1.3.6.1.4.1.16177.1.400.1.3.3.4

8.2.4 rpcCertificate

8.2.4.1 rpcCrtFile

Import, export or delete certificate to or from a file respectively

Please refer to `setCrtFileUrl` for more information on how to set the certificate file URL.

Reading this parameter will show the status of the process. A value less than 0 indicates that an error has occurred during the last process, **nop(0)** means no operation and points out that there is no process in operation and no error has occurred. A return value greater than 0 means the process is still running.

Applies to AP and STA.

<i>Enumeration</i>	validateerror (-4), deleteerror (-3), exporterror (-2), importerror (-1), nop (0), import (1), export (2), delete (3)
<i>Access</i>	readwrite
<i>OID</i>	1.3.6.1.4.1.16177.1.400.1.3.4.1

8.2.4.2 rpcCrtRefresh

Refresh Certificates

RPC to manipulate key material in the Certificate Store (CS). Also allows to restart services to use newly installed material.

- **negative values** Represents a failed operation corresponding with its positive value.
- **nop(0)** No Operation in progress.
- **all(1)** Move all available certificates from CS Location 'next' to CS Location 'current'.
- **restartNetwork(10000)** Same as **all(1)** but additionally restart the network (to use all 'current' certificates).
- **restartHttps(10001)** Restart uhttpd to use new key/certificate for https.

Note: The RPC will fail if there are pending config changes.

Applies to STA.

<i>Enumeration</i>	errorRestartHttps (-10001), errorRestartNetwork (-10000), errorAll (-1), nop (0), all (1), restartNetwork (10000), restartHttps (10001)
<i>Access</i>	readwrite
<i>OID</i>	1.3.6.1.4.1.16177.1.400.1.3.4.2

8.2.5 rpcDriver

8.2.5.1 rpcDrvTable

RPC driver module

<i>Range</i>	0 - 1
<i>OID</i>	1.3.6.1.4.1.16177.1.400.1.3.6.1

8.2.5.1.1 rpcDrvTableEntry

RPC driver module

<i>OID</i>	1.3.6.1.4.1.16177.1.400.1.3.6.1.1
------------	-----------------------------------

8.2.5.1.1.1 rpcDrvIndex

Table Entry Index

<i>Range</i>	0 - 1
<i>Access</i>	noaccess
<i>OID</i>	1.3.6.1.4.1.16177.1.400.1.3.6.1.1.1

8.2.5.1.1.2 rpcDrvName

Name of the Radio Device

Applies to AP.

<i>Type</i>	DisplayString
<i>Range</i>	1 - 255
<i>Access</i>	readonly
<i>OID</i>	1.3.6.1.4.1.16177.1.400.1.3.6.1.1.2

8.2.5.1.1.3 rpcDrvDfsSimulateRadar

Simulate Radar Detection on the Current Channel

Applies to AP.

<i>Enumeration</i>	nop (0), fire (1)
<i>Access</i>	readwrite
<i>OID</i>	1.3.6.1.4.1.16177.1.400.1.3.6.1.1.5

8.3 settings

8.3.1 setConfiguration

8.3.1.1 setCfgFileUrl

Configuration File URL

The URL defines the location of the configuration file where it will be downloaded from or uploaded to when using `rpcCfgFile`.

For import files to the device the TFTP and HTTP protocols are supported. For export from the device to a server only the TFTP protocol is supported.

Example:

- `tftp://192.168.1.1/device.cfg`
- `http://192.168.1.1/device.cfg`

Applies to AP and STA.

<i>Type</i>	DisplayString
<i>Range</i>	1 - 255
<i>Access</i>	readwrite
<i>OID</i>	1.3.6.1.4.1.16177.1.400.1.4.1.1

8.3.1.2 setCfgFileFormat

Configuration File Format

This parameter specifies which format shall be used to export the current configuration.

- **snmp(0)** A flat list of the configuration parameters in an SNMP-based format.
- **cli(1)** A hierarchical structure of the configuration parameters in the way they are used in the CLI.

Note: Because the configuration format is automatically recognised whilst importing configurations, this parameter has no influence in this case.

Applies to AP and STA.

<i>Enumeration</i>	snmp (0), cli (1)
<i>Access</i>	readwrite
<i>OID</i>	1.3.6.1.4.1.16177.1.400.1.4.1.4

8.3.2 setWireless

8.3.2.1 setWlanDeviceTable

Wireless Hardware Modules

<i>Range</i>	0 - 1
<i>OID</i>	1.3.6.1.4.1.16177.1.400.1.4.3.1

8.3.2.1.1 setWlanDeviceTableEntry

Wireless Hardware Module

<i>OID</i>	1.3.6.1.4.1.16177.1.400.1.4.3.1.1
------------	-----------------------------------

8.3.2.1.1.1 setWlanDevIndex

Table Entry Index

<i>Range</i>	0 - 1
<i>Access</i>	noaccess
<i>OID</i>	1.3.6.1.4.1.16177.1.400.1.4.3.1.1.1

8.3.2.1.1.2 setWlanDevName

Name of the Wireless Device

Applies to AP and STA.

Type	DisplayString
Range	1 - 255
Access	readonly
OID	1.3.6.1.4.1.16177.1.400.1.4.3.1.1.2

8.3.2.1.1.3 setWlanDevRfOutput

RF Output

- **interfaceDisabled(-1)** The interface is disabled. It is not possible to change this setting.
- **rfOutputOff(0)** Set the wlan interface down, stop transmitting.
- **rfOutputOn(1)** Set the wlan interface up, start transmitting.

Applies to AP and STA. 802.11n products only.

Enumeration	interfaceDisabled (-1), rfOutputOff (0), rfOutputOn (1)
Access	readwrite
OID	1.3.6.1.4.1.16177.1.400.1.4.3.1.1.3

8.3.2.1.1.4 setWlanDevFrequency

Wireless Frequency in MHz

Set and get the operating frequency of the device (radio).

Set frequency is supported on 802.11n products only.

Applies to AP and STA.

Type	Integer32
Access	readwrite
OID	1.3.6.1.4.1.16177.1.400.1.4.3.1.1.6

8.3.2.1.1.5 setWlanDevPower

Wireless Output Power

Output power as effective isotropic radiated power (EIRP) in dBm including antenna gain.

Applies to AP and STA. 802.11n products only.

Type	Integer32
Access	readwrite
OID	1.3.6.1.4.1.16177.1.400.1.4.3.1.1.8

8.3.2.2 setWlanDbgTable

Wireless Handoff Debug Parameters

Range	0 - 63
OID	1.3.6.1.4.1.16177.1.400.1.4.3.6

8.3.2.2.1 setWlanDbgTableEntry

Wireless Handoff Debug Parameters Entry

OID	1.3.6.1.4.1.16177.1.400.1.4.3.6.1
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8.3.2.2.1.1 setWlanDbgIndex

Table Entry Index

Range	0 - 63
Access	noaccess
OID	1.3.6.1.4.1.16177.1.400.1.4.3.6.1.1

8.3.2.2.1.2 setWlanDbgRatelimit

Volatile setting to enable/disable the rate limiter messages in standard syslog.

These log messages are subject to change. DO NOT PARSE!

Applies to STA. 802.11n products only.

Enumeration	disabled (0), enabled (1)
Access	readwrite
OID	1.3.6.1.4.1.16177.1.400.1.4.3.6.1.10

8.3.2.2.1.3 setWlanDbgBeacontsf

Volatile setting to enable/disable the Beacon RSSI messages in standard syslog. The TS field contains the internal TSF (mactime) instead of the system uptime.

These log messages are subject to change. DO NOT PARSE!

Applies to STA. 802.11n products only.

<i>Enumeration</i>	disabled (0), enabled (1)
<i>Access</i>	readwrite
<i>OID</i>	1.3.6.1.4.1.16177.1.400.1.4.3.6.1.12

8.3.2.2.1.4 setWlanDbgRange

Volatile setting to enable/disable the distance (range) measurement messages in standard syslog.

Note: Distance value is not in meters.

These log messages are subject to change. DO NOT PARSE!

Applies to STA. 802.11n products only.

<i>Enumeration</i>	disabled (0), enabled (1)
<i>Access</i>	readwrite
<i>OID</i>	1.3.6.1.4.1.16177.1.400.1.4.3.6.1.13

8.3.2.2.1.5 setWlanDbgReports

Volatile setting to enable/disable the periodical WLAN debug data reporting in standard syslog.

These log messages are subject to change. DO NOT PARSE!

Applies to STA.

<i>Enumeration</i>	disabled (0), enabled (1)
<i>Access</i>	readwrite
<i>OID</i>	1.3.6.1.4.1.16177.1.400.1.4.3.6.1.14

8.3.2.2.1.6 setWlanDbgIfaceName

Name of the virtual wireless interface.

Applies to AP and STA. 802.11n products only.

<i>Type</i>	DisplayString
<i>Range</i>	1 - 255
<i>Access</i>	readonly
<i>OID</i>	1.3.6.1.4.1.16177.1.400.1.4.3.6.1.2

8.3.2.2.1.7 setWlanDbgHandoff

Volatile setting to enable/disable the handoff trap.

These log messages are subject to change. DO NOT PARSE!

Applies to STA. 802.11n products only.

<i>Enumeration</i>	disabled (0), enabled (1)
<i>Access</i>	readwrite
<i>OID</i>	1.3.6.1.4.1.16177.1.400.1.4.3.6.1.3

8.3.2.2.1.8 setWlanDbgScan

Volatile setting to enable/disable the scan messages in standard syslog.

These log messages are subject to change. DO NOT PARSE!

Applies to STA. 802.11n products only.

<i>Enumeration</i>	disabled (0), enabled (1)
<i>Access</i>	readwrite
<i>OID</i>	1.3.6.1.4.1.16177.1.400.1.4.3.6.1.4

8.3.2.2.1.9 setWlanDbgMlme

Volatile setting to enable/disable the MLME messages in standard syslog.

These log messages are subject to change. DO NOT PARSE!

Applies to STA. 802.11n products only.

<i>Enumeration</i>	disabled (0), enabled (1)
<i>Access</i>	readwrite
<i>OID</i>	1.3.6.1.4.1.16177.1.400.1.4.3.6.1.5

8.3.2.2.1.10 setWlanDbgEvents

Volatile setting to enable/disable the events messages in standard syslog.

These log messages are subject to change. DO NOT PARSE!

Applies to STA. 802.11n products only.

<i>Enumeration</i>	disabled (0), enabled (1)
<i>Access</i>	readwrite
<i>OID</i>	1.3.6.1.4.1.16177.1.400.1.4.3.6.1.6

8.3.2.2.1.11 setWlanDbgBeaconrssi

Volatile setting to enable/disable the Beacon RSSI messages in commissioning syslog.

These log messages are subject to change. DO NOT PARSE!

Applies to STA. 802.11n products only.

<i>Enumeration</i>	disabled (0), enabled (1)
<i>Access</i>	readwrite
<i>OID</i>	1.3.6.1.4.1.16177.1.400.1.4.3.6.1.7

8.3.2.2.1.12 setWlanDbgAckrssi

Volatile setting to enable/disable the ACK RSSI messages in standard syslog.

These log messages are subject to change. DO NOT PARSE!

Applies to STA. 802.11n products only.

<i>Enumeration</i>	disabled (0), enabled (1)
<i>Access</i>	readwrite
<i>OID</i>	1.3.6.1.4.1.16177.1.400.1.4.3.6.1.8

8.3.2.2.1.13 setWlanDbgBeaconfiltered

Volatile setting to enable/disable the Beacon filtered RSSI messages in commissioning syslog.

These log messages are subject to change. DO NOT PARSE!

Applies to STA. 802.11n products only.

<i>Enumeration</i>	disabled (0), enabled (1)
<i>Access</i>	readwrite
<i>OID</i>	1.3.6.1.4.1.16177.1.400.1.4.3.6.1.9

8.3.3 setFirmware

8.3.3.1 setFwFileUrl

Firmware File URL

The URL defines from which location the new firmware will be downloaded when using the `rpcFwFlash`.

Supported protocols are TFTP and HTTP.

Example:

- `tftp://192.168.1.1/firmware.img`
- `http://192.168.1.1/firmware.img`

Applies to AP and STA.

Type	DisplayString
Range	1 - 255
Access	readwrite
OID	1.3.6.1.4.1.16177.1.400.1.4.5.1

8.3.3.2 setFwKeepConfig

Try to Import Configuration From the Previous Firmware Version

Enumeration	reset (0), keep (1)
Access	readwrite
OID	1.3.6.1.4.1.16177.1.400.1.4.5.2

8.3.4 setCertificate

8.3.4.1 setCrtFileUrl

Certificate File URL

The URL defines the location of the certificate file where it will be downloaded from or uploaded to.

For import files to the device the TFTP and HTTP protocols are supported. For export from the device to a server only the TFTP protocol is supported.

Example:

- tftp://192.168.1.1/uttpd.crt
- http://192.168.1.1/uttpd.crt

Applies to AP and STA.

Type	DisplayString
Range	1 - 255
Access	readwrite
OID	1.3.6.1.4.1.16177.1.400.1.4.6.1

8.3.4.2 setCrtFileSelector

Set this field to select which file should be downloaded/uploaded via the rpcCrtFile.

- **0** means legacy HTTPS. Upload/download certificate and key from the same url with a .crt and .key extension.
- **1** means HTTPS certificate
- **2** means HTTPS key
- **100** means certificate for wlan0, 101 for wlan1 and so on
- **200** means private key for wlan0, 201 for wlan1 and so on
- **300** means CA certificate for wlan0, 301 for wlan1 and so on
- **400** means CA2 certificate for wlan0, 401 for wlan1 and so on
- **500** means CRL certificate for wlan0, 501 for wlan1 and so on
- **600** means CRL2 certificate for wlan0, 601 for wlan1 and so on
- **700** means PKCS#12 container for wlan0, 701 for wlan1 and so on
- **10xxx** means upload/download the 'next' file which will be installed during next reboot or by calling rpcCrtRefresh. So a value of **10203** means next private key for wlan3.
- **20100** means certificate for ovpn0, 20101 for ovpn1 and so on
- **20200** means private key for ovpn0, 20201 for ovpn1 and so on
- **20300** means CA certificate for ovpn0, 20301 for ovpn1 and so on
- **20400** means static key for ovpn0, 20401 for ovpn1 and so on
- **30100** means CA certificate for ipsec0, 30301 for ipsec1 and so on
- **30200** means left certificate for ipsec0, 30201 for ipsec1 and so on
- **30300** means right certificate for ipsec0, 30301 for ipsec1 and so on
- **30400** means left sigkey for ipsec0, 30401 for ipsec1 and so on
- **30500** means right sigkey for ipsec0, 30501 for ipsec1 and so on
- **30600** means left private key for ipsec0, 30201 for ipsec1 and so on

For static keys the setCrtFileFormat will be ignored so they must be in PEM format.

Applies to AP and STA.

<i>Range</i>	0 - 90000
<i>Access</i>	readwrite
<i>OID</i>	1.3.6.1.4.1.16177.1.400.1.4.6.2

8.3.4.3 setCrtFileFormat

Set the certificate/key file format for `rpcCrtFile` actions

- **0** means the imported/exported certificate/key will be in the PEM format.
- **1** means the imported/exported certificate/key will be in the DER format.

Applies to AP and STA.

<i>Enumeration</i>	pem (0), der (1)
<i>Access</i>	readwrite
<i>OID</i>	1.3.6.1.4.1.16177.1.400.1.4.6.3

8.3.4.4 setCrtFilePkcs12Passphrase

Set the PKCS#12 passphrase used to import with `rpcCertFile`

This passphrase will be used during the import to decrypt the container data. The `cfgWlan802dot1xClientKeyPa` will then be used to encrypt the client key on the device.

Applies to AP and STA.

<i>Type</i>	DisplayString
<i>Range</i>	1 - 255
<i>Access</i>	readwrite
<i>OID</i>	1.3.6.1.4.1.16177.1.400.1.4.6.4

8.3.5 setSystem

8.3.5.1 setSysTime

System time as epoch

<i>Type</i>	Integer32
<i>Access</i>	readwrite
<i>OID</i>	1.3.6.1.4.1.16177.1.400.1.4.7.1

8.3.6 setTechPreview

8.3.6.1 setTechPreviewEnabled

Technical Preview disabled or enabled

The Technical Preview allows access to upcoming features that are not yet officially released.

Note: This parameter is volatile and is lost after a reboot.

Applies to AP and STA.

<i>Enumeration</i>	disabled (0), enabled (1)
<i>Access</i>	readwrite
<i>OID</i>	1.3.6.1.4.1.16177.1.400.1.4.8.1

8.4 hardware

8.4.1 hwSystem

8.4.1.1 hwSysProduct

Product Type

<i>Type</i>	DisplayString
<i>Range</i>	1 - 255
<i>Access</i>	readonly
<i>OID</i>	1.3.6.1.4.1.16177.1.400.1.5.1.1

8.4.1.2 hwSysSerial

Serial Number of the Product

<i>Type</i>	DisplayString
<i>Range</i>	1 - 255
<i>Access</i>	readonly
<i>OID</i>	1.3.6.1.4.1.16177.1.400.1.5.1.2

8.4.1.3 hwSysRevision

ERP Revision of the Product

Type	Integer32
Access	readonly
OID	1.3.6.1.4.1.16177.1.400.1.5.1.3

8.4.1.4 hwSysVersion

Version of the Product

Type	Integer32
Access	readonly
OID	1.3.6.1.4.1.16177.1.400.1.5.1.4

8.4.2 hwBaseBoard

8.4.2.1 hwBbType

Product Type of the Base Board

Type	DisplayString
Range	0 - 255
Access	readonly
OID	1.3.6.1.4.1.16177.1.400.1.5.10.1

8.4.2.2 hwBbSerial

Serial Number of the Base Board

Type	DisplayString
Range	0 - 255
Access	readonly
OID	1.3.6.1.4.1.16177.1.400.1.5.10.2

8.4.2.3 hwBbRevision

ERP Revision of the Base Board

<i>Type</i>	Integer32
<i>Access</i>	readonly
<i>OID</i>	1.3.6.1.4.1.16177.1.400.1.5.10.3

8.4.2.4 hwBbVersion

Version of the Base Board

<i>Type</i>	Integer32
<i>Access</i>	readonly
<i>OID</i>	1.3.6.1.4.1.16177.1.400.1.5.10.4

8.4.2.5 hwBbPcbId

Hardware Assembly ID

<i>Type</i>	Integer32
<i>Access</i>	readonly
<i>OID</i>	1.3.6.1.4.1.16177.1.400.1.5.10.5

8.4.2.6 hwBbAssemblyId

Hardware Assembly ID

<i>Type</i>	Integer32
<i>Access</i>	readonly
<i>OID</i>	1.3.6.1.4.1.16177.1.400.1.5.10.6

8.4.3 hwIfaceBoard

8.4.3.1 hwIfBrdAssembled

Interface Board Present or Not

<i>Enumeration</i>	inexistent (0), present (1)
<i>Access</i>	readonly
<i>OID</i>	1.3.6.1.4.1.16177.1.400.1.5.11.1

8.4.3.2 hwIfBrdType**Product Type of the Interface Board**

<i>Type</i>	DisplayString
<i>Range</i>	0 - 255
<i>Access</i>	readonly
<i>OID</i>	1.3.6.1.4.1.16177.1.400.1.5.11.2

8.4.3.3 hwIfBrdSerial**Serial Number of the Interface Board**

<i>Type</i>	DisplayString
<i>Range</i>	0 - 255
<i>Access</i>	readonly
<i>OID</i>	1.3.6.1.4.1.16177.1.400.1.5.11.3

8.4.3.4 hwIfBrdRevision**ERP Revision of the Interface Board**

<i>Type</i>	Integer32
<i>Access</i>	readonly
<i>OID</i>	1.3.6.1.4.1.16177.1.400.1.5.11.4

8.4.3.5 hwIfBrdVersion**Version of the Interface Board**

<i>Type</i>	Integer32
<i>Access</i>	readonly
<i>OID</i>	1.3.6.1.4.1.16177.1.400.1.5.11.5

8.4.3.6 hwIfBrdPcbId**Hardware Assembly ID**

<i>Type</i>	Integer32
<i>Access</i>	readonly
<i>OID</i>	1.3.6.1.4.1.16177.1.400.1.5.11.6

8.4.3.7 hwIfBrdAssemblyId

Hardware Assembly ID

<i>Type</i>	Integer32
<i>Access</i>	readonly
<i>OID</i>	1.3.6.1.4.1.16177.1.400.1.5.11.7

8.4.4 hwNetwork

8.4.4.1 hwNetEthernetTable

Ethernet Network Interfaces

<i>Range</i>	0 - 2
<i>OID</i>	1.3.6.1.4.1.16177.1.400.1.5.2.1

8.4.4.1.1 hwNetEthernetTableEntry

Ethernet Network Interface

<i>OID</i>	1.3.6.1.4.1.16177.1.400.1.5.2.1.1
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8.4.4.1.1.1 hwNetEthIndex

Table Entry Index

<i>Range</i>	0 - 2
<i>Access</i>	noaccess
<i>OID</i>	1.3.6.1.4.1.16177.1.400.1.5.2.1.1.1

8.4.4.1.1.2 hwNetEthName

Name of the Ethernet Interface

Type	DisplayString
Range	1 - 255
Access	readonly
OID	1.3.6.1.4.1.16177.1.400.1.5.2.1.1.2

8.4.4.1.1.3 hwNetEthAssembled

Ethernet Interface Present or Not

Enumeration	inexistent (0), present (1)
Access	readonly
OID	1.3.6.1.4.1.16177.1.400.1.5.2.1.1.3

8.4.4.1.1.4 hwNetEthMacAddress

Ethernet MAC Address

Type	DisplayString
Range	1 - 255
Access	readonly
OID	1.3.6.1.4.1.16177.1.400.1.5.2.1.1.4

8.4.4.1.1.5 hwNetEthOperation

Ethernet Interface Plugged or Unplugged

Enumeration	down (0), up (1)
Access	readonly
OID	1.3.6.1.4.1.16177.1.400.1.5.2.1.1.5

8.4.4.1.1.6 hwNetEthSpeed

Ethernet Speed in Mbps

Type	Integer32
Access	readonly
OID	1.3.6.1.4.1.16177.1.400.1.5.2.1.1.6

8.4.4.1.1.7 hwNetEthHwIndex

Index of MAC-Address / Interface

The physical address of the Ethernet interface of the base board, since not all products are assembled the same way this is to describe how the wiring is done.

<i>Type</i>	Integer32
<i>Access</i>	readonly
<i>OID</i>	1.3.6.1.4.1.16177.1.400.1.5.2.1.1.7

8.4.5 hwSensor

8.4.5.1 hwSensorTable

Hardware sensor information table.

<i>Range</i>	0 - 15
<i>OID</i>	1.3.6.1.4.1.16177.1.400.1.5.21.1

8.4.5.1.1 hwSensorTableEntry

Hardware Sensor Entry

<i>OID</i>	1.3.6.1.4.1.16177.1.400.1.5.21.1.1
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8.4.5.1.1.1 hwSensorIndex

Table Entry Index

<i>Range</i>	0 - 15
<i>Access</i>	noaccess
<i>OID</i>	1.3.6.1.4.1.16177.1.400.1.5.21.1.1.1

8.4.5.1.1.2 hwSensorName

Name of the Hardware Sensor

<i>Type</i>	DisplayString
<i>Range</i>	1 - 255
<i>Access</i>	readonly
<i>OID</i>	1.3.6.1.4.1.16177.1.400.1.5.21.1.1.2

8.4.5.1.1.3 hwSensorUnit

Unit of the Hardware Sensor

Type	DisplayString
Range	1 - 255
Access	readonly
OID	1.3.6.1.4.1.16177.1.400.1.5.21.1.1.3

8.4.5.1.1.4 hwSensorValue

Value of the Hardware Sensor

Type	DisplayString
Range	1 - 255
Access	readonly
OID	1.3.6.1.4.1.16177.1.400.1.5.21.1.1.4

8.4.6 hwWireless

8.4.6.1 hwWlanDeviceTable

Hardware information of the wireless LAN Devices

Range	0 - 1
OID	1.3.6.1.4.1.16177.1.400.1.5.3.1

8.4.6.1.1 hwWlanDeviceTableEntry

Wireless LAN Devices

OID	1.3.6.1.4.1.16177.1.400.1.5.3.1.1
-----	-----------------------------------

8.4.6.1.1.1 hwWlanDevIndex

Table Entry Index

Range	0 - 1
Access	noaccess
OID	1.3.6.1.4.1.16177.1.400.1.5.3.1.1.1

8.4.6.1.1.2 hwWlanDevAntennaProfileId

Antenna Profile ID

Please check the user manual for antenna details.

Type	Integer32
Access	readonly
OID	1.3.6.1.4.1.16177.1.400.1.5.3.1.1.10

8.4.6.1.1.3 hwWlanDevAntennaGain

Antenna Gain in dBi

Type	Integer32
Access	readonly
OID	1.3.6.1.4.1.16177.1.400.1.5.3.1.1.11

8.4.6.1.1.4 hwWlanDevCableLoss

Cable Loss in dB

Type	Integer32
Access	readonly
OID	1.3.6.1.4.1.16177.1.400.1.5.3.1.1.12

8.4.6.1.1.5 hwWlanDevAssembled

Wireless Device Present or Not

Enumeration	inexistent (0), present (1)
Access	readonly
OID	1.3.6.1.4.1.16177.1.400.1.5.3.1.1.2

8.4.6.1.1.6 hwWlanDevType

Type of the Wireless Device

Type	DisplayString
Range	1 - 255
Access	readonly
OID	1.3.6.1.4.1.16177.1.400.1.5.3.1.1.3

8.4.6.1.1.7 hwWlanDevSerial

Serial Number / Customer Field

Type	DisplayString
Range	1 - 255
Access	readonly
OID	1.3.6.1.4.1.16177.1.400.1.5.3.1.1.4

8.4.6.1.1.8 hwWlanDevRevision**ERP Revision of the RF Board**

Type	Integer32
Access	readonly
OID	1.3.6.1.4.1.16177.1.400.1.5.3.1.1.5

8.4.6.1.1.9 hwWlanDevVersion**Version of the RF board**

Type	Integer32
Access	readonly
OID	1.3.6.1.4.1.16177.1.400.1.5.3.1.1.6

8.4.6.1.1.10 hwWlanDevPcbId**Hardware Assembly ID**

Type	Integer32
Access	readonly
OID	1.3.6.1.4.1.16177.1.400.1.5.3.1.1.7

8.4.6.1.1.11 hwWlanDevAssemblyId**Hardware Assembly ID**

Type	Integer32
Access	readonly
OID	1.3.6.1.4.1.16177.1.400.1.5.3.1.1.8

8.4.6.1.1.12 hwWlanDevMacAddress**MAC Address**

Type	DisplayString
Range	1 - 255
Access	readonly
OID	1.3.6.1.4.1.16177.1.400.1.5.3.1.1.9

8.4.6.2 hwWlanGlobal**8.4.6.2.1 hwWlanGlbRegulatoryRegionId**

Regulatory Region ID

Applies to AP and STA.

<i>Type</i>	DisplayString
<i>Range</i>	1 - 255
<i>Access</i>	readonly
<i>OID</i>	1.3.6.1.4.1.16177.1.400.1.5.3.2.2

8.4.7 hwCellular**8.4.7.1 hwCellAssembled****Cellular Module Assembled**

<i>Enumeration</i>	inexistent (0), present (1)
<i>Access</i>	readonly
<i>OID</i>	1.3.6.1.4.1.16177.1.400.1.5.50.1

8.4.7.2 hwCellType**Cellular Module Type**

<i>Type</i>	DisplayString
<i>Range</i>	0 - 255
<i>Access</i>	readonly
<i>OID</i>	1.3.6.1.4.1.16177.1.400.1.5.50.2

8.4.7.3 hwCellSerial**Cellular Module Serial Number**

<i>Type</i>	DisplayString
<i>Range</i>	0 - 255
<i>Access</i>	readonly
<i>OID</i>	1.3.6.1.4.1.16177.1.400.1.5.50.3

8.4.7.4 hwCellImei**Cellular Module IMEI Number**

<i>Type</i>	DisplayString
<i>Range</i>	0 - 255
<i>Access</i>	readonly
<i>OID</i>	1.3.6.1.4.1.16177.1.400.1.5.50.4

8.4.7.5 hwCellFwVersion**Cellular Module Firmware Version**

<i>Type</i>	DisplayString
<i>Range</i>	0 - 255
<i>Access</i>	readonly
<i>OID</i>	1.3.6.1.4.1.16177.1.400.1.5.50.5

8.4.8 hwGnss**8.4.8.1 hwGnssAssembled****GNSS Module Assembled**

<i>Enumeration</i>	inexistent (0), present (1)
<i>Access</i>	readonly
<i>OID</i>	1.3.6.1.4.1.16177.1.400.1.5.51.1

8.4.8.2 hwGnssType**GNSS Module Type**

<i>Type</i>	DisplayString
<i>Range</i>	0 - 255
<i>Access</i>	readonly
<i>OID</i>	1.3.6.1.4.1.16177.1.400.1.5.51.2

8.4.8.3 hwGnssSerial

GNSS Module Serial Number

Type	DisplayString
Range	0 - 255
Access	readonly
OID	1.3.6.1.4.1.16177.1.400.1.5.51.3

8.5 software

8.5.1 swFirmware

8.5.1.1 swFwName

Firmware Name

Type	DisplayString
Range	1 - 255
Access	readonly
OID	1.3.6.1.4.1.16177.1.400.1.6.2.1

8.5.1.2 swFwVersion

Firmware Version

Type	DisplayString
Range	1 - 255
Access	readonly
OID	1.3.6.1.4.1.16177.1.400.1.6.2.2

8.5.1.3 swFwRevision

Firmware Revision

Type	DisplayString
Range	1 - 255
Access	readonly
OID	1.3.6.1.4.1.16177.1.400.1.6.2.3

8.5.2 swBootloader

8.5.2.1 swBootName

Name of the Bootloader

Applies to AP and STA.

<i>Type</i>	DisplayString
<i>Range</i>	1 - 255
<i>Access</i>	readonly
<i>OID</i>	1.3.6.1.4.1.16177.1.400.1.6.20.1

8.5.2.2 swBootVersion

Version of the Bootloader

Applies to AP and STA.

<i>Type</i>	DisplayString
<i>Range</i>	1 - 255
<i>Access</i>	readonly
<i>OID</i>	1.3.6.1.4.1.16177.1.400.1.6.20.2

8.5.2.3 swBootBuildDate

Date when the Bootloader was Built

Applies to AP and STA.

<i>Type</i>	DisplayString
<i>Range</i>	1 - 255
<i>Access</i>	readonly
<i>OID</i>	1.3.6.1.4.1.16177.1.400.1.6.20.3

8.5.3 swSystem

8.5.3.1 swSysRebootReason

System Reboot Reason

<i>Enumeration</i>	coldstart (0), warmstart (1), watchdog (2), oops (3), unknown (9)
<i>Access</i>	readonly
<i>OID</i>	1.3.6.1.4.1.16177.1.400.1.6.3.1

8.5.3.2 swSysMessageTable

System messages (e.g. Errors, Warnings)

<i>Range</i>	0 - 63
<i>OID</i>	1.3.6.1.4.1.16177.1.400.1.6.3.100

8.5.3.2.1 swSysMessageTableEntry

System message entry

<i>OID</i>	1.3.6.1.4.1.16177.1.400.1.6.3.100.1
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8.5.3.2.1.1 swSysMsgIndex

Table Entry Index

<i>Range</i>	0 - 63
<i>Access</i>	noaccess
<i>OID</i>	1.3.6.1.4.1.16177.1.400.1.6.3.100.1.1

8.5.3.2.1.2 swSysMsgPriority

Message Priority/Level

<i>Enumeration</i>	emergency (0), alert (1), critical (2), error (3), warning (4), notice (5), info (6), debug (7)
<i>Access</i>	readonly
<i>OID</i>	1.3.6.1.4.1.16177.1.400.1.6.3.100.1.2

8.5.3.2.1.3 swSysMsgCode

Message Code

Type	Integer32
Access	readonly
OID	1.3.6.1.4.1.16177.1.400.1.6.3.100.1.3

8.5.3.2.1.4 swSysMsgText

Message

Type	DisplayString
Range	1 - 255
Access	readonly
OID	1.3.6.1.4.1.16177.1.400.1.6.3.100.1.4

8.5.3.3 swSysBootStatus

The boot status indicates whether the booting sequence of the device has been completed.

Enumeration	done (0), booting (1)
Access	readonly
OID	1.3.6.1.4.1.16177.1.400.1.6.3.2

8.5.4 swConfiguration

8.5.4.1 swCfgChangesCount

Number of not yet Applied Device Configuration Changes

Applies to AP and STA.

Type	Integer32
Access	readonly
OID	1.3.6.1.4.1.16177.1.400.1.6.30.1

8.5.5 swOperatingSystem

8.5.5.1 swOsName

Operating System Name

Type	DisplayString
Range	1 - 255
Access	readonly
OID	1.3.6.1.4.1.16177.1.400.1.6.4.1

8.5.5.2 swOsVersion

Operating System Version

Type	DisplayString
Range	1 - 255
Access	readonly
OID	1.3.6.1.4.1.16177.1.400.1.6.4.2

8.5.5.3 swOsRevision

Operating System Revision

Type	DisplayString
Range	1 - 255
Access	readonly
OID	1.3.6.1.4.1.16177.1.400.1.6.4.3

8.5.5.4 swOsUptime

Uptime of the Operating System

This is the time in TimeTicks (hundredth of a second) the device has been up and running since the last reboot.

Example:

456732 means up since 4567.32 seconds

Applies to AP and STA.

<i>Type</i>	TimeTicks
<i>Access</i>	readonly
<i>OID</i>	1.3.6.1.4.1.16177.1.400.1.6.4.4

8.5.6 swDriver

8.5.6.1 swDrvDfsTable

DFS Driver Statistics

<i>Range</i>	0 - 1
<i>OID</i>	1.3.6.1.4.1.16177.1.400.1.6.5.1

8.5.6.1.1 swDrvDfsTableEntry

DFS Driver Statistics

<i>OID</i>	1.3.6.1.4.1.16177.1.400.1.6.5.1.1
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8.5.6.1.1.1 swDrvDfsIndex

Table Entry Index

<i>Range</i>	0 - 1
<i>Access</i>	noaccess
<i>OID</i>	1.3.6.1.4.1.16177.1.400.1.6.5.1.1.1

8.5.6.1.1.2 swDrvDfsName

Name of The Wireless Device

Applies to AP and STA. 802.11n products only.

<i>Type</i>	DisplayString
<i>Range</i>	1 - 255
<i>Access</i>	readonly
<i>OID</i>	1.3.6.1.4.1.16177.1.400.1.6.5.1.1.2

8.5.6.1.1.3 swDrvDfsPulsesDetected

Pulses Detected by The Wireless Device

Applies to AP. 802.11n products only.

Type	Integer32
Access	readonly
OID	1.3.6.1.4.1.16177.1.400.1.6.5.1.1.3

8.5.6.1.1.4 swDrvDfsPulsesProcessed

Pulses Processed by The Wireless Device

Applies to AP. 802.11n products only.

Type	Integer32
Access	readonly
OID	1.3.6.1.4.1.16177.1.400.1.6.5.1.1.4

8.5.6.1.1.5 swDrvDfsRadarDetected

Radar Sequences Detected by The Wireless Device

Applies to AP. 802.11n products only.

Type	Integer32
Access	readonly
OID	1.3.6.1.4.1.16177.1.400.1.6.5.1.1.5

8.5.6.2 swDrvCntWlanMacTable

Wireless MAC-Layer Statistics

Range	0 - 63
OID	1.3.6.1.4.1.16177.1.400.1.6.5.4

8.5.6.2.1 swDrvCntWlanMacTableEntry

Wireless MAC-Layer Statistics

OID	1.3.6.1.4.1.16177.1.400.1.6.5.4.1
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8.5.6.2.1.1 swDrvCntWlanMacIndex

Table Entry Index

<i>Range</i>	0 - 63
<i>Access</i>	noaccess
<i>OID</i>	1.3.6.1.4.1.16177.1.400.1.6.5.4.1.1

8.5.6.2.1.2 swDrvCntWlanMacRxHandlersDrop

MAC Debug Entry

Applies to AP and STA. 802.11n products only.

<i>Type</i>	Integer32
<i>Access</i>	readonly
<i>OID</i>	1.3.6.1.4.1.16177.1.400.1.6.5.4.1.10

8.5.6.2.1.3 swDrvCntWlanMacRxHandlersQueued

MAC Debug Entry

Applies to AP and STA. 802.11n products only.

<i>Type</i>	Integer32
<i>Access</i>	readonly
<i>OID</i>	1.3.6.1.4.1.16177.1.400.1.6.5.4.1.11

8.5.6.2.1.4 swDrvCntWlanMacRxHandlersDropNullfunc

MAC Debug Entry

Applies to AP and STA. 802.11n products only.

<i>Type</i>	Integer32
<i>Access</i>	readonly
<i>OID</i>	1.3.6.1.4.1.16177.1.400.1.6.5.4.1.12

8.5.6.2.1.5 swDrvCntWlanMacRxHandlersDropDefrag

MAC Debug Entry

Applies to AP and STA. 802.11n products only.

Type	Integer32
Access	readonly
OID	1.3.6.1.4.1.16177.1.400.1.6.5.4.1.13

8.5.6.2.1.6 swDrvCntWlanMacRxHandlersDropShort**MAC Debug Entry**

Applies to AP and STA. 802.11n products only.

Type	Integer32
Access	readonly
OID	1.3.6.1.4.1.16177.1.400.1.6.5.4.1.14

8.5.6.2.1.7 swDrvCntWlanMacTxExpandSkbHead**MAC Debug Entry**

Applies to AP and STA. 802.11n products only.

Type	Integer32
Access	readonly
OID	1.3.6.1.4.1.16177.1.400.1.6.5.4.1.15

8.5.6.2.1.8 swDrvCntWlanMacTxExpandSkbHeadCloned**MAC Debug Entry**

Applies to AP and STA. 802.11n products only.

Type	Integer32
Access	readonly
OID	1.3.6.1.4.1.16177.1.400.1.6.5.4.1.16

8.5.6.2.1.9 swDrvCntWlanMacRxExpandSkbHead**MAC Debug Entry**

Applies to AP and STA. 802.11n products only.

Type	Integer32
Access	readonly
OID	1.3.6.1.4.1.16177.1.400.1.6.5.4.1.17

8.5.6.2.1.10 swDrvCntWlanMacRxExpandSkbHead2

MAC Debug Entry

Applies to AP and STA. 802.11n products only.

Type	Integer32
Access	readonly
OID	1.3.6.1.4.1.16177.1.400.1.6.5.4.1.18

8.5.6.2.1.11 swDrvCntWlanMacRxHandlersFragments

MAC Debug Entry

Applies to AP and STA. 802.11n products only.

Type	Integer32
Access	readonly
OID	1.3.6.1.4.1.16177.1.400.1.6.5.4.1.19

8.5.6.2.1.12 swDrvCntWlanMacName

Name of The Wireless Device

Applies to AP and STA. 802.11n products only.

Type	DisplayString
Range	1 - 255
Access	readonly
OID	1.3.6.1.4.1.16177.1.400.1.6.5.4.1.2

8.5.6.2.1.13 swDrvCntWlanMacTxstatusDrop

MAC Debug Entry

Applies to AP and STA. 802.11n products only.

Type	Integer32
Access	readonly
OID	1.3.6.1.4.1.16177.1.400.1.6.5.4.1.20

8.5.6.2.1.14 swDrvCntWlanMacTxHandlersDrop

MAC Debug Entry

Applies to AP and STA. 802.11n products only.

Type	Integer32
Access	readonly
OID	1.3.6.1.4.1.16177.1.400.1.6.5.4.1.3

8.5.6.2.1.15 swDrvCntWlanMacTxHandlersQueued

MAC Debug Entry

Applies to AP and STA. 802.11n products only.

Type	Integer32
Access	readonly
OID	1.3.6.1.4.1.16177.1.400.1.6.5.4.1.4

8.5.6.2.1.16 swDrvCntWlanMacTxHandlersDropUnencrypted

MAC Debug Entry

Applies to AP and STA. 802.11n products only.

Type	Integer32
Access	readonly
OID	1.3.6.1.4.1.16177.1.400.1.6.5.4.1.5

8.5.6.2.1.17 swDrvCntWlanMacTxHandlersDropFragment

MAC Debug Entry

Applies to AP and STA. 802.11n products only.

Type	Integer32
Access	readonly
OID	1.3.6.1.4.1.16177.1.400.1.6.5.4.1.6

8.5.6.2.1.18 swDrvCntWlanMacTxHandlersDropWep

MAC Debug Entry

Applies to AP and STA. 802.11n products only.

Type	Integer32
Access	readonly
OID	1.3.6.1.4.1.16177.1.400.1.6.5.4.1.7

8.5.6.2.1.19 swDrvCntWlanMacTxHandlersDropNotAssoc

MAC Debug Entry

Applies to AP and STA. 802.11n products only.

Type	Integer32
Access	readonly
OID	1.3.6.1.4.1.16177.1.400.1.6.5.4.1.8

8.5.6.2.1.20 swDrvCntWlanMacTxHandlersDropUnauthPort

MAC Debug Entry

Applies to AP and STA. 802.11n products only.

Type	Integer32
Access	readonly
OID	1.3.6.1.4.1.16177.1.400.1.6.5.4.1.9

8.5.6.3 swDrvCntWlanWmmTable

WMM statistics

Range	0 - 4
OID	1.3.6.1.4.1.16177.1.400.1.6.5.6

8.5.6.3.1 swDrvCntWlanWmmTableEntry

WMM statistics

OID	1.3.6.1.4.1.16177.1.400.1.6.5.6.1
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8.5.6.3.1.1 swDrvCntWlanWmmTableIndex

Table Entry Index

Range	0 - 4
Access	noaccess
OID	1.3.6.1.4.1.16177.1.400.1.6.5.6.1.1

8.5.6.3.1.2 swDrvCntWlanWmmName

Name of The Queue

Applies to AP and STA. 802.11n products only.

Type	DisplayString
Range	1 - 255
Access	readonly
OID	1.3.6.1.4.1.16177.1.400.1.6.5.6.1.2

8.5.6.3.1.3 swDrvCntWlanWmmTx

Number of Frames Sent in his Queue

Applies to AP and STA. 802.11n products only.

Type	Integer32
Access	readonly
OID	1.3.6.1.4.1.16177.1.400.1.6.5.6.1.3

8.5.6.3.1.4 swDrvCntWlanWmmRx

Number of Frames Received in This Queue

Applies to AP and STA. 802.11n products only.

Type	Integer32
Access	readonly
OID	1.3.6.1.4.1.16177.1.400.1.6.5.6.1.4

8.5.6.3.1.5 swDrvCntWlanWmmShortRetries

Number of Retries for Frames Shorter Than RTS

Applies to AP and STA. 802.11n products only.

Type	Integer32
Access	readonly
OID	1.3.6.1.4.1.16177.1.400.1.6.5.6.1.5

8.5.6.3.1.6 swDrvCntWlanWmmLongRetries

Number of Retries for Frames Longer Than RTS

Applies to AP and STA. 802.11n products only.

Type	Integer32
Access	readonly
OID	1.3.6.1.4.1.16177.1.400.1.6.5.6.1.6

8.5.6.3.1.7 swDrvCntWlanWmmExceededRetries

Number of Failed Transmissions Due to Exceeding of The Retry Limit

Applies to AP and STA. 802.11n products only.

Type	Integer32
Access	readonly
OID	1.3.6.1.4.1.16177.1.400.1.6.5.6.1.7

8.5.6.4 swDrvConStatWlanIf

Volatile wlan interface selector for swDrvConStatTable.

Changes made here will be lost upon reconfiguration or a reboot. Use cfgWlanGlb1ConnectionStatusWlanInt to set a persistent value which is used during initialisation.

Applies to AP and STA.

Type	DisplayString
Range	4 - 17
Access	readwrite
OID	1.3.6.1.4.1.16177.1.400.1.6.5.7

8.5.6.5 swDrvConStatTable

Connection Status Information.

Range	0 - 9
OID	1.3.6.1.4.1.16177.1.400.1.6.5.8

8.5.6.5.1 swDrvConStatTableEntry

Connection Status Information per Station.

OID	1.3.6.1.4.1.16177.1.400.1.6.5.8.1
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8.5.6.5.1.1 swDrvConStatIndex

Table Entry Index

<i>Range</i>	0 - 9
<i>Access</i>	noaccess
<i>OID</i>	1.3.6.1.4.1.16177.1.400.1.6.5.8.1.1

8.5.6.5.1.2 swDrvConStatTxBrType

Station TX Bitrate Type

Applies to AP and STA.

<i>Type</i>	DisplayString
<i>Range</i>	1 - 255
<i>Access</i>	readonly
<i>OID</i>	1.3.6.1.4.1.16177.1.400.1.6.5.8.1.10

8.5.6.5.1.3 swDrvConStatTxBrValue

Station TX Bitrate Value in Mbps

Multiplied by 10 if swDrvConStatRxBrType is 'legacy' or '11AC'.

Applies to AP and STA.

<i>Type</i>	Integer32
<i>Access</i>	readonly
<i>OID</i>	1.3.6.1.4.1.16177.1.400.1.6.5.8.1.11

8.5.6.5.1.4 swDrvConStatTxBytes

Station TX Bytes

Applies to AP and STA. 802.11n products only.

<i>Type</i>	Integer32
<i>Access</i>	readonly
<i>OID</i>	1.3.6.1.4.1.16177.1.400.1.6.5.8.1.12

8.5.6.5.1.5 swDrvConStatTxPackets

Station TX Packets

Applies to AP and STA. 802.11n products only.

Type	Integer32
Access	readonly
OID	1.3.6.1.4.1.16177.1.400.1.6.5.8.1.13

8.5.6.5.1.6 swDrvConStatSigChain0

Station Signal Chain 0 in dBm

Applies to AP and STA. 802.11n products only.

Type	Integer32
Access	readonly
OID	1.3.6.1.4.1.16177.1.400.1.6.5.8.1.14

8.5.6.5.1.7 swDrvConStatSigChain1

Station Signal Chain 1 in dBm

Applies to AP and STA. 802.11n products only.

Type	Integer32
Access	readonly
OID	1.3.6.1.4.1.16177.1.400.1.6.5.8.1.15

8.5.6.5.1.8 swDrvConStatSigChain2

Station Signal Chain 2 in dBm

Applies to AP and STA. 802.11n products only.

Type	Integer32
Access	readonly
OID	1.3.6.1.4.1.16177.1.400.1.6.5.8.1.16

8.5.6.5.1.9 swDrvConStatSigAvgChain0

Station Signal Average Chain 0 in dBm

Applies to AP and STA. 802.11n products only.

Type	Integer32
Access	readonly
OID	1.3.6.1.4.1.16177.1.400.1.6.5.8.1.17

8.5.6.5.1.10 swDrvConStatSigAvgChain1

Station Signal Average Chain 1 in dBm

Applies to AP and STA. 802.11n products only.

Type	Integer32
Access	readonly
OID	1.3.6.1.4.1.16177.1.400.1.6.5.8.1.18

8.5.6.5.1.11 swDrvConStatSigAvgChain2

Station Signal Average Chain 2 in dBm

Applies to AP and STA. 802.11n products only.

Type	Integer32
Access	readonly
OID	1.3.6.1.4.1.16177.1.400.1.6.5.8.1.19

8.5.6.5.1.12 swDrvConStatWlanName

WLAN Interface Name

Applies to AP and STA.

Type	DisplayString
Range	4 - 5
Access	readonly
OID	1.3.6.1.4.1.16177.1.400.1.6.5.8.1.2

8.5.6.5.1.13 swDrvConStatTxRetries

Station TX Retries

Applies to AP and STA.

Type	Integer32
Access	readonly
OID	1.3.6.1.4.1.16177.1.400.1.6.5.8.1.20

8.5.6.5.1.14 swDrvConStatTxFailed

Station TX Failed

Applies to AP and STA. 802.11n products only.

Type	Integer32
Access	readonly
OID	1.3.6.1.4.1.16177.1.400.1.6.5.8.1.21

8.5.6.5.1.15 swDrvConStatCacheNo

Station Dump Cache Access Number

The cache gets refreshed if it is older than 5 seconds.

Applies to AP and STA. 802.11n products only.

Type	Integer32
Access	readonly
OID	1.3.6.1.4.1.16177.1.400.1.6.5.8.1.22

8.5.6.5.1.16 swDrvConStatSigCombined

Station Signal Combined of All Active Chains in dBm

Applies to AP and STA.

Type	Integer32
Access	readonly
OID	1.3.6.1.4.1.16177.1.400.1.6.5.8.1.23

8.5.6.5.1.17 swDrvConStatSigAvgCombined

Station Signal Average Combined of All Active Chains in dBm

Applies to AP and STA.

Type	Integer32
Access	readonly
OID	1.3.6.1.4.1.16177.1.400.1.6.5.8.1.24

8.5.6.5.1.18 swDrvConStatMacName

Station MAC Address

In AP mode this is the MAC address of the connected client (STA). In client (STA) mode this is the MAC address of the AP to which the client is connected.

Applies to AP and STA.

Type	DisplayString
Range	1 - 17
Access	readonly
OID	1.3.6.1.4.1.16177.1.400.1.6.5.8.1.3

8.5.6.5.1.19 swDrvConStatRxBrExtra

Station RX Bitrate Extra

Applies to AP and STA.

Type	DisplayString
Range	1 - 255
Access	readonly
OID	1.3.6.1.4.1.16177.1.400.1.6.5.8.1.4

8.5.6.5.1.20 swDrvConStatRxBrType

Station RX Bitrate Type

Applies to AP and STA.

Type	DisplayString
Range	1 - 255
Access	readonly
OID	1.3.6.1.4.1.16177.1.400.1.6.5.8.1.5

8.5.6.5.1.21 swDrvConStatRxBrValue

Station RX Bitrate Value in Mbps

Multiplied by 10 if swDrvConStatRxBrType is 'legacy' or '11AC'.

Applies to AP and STA.

Type	Integer32
Access	readonly
OID	1.3.6.1.4.1.16177.1.400.1.6.5.8.1.6

8.5.6.5.1.22 swDrvConStatRxBytes

Station RX Bytes

Applies to AP and STA. 802.11n products only.

Type	Integer32
Access	readonly
OID	1.3.6.1.4.1.16177.1.400.1.6.5.8.1.7

8.5.6.5.1.23 swDrvConStatRxPackets

Station RX Packets

Applies to AP and STA. 802.11n products only.

Type	Integer32
Access	readonly
OID	1.3.6.1.4.1.16177.1.400.1.6.5.8.1.8

8.5.6.5.1.24 swDrvConStatTxBrExtra

Station TX Bitrate Extra

Applies to AP and STA.

Type	DisplayString
Range	1 - 255
Access	readonly
OID	1.3.6.1.4.1.16177.1.400.1.6.5.8.1.9

8.5.6.6 swDrvCntWlanTable

Wireless Dev Counters

Range	0 - 63
OID	1.3.6.1.4.1.16177.1.400.1.6.5.9

8.5.6.6.1 swDrvCntWlanTableEntry

Wireless Dev Counters

OID	1.3.6.1.4.1.16177.1.400.1.6.5.9.1
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8.5.6.6.1.1 swDrvCntWlanIndex

Table Entry Index

<i>Range</i>	0 - 63
<i>Access</i>	noaccess
<i>OID</i>	1.3.6.1.4.1.16177.1.400.1.6.5.9.1.1

8.5.6.6.1.2 swDrvCntWlanChannelActive

Time in ms during which the device has been on the current channel.

On STA when it's no associated the current channel is the lowest channel of selected country code.

Applies to AP and STA. 802.11n products only.

<i>Type</i>	Counter32
<i>Access</i>	readonly
<i>OID</i>	1.3.6.1.4.1.16177.1.400.1.6.5.9.1.10

8.5.6.6.1.3 swDrvCntWlanChannelBusy

Time in ms during which the medium has been considered busy on the current channel.

Applies to AP and STA. 802.11n products only.

<i>Type</i>	Counter32
<i>Access</i>	readonly
<i>OID</i>	1.3.6.1.4.1.16177.1.400.1.6.5.9.1.11

8.5.6.6.1.4 swDrvCntWlanChannelTransmit

Time in ms during which the device has been in transmit mode on the current channel.

Applies to AP and STA. 802.11n products only.

<i>Type</i>	Counter32
<i>Access</i>	readonly
<i>OID</i>	1.3.6.1.4.1.16177.1.400.1.6.5.9.1.12

8.5.6.6.1.5 swDrvCntWlanChannelReceive

Time in ms during which the device has been in receive mode on the current channel.

Applies to AP and STA. 802.11n products only.

<i>Type</i>	Counter32
<i>Access</i>	readonly
<i>OID</i>	1.3.6.1.4.1.16177.1.400.1.6.5.9.1.13

8.5.6.6.1.6 swDrvCntWlanChannelNoise

Measured channel noise in 1 dB steps.

Note: This value is not an absolute power level but the internal representation of the measured noise floor.

Applies to AP and STA. 802.11n products only.

Type	Integer32
Access	readonly
OID	1.3.6.1.4.1.16177.1.400.1.6.5.9.1.14

8.5.6.6.1.7 swDrvCntWlanName

Name of the wireless Dev

Type	DisplayString
Range	1 - 255
Access	readonly
OID	1.3.6.1.4.1.16177.1.400.1.6.5.9.1.2

8.5.6.6.1.8 swDrvCntWlanEapAuthStartedFT

EAP sessions started through FT.

Applies to AP. 802.11n products only.

Type	Counter32
Access	readonly
OID	1.3.6.1.4.1.16177.1.400.1.6.5.9.1.20

8.5.6.6.1.9 swDrvCntWlanEapAuthStartedFILS

EAP sessions started through FILS.

Applies to AP. 802.11n products only.

Type	Counter32
Access	readonly
OID	1.3.6.1.4.1.16177.1.400.1.6.5.9.1.21

8.5.6.6.1.10 swDrvCntWlanEapAuthStartedPKMSA

EAP sessions started through PKMSA.

Applies to AP. 802.11n products only.

Type	Counter32
Access	readonly
OID	1.3.6.1.4.1.16177.1.400.1.6.5.9.1.22

8.5.6.6.1.11 swDrvCntWlanAssocSuccess

Number of successful associations. 802.11n products only.

Type	Counter32
Access	readonly
OID	1.3.6.1.4.1.16177.1.400.1.6.5.9.1.4

8.5.6.6.1.12 swDrvCntWlanAssocFailure

Number of unsuccessful associations. 802.11n products only.

Type	Counter32
Access	readonly
OID	1.3.6.1.4.1.16177.1.400.1.6.5.9.1.5

8.5.6.6.1.13 swDrvCntWlanAssocFailureMaxSta

Number of times the maximum number of stations has been exceeded.

Applies to AP. 802.11n products only.

Type	Counter32
Access	readonly
OID	1.3.6.1.4.1.16177.1.400.1.6.5.9.1.6

8.5.6.6.1.14 swDrvCntWlanNumAssocSta

Number of associated STA.

Applies to AP. 802.11n products only.

Type	Integer32
Access	readonly
OID	1.3.6.1.4.1.16177.1.400.1.6.5.9.1.7

8.5.6.6.1.15 swDrvCntWlanEapAuthStarted

Number of EAP authentication sessions started since the AP has been started.

Applies to AP. 802.11n products only.

<i>Type</i>	Counter32
<i>Access</i>	readonly
<i>OID</i>	1.3.6.1.4.1.16177.1.400.1.6.5.9.1.8

8.5.6.6.1.16 swDrvCntWlanEapAuthFailed

Number of EAP authentication sessions that have failed since the AP has been started.

Applies to AP. 802.11n products only.

<i>Type</i>	Counter32
<i>Access</i>	readonly
<i>OID</i>	1.3.6.1.4.1.16177.1.400.1.6.5.9.1.9

8.5.7 swCellular

8.5.7.1 swCellTable

Cellular Status Information Table

This table is in a one-to-one relation to the `cfgNetWwanTable` where the indexes of the respective entries match.

Applies to cellular products only.

<i>Range</i>	0 - 0
<i>OID</i>	1.3.6.1.4.1.16177.1.400.1.6.50.1

8.5.7.1.1 swCellTableEntry

Cellular Status Information Entry

Applies to cellular products only.

<i>OID</i>	1.3.6.1.4.1.16177.1.400.1.6.50.1.1
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8.5.7.1.1.1 swCellIndex

Table Entry Index

<i>Range</i>	0 - 0
<i>Access</i>	noaccess
<i>OID</i>	1.3.6.1.4.1.16177.1.400.1.6.50.1.1.1

8.5.7.1.1.2 swCellConnectionStatus

Connection Status

- **unknown(0)**
- **disabled(1)**
- **simSwitching(2)**: Switching the SIM card.
- **netDown(3)**: Network is down.
- **resetting(4)**
- **netUp(5)**: Network is up.
- **connecting(6)**
- **connected(7)**
- **disconnected(8)**
- **invalid(9)**

Applies to cellular products only.

<i>Enumeration</i>	unknown (0), disabled (1), simSwitching (2), netDown (3), resetting (4), netUp (5), connecting (6), connected (7), disconnected (8), invalid (9)
<i>Access</i>	readonly
<i>OID</i>	1.3.6.1.4.1.16177.1.400.1.6.50.1.1.10

8.5.7.1.1.3 swCellConnectionMessage

Connection Status Message

Applies to cellular products only.

<i>Type</i>	DisplayString
<i>Range</i>	0 - 255
<i>Access</i>	readonly
<i>OID</i>	1.3.6.1.4.1.16177.1.400.1.6.50.1.1.11

8.5.7.1.1.4 swCellSignalType

Signal Type

unknown(0): Unknown signal type **wcdma(3):** UMTS (3G) **lte(4):** LTE (4G)

Applies to cellular products only.

<i>Enumeration</i>	unknown (0), wcdma (3), lte (4)
<i>Access</i>	readonly
<i>OID</i>	1.3.6.1.4.1.16177.1.400.1.6.50.1.1.12

8.5.7.1.1.5 swCellSignalRssi

Received Signal Strength Indication

Given in dB.

Applies to cellular products only.

<i>Type</i>	Integer32
<i>Access</i>	readonly
<i>OID</i>	1.3.6.1.4.1.16177.1.400.1.6.50.1.1.13

8.5.7.1.1.6 swCellSignalRsrq

Reference Signal Received Quality

Given in dB and only applies to signals of type **lte(1)**.

Applies to cellular products only.

<i>Type</i>	Integer32
<i>Access</i>	readonly
<i>OID</i>	1.3.6.1.4.1.16177.1.400.1.6.50.1.1.14

8.5.7.1.1.7 swCellSignalRsrp

Reference Signal Received Power

Given in dBm and only applies to signals of type **lte(1)**.

Applies to cellular products only.

<i>Type</i>	Integer32
<i>Access</i>	readonly
<i>OID</i>	1.3.6.1.4.1.16177.1.400.1.6.50.1.1.15

8.5.7.1.1.8 swCellSignalSinr

Signal-To-Interference-Plus-Noise Ratio

The signal-to-interference-plus-noise ratio (SINR) in dB is calculated from the value v of this entry using the following equation:

$$\text{SINR} = (v / 5) - 20$$

Example:

- **0**: SINR = -20dB
- **100**: SINR = 0dB
- **250**: SINR = 30dB

This only applies to signals of type **lte(1)**.

Applies to cellular products only.

Type	Integer32
Access	readonly
OID	1.3.6.1.4.1.16177.1.400.1.6.50.1.1.16

8.5.7.1.1.9 swCellSignalRscp

Received Signal Code Power

Given in dBm and only applies to signals of type **wcdma(0)**.

Applies to cellular products only.

Type	Integer32
Access	readonly
OID	1.3.6.1.4.1.16177.1.400.1.6.50.1.1.17

8.5.7.1.1.10 swCellSignalEcio

Carrier to Noise Ratio

Given in dB and only applies to signals of type **wcdma(0)**.

Applies to cellular products only.

Type	Integer32
Access	readonly
OID	1.3.6.1.4.1.16177.1.400.1.6.50.1.1.18

8.5.7.1.1.11 swCellWwanName

Name of the Cellular Network Interface

This is name of the corresponding network interface.

Applies to cellular products only.

Type	DisplayString
Range	4 - 5
Access	readonly
OID	1.3.6.1.4.1.16177.1.400.1.6.50.1.1.2

8.5.7.1.1.12 swCellSimSlot

Active SIM Slot

- slot1(1)
- slot2(2)

Applies to cellular products only.

Enumeration	slot1 (1), slot2 (2)
Access	readonly
OID	1.3.6.1.4.1.16177.1.400.1.6.50.1.1.3

8.5.7.1.1.13 swCellSimStatus

SIM Status

This entry shows the status of the SIM card in the active slot:

noSim(0): No SIM card is inserted. **ready(1):** The SIM card is ready for operation. **pinReq(2):** Requesting the PIN of the SIM card. **pukReq(3):** Requesting the PUK of the SIM card.

Applies to cellular products only.

Enumeration	noSim (0), ready (1), pinReq (2), pukReq (3)
Access	readonly
OID	1.3.6.1.4.1.16177.1.400.1.6.50.1.1.4

8.5.7.1.1.14 swCellSimPrimary

Is SIM Primary?

Is the SIM card of the primary slot used?

- **no(0)**
- **yes(1)**

Applies to cellular products only.

<i>Enumeration</i>	no (0), yes (1)
<i>Access</i>	readonly
<i>OID</i>	1.3.6.1.4.1.16177.1.400.1.6.50.1.1.5

8.5.7.1.1.15 swCellSimRoaming

Is SIM Roaming?

Is the active SIM card currently roaming?

- **no(0)**
- **yes(1)**

Applies to cellular products only.

<i>Enumeration</i>	no (0), yes (1)
<i>Access</i>	readonly
<i>OID</i>	1.3.6.1.4.1.16177.1.400.1.6.50.1.1.6

8.5.7.1.1.16 swCellServiceName

Service Provider Name

Applies to cellular products only.

<i>Type</i>	DisplayString
<i>Range</i>	0 - 255
<i>Access</i>	readonly
<i>OID</i>	1.3.6.1.4.1.16177.1.400.1.6.50.1.1.7

8.5.7.1.1.17 swCellServiceMcc

Mobile Country Code

Mobile Country Code (MCC) are defined by the ITU-T Recommendation E.212.

Applies to cellular products only.

<i>Type</i>	Integer32
<i>Access</i>	readonly
<i>OID</i>	1.3.6.1.4.1.16177.1.400.1.6.50.1.1.8

8.5.7.1.1.18 swCellServiceMnc

Mobile Network Code

Mobile Network Code (MNC) are defined by the ITU-T Recommendation E.212.

Applies to cellular products only.

<i>Type</i>	Integer32
<i>Access</i>	readonly
<i>OID</i>	1.3.6.1.4.1.16177.1.400.1.6.50.1.1.9

8.5.8 swNlm

8.5.8.1 swNlmMonitorTable

NLM Monitor Status Table

<i>Range</i>	0 - 255
<i>OID</i>	1.3.6.1.4.1.16177.1.400.1.6.52.1

8.5.8.1.1 swNlmMonitorTableEntry

NLM Monitor Status Table Entry

<i>OID</i>	1.3.6.1.4.1.16177.1.400.1.6.52.1.1
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8.5.8.1.1.1 swNlmMonIndex

Table Entry Index

<i>Range</i>	0 - 255
<i>Access</i>	noaccess
<i>OID</i>	1.3.6.1.4.1.16177.1.400.1.6.52.1.1.1

8.5.8.1.1.2 swNlmMonState

Status Of Monitor

This value is cached and updates at most once a second.

<i>Enumeration</i>	down (0), up (1), disabled (2)
<i>Access</i>	readonly
<i>OID</i>	1.3.6.1.4.1.16177.1.400.1.6.52.1.1.2

8.5.9 swRdm

8.5.9.1 swRdmMaxEirp

Maximal equivalent isotropically radiated power (EIRP) in dBm.

This value shows the maximal aggregated transmit power over all configured chains.

Applies to AP.

<i>Type</i>	Integer32
<i>Access</i>	readonly
<i>OID</i>	1.3.6.1.4.1.16177.1.400.1.6.6.1

8.5.9.2 swRdmMaxApp

Maximal antenna port power in dBm.

This value shows the maximal transmit power of a single chain.

Applies to AP.

<i>Type</i>	Integer32
<i>Access</i>	readonly
<i>OID</i>	1.3.6.1.4.1.16177.1.400.1.6.6.2

9 WESTERMO-SW6-BRIDGE-MIB

9.1 rstp

9.1.1 configuration

9.1.1.1 cfgRstpBridge

9.1.1.1.1 cfgRstpBridgeEnabled

Enable RSTP on bridge.

Applies to AP and STA.

<i>Enumeration</i>	disabled (0), enabled (1)
<i>Access</i>	readwrite
<i>OID</i>	1.3.6.1.4.1.16177.1.400.2.8.1.1.1

9.1.1.1.2 cfgRstpBridgePriority

The bridge's relative priority value for determining the root bridge (the upper 16 bits of the bridge-id). A bridge with the lowest bridge-id is elected the root. By default, the priority is 0x8000 (32768). This value needs to be a multiple of 4096, otherwise it's rounded to the nearest inferior one.

Applies to AP and STA.

<i>Range</i>	0 - 61440
<i>Access</i>	readwrite
<i>OID</i>	1.3.6.1.4.1.16177.1.400.2.8.1.1.2

9.1.1.1.3 cfgRstpBridgeHelloTime

Hello message send interval

The interval in seconds between transmissions of hello messages by designated ports. With RSTP this value is 2 seconds.

Applies to AP and STA.

<i>Range</i>	2 - 2
<i>Access</i>	readonly
<i>OID</i>	1.3.6.1.4.1.16177.1.400.2.8.1.1.3

9.1.1.1.4 **cfgRstpBridgeForwardDelay**

The delay used by STP bridges to transition root and designated ports to forwarding. The default value is 15.

Applies to AP and STA.

<i>Range</i>	4 - 30
<i>Access</i>	readwrite
<i>OID</i>	1.3.6.1.4.1.16177.1.400.2.8.1.1.4

9.1.1.1.5 **cfgRstpBridgeMaxAge**

The maximum age of the information transmitted by the bridge when it is the root bridge. The default value is 20.

Applies to AP and STA.

<i>Range</i>	6 - 40
<i>Access</i>	readwrite
<i>OID</i>	1.3.6.1.4.1.16177.1.400.2.8.1.1.5

9.1.1.1.6 **cfgRstpBridgeTransmitHoldCount**

The transmit hold count used by the port transmit state machine to limit transmission rate. The default value is 6.

Applies to AP and STA.

<i>Range</i>	1 - 10
<i>Access</i>	readwrite
<i>OID</i>	1.3.6.1.4.1.16177.1.400.2.8.1.1.6

9.1.1.2 **cfgRstpPort**

9.1.1.2.1 cfgRstpPortTable

RSTP port configuration table.

<i>Range</i>	0 - 18
<i>OID</i>	1.3.6.1.4.1.16177.1.400.2.8.1.2.1

9.1.1.2.1.1 cfgRstpPortTableEntry

RSTP port configuration table entry.

<i>OID</i>	1.3.6.1.4.1.16177.1.400.2.8.1.2.1.1
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cfgRstpPortIndex

Entry index of port forward table.

<i>Range</i>	0 - 18
<i>Access</i>	noaccess
<i>OID</i>	1.3.6.1.4.1.16177.1.400.2.8.1.2.1.1.1

cfgRstpPortEnabled

Enable RSTP on port.

Applies to AP and STA.

<i>Enumeration</i>	disabled (0), enabled (1)
<i>Access</i>	readwrite
<i>OID</i>	1.3.6.1.4.1.16177.1.400.2.8.1.2.1.1.2

cfgRstpPortName

Port name to apply settings, e.g. eth0 or wlan0.

Applies to AP and STA.

<i>Type</i>	DisplayString
<i>Range</i>	1 - 255
<i>Access</i>	readonly
<i>OID</i>	1.3.6.1.4.1.16177.1.400.2.8.1.2.1.1.3

cfgRstpPortPriority

The port's relative priority value for determining the root port, in multiples of 16. By default, the port priority is 0x80 (128). Any value in the lower 4 bits is rounded off. The significant upper 4 bits become the upper 4 bits of the port-id. A port with the lowest port-id is elected as the root.

Applies to AP and STA.

<i>Range</i>	0 - 240
<i>Access</i>	readwrite
<i>OID</i>	1.3.6.1.4.1.16177.1.400.2.8.1.2.1.1.4

cfgRstpPortPathCost

The port path cost. The port's contribution, when it is the root port, to the root path cost for the bridge. By default the cost is automatically calculated from the port's speed. If -1 is defined, the port path cost is automatically calculated.

Data rate	RSTP cost (802.1W-2004, default value)
4 Mbit/s	5,000,000
10 Mbit/s	2,000,000
16 Mbit/s	1,250,000
100 Mbit/s	200,000
1 Gbit/s	20,000
2 Gbit/s	10,000
10 Gbit/s	2,000

Applies to AP and STA.

<i>Range</i>	-1 - 5000000
<i>Access</i>	readwrite
<i>OID</i>	1.3.6.1.4.1.16177.1.400.2.8.1.2.1.1.5

cfgRstpPortAutoEdge

The auto edge port parameter allows the automatic detection of edge ports.

Ports can be configured as edge ports to facilitate rapid changes to the forwarding state when connected to endpoints.

If enabled, the port will look for BPDUs; if there are none it begins forwarding packets. It is recommended to disable auto-edge for non-edge ports.

Applies to AP and STA.

<i>Enumeration</i>	disabled (0), enabled (1)
<i>Access</i>	readwrite
<i>OID</i>	1.3.6.1.4.1.16177.1.400.2.8.1.2.1.1.6

10 WESTERMO-SW6-FIREWALL-MIB

10.1 firewall

10.1.1 configuration

10.1.1.1 cfgFwEnabled

Firewall disabled or enabled.

Applies to AP and STA.

<i>Enumeration</i>	disabled (0), enabled (1)
<i>Access</i>	readwrite
<i>OID</i>	1.3.6.1.4.1.16177.1.400.2.1.1.1

10.1.1.2 cfgFwNat

10.1.1.2.1 cfgFwNatPortForwardTable

Firewall Port Forward Rules Table (DNAT)

<i>Range</i>	0 - 255
<i>OID</i>	1.3.6.1.4.1.16177.1.400.2.1.1.2.1

10.1.1.2.1.1 cfgFwNatPortForwardTableEntry

Firewall Port Forward Rules Table (DNAT)

<i>OID</i>	1.3.6.1.4.1.16177.1.400.2.1.1.2.1.1
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cfgFwNatPrtFwdIndex

Entry Index of Port Forward Table

<i>Range</i>	0 - 255
<i>Access</i>	noaccess
<i>OID</i>	1.3.6.1.4.1.16177.1.400.2.1.1.2.1.1.1

cfgFwNatPrtFwdDestinationPortEnd

Destination End Port to Redirect

When forwarding multiple port, this value is the end of the range. Set to -1 if no range is forwarded. Can only be used with TCP, UDP or TCP/UDP.

<i>Range</i>	-1 - 65535
<i>Access</i>	readwrite
<i>OID</i>	1.3.6.1.4.1.16177.1.400.2.1.1.2.1.1.10

cfgFwNatPrtFwdRedirectDestinationAddress

Redirect Traffic to This Redirection Destination Address

<i>Type</i>	IpAddress
<i>Access</i>	readwrite
<i>OID</i>	1.3.6.1.4.1.16177.1.400.2.1.1.2.1.1.11

cfgFwNatPrtFwdRedirectDestinationPort

Redirect Traffic to This Destination Port

Can only be used with TCP, UDP or TCP/UDP.

<i>Range</i>	-1 - 65535
<i>Access</i>	readwrite
<i>OID</i>	1.3.6.1.4.1.16177.1.400.2.1.1.2.1.1.12

cfgFwNatPrtFwdComment

User Comment

This parameter has no operational function. It allows to store a comment about the use of this port forward.

<i>Type</i>	DisplayString
<i>Access</i>	readwrite
<i>OID</i>	1.3.6.1.4.1.16177.1.400.2.1.1.2.1.1.13

cfgFwNatPrtFwdEnabled

Disable or Enable this Rule

<i>Enumeration</i>	disabled (0), enabled (1)
<i>Access</i>	readwrite
<i>OID</i>	1.3.6.1.4.1.16177.1.400.2.1.1.2.1.1.2

cfgFwNatPrtFwdInterface

Name of the Network Interface on Which the Rule Applies

Defines on which interface traffic is coming in. Groups of interfaces can be matched by adding the character '+' at the end. E.g. eth+ to match the interfaces eth0, eth1 and eth2. To match all interfaces use the character '+' alone.

<i>Type</i>	DisplayString
<i>Range</i>	1 - 255
<i>Access</i>	readwrite
<i>OID</i>	1.3.6.1.4.1.16177.1.400.2.1.1.2.1.1.3

cfgFwNatPrtFwdProtocol

Choose Which IP Protocol the Rule Matches

Allowed protocols are:

- **any(0):** Any ip protocol
- **udp(1):** Only UDP protocol
- **tcp(2):** Only TCP protocol
- **udptcp(3):** UDP and TCP protocol

<i>Enumeration</i>	any (0), udp (1), tcp (2), udptcp (3)
<i>Access</i>	readwrite
<i>OID</i>	1.3.6.1.4.1.16177.1.400.2.1.1.2.1.1.4

cfgFwNatPrtFwdSourceAddress

Source Address to Match

This can be a specific ip address or a range in CIDR notation. Set to 0.0.0.0/0 to match all inbound traffic. Set to 172.17.29.7/32 to match the specific IP 172.17.29.7 You can use ! to invert the sense of the rule: E.g. !192.168.0.0/24

Note: Usually you want 0.0.0.0/0.

Type	DisplayString
Range	9 - 19
Access	readwrite
OID	1.3.6.1.4.1.16177.1.400.2.1.1.2.1.1.5

cfgFwNatPrtFwdSourcePortStart

Source Start Port to Match

Specify the port or start of a port range from which a connection originates. Can only be used with TCP, UDP or TCP/UDP. Leave this on -1 to disable. You can use ! to invert the sense of the rule: E.g. !80. When used in a range, the inversion applies to the range.

Note: Usually you want this disabled.

Type	DisplayString
Range	1 - 6
Access	readwrite
OID	1.3.6.1.4.1.16177.1.400.2.1.1.2.1.1.6

cfgFwNatPrtFwdSourcePortEnd

Destination End Port to Match

When matching multiple port, this value is the end of the range. Set to -1 if no range is to be matched. Can only be used with TCP, UDP or TCP/UDP.

Note: Usually you want this disabled.

Range	-1 - 65535
Access	readwrite
OID	1.3.6.1.4.1.16177.1.400.2.1.1.2.1.1.7

cfgFwNatPrtFwdDestinationAddress

Destination Address to Redirect

This can be a specific ip address or a range in CIDR notation. Set to 0.0.0.0/0 to match all inbound traffic on the interface specified in cfgFwNatPrtFwdInterface. You can use ! to invert the sense of the rule: E.g. !192.168.0.0/24. When using static IPs set this to the configured address of the respective interface or alias you want to forward.

Be aware, that setting 0.0.0.0/0 will redirect everything arriving on the configured interface, even if not sent to the device itself.

Note: Leave this on 0.0.0.0/0 when using DHCP.

<i>Type</i>	DisplayString
<i>Range</i>	9 - 19
<i>Access</i>	readwrite
<i>OID</i>	1.3.6.1.4.1.16177.1.400.2.1.1.2.1.1.8

cfgFwNatPrtFwdDestinationPortStart

Destination Start Port to Redirect

Specify the port or start of a port range for the destination. You can use ! to invert the sense of the rule: E.g. !80. When used in a range, the inversion applies to the range. Can only be used with TCP, UDP or TCP/UDP.

<i>Type</i>	DisplayString
<i>Range</i>	1 - 20
<i>Access</i>	readwrite
<i>OID</i>	1.3.6.1.4.1.16177.1.400.2.1.1.2.1.1.9

10.1.1.2.2 cfgFwNatOutboundTable

Firewall Outbound NAT Rules Table (SNAT)

<i>Range</i>	0 - 255
<i>OID</i>	1.3.6.1.4.1.16177.1.400.2.1.1.2.2

10.1.1.2.2.1 cfgFwNatOutboundTableEntry

Firewall Outbound NAT Rules Table (SNAT)

<i>OID</i>	1.3.6.1.4.1.16177.1.400.2.1.1.2.2.1
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cfgFwNatOutIndex

Table Entry Index

<i>Range</i>	0 - 255
<i>Access</i>	noaccess
<i>OID</i>	1.3.6.1.4.1.16177.1.400.2.1.1.2.2.1.1

cfgFwNatOutDestinationPortEnd

Destination End Port to Match

When forwarding multiple port, this value is the end of the range. Set to -1 if no range is forwarded. Can only be used with TCP, UDP or TCP/UDP.

Note: Usually you want this disabled.

<i>Range</i>	-1 - 65535
<i>Access</i>	readwrite
<i>OID</i>	1.3.6.1.4.1.16177.1.400.2.1.1.2.2.1.10

cfgFwNatOutSourceRewriteAddress

Set the Address With Which Outbound Traffic Shall be Rewritten

In case you are using DHCP leave this on 0.0.0.0.

Note: If you are not rewriting the source to a specific aliases you can set this to 0.0.0.0 to automatically rewrite to the configured primary (first) address of the interface.

<i>Type</i>	IpAddress
<i>Access</i>	readwrite
<i>OID</i>	1.3.6.1.4.1.16177.1.400.2.1.1.2.2.1.11

cfgFwNatOutSourceRewritePort

Set the Source Port With Which Outbound Traffic Shall be Rewritten

Can only be used with TCP, UDP or TCP/UDP. Set to -1 to disable source port rewrite.

Note: Usually you want this disabled.

<i>Range</i>	-1 - 65535
<i>Access</i>	readwrite
<i>OID</i>	1.3.6.1.4.1.16177.1.400.2.1.1.2.2.1.12

cfgFwNatOutComment

User Comment

This parameter has no operational function. It allows to store a comment about the use of this outbound rule.

<i>Type</i>	DisplayString
<i>Access</i>	readwrite
<i>OID</i>	1.3.6.1.4.1.16177.1.400.2.1.1.2.2.1.13

cfgFwNatOutEnabled

Disable or Enable this Rule

<i>Enumeration</i>	disabled (0), enabled (1)
<i>Access</i>	readwrite
<i>OID</i>	1.3.6.1.4.1.16177.1.400.2.1.1.2.2.1.2

cfgFwNatOutInterface

Name of the Network Interface on Which the Rule Applies

Matches traffic leaving on this interface. Needs to be set to an interface name if you are using DHCP. Set to -1 if you don't know on which interface traffic will be leaving. Match the traffic with cfgFwNatOutDestinationAddress instead. You can use ! to invert the sense of the rule. E.g. !wlan0.

<i>Type</i>	DisplayString
<i>Range</i>	1 - 255
<i>Access</i>	readwrite
<i>OID</i>	1.3.6.1.4.1.16177.1.400.2.1.1.2.2.1.3

cfgFwNatOutProtocol

Choose Which IP Protocol the Rule Matches

Allowed protocols are:

- **any(0):** Any ip protocol
- **udp(1):** Only UDP protocol
- **tcp(2):** Only TCP protocol
- **udptcp(3):** UDP and TCP protocol

<i>Enumeration</i>	any (0), udp (1), tcp (2), udptcp (3)
<i>Access</i>	readwrite
<i>OID</i>	1.3.6.1.4.1.16177.1.400.2.1.1.2.2.1.4

cfgFwNatOutSourceAddress

Source Address to Match

This can be a specific ip address or a range in CIDR notation. Set to 0.0.0.0/0 to match all inbound traffic. Set to 172.17.29.7/32 to match the specific IP 172.17.29.7. You can use ! to invert the sense of the rule: E.g. !192.168.0.0/24.

Note: Usually you want 0.0.0.0/0.

Type	DisplayString
Range	9 - 19
Access	readwrite
OID	1.3.6.1.4.1.16177.1.400.2.1.1.2.2.1.5

cfgFwNatOutSourcePortStart

Source Start Port to Match

Specify the port or start of a port range from which a connection originates. Can only be used with TCP, UDP or TCP/UDP. Leave this on -1 to disable. You can use ! to invert the sense of the rule: E.g. !80. When used in a range, the inversion applies to the range.

Note: Usually you want this disabled.

Type	DisplayString
Range	1 - 6
Access	readwrite
OID	1.3.6.1.4.1.16177.1.400.2.1.1.2.2.1.6

cfgFwNatOutSourcePortEnd

Destination End Port to Match

When matching multiple port, this value is the end of the range. Set to -1 if no range is to be matched. Can only be used with TCP, UDP or TCP/UDP.

Note: Usually you want this disabled.

Range	-1 - 65535
Access	readwrite
OID	1.3.6.1.4.1.16177.1.400.2.1.1.2.2.1.7

cfgFwNatOutDestinationAddress

Destination Address to Match

This can be a specific ip address or a range in CIDR notation. Set to 0.0.0.0/0 to match all outbound traffic on the interface specified in cfgFwNatOutInterface. You can use ! to invert the sense of the rule: E.g. !192.168.0.0/24.

Type	DisplayString
Range	9 - 19
Access	readwrite
OID	1.3.6.1.4.1.16177.1.400.2.1.1.2.2.1.8

cfgFwNatOutDestinationPortStart

Destination Start Port to Match

Specify the port or start of a port range for the destination. Can only be used with TCP, UDP or TCP/UDP. You can use ! to invert the sense of the rule: E.g. !80. When used in a range, the inversion applies to the range.

Note: Usually you want this disabled.

Type	DisplayString
Range	1 - 6
Access	readwrite
OID	1.3.6.1.4.1.16177.1.400.2.1.1.2.2.1.9

10.1.1.3 cfgFwL2IpFilter

10.1.1.3.1 cfgFwL2IpFilterEnabled

Globally Disable or Enable the L2 IP Filter Option of all Bridges

This filter will only apply on IP frames and will not affect any other frames.

Enumeration	disabled (0), enabled (1)
Access	readwrite
OID	1.3.6.1.4.1.16177.1.400.2.1.1.3.1

10.1.1.3.2 cfgFwL2IpFilterDefaultAction

Set the Default Action of all Bridges When Filtering is Enabled

Do not lock yourself out when the default action is 'drop'.

Enumeration	accept (0), drop (1)
Access	readwrite
OID	1.3.6.1.4.1.16177.1.400.2.1.1.3.2

10.1.1.3.3 cfgFwL2IpFilterTable

L2 IP Filter

Range	0 - 255
OID	1.3.6.1.4.1.16177.1.400.2.1.1.3.3

10.1.1.3.3.1 cfgFwL2IpFilterTableEntry

L2 IP Filter

<i>OID</i>	1.3.6.1.4.1.16177.1.400.2.1.1.3.3.1
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cfgFwL2IpFiltrIndex

Entry Index of Table

<i>Range</i>	0 - 255
<i>Access</i>	noaccess
<i>OID</i>	1.3.6.1.4.1.16177.1.400.2.1.1.3.3.1.1

cfgFwL2IpFiltrSourcePort

The Source Port On Which The Rule Matches

Set to -1, to not use the source port to match.

When `cfgFwL2IpFiltrEthertype` is set to 0800 and `cfgFwL2IpFiltrProtocol` is 6 (TCP) or 17 (UDP), this field 'may be used to match a specific source port.

<i>Range</i>	-1 - 65535
<i>Access</i>	readwrite
<i>OID</i>	1.3.6.1.4.1.16177.1.400.2.1.1.3.3.1.10

cfgFwL2IpFiltrDestinationPort

The Destination Port On Which The Rule Matches

Set to -1, to not use the destination port to match.

When `cfgFwL2IpFiltrEthertype` is set to 0800 and `cfgFwL2IpFiltrProtocol` is 6 (TCP) or 17 (UDP), this field 'may be used to match a specific source port.

<i>Range</i>	-1 - 65535
<i>Access</i>	readwrite
<i>OID</i>	1.3.6.1.4.1.16177.1.400.2.1.1.3.3.1.11

cfgFwL2IpFiltrSourceMac

The Source MAC Addresses On Which The Rule Matches

This is a MAC address and a mask in the form of `xx:xx:xx:xx:xx:xx/yy:yy:yy:yy:yy:yy`.

Examples:

- 00:00:00:00:00:00/00:00:00:00:00:00 match any source
- 00:14:5a:02:04:4c/ff:ff:ff:ff:ff:ff exact match of the source address 00:14:5a:02:04:4c
- 00:14:5a:00:00:00/ff:ff:ff:00:00:00 match any source addresses with the vendor OUI 00:14:5a

Type	DisplayString
Range	35 - 35
Access	readwrite
OID	1.3.6.1.4.1.16177.1.400.2.1.1.3.3.1.12

cfgFwL2IpFiltrDestinationMac

The Destination MAC Addresses On Which The Rule Matches

This is a MAC address and a mask in the form of xx:xx:xx:xx:xx:xx/yy:yy:yy:yy:yy:yy.

Examples:

- 00:00:00:00:00:00/00:00:00:00:00:00 match any destination
- 01:00:00:00:00:00/01:00:00:00:00:00 match all frames with a multicast destination (including broadcast)
- 00:14:5a:02:04:4c/ff:ff:ff:ff:ff:ff exact match of the destination address 00:14:5a:02:04:4c
- 00:14:5a:00:00:00/ff:ff:ff:00:00:00 match any destination addresses with the vendor OUI 00:14:5a

Type	DisplayString
Range	35 - 35
Access	readwrite
OID	1.3.6.1.4.1.16177.1.400.2.1.1.3.3.1.13

cfgFwL2IpFiltrEthertype

The Ethertype On Which The Rule Matches

This is an ethertype in the form of XXXX where each X is a hexadecimal character [0-9A-Fa-f].

Default is 0800, which represents IPv4.

Set to 0000, to match any ethertype.

Examples:

- 0000: any

- 0800: IPv4
- 0806: ARP
- 8035: RARP
- 86DD: IPv6
- 8847: MPLS unicast
- 8848: MPLS multicast
- 8892: Profinet

For a full list of IANA assigned ethertypes see <https://www.iana.org/assignments/ieee-802-numbers/ieee-802-numbers.xhtml>

<i>Type</i>	DisplayString
<i>Range</i>	4 - 4
<i>Access</i>	readwrite
<i>OID</i>	1.3.6.1.4.1.16177.1.400.2.1.1.3.3.1.14

cfgFwL2IpFiltrVlan

The VLAN On Which The Rule Matches

Set to -1, to not use the VLAN to match.

Match on the VLAN number in the 802.1Q header when it is present.

NOTE: Frames that ingress on an access ports are not considered part of the assigned vlan yet.

<i>Range</i>	-1 - 4095
<i>Access</i>	readwrite
<i>OID</i>	1.3.6.1.4.1.16177.1.400.2.1.1.3.3.1.15

cfgFwL2IpFiltrInputInterface

The Ingress Interface On Which The Rule Matches

The name of the interface on which a frame ingresses.

Set to any, to match any interface.

Examples:

- any
- wlan0
- eth0

Type	DisplayString
Range	1 - 15
Access	readwrite
OID	1.3.6.1.4.1.16177.1.400.2.1.1.3.3.1.16

cfgFwL2IpFiltrEnabled

Rule Disabled or Enabled

Enumeration	disabled (0), enabled (1)
Access	readwrite
OID	1.3.6.1.4.1.16177.1.400.2.1.1.3.3.1.2

cfgFwL2IpFiltrBridge

Bridge on Which the Rule is Applied

Range	-1 - 255
Access	readwrite
OID	1.3.6.1.4.1.16177.1.400.2.1.1.3.3.1.3

cfgFwL2IpFiltrAction

Action to Perform

Enumeration	accept (0), drop (1)
Access	readwrite
OID	1.3.6.1.4.1.16177.1.400.2.1.1.3.3.1.4

cfgFwL2IpFiltrPriority

Filter Priority

When multiple rules match, the rule with the highest priority will be applied.

Multiple rules with the same priority have undefined behaviour, thus are not allowed.

Type	Integer32
Access	readwrite
OID	1.3.6.1.4.1.16177.1.400.2.1.1.3.3.1.5

cfgFwL2IpFiltrSource

Source Network/IP On Which The Rule Matches (CIDR Notation)

When `cfgFwL2IpFilterMode` is set to **full(1)**, then this field only has an effect when `cfgFwL2IpFltrEthertype` is set to 0800, 0806 or 8035.

When `cfgFwL2IpFltrEthertype` is set to 0800 this is an IP or network.

When `cfgFwL2IpFltrEthertype` is set to 0806 or 8035 it is the source address or a range of source addresses of an ARP/RARP frame (ARP_SPA).

<i>Type</i>	DisplayString
<i>Range</i>	9 - 19
<i>Access</i>	readwrite
<i>OID</i>	1.3.6.1.4.1.16177.1.400.2.1.1.3.3.1.6

cfgFwL2IpFltrDestination

Destination Network/IP On Which The Rule Matches (CIDR Notation)

When `cfgFwL2IpFilterMode` is set to **full(1)**, then this field only has an effect when `cfgFwL2IpFltrEthertype` is set to 0800, 0806 or 8035.

When `cfgFwL2IpFltrEthertype` is set to 0800 this is an IP or network.

When `cfgFwL2IpFltrEthertype` is set to 0806 or 8035 it is the target address or a range of target addresses of an ARP/RARP frame (ARP_TPA).

<i>Type</i>	DisplayString
<i>Range</i>	9 - 19
<i>Access</i>	readwrite
<i>OID</i>	1.3.6.1.4.1.16177.1.400.2.1.1.3.3.1.7

cfgFwL2IpFltrComment

User Comment

This parameter has no operational function. It allows to store a comment about the use of this I2 filter rule.

<i>Type</i>	DisplayString
<i>Access</i>	readwrite
<i>OID</i>	1.3.6.1.4.1.16177.1.400.2.1.1.3.3.1.8

cfgFwL2IpFltrProtocol

IP Protocol On Which The Rule Matches

Set to -1, to not use the protocol to match.

When `cfgFwL2IpFltrEthertype` is set to 0800, this field may be used to match a specific IP protocol.

Examples:

- -1: any protocol
- 1: ICMP
- 2: IGMP
- 6: TCP
- 17: UDP
- 50: ESP (IPsec)
- 51: AH (IPsec)
- 112: VRRP / CARP

For a full list of available protocols see https://en.wikipedia.org/wiki/List_of_IP_protocol_numbers

When `cfgFwL2IpFltrEthertype` is set to 0806 or 8035, this field may be used to match the OP code of an ARP frame.

Examples:

- -1: any OP code
- 1: request
- 2: response
- 3: request reverse
- 4: response reverse

<i>Range</i>	-1 - 255
<i>Access</i>	readwrite
<i>OID</i>	1.3.6.1.4.1.16177.1.400.2.1.1.3.3.1.9

10.1.1.3.4 `cfgFwL2IpFilterMode`

Filter Mode

The filtering bridge provides two modes of operation:

- **ip(0)**: A simplified mode to filter IP traffic. All non-IP traffic is allowed.
- **full(1)**: The full mode allows to filter on any kind of header information present.

<i>Enumeration</i>	ip (0), full (1)
<i>Access</i>	readwrite
<i>OID</i>	1.3.6.1.4.1.16177.1.400.2.1.1.3.4

10.1.1.4 cfgFwFilter

10.1.1.4.1 cfgFwFltDefaultPolicyInput

The Default Filter Policy on the Input Path

<i>Enumeration</i>	drop (0), accept (1)
<i>Access</i>	readwrite
<i>OID</i>	1.3.6.1.4.1.16177.1.400.2.1.1.4.1

10.1.1.4.2 cfgFwFilterRulesTable

Firewall Filter Rules Table

<i>Range</i>	0 - 255
<i>OID</i>	1.3.6.1.4.1.16177.1.400.2.1.1.4.10

10.1.1.4.2.1 cfgFwFilterRulesTableEntry

Firewall Filter Rules Table

<i>OID</i>	1.3.6.1.4.1.16177.1.400.2.1.1.4.10.1
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cfgFwFltRIndex

Table Entry Index

<i>Range</i>	0 - 255
<i>Access</i>	noaccess
<i>OID</i>	1.3.6.1.4.1.16177.1.400.2.1.1.4.10.1.1

cfgFwFltRSourcePortEnd

Source End Port to Match

When matching multiple ports, this value is the end of the range. Can only be used with tcp or udp.

Set to -1 when no range is to be matched.

<i>Range</i>	-1 - 65535
<i>Access</i>	readwrite
<i>OID</i>	1.3.6.1.4.1.16177.1.400.2.1.1.4.10.1.10

cfgFwFltRDestinationAddress

Destination Address to Match

This can be a specific ip address or a range in CIDR notation. Set to 0.0.0.0/0 to match all destinations. Set to 172.17.29.7/32 to match the specific IP 172.17.29.7. You can use ! to invert the sense of the rule: E.g. !192.168.0.0/24

Set to -1 to not use this parameter.

Type	DisplayString
Range	1 - 20
Access	readwrite
OID	1.3.6.1.4.1.16177.1.400.2.1.1.4.10.1.11

cfgFwFltRDestinationPortStart

Destination Start Port to Match

Specify the port or start of a port range to which a connection is going. Can only be used with tcp or udp. You can use ! to invert the sense of the rule: E.g. !80. When used in a range, the inversion applies to the range.

Set to -1 to not use this parameter.

Type	DisplayString
Range	1 - 20
Access	readwrite
OID	1.3.6.1.4.1.16177.1.400.2.1.1.4.10.1.12

cfgFwFltRDestinationPortEnd

Destination End Port to Match

When matching multiple ports, this value is the end of the range. Can only be used with tcp or udp.

Set to -1 when no range is to be matched.

Range	-1 - 65535
Access	readwrite
OID	1.3.6.1.4.1.16177.1.400.2.1.1.4.10.1.13

cfgFwFltRComment

User Comment

This parameter has no operational function. It allows to store a comment about the use of this filter rule.

<i>Type</i>	DisplayString
<i>Access</i>	readwrite
<i>OID</i>	1.3.6.1.4.1.16177.1.400.2.1.1.4.10.1.14

cfgFwFltREnabled

Disable or Enable This Rule

<i>Enumeration</i>	disabled (0), enabled (1)
<i>Access</i>	readwrite
<i>OID</i>	1.3.6.1.4.1.16177.1.400.2.1.1.4.10.1.2

cfgFwFltRChain

Chain on Which Action is Performed

<i>Enumeration</i>	none (0), input (1), forward (2), output (3)
<i>Access</i>	readwrite
<i>OID</i>	1.3.6.1.4.1.16177.1.400.2.1.1.4.10.1.3

cfgFwFltRAction

Action to be Performed

<i>Enumeration</i>	drop (0), accept (1)
<i>Access</i>	readwrite
<i>OID</i>	1.3.6.1.4.1.16177.1.400.2.1.1.4.10.1.4

cfgFwFltRInputInterface

Name of the Input Interface to Match

Groups of interfaces can be matched by adding the character '+' at the end. E.g. eth+ to match the interfaces eth0, eth1 and eth2. To match all interfaces use the character '+' alone. Set to -1 to not use this parameter.

<i>Type</i>	DisplayString
<i>Range</i>	1 - 16
<i>Access</i>	readwrite
<i>OID</i>	1.3.6.1.4.1.16177.1.400.2.1.1.4.10.1.5

cfgFwFltROutputInterface

Name of the Output Interface to Match

Groups of interfaces can be matched by adding the character '+' at the end. E.g. eth+ to match the interfaces eth0, eth1 and eth2. To match all interfaces use the character '+' alone.

Set to -1 to not use this parameter.

This parameter is ignored for rules on the input chain.

Type	DisplayString
Range	1 - 16
Access	readwrite
OID	1.3.6.1.4.1.16177.1.400.2.1.1.4.10.1.6

cfgFwFltRProtocol

Choose Which IP Protocol the Rule Matches

For a list of the currently existing protocols see: https://en.wikipedia.org/wiki/List_of_IP_protocol_numbers

Examples:

- **any(0)**: Match any ip protocol
- **icmp(1)**
- **igmp(2)**
- **tcp(6)**
- **udp(17)**
- **gre(47)**
- **esp(50)**
- **ah(51)**
- **ospf(89)**
- **vrrp / carp(112)**
- **l2tp(115)**

Range	0 - 255
Access	readwrite
OID	1.3.6.1.4.1.16177.1.400.2.1.1.4.10.1.7

cfgFwFltRSourceAddress

Source Address to Match

This can be a specific ip address or a range in CIDR notation. Set to 0.0.0.0/0 to match all sources. Set to 172.17.29.7/32 to match the specific IP 172.17.29.7. You can use ! to invert the sense of the

rule: E.g. !192.168.0.0/24

Set to -1 to not use this parameter.

Type	DisplayString
Range	1 - 20
Access	readwrite
OID	1.3.6.1.4.1.16177.1.400.2.1.1.4.10.1.8

cfgFwFltRSourcePortStart

Source Start Port to Match

Specify the port or start of a port range from which a connection originates. Can only be used with tcp or udp. You can use ! to invert the sense of the rule: E.g. !80. When used in a range, the inversion applies to the range.

Set to -1 to not use this parameter.

Type	DisplayString
Range	1 - 20
Access	readwrite
OID	1.3.6.1.4.1.16177.1.400.2.1.1.4.10.1.9

10.1.1.4.3 cfgFwFltDefaultPolicyForward

The Default Filter Policy on the Forward Path

Enumeration	drop (0), accept (1)
Access	readwrite
OID	1.3.6.1.4.1.16177.1.400.2.1.1.4.2

10.1.1.4.4 cfgFwFltDefaultPolicyOutput

The Default Filter Policy on the Output Path

Enumeration	drop (0), accept (1)
Access	readwrite
OID	1.3.6.1.4.1.16177.1.400.2.1.1.4.3

11 WESTERMO-SW6-GNSS-MIB

11.1 gnss

11.1.1 configuration

11.1.1.1 cfgGnss

11.1.1.1.1 cfgGnssGpsd

11.1.1.1.1.1 cfgGnssGpsdEnabled

GPS Service Daemon Disabled or Enabled

If the GPS service daemon `gpsd` is **enabled(1)**, it makes location data available through TCP/IP.

Applies to cellular products only.

<i>Enumeration</i>	disabled (0), enabled (1)
<i>Access</i>	readwrite
<i>OID</i>	1.3.6.1.4.1.16177.1.400.2.10.1.1.1.1

11.1.1.1.1.2 cfgGnssGpsdAddress

Bind Address

The `gpsd` can only bind to the localhost or to all addresses. This means, that it will listen to all addresses, when this address is set to a value other than `127.0.0.1:<port>`.

The default bind address is `0.0.0.0:2974`.

Examples:

- **0.0.0.0:2974**

- **127.0.0.1:1234**

Applies to cellular products only.

Type	DisplayString
Range	1 - 255
Access	readwrite
OID	1.3.6.1.4.1.16177.1.400.2.10.1.1.1.2

11.1.1.1.2 cfgGnssDevice

11.1.1.1.2.1 cfgGnssDevSatelliteSystems

Satellite Systems

This entry allows the selection of different satellite systems, whereby the following combinations are possible:

- **gps(1)**: GPS
- **glonass(2)**: Glonass
- **gpsGlo(3)**: GPS and Glonass
- **beidou(4)**: BeiDou
- **gpsBei(5)**: GPS and BeiDou
- **gloBei(6)**: Glonass and BeiDou
- **galileo(8)**: Galileo
- **gpsGal(9)**: GPS and Galileo
- **gloGal(10)**: Glonass and Galileo
- **gpsGloGal(11)**: GPS, Glonass and Galileo
- **beiGal(12)**: BeiDou and Galileo
- **gpsBeiGal(13)**: GPS, BeiDou and Glonass

This entry can also be interpreted as bitmask, with each bit reflecting a satellite system as follows:

- Bit 1: GPS
- Bit 2: Glonass
- Bit 3: BeiDou
- Bit 4: Galileo

However, only the combinations of the enumeration defined above are valid.

Applies to cellular products only.

<i>Enumeration</i>	gps (1), glonass (2), gpsGlo (3), beidou (4), gpsBei (5), gloBei (6), galileo (8), gpsGal (9), gloGal (10), gpsGloGal (11), beiGal (12), gpsBeiGal (13)
<i>Access</i>	readwrite
<i>OID</i>	1.3.6.1.4.1.16177.1.400.2.10.1.1.2.1

11.1.1.1.2.2 cfgGnssDevMeasurementPeriod

Measurement Period

The measurement period T defines the time interval in milliseconds between two consecutive localisations. The minimal time is limited to 50 ms. Certain GNSS hardware may silently adjust the measurement period to even higher values.

The sampling rate f is calculated as follows:

$$f = 1000 / T$$

Examples:

- **10000**: $f = 0.1$ Hz
- **1000**: $f = 1$ Hz
- **100**: $f = 10$ Hz

Applies to cellular products only.

<i>Type</i>	Integer32
<i>Access</i>	readwrite
<i>OID</i>	1.3.6.1.4.1.16177.1.400.2.10.1.1.2.2

11.1.1.1.2.3 cfgGnssDevMessages

cfgGnssDevMsgsNmeaTable

NMEA Sentences

The National Marine Electronics Association (NMEA) defined in the NMEA-0183 standard how data are transmitted in a sentences from one talker to multiple listeners.

Applies to cellular products only.

<i>Range</i>	0 - 12
<i>OID</i>	1.3.6.1.4.1.16177.1.400.2.10.1.1.2.3.1

cfgGnssDevMsgsNmeaTableEntry

NMEA Sentence

Applies to cellular products only.

<i>OID</i>	1.3.6.1.4.1.16177.1.400.2.10.1.1.2.3.1.1
------------	--

cfgGnssDevMsgsNmeaTableIndex

Table Entry Index

<i>Range</i>	0 - 12
<i>Access</i>	noaccess
<i>OID</i>	1.3.6.1.4.1.16177.1.400.2.10.1.1.2.3.1.1.1

cfgGnssDevMsgsNmeaType

NMEA Sentence Type

The following NMEA sentence types are supported:

- GxGGA (61440)
- GxGLL (61441)
- GxGSA (61442)
- GxGSV (61443)
- GxRMC (61444)
- GxVTG (61445)
- GxGRS (61446)
- GxGST (61447)
- GxZDA (61448)
- GxGBS (61449)
- GxDTM (61450)
- GxGNS (61453)
- GxVLW (61455)

Applies to cellular products only.

<i>Type</i>	Integer32
<i>Access</i>	readwrite
<i>OID</i>	1.3.6.1.4.1.16177.1.400.2.10.1.1.2.3.1.1.2

cfgGnssDevMsgsNmeaRate

Sentence Rate

The sentence rate r multiplies the measurement period T specified in `cfgGnssDevMeasurementPeriod` for the corresponding NMEA sentence type. It allows to define individual sampling rates for each sentence type or disabling the them by setting their rate to 0.

The individual sampling rate f is calculated as follows:

$$f = 1000 / (r * T)$$

Examples:

- 1 ($T = 1000$ ms): $f = 1$ Hz
- 2 ($T = 1000$ ms): $f = 0.5$ Hz
- 5 ($T = 100$ ms): $f = 2$ Hz

Applies to cellular products only.

Type	Integer32
Access	readwrite
OID	1.3.6.1.4.1.16177.1.400.2.10.1.1.2.3.1.1.3

cfgGnssDevMsgsNmeaEnabled

Enable or Disable NMEA Sentences

Applies to cellular products only.

Enumeration	disabled (0), enabled (1)
Access	readwrite
OID	1.3.6.1.4.1.16177.1.400.2.10.1.1.2.3.1.1.4

cfgGnssDevMsgsUbxTable

UBX Messages

UBX is a proprietary protocol from the company u-blox for exchanging GNSS data.

Applies to cellular products only.

Range	0 - 127
OID	1.3.6.1.4.1.16177.1.400.2.10.1.1.2.3.2

cfgGnssDevMsgsUbxTableEntry

UBX Message

Applies to cellular products only.

<i>OID</i>	1.3.6.1.4.1.16177.1.400.2.10.1.1.2.3.2.1
------------	--

cfgGnssDevMsgsUbxTableIndex

Table Entry Index

<i>Range</i>	0 - 127
<i>Access</i>	noaccess
<i>OID</i>	1.3.6.1.4.1.16177.1.400.2.10.1.1.2.3.2.1.1

cfgGnssDevMsgsUbxType

UBX Message Type

The following UBX message types are supported:

- NAV-POSECEF (257)
- NAV-STATUS (259)
- NAV-PVT (263)
- NAV-VELECEF(273)
- NAV-TIMEUTC (289)
- NAV-SAT (309)

Applies to cellular products only.

<i>Range</i>	0 - 65535
<i>Access</i>	readwrite
<i>OID</i>	1.3.6.1.4.1.16177.1.400.2.10.1.1.2.3.2.1.2

cfgGnssDevMsgsUbxRate

Message Rate

The message rate r multiplies the measurement period T specified in `cfgGnssDevMeasurementPeriod` for the corresponding UBX message type. It allows to define individual sampling rates for each message type or disabling the them by setting their rate to 0.

The individual sampling rate f is calculated as follows:

$$f = 1000 / (r * T)$$

Examples:

- 1 (T = 1000 ms): f = 1 Hz
- 2 (T = 1000 ms): f = 0.5 Hz
- 5 (T = 100 ms): f = 2 Hz

Applies to cellular products only.

Type	Integer32
Access	readwrite
OID	1.3.6.1.4.1.16177.1.400.2.10.1.1.2.3.2.1.3

cfgGnssDevMsgsUbxEnabled

Enable or Disable UBX Messages

Applies to cellular products only.

Enumeration	disabled (0), enabled (1)
Access	readwrite
OID	1.3.6.1.4.1.16177.1.400.2.10.1.1.2.3.2.1.4

11.1.2 software

11.1.2.1 swGnss

11.1.2.1.1 swGnssTime

GNSS Time

The time is given in the Coordinated Universal Time (UTC) according to the ISO 8601 standard.

Example:

- 2020-06-19T10:55:50.000Z

Applies to cellular products only.

Type	DisplayString
Range	1 - 255
Access	readonly
OID	1.3.6.1.4.1.16177.1.400.2.10.2.1.1

11.1.2.1.2 swGnssMode

GNSS Navigation Mode

Possible navigation modes are:

- **na(0)**: Not available
- **none(1)**: None
- **twoD(2)**: 2D
- **threeD(3)**: 3D

Applies to cellular products only.

<i>Enumeration</i>	na (0), none (1), twoD (2), threeD (3)
<i>Access</i>	readonly
<i>OID</i>	1.3.6.1.4.1.16177.1.400.2.10.2.1.2

11.1.2.1.3 swGnssLon

GNSS Longitude

Longitude of the current location in decimal degrees (DD).

Example:

- 8.825446333

Applies to cellular products only.

<i>Type</i>	DisplayString
<i>Range</i>	1 - 255
<i>Access</i>	readonly
<i>OID</i>	1.3.6.1.4.1.16177.1.400.2.10.2.1.3

11.1.2.1.4 swGnssLat

GNSS Latitude

Latitude of the current location in decimal degrees (DD).

- 47.268953167

Applies to cellular products only.

<i>Type</i>	DisplayString
<i>Range</i>	1 - 255
<i>Access</i>	readonly
<i>OID</i>	1.3.6.1.4.1.16177.1.400.2.10.2.1.4

11.1.2.1.5 swGnssAlt

GNSS Altitude

Altitude of the current location in meters.

Applies to cellular products only.

<i>Type</i>	DisplayString
<i>Range</i>	1 - 255
<i>Access</i>	readonly
<i>OID</i>	1.3.6.1.4.1.16177.1.400.2.10.2.1.5

12 WESTERMO-SW6-ICL-MIB

12.1 icl

12.1.1 configuration

12.1.1.1 cfgIcl

12.1.1.1.1 cfgIclEnabled

Enable Inter-Carriage Link application.

Applies to AP. 802.11n products only.

<i>Enumeration</i>	disabled (0), enabled (1)
<i>Access</i>	readwrite
<i>OID</i>	1.3.6.1.4.1.16177.1.400.2.5.1.1.1

12.1.1.1.2 cfgIclConnectionDelay

Connection delay after a potential ICL partner was first detected.

This value in conjunction with cfgIclCycleTime defines how extensively a potential ICL partner is monitored and analyzed before a connection is established.

Applies to AP. 802.11n products only.

<i>Range</i>	0 - 600
<i>Access</i>	readwrite
<i>OID</i>	1.3.6.1.4.1.16177.1.400.2.5.1.1.2

12.1.1.1.3 cfgIclConnectionThreshold

This value defines the minimum signal level necessary for the ICL application to start evaluating a potential ICL partner.

Applies to AP. 802.11n products only.

<i>Range</i>	-90 - 0
<i>Access</i>	readwrite
<i>OID</i>	1.3.6.1.4.1.16177.1.400.2.5.1.1.3

12.1.1.1.4 cfgIclDisconnectionDelay

Disconnection delay in seconds defines how quickly a connected ICL pair resets to scanning mode after the current ICL partner reaches a low signal level or gets disconnected.

Applies to AP. 802.11n products only.

<i>Range</i>	1 - 600
<i>Access</i>	readwrite
<i>OID</i>	1.3.6.1.4.1.16177.1.400.2.5.1.1.4

12.1.1.1.5 cfgIclDisconnectionThreshold

This value defines the minimum signal level necessary for a ICL pair to stay connected. If the signal level drops below this level for longer than in cfgIclDisconnectionDelay specified, the ICL application will revert the device to access point and resume scans for a new partner.

Applies to AP. 802.11n products only.

<i>Range</i>	-90 - 0
<i>Access</i>	readwrite
<i>OID</i>	1.3.6.1.4.1.16177.1.400.2.5.1.1.5

12.1.1.1.6 cfgIclInterfaceName

This value describes the interface the ICL Application will use for its services.

Applies to AP. 802.11n products only.

<i>Type</i>	DisplayString
<i>Range</i>	1 - 255
<i>Access</i>	readwrite
<i>OID</i>	1.3.6.1.4.1.16177.1.400.2.5.1.1.6

12.1.1.1.7 cfgIclCycleTime

Interval of background scans in seconds.

This value in conjunction with `cfgIclConnectionDelay` defines how extensively a potential ICL partner is monitored and analyzed before a connection is established.

Applies to AP. 802.11n products only.

<i>Range</i>	2 - 60
<i>Access</i>	readwrite
<i>OID</i>	1.3.6.1.4.1.16177.1.400.2.5.1.1.7

12.1.1.1.8 `cfgIclBlacklistTime`

Duration of blacklisting in seconds.

Applies to AP. 802.11n products only.

<i>Range</i>	1 - 3600
<i>Access</i>	readwrite
<i>OID</i>	1.3.6.1.4.1.16177.1.400.2.5.1.1.8

12.1.1.1.9 `cfgIclSuspended`

Initial state of ICL when it starts up.

Applies to AP. 802.11n products only.

<i>Enumeration</i>	resumed (0), suspended (1)
<i>Access</i>	readwrite
<i>OID</i>	1.3.6.1.4.1.16177.1.400.2.5.1.1.9

12.1.2 `rpc`

12.1.2.1 `rpclcl`

12.1.2.1.1 `rpclclForceDisconnect`

Force the device to disconnect from the current ICL partner and resume background scanning for a new partner.

<i>Enumeration</i>	nop (0), disconnect (1)
<i>Access</i>	readwrite
<i>OID</i>	1.3.6.1.4.1.16177.1.400.2.5.3.1.1

12.1.2.1.2 rpclclClearBlacklist

Clear all currently blacklisted entries.

<i>Enumeration</i>	nop (0), clear (1)
<i>Access</i>	readwrite
<i>OID</i>	1.3.6.1.4.1.16177.1.400.2.5.3.1.2

12.1.3 settings

12.1.3.1 setlcl

12.1.3.1.1 setlclSuspended

Suspend or resume ICL operation.

When suspended, ICL brings down the wireless interface and pauses operation. It remains silent until it is resumed.

<i>Enumeration</i>	resumed (0), suspended (1)
<i>Access</i>	readwrite
<i>OID</i>	1.3.6.1.4.1.16177.1.400.2.5.4.1.1

12.1.4 software

12.1.4.1 swlcl

12.1.4.1.1 swlclStatus

Current status of ICL application.

- **scanning(1):** Scanning indicates Access Point mode with background scanning activated.
- **connected(2):** Connected indicates a connection with an ICL partner is established and background scanning is disabled.
- **suspended(3):** Suspended indicates that ICL is currently suspended.

<i>Enumeration</i>	disabled (0), scanning (1), connected (2), suspended (3)
<i>Access</i>	readonly
<i>OID</i>	1.3.6.1.4.1.16177.1.400.2.5.6.1.1

13 WESTERMO-SW6-NWM-MIB

13.1 nwm

13.1.1 configuration

13.1.1.1 cfgHttpRequest

13.1.1.1.1 cfgHttpRprtServerUrl

URL Of The Remote Server Where AFM, AFC And IDF Reports Are Sent To

Format: http://<ipv4>[:<port>] [/<path>]

<port> is optional and defaults to 80 when not specified. <path> is optional and defaults to / when not specified.

Applies to AP. 802.11n products only.

<i>Type</i>	DisplayString
<i>Range</i>	1 - 255
<i>Access</i>	readwrite
<i>OID</i>	1.3.6.1.4.1.16177.1.400.2.3.1.1.1

13.1.1.2 cfgChannelManager

13.1.1.2.1 cfgChMgrUsableFrequencyList

Configure which frequency list is to be used as 'list of usable frequencies' by the ChannelManager. A value of -1 means no list defined using all frequencies available in current country.

Applies to AP. 802.11n products only.

<i>Range</i>	-1 - 23
<i>Access</i>	readwrite
<i>OID</i>	1.3.6.1.4.1.16177.1.400.2.3.1.3.2

13.1.1.2.2 cfgChMgrDfsUseNvram

When 'enabled' the DFS states will be load / stored from / to the non-volatile ram device.

Applies to AP. 802.11n products only.

<i>Enumeration</i>	disabled (0), enabled (1)
<i>Access</i>	readwrite
<i>OID</i>	1.3.6.1.4.1.16177.1.400.2.3.1.3.3

13.1.1.3 cfgNwm

13.1.1.3.1 cfgNwmEnabled

Enable the Wireless Manager.

Applies to AP. 802.11n products only.

<i>Enumeration</i>	disabled (0), enabled (1)
<i>Access</i>	readwrite
<i>OID</i>	1.3.6.1.4.1.16177.1.400.2.3.1.4.1

13.1.1.4 cfgldf

13.1.1.4.1 cfgldfEnabled

Enable Interference Detection Function.

IDF is only supported for devices with two radios.

Applies to AP. 802.11n products only.

<i>Enumeration</i>	disabled (0), enabled (1)
<i>Access</i>	readwrite
<i>OID</i>	1.3.6.1.4.1.16177.1.400.2.3.1.5.1

13.1.1.4.2 cfgldfScanWorkTable

Table of scan works items.

Applies to AP. 802.11n products only.

<i>Range</i>	0 - 31
<i>OID</i>	1.3.6.1.4.1.16177.1.400.2.3.1.5.10

13.1.1.4.2.1 cfgldfScanWorkTableEntry

Scan work list.

Applies to AP. 802.11n products only.

<i>OID</i>	1.3.6.1.4.1.16177.1.400.2.3.1.5.10.1
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cfgldfScanWorkIndex

Entry index of Table.

Applies to AP. 802.11n products only.

<i>Range</i>	0 - 31
<i>Access</i>	noaccess
<i>OID</i>	1.3.6.1.4.1.16177.1.400.2.3.1.5.10.1.1

cfgldfScanWorkFreq

Center frequency in MHz of the channel to scan.

Set center frequency to 0 to disable this scan work item and all following items.

Applies to AP. 802.11n products only.

<i>Range</i>	0 - 6100
<i>Access</i>	readwrite
<i>OID</i>	1.3.6.1.4.1.16177.1.400.2.3.1.5.10.1.2

cfgldfScanWorkAction

Scan Work Action of the interference function.

The following scan work actions are available:

none(0)

Scan work item and all following items of the table are ignored.

spectral(2)

Spectral data are collected from Antenna port 3 and spectral statistics is generated at the end of the scan work item. During a Spectral Scan Work the raw spectral data can be retrieved from the AP. Please refer to the Software User Manual for more information.

radar(3)

Runs the radar detection engine and counts all radar sequences detected by the monitor wireless device on Antenna port 3.

wifi(4)

Wifi data are collected from Antenna port 3 and wifi statistics is generated at the end of the scan work item.

Applies to AP. 802.11n products only.

<i>Enumeration</i>	none (0), spectral (2), radar (3), wifi (4)
<i>Access</i>	readwrite
<i>OID</i>	1.3.6.1.4.1.16177.1.400.2.3.1.5.10.1.3

cfgldfScanWorkSeconds

Duration of the Scan Work Item in seconds.

At the end of the Scan Work a JSON formatted Report is generated and buffered. Set scan work time to 0 to disable this scan work item and all following items.

Applies to AP. 802.11n products only.

<i>Range</i>	0 - 86400
<i>Access</i>	readwrite
<i>OID</i>	1.3.6.1.4.1.16177.1.400.2.3.1.5.10.1.4

13.1.1.4.3 cfgldfInterval

Report interval in seconds.

This value defines the interval when all available Reports are sent to the URL defined in cfgldfHttpReportUrl.

Applies to AP. 802.11n products only.

<i>Range</i>	1 - 86400
<i>Access</i>	readwrite
<i>OID</i>	1.3.6.1.4.1.16177.1.400.2.3.1.5.3

13.1.1.4.4 cfgldfName

IDF Name.

Can be used to set an unique identifier for the reports.

Applies to AP. 802.11n products only.

<i>Type</i>	DisplayString
<i>Range</i>	1 - 255
<i>Access</i>	readwrite
<i>OID</i>	1.3.6.1.4.1.16177.1.400.2.3.1.5.4

13.1.1.4.5 cfgldfTrigger

13.1.1.4.5.1 cfgldfTrigRadarCntTh

IDF trigger radar_count.

Defines the number of radar events to trigger a report.

Applies to AP. 802.11n products only.

<i>Type</i>	Integer32
<i>Access</i>	readwrite
<i>OID</i>	1.3.6.1.4.1.16177.1.400.2.3.1.5.5.1

13.1.1.4.5.2 cfgldfTrigChanLoadTh

IDF trigger channel_load.

Defines the channel load ratio in percent to trigger a report.

Applies to AP. 802.11n products only.

<i>Range</i>	0 - 100
<i>Access</i>	readwrite
<i>OID</i>	1.3.6.1.4.1.16177.1.400.2.3.1.5.5.2

13.1.1.4.5.3 cfgldfTrigAlienLoadTh

IDF trigger alien_load.

Defines the ratio of alien channel load in percent to trigger a report.

Applies to AP. 802.11n products only.

<i>Range</i>	0 - 100
<i>Access</i>	readwrite
<i>OID</i>	1.3.6.1.4.1.16177.1.400.2.3.1.5.5.3

13.1.1.4.5.4 cfgldfTrigDomLoadTh

IDF trigger domestic_load.

Defines the ratio of domestic channel load in percent to trigger a report.

Applies to AP. 802.11n products only.

<i>Range</i>	0 - 100
<i>Access</i>	readwrite
<i>OID</i>	1.3.6.1.4.1.16177.1.400.2.3.1.5.5.4

13.1.1.4.5.5 cfgldfTrigAlienMaxRssiTh

IDF trigger alien_avg_rssi.

Defines the average RSSI of alien traffic to trigger a report.

Applies to AP. 802.11n products only.

<i>Range</i>	0 - 127
<i>Access</i>	readwrite
<i>OID</i>	1.3.6.1.4.1.16177.1.400.2.3.1.5.5.5

13.1.1.4.5.6 cfgldfTrigDomMaxRssiTh

IDF trigger domestic_avg_rssi.

Defines the average RSSI of domestic traffic to trigger a report.

Applies to AP. 802.11n products only.

<i>Range</i>	0 - 127
<i>Access</i>	readwrite
<i>OID</i>	1.3.6.1.4.1.16177.1.400.2.3.1.5.5.6

13.1.1.5 cfgChannelCleaner

13.1.1.5.1 cfgChanCleanUsableFrequencyList

Configure which frequency list `cfgWlanIndex` is to be used as 'list of usable frequencies' by ChannelCleaner.

Applies to AP. 802.11n products only.

<i>Range</i>	-1 - 23
<i>Access</i>	readwrite
<i>OID</i>	1.3.6.1.4.1.16177.1.400.2.3.1.6.10

13.1.1.5.2 `cfgChanCleanDfsUseNvram`

When 'enabled' the DFS states will be load/stored from/to the non-volatile ram device.

Applies to AP. 802.11n products only.

<i>Enumeration</i>	disabled (0), enabled (1)
<i>Access</i>	readwrite
<i>OID</i>	1.3.6.1.4.1.16177.1.400.2.3.1.6.9

13.1.1.6 `cfgAfm`

13.1.1.6.1 `cfgAfmEnabled`

Enable the Area Frequency Manager on the local device.

Applies to AP. 802.11n products only.

<i>Enumeration</i>	disabled (0), enabled (1)
<i>Access</i>	readwrite
<i>OID</i>	1.3.6.1.4.1.16177.1.400.2.3.1.7.1

13.1.1.6.2 `cfgAfmRedundant`

13.1.1.6.2.1 `cfgAfmRedundantIp`

IP address of the redundant Area Frequency Manager.

Set to '0.0.0.0' to disable.

Applies to AP. 802.11n products only.

Type	IpAddress
Access	readwrite
OID	1.3.6.1.4.1.16177.1.400.2.3.1.7.10.1

13.1.1.6.2.2 cfgAfmRedundantName

Name of the redundant Area Frequency Manager.

Applies to AP. 802.11n products only.

Type	DisplayString
Access	readwrite
OID	1.3.6.1.4.1.16177.1.400.2.3.1.7.10.2

13.1.1.6.3 cfgAfmName

Name of the Area Frequency Manager on the local device.

Applies to AP. 802.11n products only.

Type	DisplayString
Range	1 - 255
Access	readwrite
OID	1.3.6.1.4.1.16177.1.400.2.3.1.7.2

13.1.1.6.4 cfgAfmNeighbourTable

A table of the two neighbour (adjacent) Area Frequency Manager.

Currently only up to two neighbour AFMs are supported.

The area index of the neighbour AFM must be less or greater than the index of this area.

Applies to AP. 802.11n products only.

Range	0 - 4
OID	1.3.6.1.4.1.16177.1.400.2.3.1.7.20

13.1.1.6.4.1 cfgAfmNeighbourTableEntry

Neighbour AFM list.

Applies to AP. 802.11n products only.

OID	1.3.6.1.4.1.16177.1.400.2.3.1.7.20.1
-----	--------------------------------------

cfgAfmNeighbourTableIndex

Entry index of Table.

Applies to AP. 802.11n products only.

<i>Range</i>	0 - 4
<i>Access</i>	noaccess
<i>OID</i>	1.3.6.1.4.1.16177.1.400.2.3.1.7.20.1.1

cfgAfmNeighbourIp

IP address of the neighbour (adjacent) Area Frequency Manager.

Applies to AP. 802.11n products only.

<i>Type</i>	IpAddress
<i>Access</i>	readwrite
<i>OID</i>	1.3.6.1.4.1.16177.1.400.2.3.1.7.20.1.2

cfgAfmNeighbourName

Name of the neighbour (adjacent) Area Frequency Manager.

Applies to AP. 802.11n products only.

<i>Type</i>	DisplayString
<i>Access</i>	readwrite
<i>OID</i>	1.3.6.1.4.1.16177.1.400.2.3.1.7.20.1.3

13.1.1.6.5 cfgAfmIndex

Index of the Area which the local running Area Frequency Manager manages.

Applies to AP. 802.11n products only.

<i>Type</i>	Integer32
<i>Access</i>	readwrite
<i>OID</i>	1.3.6.1.4.1.16177.1.400.2.3.1.7.3

13.1.1.6.6 cfgAfmAfcTable

A table of all Area Frequency Clients controlled by the Area Frequency Manager running on this device.

Applies to AP. 802.11n products only.

<i>Range</i>	0 - 63
<i>OID</i>	1.3.6.1.4.1.16177.1.400.2.3.1.7.30

13.1.1.6.6.1 cfgAfmAfcTableEntry

AFC list.

Applies to AP. 802.11n products only.

<i>OID</i>	1.3.6.1.4.1.16177.1.400.2.3.1.7.30.1
------------	--------------------------------------

cfgAfmAfcTableIndex

Entry index of Table.

Applies to AP. 802.11n products only.

<i>Range</i>	0 - 63
<i>Access</i>	noaccess
<i>OID</i>	1.3.6.1.4.1.16177.1.400.2.3.1.7.30.1.1

cfgAfmAfcName

Name of the Area Frequency Client to control.

Applies to AP. 802.11n products only.

<i>Type</i>	DisplayString
<i>Range</i>	1 - 255
<i>Access</i>	readwrite
<i>OID</i>	1.3.6.1.4.1.16177.1.400.2.3.1.7.30.1.2

cfgAfmAfcIp

IP address of the Area Frequency Client to control.

Applies to AP. 802.11n products only.

<i>Type</i>	IpAddress
<i>Access</i>	readwrite
<i>OID</i>	1.3.6.1.4.1.16177.1.400.2.3.1.7.30.1.3

13.1.1.6.7 cfgAfmAreaSize

Number of segments controlled by this AFM.

Applies to AP. 802.11n products only.

<i>Range</i>	1 - 32
<i>Access</i>	readwrite
<i>OID</i>	1.3.6.1.4.1.16177.1.400.2.3.1.7.5

13.1.1.6.8 cfgAfmPrimary

Enable to set this AFM as primary in this area; set disable for stand-by.

Applies to AP. 802.11n products only.

<i>Enumeration</i>	disabled (0), enabled (1)
<i>Access</i>	readwrite
<i>OID</i>	1.3.6.1.4.1.16177.1.400.2.3.1.7.6

13.1.1.6.9 cfgAfmReportEnabled

When enabled(1) the AFM sends reports to the configured server URL (see cfgLdfServerUrl).

Please refer to Software 6 Interface Description for more information.

Applies to AP. 802.11n products only.

<i>Enumeration</i>	disabled (0), enabled (1)
<i>Access</i>	readwrite
<i>OID</i>	1.3.6.1.4.1.16177.1.400.2.3.1.7.7

13.1.1.6.10 cfgAfmVisibility

Visibility of segments

The visibility must be greater or equal as the absolute segment offset configured in any `cfgAfcNeighbourOffset` within the line.

`cfgAfmVisibility >= abs(cfgAfcNeighbourOffset)`

Applies to AP. 802.11n products only.

<i>Range</i>	1 - 1024
<i>Access</i>	readwrite
<i>OID</i>	1.3.6.1.4.1.16177.1.400.2.3.1.7.8

13.1.1.7 cfgAfc

13.1.1.7.1 cfgAfcEnabled

Enable the Area Frequency Client on the local device.

Applies to AP. 802.11n products only.

<i>Enumeration</i>	disabled (0), enabled (1)
<i>Access</i>	readwrite
<i>OID</i>	1.3.6.1.4.1.16177.1.400.2.3.1.8.1

13.1.1.7.2 cfgAfcAfmTable

A table of Area Frequency Managers which are allowed to control this Area Frequency Client (AFC).

Applies to AP. 802.11n products only.

<i>Range</i>	0 - 4
<i>OID</i>	1.3.6.1.4.1.16177.1.400.2.3.1.8.10

13.1.1.7.2.1 cfgAfcAfmTableEntry

AFC list.

Applies to AP. 802.11n products only.

<i>OID</i>	1.3.6.1.4.1.16177.1.400.2.3.1.8.10.1
------------	--------------------------------------

cfgAfcAfmTableIndex

Entry index of Table.

Applies to AP. 802.11n products only.

<i>Range</i>	0 - 4
<i>Access</i>	noaccess
<i>OID</i>	1.3.6.1.4.1.16177.1.400.2.3.1.8.10.1.1

cfgAfcAfmName

Name of the connecting Area Frequency Manager.

Applies to AP. 802.11n products only.

Type	DisplayString
Range	1 - 255
Access	readwrite
OID	1.3.6.1.4.1.16177.1.400.2.3.1.8.10.1.2

cfgAfcAfmIp

IP address of the connecting Area Frequency Manager.

Applies to AP. 802.11n products only.

Type	IpAddress
Access	readwrite
OID	1.3.6.1.4.1.16177.1.400.2.3.1.8.10.1.3

13.1.1.7.3 cfgAfcName

Name of the Area Frequency Client on the local device.

Applies to AP. 802.11n products only.

Type	DisplayString
Range	1 - 255
Access	readwrite
OID	1.3.6.1.4.1.16177.1.400.2.3.1.8.2

13.1.1.7.4 cfgAfcIndex

Index of the Area Frequency Client within the local Area.

Applies to AP. 802.11n products only.

Range	0 - 31
Access	readwrite
OID	1.3.6.1.4.1.16177.1.400.2.3.1.8.3

13.1.1.7.5 cfgAfcNeighbourOffsetTable

Table of up to 24 segment offsets relevant for neighbour list.

Applies to AP. 802.11n products only.

<i>Range</i>	0 - 23
<i>OID</i>	1.3.6.1.4.1.16177.1.400.2.3.1.8.40

13.1.1.7.5.1 cfgAfcNeighbourOffsetTableEntry

Neighbour segment offset list

<i>OID</i>	1.3.6.1.4.1.16177.1.400.2.3.1.8.40.1
------------	--------------------------------------

cfgAfcNeighbourOffsetTableIndex

Entry index of Table.

Applies to AP. 802.11n products only.

<i>Range</i>	0 - 23
<i>Access</i>	noaccess
<i>OID</i>	1.3.6.1.4.1.16177.1.400.2.3.1.8.40.1.1

cfgAfcNeighbourOffset

Segment offset to be considered for neighbour list.

Applies to AP. 802.11n products only.

<i>Range</i>	-15 - 15
<i>Access</i>	readwrite
<i>OID</i>	1.3.6.1.4.1.16177.1.400.2.3.1.8.40.1.2

13.1.1.7.6 cfgAfcBackupFreq

Backup frequency which the Area Frequency Client will use if the Area Frequency Manager is not present.

Applies to AP. 802.11n products only.

<i>Range</i>	0 - 6100
<i>Access</i>	readwrite
<i>OID</i>	1.3.6.1.4.1.16177.1.400.2.3.1.8.5

13.1.1.7.7 cfgAfcReportEnabled

When enabled(1) the AFC sends reports to the configured server URL (see cfgHttpRprtServerUrl).

Please refer to Software 6 Interface Description for more information.

Applies to AP. 802.11n products only.

<i>Enumeration</i>	disabled (0), enabled (1)
<i>Access</i>	readwrite
<i>OID</i>	1.3.6.1.4.1.16177.1.400.2.3.1.8.6

13.1.2 rpc

13.1.2.1 rpcChannelManager

13.1.2.1.1 rpcChMgrHttpReport

Requests a HTTP report from the Channel Manager.

Applies to AP. 802.11n products only.

<i>Enumeration</i>	nop (0), freqstate (1), channels (2)
<i>Access</i>	readwrite
<i>OID</i>	1.3.6.1.4.1.16177.1.400.2.3.3.1.1

13.1.2.2 rpcNwm

13.1.2.2.1 rpcNwmHttpReport

Requests a HTTP report from the Wireless Manager.

Applies to AP. 802.11n products only.

<i>Enumeration</i>	nop (0), status (1), freqstate (2)
<i>Access</i>	readwrite
<i>OID</i>	1.3.6.1.4.1.16177.1.400.2.3.3.2.1

13.1.2.3 rpcNvram

13.1.2.3.1 rpcNvramFreqStatesReset

Resets the state of DFS frequencies in the Available Channel List which is stored in non-volatile memory.

The device performs a reboot after resetting the state of the frequencies. The behaviour is similar to a factory reset but does only reset the Available Channel List.



Applies to AP. 802.11n products only.

<i>Enumeration</i>	reset (0)
<i>Access</i>	readwrite
<i>OID</i>	1.3.6.1.4.1.16177.1.400.2.3.3.3.1

14 WESTERMO-SW6-PWN-MIB

14.1 pwn

14.1.1 configuration

14.1.1.1 cfgWireless

14.1.1.1.1 cfgWlanBandsteering

14.1.1.1.1.1 cfgWlanBsteerEnabled

Enable Band Steering.

Band steering is a technique used in dual-band (2G4 and 5G) wireless setups to encourage dual-band STAs to use the less-congested band.

Applies to AP. 802.11ac products only.

<i>Enumeration</i>	disabled (0), enabled (1)
<i>Access</i>	readwrite
<i>OID</i>	1.3.6.1.4.1.16177.1.400.2.9.1.1.1.1

14.1.1.1.1.2 cfgWlanBsteerMatchingSsid

List of Matching SSIDs.

This is a comma and/or space separated list of SSIDs. For each SSID, there shall be a matching SSID in the 2G4 and 5G band so that the band steering process is able to work and decide which STA shall be connected in which band.

Applies to AP. 802.11ac products only.

Type	DisplayString
Range	1 - 255
Access	readwrite
OID	1.3.6.1.4.1.16177.1.400.2.9.1.1.1.2

15 WebAPI Detailed Specification

- This appendix specifies the WebAPI interface.
- Top level URL for the API is `/cgi-bin/luci/api`

15.1 Authentication API

- Used for authentication (login / logout).
- Implemented as JSON RPC version 2.0 API with `/cgi-bin/luci/api/auth` as its top level URL.
- Further requests have to include the token as cookie with name "sysauth".

15.1.1 RPC Methods

15.1.1.1 Login

- method: `login(username, password)`

Example POST request

```
{
  "id": 1, "jsonrpc": "2.0",
  "method": "login",
  "params": {
    "username": "webadmin",
    "password": "10SoaL_98"
  }
}
```

Example response

```
{
  "id": 1, "jsonrpc": "2.0",
  "result": { "token": "4855d491f8dbde3628edc4e5be4861fe" }
}
```

15.1.1.2 Logout

- method: `logout()`

Example POST request:

```
{
  "id": 1, "jsonrpc": "2.0",
  "method": "logout"
}
```

Example response

```
{
  "id": 1, "jsonrpc": "2.0",
  "result": { "message": "Logout successful" }
}
```

15.2 Files API

- Top level URL for the Files API is `/cgi-bin/luci/api/files`
- Using multi part form protocol (i.e. `Content-Type: multipart/form-data;`)
- Additional form data might be supported
- When a system upgrade is running, any call to the Files API will result in an error -32003
 - Example: `{"error": {"code": -32003, "message": "System upgrade running"}}`

15.2.1 Device configuration file import / export

- URL: `/cgi-bin/luci/api/files/config`

- HTTP GET request for configuration file export
- Multi part HTTP POST request for configuration file import
 - Name field for configuration file part is `config`.
 - Optional form value `reset` (default: `false`). If `reset` is `true`, then the device configuration will be reset before importing.
 - if the import is successful, the answer will be a JSON object with the member `"result"` which is itself an object with the member `"nof_changes"` (number of changes).
 - Example: `{"result": {"nof_changes": "1"}}`
- if the import fails, the answer will be a JSON object with the member `"error"` which is itself an object with the members `"code"` (being `-32001`) and `"message"` (e.g. `"Import failed"`)
 - Example: `"error": {"code": -32001, "message": "Import failed"}`

15.2.2 Syslog export

- URL: `/cgi-bin/luci/api/files/syslog`
- HTTP GET request for syslog export

15.2.3 System messages export

- URL: `/cgi-bin/luci/api/files/sys_messages`
- HTTP GET request for system message export

15.2.4 Support File export

- URL: `/cgi-bin/luci/api/files/support`
- HTTP GET request for support file export

15.2.5 Firmware upload and upgrade

- URL: `/cgi-bin/luci/api/files/firmware`
- Multi part HTTP POST request for firmware image upload and upgrade
 - Name fields for file parts are:
 - * `image` for the firmware image file
 - * `custom-config` for the custom configuration file
 - Optional form value `config_mode`
 - * `reset`: Reset to default configuration
 - * `keep`: Keep current configuration (this is the default)
 - * `custom`: Apply custom configuration after upgrade
 - Additional custom configuration file must be provided
 - if the uploaded firmware package is valid and the firmware upgrade is successfully started, the answer will be a JSON object with the member "result" which is itself an object with the member "message" (e.g. `{"result": {"message": "System upgrade started"}}`). At end of upgrade, the device will reboot. To see if upgrade is done: Poll [swOsUptime.0](#)
 - * It will count up until reboot,
 - * then timeout during reboot ("connection timeout, no route to host")
 - * and finally request new login (http status 403 "Forbidden") after reboot.
 - if the firmware upload and upgrade fails, the answer will be a JSON object with the member "error" which is itself an object with the members "code" (being -32002) and "message" (e.g. "Invalid firmware package")
 - * Example: `{"error": {"code": -32002, "message": "Invalid firmware package"}}`

15.3 Device Configuration API

- Implemented as JSON RPC version 2.0 API with `/cgi-bin/luci/api/mib` as it's top level URL.
- Same well-known interface as used by the SNMP and CLI API.

15.3.1 RPC Methods

15.3.1.1 RPC Methods Errors

Number	Category	Message
-1	General Error	E.g. "general error" or "set failed"
-2	Not Available	E.g. "element not available"

15.3.1.2 Get MIB items / device configuration

- method: `get(Array of ConfigurationItemKey)`
 - ConfigurationItem: Supported are all items which are defined in the available MIBs of the device.
- Note:
 - if "params" is not given, then return list of all available configuration items and it's values.

Example POST request

```
{
  "id": 1, "jsonrpc": "2.0",
  "method": "get",
  "params": {
    "items": [
      {"key": "WESTERMO-SW6-MIB::cfgSysHostname.0"},
      {"key": "WESTERMO-SW6-MIB::cfgSysTimezone.0"},
      {"key": "WESTERMO-SW6-MIB::swCfgChangesCount.0"},
      {"key": "WESTERMO-SW6-MIB::cfgSysHostname"}
    ]
  }
}
```

Example response

```
{
  "id": 1, "jsonrpc": "2.0",
  "result": {
    "items": [
      {"key": "WESTERMO-SW6-MIB::cfgSysHostname.0", "value": "Rmodem", "type": "
        ↳ string"},
      {"key": "WESTERMO-SW6-MIB::cfgSysTimezone.0", "value": "UTC", "type": "
        ↳ string"},
      {"key": "WESTERMO-SW6-MIB::swCfgChangesCount.0", "value": 0, "type": "
        ↳ int"}
    ],
    "errors": [
      {"key": "WESTERMO-SW6-MIB::cfgSysHostname", "error_no": -1, "error_msg": "
        ↳ general error"}
    ]
  }
}
```

15.3.1.3 Set MIB items

- method: set(Array of ConfigurationItem)
 - ConfigurationItem: Supported are all items which are defined in the available MIBs of the device.
- result:
 - items: Array of ConfigurationItemValues which have been set successfully.
 - errors: Array of ConfigurationItemValues with invalid keys (this element is only present if invalid keys have been detected).
- **Note:** Changes will not be applied as long as you do not set rpcCfgApply.0 itself.

Example POST request

```
{
  "id": 1, "jsonrpc": "2.0",
  "method": "set",
  "params": {
    "items": [
      {"key": "WESTERMO-SW6-MIB::cfgSysHostname.0", "value": "RM1"},
      {"key": "WESTERMO-SW6-MIB::cfgSysHostname", "value": "foo"},
    ]
  }
}
```

```
    {"key": "WESTERMO-SW6-MIB::cfgLogFileEnabled.0", "value": 1}
  ]
}
```

Example response

```
{
  "id": 1, "jsonrpc": "2.0",
  "result": {
    "items": [
      {"key": "WESTERMO-SW6-MIB::cfgSysHostname.0", "value": "RM1"},
      {"key": "WESTERMO-SW6-MIB::cfgLogFileEnabled.0", "value": 1}
    ],
    "errors": [
      {"key": "WESTERMO-SW6-MIB::cfgSysHostname", "value": "foo", "error_no":
        ↪ -1, "error_msg": "set failed"}
    ]
  }
}
```

16 Message Codes

Variable text, inserted at the time the message is created, is displayed using the place holder '<val>'.

16.0.0.1 Code: [0]

```
INFO 0 <val>
```

Value is either 'Reset errors by WebGUI' or 'Manual Error Reset'.

16.0.0.2 Code: [100]

```
ERROR 100 SYS_MON: Voltage sensor '<val>' is out of range <val> / [<val>, <val>  
↪ >]
```

This trap is sent when supply voltage or one of the internal voltages are outside specified limits which are hardware dependent.

16.0.0.3 Code: [101]

```
ERROR 101 SYS_MON: Temperature sensor '<val>' is out of range <val> / [<val>,  
↪ <val>]
```

This trap is sent when internally measured temperature is outside specified limits which are hardware dependent.

16.0.0.4 Code: [105]

```
CRITICAL 105 SYS_MON: Failure by reading value <val>. This value isn't  
↪ monitored anymore
```

This trap is sent when internally a value couldn't read.

16.0.0.5 Code: [130]

```
INFO 130 <val>|<val>|<val>||<val>|<val>|<val>|<val>|<val>|<val>||<val>|<val>|<
↳ val>|<val>|<val>|<val>|<val>|<val>||<val>|<val>|<val>|<val>
```

This message is sent periodically containing current 'iw <iface> station dump'. Format is 'interface|mac|inactive time||rx bytes|rx packets|tx bytes|tx packets|tx retries|tx failed||signal combined|avg signal combined|signal ch0|avg signal ch0|signal ch1|avg signal ch1|signal ch2|avg signal ch2||rx bitrate mode|rx bitrate value|tx bitrate mode|tx bitrate value'.

16.0.0.6 Code: [131]

```
INFO 131 <val>|<val>|<val>|<val>|<val>|<val>|<val>|<val>|<val>|<val>|<val>|<val>
```

This message is sent periodically containing counters of the wireless interface. Format: 'NofStations|TxPackets|RetryCount|6Mbps|9Mbps|..|54Mbps'.

16.0.0.7 Code: [200]

```
NOTICE 200 Device has restarted because of <val>
```

This message is sent at boot-up process. RESET CAUSE can be either coldstart, warmstart, watchdog and oops.

16.0.0.8 Code: [201]

```
NOTICE 201 System startup
```

This message is sent when the system starts up.

16.0.0.9 Code: [202]

```
NOTICE 202 Firmware update started
```

This message is sent when a firmware update is initiated.

16.0.0.10 Code: [203]

```
NOTICE 203 System reboot
```

This message is sent when a system reboot is issued.

16.0.0.11 Code: [205]

```
NOTICE 205 Factory Reset confirmed, System configuration changed
```

This message is sent whenever the system configuration is changed.

16.0.0.12 Code: [206]

```
NOTICE 206 Reserved
```

16.0.0.13 Code: [207]

```
WARNING 207 Invalid upgrade image for this platform.
```

16.0.0.14 Code: [208]

```
ERROR 208 Corrupt firmware package!
```

16.0.0.15 Code: [209]

```
ERROR 209 Transfer firmware to device failed!
```

16.0.0.16 Code: [210]

```
WARNING 210 Decryption of the upgrade image failed!
```

16.0.0.17 Code: [220]

```
NOTICE 220 Start writing device identification memory!
```

16.0.0.18 Code: [221]

```
NOTICE 221 Device identification memory successfully updated!
```

16.0.0.19 Code: [222]

```
CRITICAL 222 Failed to update device identification memory!
```

16.0.0.20 Code: [300]

```
INFO 300 NTP: time synchronization failed!
```

This message is generated when ntp client is configured in unicast mode and it failed to connect to NTP server.

16.0.0.21 Code: [310]

```
NOTICE 310 Reserved
```

16.0.0.22 Code: [320]

```
ERROR 320 BIST: Daemon '<val>' isn't running, recover it
```

This message is generated when process with name process is not running and has to be restarted by the bist

16.0.0.23 Code: [321]

```
ERROR 321 BIST: Daemon watchdog is not running - force restart
```

This message is generated when watchdog process is not running. System reboots afterwards.

16.0.0.24 Code: [322]

```
CRITICAL 322 Critical hardware failure: <val>
```

This message is generated when a critical hardware failure was detected during boot up. Please read the user manual about the "Support File".

16.0.0.25 Code: [323]

```
ERROR 323 Kernel log(s) found
```

This message is generated when a kernel log(s) was found during boot up. Please read the user manual about the "Technical Support File".

16.0.0.26 Code: [330]

```
INFO 330 BBMON: Backbone is down.
```

This message is generated when the connection to the backbone is detected as lost.

16.0.0.27 Code: [331]

```
INFO 331 BBMON: Backbone is up.
```

This message is generated when the connection to the backbone is regained.

16.0.0.28 Code: [332]

```
INFO 332 BBMON: <val> target <val> is down.
```

This message is generated when the icmp target on the backbone is detected as down, or the phy loses its link.

16.0.0.29 Code: [333]

```
INFO 333 BBMON: <val> target <val> is up.
```


This message is generated when the icmp target on the backbone is detected as up, or the phy gains its link when gateway monitoring is disabled.

16.0.0.30 Code: [350]

```
INFO 350 ICL: connection established (mode: <val>)
```

This message is generated when ICL establishes a connection to a partner. mode=AP/STA

16.0.0.31 Code: [351]

```
INFO 351 ICL: connection lost (mode: <val>)
```

This message is generated when ICL lost connection to the partner. mode=AP/STA

16.0.0.32 Code: [360]

```
INFO 360 CELLULAR: Link is up (iface: '<val>', slot: <val>)
```

Arguments are <iface>, the interface name; and <slot>, the number of the active SIM slot.

16.0.0.33 Code: [361]

```
INFO 361 CELLULAR: Link is down (iface: '<val>', slot <val>, reason: '<val>')
```

Arguments are <iface>, the interface name; <slot>, the number of the active SIM slot; and a <reason> (e.g. 'low RSSI').

16.0.0.34 Code: [400]

```
INFO 400 <val>: Station is associated
```

This message is used to trigger led status

16.0.0.35 Code: [401]

```
INFO 401 <val>: Station is disassociated, Reason: <val>
```

This message is used to trigger led status

16.0.0.36 Code: [402]

```
INFO 402 <val>: Station is authorized
```

This message is used to trigger various daemons.

16.0.0.37 Code: [403]

```
NOTICE 403 WLAN: Authentication failure
```

16.0.0.38 Code: [404]

```
NOTICE 404 <val>: Association failure
```

16.0.0.39 Code: [405]

```
NOTICE 405 Reserved
```

16.0.0.40 Code: [406]

```
NOTICE 406 WLAN: Max number of station exceeded
```

16.0.0.41 Code: [407]

```
ERROR 407 Reserved
```

16.0.0.42 Code: [408]

```
INFO 408 WLAN: Failed to connect to <val>: <val>
```

Format: WLAN: Failed to connect to <mac>: [timed out|(code) error message]

16.0.0.43 Code: [413]

```
WARNING 413 Switched to secondary SSID |<val>|
```

16.0.0.44 Code: [415]

```
NOTICE 415 WLAN: Station couldn't find better AP after <val> consecutive scans
```

16.0.0.45 Code: [421]

```
NOTICE 421 <val>: CSA <val> => <val>
```

Format: '<iface>: <old freq> => <new freq>'

16.0.0.46 Code: [430]

```
NOTICE 430 TS|<val>|<val>|RSSI_BCN|<val>
```

Format: 'TS|<timestamp>|<bssid>|RSSI BCN|<rsssi>'

16.0.0.47 Code: [431]

```
NOTICE 431 TS|<val>|<val>|RSSI_BCN|<val>|<val>|<val>|<val>
```

Format: 'TS|<timestamp>|<bssid>|RSSI BCN|<raw rssi>|<cur filer rssi>|<short filter rssi>|<long filter rssi>'

16.0.0.48 Code: [432]

```
NOTICE 432 Reserved
```

16.0.0.49 Code: [433]

```
INFO 433 WLAN: Low RSSI event: <val>
```

This event is used to trigger various daemons.

16.0.0.50 Code: [434]

```
NOTICE 434 <val>: Handoff:|<val>|<val>|<val>|<val>|<val>|<val>|<val>|
```

Format: '<iface>: Handoff:|<reason>|<count>|<prev bssid>|<cur bssid>|<ssid mgmt>|<ho time>|<offchan scan time>|'

16.0.0.51 Code: [435]

```
ERROR 435 Radius <val> server <val>:<val> not available (reason <val>)
```

AP: Radius server not available/invalid shared secret/connection rejected. Format: Radius <authentication, accounting> server <ip>:<port> not available (reason: <not available>, <shared secret>, <rejected>)

16.0.0.52 Code: [436]

```
INFO 436 EAP-TLS: Start session
```

STA: Full EAP-(T)TLS authentication has been started

16.0.0.53 Code: [437]

```
ERROR 437 Certificate <val> is <val>
```

STA: T(TLS) certificate errors. Format: Certificate <ca, client> is <wrong, missing, expired>

16.0.0.54 Code: [438]

```
ERROR 438 TTLS: Invalid username and/or password
```

STA: TTLS Username and/or password is not recognized by radius server

16.0.0.55 Code: [440]

```
WARNING 440 EAP: Auth. session failure: EAP ID: <val>, Reason: <val>
```

EAP ID: RADIUS Server IP address; reason: Failure reason code

Failure reason codes: initialize(1), disconnected(2), connecting(3), authenticating(4), authenticated(5), aborting(6), held(7), forceAuth(8), forceUnauth(9), restart(10)

16.0.0.56 Code: [441]

```
WARNING 441 RADIUS: Access-Reject: RADIUS Server: <val>
```

val: <IpAddress>

16.0.0.57 Code: [442]

```
WARNING 442 RADIUS: Response Auth. failed: RADIUS Server: <val>
```

val: <IpAddress>

16.0.0.58 Code: [443]

```
INFO 443 RADIUS: Switching to <val> server
```

val: 'PRIMARY' or 'SECONDARY'

16.0.0.59 Code: [444]

```
INFO 444 Certmgmt: Certificate with ID <val> expiration. Serial: <val>; Exp.  
    ↳ date: <val>
```

val: <ID>, val: <serial>, val: <ExpirationDate>

16.0.0.60 Code: [445]

```
INFO 445 Certmgmt: CRL with ID <val> expiration. Exp.date: <val>
```

val: <ID>, val: <ExpirationDate>

16.0.0.61 Code: [446]

```
INFO 446 SCEP: CA certificate retrieval (<val>): <val>
```

val: <SCEP IDX>, val: 'SUCCESS' or 'FAILURE (err: <code>)'

16.0.0.62 Code: [447]

```
INFO 447 SCEP: Client certificate enrollment (<val>): <val>
```

val: <SCEP IDX>, val: 'SUCCESS' or 'FAILURE (err: <code>)'

16.0.0.63 Code: [448]

```
INFO 448 SCEP: Client certificate re-enrollment (<val>): <val>
```

val: <SCEP IDX>, val: 'SUCCESS' or 'FAILURE (err: <code>)'

16.0.0.64 Code: [449]

```
INFO 449 Certmgmt: CRL download for CA ID <val>: <val>
```

val: <ID>, val: 'SUCCESS' or 'FAILURE (err: <code>)'

16.0.0.65 Code: [450]

```
NOTICE 450 Forced deauthentication of all stations
```

16.0.0.66 Code: [451]

```
WARNING 451 Noisefloor above limit
```

16.0.0.67 Code: [452]

```
WARNING 452 Noisefloor below limit
```

16.0.0.68 Code: [460]

```
NOTICE 460 Authorization timed out
```

DEBUG: Used to make conditional assessment of long handoff limits for frame loss cases

16.0.0.69 Code: [461]

```
NOTICE 461 Association from <val> denied
```

DEBUG: The SM struct is NULL but station tries to reassociate

16.0.0.70 Code: [462]

```
NOTICE 462 WPA Key Group Fatal Error
```

DEBUG: The group key state machine has a fatal error

16.0.0.71 Code: [463]

```
INFO 463 SCEP: re-enroll (<val>): Transaction ID: <val>; DN: <val>; CA DN: <  
  ↳ val>
```

val: <SCEP IDX>, val: <TRANSACTION ID>, val: SUBJECT ID, val: <CA DN>

16.0.0.72 Code: [465]

```
INFO 465 SCEP: certificate polling (<val>): retry in <val> minutes
```

val: <SCEP IDX>, val: <minutes>

16.0.0.73 Code: [466]

```
INFO 466 Import <val> from <val>
```

val: <type>, val: <url>

16.0.0.74 Code: [467]

```
INFO 467 SCEP: Retrieving CA certificate (<val>): CA ID: <val>
```

val: <SCEP IDX>, val: <CA ID>

16.0.0.75 Code: [468]

```
INFO 468 Certificate import: <val>
```

val: 'SUCCESS' or 'FAILURE <errorCode>'

16.0.0.76 Code: [469]

```
INFO 469 Certmgmt: Import CRL for CA ID <val> from <val>
```

val: <ID>, val: <HTTPAddr>

16.0.0.77 Code: [470]

```
INFO 470 SCEP: enroll (<val>): Transaction ID: <val>; DN: <val>; CA DN: <val>
```

val: <SCEP IDX>, val: <TRANSACTION ID>, val: SUBJECT ID, val: <CA DN>

16.0.0.78 Code: [471]

```
INFO 471 Certmgmt: CRL with ID <val> stored. ISSUE_DATE: <val>; ISSUER_DN: <
↳ val>
```

val: <ID>, val: <date>, val: <CRL ISSUER DN>

16.0.0.79 Code: [472]

```
INFO 472 AP_TIME: invalid AP_TIME from <val>: ap_time (<val>) / systime (<val
↳ >)
```

val: <AP-MAC>, val: AP TIME, val: <SYS TIME>

16.0.0.80 Code: [473]

```
WARNING 473 RADIUS: retransmit to auth server <val>
```

val: <RadiusAuthServerIPAddr>

16.0.0.81 Code: [475]

```
WARNING 475 EAP-TLS: session failed, code=<val>
```

val: <FailureReasonCode>

16.0.0.82 Code: [476]

```
INFO 476 EAP-TLS: successfully closed session
```

16.0.0.83 Code: [477]

```
INFO 477 EAP: Auth. Session started. EAP ID: <val>
```

val: <EAP ID>

16.0.0.84 Code: [478]

```
INFO 478 EAP: Auth. Session success. EAP ID: <val>
```

val: <EAP ID>

16.0.0.85 Code: [479]

```
INFO 479 Certmgmt: Certificate with ID <val> stored. DN: <val>; SN: <val>;  
    ↳ ISSUER: <val>
```

val: <ID>, val: <CERT DN>, val: <CERT SN>, val: <ISSUER DN>

16.0.0.86 Code: [480]

```
INFO 480 Certmgmt: Certificate with ID <val> (DN: <val>; ISSUER: <val>)  
    ↳ replaced by new certificate (DN: <val>; SN: <val>; ISSUER: <val>)
```

val: <ID>, val: <SUBJECT DN>, val: <ISSUER DN>, val: <NEW SUBJECT DN>, val: <NEW SERIAL>, val: <NEW ISSUER DN>

16.0.0.87 Code: [481]

```
INFO 481 Certificate (DN: <val>; SN: <val>) is set as certificate for TLS  
    ↳ handshake for 802.1X NAC interface.
```

Certificate (DN: <subject>; SN: <serial number>) is set as certificate for TLS handshake for 802.1X NAC interface.

16.0.0.88 Code: [482]

```
INFO 482 Certificate (DN: <val>; SN: <val>) is set as certificate for TLS  
    ↳ handshake for HTTPS server interface.
```

Certificate (DN: <subject>; SN: <serial number>) is set as certificate for TLS handshake for HTTPS server interface.

16.0.0.89 Code: [483]

```
INFO 483 Certificate (DN: <val>; SN: <val>) is set as trust anchor for 802.1X  
    ↪ NAC interface.
```

Certificate (DN: <subject>; SN: <serial number>) is set as trust anchor for 802.1X NAC interface.

16.0.0.90 Code: [484]

```
INFO 484 Certificate (DN: <val>; SN <val>) set as trust anchor for LDAPS  
    ↪ client.
```

Certificate (DN: <subject>; SN <serial number>) set as trust anchor for LDAPS client.

16.0.0.91 Code: [485]

```
INFO 485 Certmgmt: Key with ID <val> stored
```

val: <ID>

16.0.0.92 Code: [486]

```
INFO 486 Certificate (DN: <val>; SN <val>) set as trust anchor for HTTPS  
    ↪ client.
```

Certificate (DN: <subject>; SN <serial number>) set as trust anchor for HTTPS client.

16.0.0.93 Code: [487]

```
INFO 487 Certmgmt: CRL with ID <val> (ISSUER: <val>) replaced by new CRL (  
    ↪ ISSUER: <val>)
```

val: <ID>, val: <ISSUER DN>, val: <NEW ISSUER DN>

16.0.0.94 Code: [490]

```
WARNING 490 RADIUS: retransmit to acct server <val>
```

val: <RadiusAcctServerIPAddr>

16.0.0.95 Code: [500]

```
CRITICAL 500 CONFIG: Unable to connect to IPC system (ubus)!
```

16.0.0.96 Code: [501]

```
CRITICAL 501 CONFIG: Unable to read from UCI!
```

16.0.0.97 Code: [510]

```
ERROR 510 CONFIG: Invalid configuration, reverting to previous configuration!
```

16.0.0.98 Code: [511]

```
ERROR 511 CONFIG: Unable to save new configuration!
```

16.0.0.99 Code: [512]

```
CRITICAL 512 CONFIG: Unable to apply previous configuration!
```

16.0.0.100 Code: [513]

```
WARNING 513 CONFIG: Unable to set config parameter: <val>
```

Warning for configuration import.

16.0.0.101 Code: [514]

```
ERROR 514 CONFIG: <val>
```

Error detected during configuration validation.

16.0.0.102 Code: [515]

```
ERROR 515 CONFIG: Invalid WLAN operation mode: <val>
```

Some devices may not support all operation modes. Please check the datasheet of the device.

16.0.0.103 Code: [516]

```
WARNING 516 Country code <val> not supported on this product/revision!
```

Use this message when country code is not supported by this product/revision

16.0.0.104 Code: [517]

```
ERROR 517 <val> is not supported on this product!
```

Some devices may not support all functions. Please check the manual of the device.

16.0.0.105 Code: [518]

```
WARNING 518 CONFIG: Invalid Certificate Chain or Private Key configured. Using  
↪ default self-signed Certificate/Key.
```

When an invalid Certificate Chain or Private Key is configured, the HTTPS server falls back to its own self-signed Certificate/Key.

16.0.0.106 Code: [530]

```
ERROR 530 PENDING CHANGES: <val>
```

A firmware upgrade is not allowed if there are pending changes in the configuration.

16.0.0.107 Code: [531]

ERROR 531 System upgrade to a major version other than 6 is not supported.

16.0.0.108 Code: [532]

ERROR 532 'Keep config' on a system downgrade is not supported.

16.0.0.109 Code: [533]

CRITICAL 533 Config manipulation for an upgrade failed. This is not
↳ correctable situation. Do a factory reset.

16.0.0.110 Code: [534]

ERROR 534 Firmware up/downgrade not possible because: <val>

Message when a firmware up/down-grade is not possible. Details why it's not possible.

16.0.0.111 Code: [540]

ERROR 540 Failed to verify key or certificate file.

16.0.0.112 Code: [580]

ERROR 580 CONFIG FILE: Transfer failed!

16.0.0.113 Code: [581]

WARNING 581 CONFIG FILE: <val>

Configuration file import warning: Config version, Product, Firmware Name or Version mismatch.

16.0.0.114 Code: [582]

ERROR 582 CONFIG FILE: Unable to read or parse!

16.0.0.115 Code: [583]

ERROR 583 CONFIG FILE: Incorrect encryption key!

16.0.0.116 Code: [585]

ERROR 585 Certificate import/export failed!

16.0.0.117 Code: [600]

ERROR 600 AFC <val>: AFM did not declare after <val> seconds

16.0.0.118 Code: [601]

ERROR 601 AFM <val>: adjacent AFM did not declare after <val> seconds

Not used, since role of AFM is unknown at startup

16.0.0.119 Code: [602]

ERROR 602 AFM <val>: redundant AFM did not declare after <val> seconds

Not used, since role of AFM is unknown at startup

16.0.0.120 Code: [603]

ERROR 603 AFC <val>: invalid AFM <val> tried to declare

16.0.0.121 Code: [604]

```
ERROR 604 AFM <val>: could not connect to AFC <val>
```

16.0.0.122 Code: [605]

```
ERROR 605 <val>: could not send report to <val>
```

16.0.0.123 Code: [630]

```
INFO 630 AFM <val>: AFC <val> set to operational frequency <val> with penalty  
↪ <val>
```

16.0.0.124 Code: [631]

```
INFO 631 AFM <val>: becoming active for area <val>
```

16.0.0.125 Code: [632]

```
ERROR 632 AFM <val>: invalid AFM configured: <val>
```

This unifies all AFM/RAFM/AAFM configuration errors

16.0.0.126 Code: [650]

```
INFO 650 RADIUS: <val> authentication server is up (server <val>:<val>)
```

This message is sent if the Authenticator declares a RADIUS server as up

16.0.0.127 Code: [651]

```
INFO 651 RADIUS: <val> authentication server is down (server <val>:<val>)
```

This message is sent if the Authenticator declares a RADIUS server as down

16.0.0.128 Code: [700]

```
ERROR 700 NET: Configuration failed!
```

This message is sent if the network couldn't be set up. Possible reason are wrong protocol, missing or invalid netmask or ip address.

16.0.0.129 Code: [701]

```
WARNING 701 NET: Configuration failed, but try to continue anyway.
```

This message is sent if the network couldn't be set up. Possible reason are the interface we try to configure does not exist.

16.0.0.130 Code: [710]

```
ERROR 710 NET: Unable to set the default gateway!
```

This message is sent if the default gateway couldn't be set properly. This happens if the destination can not be reached, or no matching subnet exist.

16.0.0.131 Code: [711]

```
WARNING 711 NET: Unable to set the default gateway!
```

This message is sent if the default gateway couldn't be set properly. This can happen if the default gateway is already set by DHCP.

16.0.0.132 Code: [712]

```
ERROR 712 NET: Unable to set a static route!
```

This message is sent when an static route couldn't be set.

16.0.0.133 Code: [713]

```
WARNING 713 NET: Unable to set a static route, but apply process continued.
```

This message is sent when an static route couldn't be set.

16.0.0.134 Code: [720]

```
ERROR 720 NET: Wireless configuration failed!
```

This message is sent when the configuration manager is not able to set up a wireless interface.

16.0.0.135 Code: [721]

```
ERROR 721 NET: A wireless interface can not be bridged to an eth without 4addr  
↔ mode or l2nat!
```

This message is sent when the configuration manager is not able to set up a wireless interface because of missing 4addr mode or l2nat.

16.0.0.136 Code: [722]

```
ERROR 722 NET: 4addr mode and Layer 2 NAT can not be enabled at the same time!
```

This message is sent when the configuration manager is not able to set up a wireless interface because 4addr mode and L2 NAT is enabled at the same time.

16.0.0.137 Code: [730]

```
ERROR 730 NET: Creation of IP address failed! Probably the IP/netmask is  
↔ invalid.
```

This message is sent if an ip couldn't be set up. Possible reason are missing or invalid netmask or ip address.

16.0.0.138 Code: [731]

```
WARNING 731 NET: The interface <val> to create the IP address on doesn't exist.  
↩
```

This message is sent if the parent interface for an ip doesn't exist.

16.0.0.139 Code: [732]

```
WARNING 732 NET: The interface <val> is not a valid interface to create an IP  
↩ address on.
```

This message is sent if the parent interface for an ip is not valid (e.g. it's bridged).

16.0.0.140 Code: [740]

```
ERROR 740 NET: Creation of VLAN failed!
```

This message is sent if the creation of a vlan failed.

16.0.0.141 Code: [741]

```
WARNING 741 NET: Parent interface missing for VLAN.
```

This message is sent if the creation of a vlan failed because the parent doesn't exist.

16.0.0.142 Code: [742]

```
ERROR 742 NET: Creation of MacVLAN <val> failed!
```

This message is sent if the creation of a macvlan failed.

16.0.0.143 Code: [743]

```
ERROR 743 NET: Setting MAC-address <val> on Interface <val> failed!
```

This message is sent if a MAC address for an interface could not be set.

16.0.0.144 Code: [750]

```
WARNING 750 NET: Adding a QoS rule failed.
```

16.0.0.145 Code: [760]

```
WARNING 760 NET: Bridges which forward link local traffic can not have more  
↳ than 2 interfaces (bridge index >=100)
```

16.0.0.146 Code: [770]

```
ERROR 770 NET: Error during network init: <val>
```

This message is sent when an error occurs during network initialisation. It contains additional information regarding what went wrong.

16.0.0.147 Code: [771]

```
WARNING 771 NET: Runtime Error: <val>
```

This message is sent when a network error occurs during operation.. It contains additional information regarding what went wrong.

16.0.0.148 Code: [772]

```
ERROR 772 WIFI: Error during wireless init: <val>
```

This message is sent when an error occurs during wireless initialisation. It contains additional information regarding what went wrong.

16.0.0.149 Code: [800]

```
INFO 800 DFS: Starting CAC on <val> MHz.
```

This message is sent when a CAC or Off-Channel CAC is started.

16.0.0.150 Code: [801]

```
INFO 801 DFS: Radar found on <val> MHz.
```

This message is sent when a radar pattern during In-Service Monitoring, CAC or Off-Channel CAC is detected.

16.0.0.151 Code: [802]

```
INFO 802 DFS: Channel on <val> MHz becomes Available.
```

This message is sent when a channel on the given frequency becomes Available after a CAC or Off-Channel CAC.

16.0.0.152 Code: [803]

```
INFO 803 DFS: Channel on <val> MHz becomes Usable again.
```

This message is sent when a channel on the given frequency becomes Usable after the NOP time.

16.0.0.153 Code: [804]

```
INFO 804 DFS: Starting In-Service Monitoring on <val> MHz.
```

This message is sent when the In-Service Monitoring for the Operating Channel on the given frequency.

16.0.0.154 Code: [805]

```
INFO 805 DFS: All initial CACs done.
```

This message is sent when all DFS frequencies have passed the initial CAC.

16.0.0.155 Code: [810]

```
ERROR 810 Cellular interface disabled, reason: <val>
```

This message is sent when the cellular interface was disabled due to HW reasons or wrong PIN configuration.

16.0.0.156 Code: [820]

```
INFO 820 TS|<val>||CELL|<val>||SIGNAL|<val>
```

This message is sent when the cellular debug message for signal is enabled. The output format for LTE and WCDMA is different.

Format LTE: 'TS|<timestamp>||CELL|<mcc>|<mnc>|<cellid>|<pcid>|<tac>|<freq band ind>|<earfcn>|<dl bandwidth>|<ul bandwidth>|<is tdd>|<state>|<srxlev>||SIGNAL|<type>|<rssi>|<rsrq>|<rsrp>|<sinr>'

Format WCDMA: 'TS|<timestamp>||CELL|<mcc>|<mnc>|<cellid>|<rac>|<psc>|<uarfcn>|<state>|SIGNAL|<type>|<rsrp>|<ecio>'

16.0.0.157 Code: [821]

```
INFO 821 TS|<val>||SIM|<val>|<val>|<val>|<val>|<val>|<val>|<val>||STATE|<val>  
↔ >|<val>|<val>|<val>
```

This message is sent when the cellular debug message for connection is enabled.

Format: 'TS|<timestamp>||SIM|<sim active slot>|<primary>|<roaming>|<pin status>|<mcc>|<mnc>|<imsi>|STATE|<failure count>|<state timeout>|<prev state>|<new state>'

16.0.0.158 Code: [900]

```
INFO 900 CARP: instance <val> is now a BACKUP
```

This message is sent when a CARP instance became a backup.

16.0.0.159 Code: [901]

```
INFO 901 CARP: instance <val> is now a MASTER
```

This message is sent when a CARP instance became a master.

16.0.0.160 Code: [910]

```
ERROR 910 CARP: The syncinterface <val> does not have an unique IP.
```

This message is sent when an error occurs during carp-init.

16.0.0.161 Code: [911]

```
ERROR 911 CARP: No valid ipaddr for the primary IP of the carp instance <val>  
    ↪ found.
```

This message is sent when an error occurs during carp-init.

16.0.0.162 Code: [2700]

```
INFO 2700 IF-MIB: linkUp (<val>, <val>, <val>)
```

IF-MIB::linkUp: Provide ifIndex, ifAdminStatus and ifOperStatus

16.0.0.163 Code: [2701]

```
INFO 2701 IF-MIB: linkDown (<val>, <val>, <val>)
```

IF-MIB::linkDown: Provide ifIndex, ifAdminStatus and ifOperStatus

16.0.0.164 Code: [2710]

```
INFO 2710 BRIDGE-MIB: newRoot (<val>)
```

BRIDGE-MIB::newRoot: Provide ifIndex or 0

16.0.0.165 Code: [2711]

```
INFO 2711 BRIDGE-MIB: topologyChange (<val>)
```

BRIDGE-MIB::topologyChange: Provide ifIndex or 0

16.0.0.166 Code: [3021]

CRITICAL 3021 TLS Client: Fatal alert error appears. Connection closed.

TLS Client: Fatal alert error appears. Connection closed.

16.0.0.167 Code: [3022]

CRITICAL 3022 TLS Server: Fatal alert error appears. Connection closed.

TLS Server: Fatal alert error appears. Connection closed.

17 CLI Commands

17.1 apply - Apply all pending configuration changes

USAGE: apply

The apply command applies all pending configuration changes. To show a list of all pending changes use the changes command.

EXAMPLES:

- * Apply all pending changes:
 apply

17.2 changes - Show a list of changed configuration parameters

USAGE: changes

The changes command shows a list of changed configuration parameters. All listed parameters are still pending, to apply these parameters use the apply command.

EXAMPLES:

- * Show a list of all pending changes:
 changes

17.3 configure - Manage the global configuration

USAGE: configure [COMMAND [ARGS...]]

The configuration tool provides access to the global configuration. It is grouped into COMMANDs and their respective arguments ARGS. If no COMMAND is specified, it defaults to the 'terminal' command.

COMMANDS:

 terminal

Enter the configuration in the terminal.

EXAMPLES:

```
* Enter the configuration:
    configure
```

17.4 date - Display the current system time

USAGE: date

The date program may be used to display the current system time.

EXAMPLES:

```
* Display the current system time:
    date
```

17.5 dmesg - Print the kernel ring buffer

USAGE: dmesg

The dmesg tool is used to examine the bootup messages of the kernel.

EXAMPLES:

```
* Print the bootup messages:
    dmesg
```

17.6 get - Show the value of a configuration parameter

USAGE: get [MIB::]PARAMETER

Get the given configuration PARAMETER. If you want to get parameters which are not part of the default MIB of the product, you must specify the MIB.

Be aware that the get command always returns the current value. This may differ from the applied value. Use the changes command to review changed values.

EXAMPLES:

```
* Get the hostname:
    get cfgSysHostname.0
* Get the base MAC address of the IEEE 802.1d bridge:
```

```
get BRIDGE-MIB::dot1dBaseBridgeAddress.0
```

17.7 gpsmon - GPS packet monitor and control utility

USAGE: `gpsmon [-hVn] IP:PORT`

`gpsmon` is a monitor that watches packets coming from a GPS and displays them along with diagnostic information.

OPTIONS:

- `-h` Show detailed options
- `-V` Show version of `gpsmon`
- `-n` Force `gpsmon` to request NMEA0183 packets instead of the raw data stream from `gpsd`.

EXAMPLES:

- * Bind to given ip:port
`gpsmon 0.0.0.0:2947`

17.8 grep - Print lines matching a pattern

USAGE: `grep [OPTIONS] PATTERN [FILE...]`

`Grep` searches the named input FILES for lines containing a match to the given PATTERN. If no files are specified, or if the file "-" is given, `grep` searches standard input. By default, `grep` prints the matching lines.

OPTIONS:

- `-i` Ignore case
- `-v` Select non-matching lines
- `-h` Show a more detailed help text.

EXAMPLES:

- * Filter the output of `logread` for handoff messages while ignoring upper and lower case:
`logread -f | grep -i handoff`
- * Show all system log messages except for those including the keyword `RSSI`:
`logread -f | grep -v RSSI`

17.9 help - Show a list of all CLI commands

USAGE: help [COMMAND]

The help command show a list of all Command Line Interface (CLI) commands. If a COMMAND is specified it shows a more detailed help text of the command.

EXAMPLES:

- * Show all commands:
 help
- * Show the help text of iperf:
 help iperf

17.10 ip - Show or manipulate network devices

USAGE: ip [OPTIONS] OBJECT {COMMAND | help}

The ip tool allows the user to run COMMANDs on different network OBJECTs. The set of possible actions depends on the object type. As a general rule, it is possible to add, delete and show (or list) objects. The help command is available for all objects. It prints out a list of available commands and argument syntax conventions.

The most important OBJECTs are: address, link, route, neigh and monitor.

OPTIONS:

- br[ief] Print only basic information in a tabular format.

EXAMPLES:

- * Show more detailed help of the route object:
 ip route help
- * Show a brief overview of all network addresses:
 ip -br address
- * Show a brief overview of all network links:
 ip -br link
- * Show the neighbour table (ARP table):
 ip neigh
- * Show the routing table:
 ip route

17.11 iperf - Perform network throughput tests

USAGE: iperf [-u] [-s | -c SERVER] [OPTIONS]

Iperf is a tool for performing network throughput measurements. It can test

either TCP or UDP throughput. To perform an iperf test the user must establish both a server and a client.

OPTIONS:

- u Use UDP rather than TCP.
- s Run in server mode.
- c SERVER Run in client mode, connecting to SERVER.
- i N Pause N seconds between periodic bandwidth reports.

Client specific options:

- b n[KM] Set target bandwidth to n bits/s (default 1Mbit/s). This option requires UDP mode (-u).
- t N Time in seconds to transmit for (default 10 secs).

EXAMPLES:

- * Start a server to discard traffic:
 iperf -s
- * Start a client connecting to 192.168.1.1 to generate traffic for 5 seconds:
 iperf -c 192.168.1.1 -t 5
- * Start an UDP server on port 6001 the report every second:
 iperf -u -s -p 6001 -i 1
- * Start an UDP client connecting to 192.168.1.20 on port 6001 with 100Mbit/s:
 iperf -c 192.168.1.20 -p 6001 -u -i 1 -b 100M

17.12 iw - Show or manipulate wireless devices

USAGE: iw [OPTIONS] {help [COMMAND] | {dev | phy | reg } COMMAND }

The iw tool allows the user to run COMMANDs on different wireless objects (dev, phy, reg). The set of possible actions depends on the object type. The help command will print all supported commands.

OPTIONS:

- debug Enable netlink debugging
- version Show the version of iw

EXAMPLES:

- * Show a help text for the event command:
 iw help event
- * List all wireless interfaces:
 iw dev
- * List all physical interfaces:
 iw phy
- * Show all connected clients to wlan0:
 iw wlan0 station dump
- * Show wireless events and their relative timestamp:
 iw event -

17.13 logread - Show system log messages

USAGE: logread [OPTIONS]

Logread shows system log messages from the internal buffer.

OPTIONS:

-f Follow the log messages.
-t Add an extra timestamp.
-h Show a more detailed help text.

EXAMPLES:

* Show the system log messages (syslog) as they are created:
logread -f
* Show the syslog messages with their timestamp information:
logread -t

17.14 ping - Ping network hosts

USAGE: ping [OPTIONS] HOST

Send ICMP ECHO_REQUEST packets to network hosts.

OPTIONS:

-h Show a more detailed help text.

EXAMPLES:

* Ping 192.168.1.1 until the ping command is terminated by pressing CTRL-C:
ping 192.168.1.1

17.15 reset - Reset configuration parameters

USAGE: reset

The reset command resets all configuration parameters to their default values. The changes made by reset might be reviewed using the changes command.

EXAMPLES:

```
* Reset all configuration parameters:
    reset
```

17.16 revert - Revert all pending changes

USAGE: revert

All pending configuration changes can be undone by using the revert command. This is not valid for already applied changes.

EXAMPLES:

```
* Revert all pending configuration changes:
    revert
```

17.17 set - Set the value of a configuration parameter

USAGE: set [MIB::]PARAMETER VALUE

Change the VALUE of a given configuration PARAMETER. If you want to set parameters which are not part of the default MIB of the product, you must specify the MIB.

Note the changes only take effect after the apply command is issued.

EXAMPLES:

```
* Set a new hostname:
    set cfgSysHostname.0 new-hostname
```

17.18 ssh - A secure shell client

USAGE: ssh [OPTIONS] [USER@]HOST[/PORT] [COMMAND]

The ssh client allows to connect to a remote ssh server.

EXAMPLES:

```
* Connect as user root to 192.168.1.20:
    ssh root@192.168.1.20
```

17.19 tcpdump - Dump traffic on a network

USAGE: `tcpdump [-n] [-i IFACE] [EXPRESSION]`

Tcpdump prints out a description of the contents of packets on a network interface that match the boolean EXPRESSION; the description is preceded by a time stamp, printed, by default, as hours, minutes, seconds, and fractions of a second since midnight.

OPTIONS:

- `-n` Don't convert addresses to names.
- `-i` Listen on interface. If unspecified, tcpdump searches the system interface list for the lowest numbered, configured up interface.
- `-h` Show a more detailed help text.

EXAMPLES:

- * Do not resolve names and print all traffic on interface eth0:
`tcpdump -n -i eth0`
- * Show only ICMP packets on eth0:
`tcpdump -n -i eth0`
- * Show all packets on eth0 except for port 22 (SSH):
`tcpdump -i eth0 -n not port 22`

17.20 traceroute - Traceroute network hosts

USAGE: `traceroute [OPTIONS] HOST`

Trace the path through a network to a host.

OPTIONS:

- `-h` Show a more detailed help text.

EXAMPLES:

- * Traces to path to 192.168.1.1
`traceroute 192.168.1.1`

17.21 watch - Execute a program periodically

USAGE: `watch [-n SEC] COMMAND`

The watch tool runs a COMMAND repeatedly, displaying its output. This allows to watch the program output change over time. By default, the command is run every 2 seconds. This interval may be changes by the `-n` option.

OPTIONS:

-n Repetition interval in seconds (default 2)

EXAMPLES:

* Check every second for connected stations:
 watch -n 1 iw wlan0 station dump

17.22 wg - Wireguard management tool

USAGE: wg

The wireguard command allows to display the status of wireguard and generate keys.

EXAMPLES:

* Show the current wireguard configuration
 wg
* Generate a wireguard private key
 wg genkey
* Generate a wireguard private shared key
 wg genpsk